Journal of Non-Crystalline Solids: X 8 (2020) 100054

Contents lists available at ScienceDirect

Journal of Non-Crystalline Solids: X

journal homepage: www.journals.elsevier.com/journal-of-non-crystalline-solids-x

Review

Ionic glasses: Structure, properties and classification

Courtney Calahoo, Lothar Wondraczek*

Otto Schott Institute of Materials Research, University of Jena, Fraunhoferstrasse 6, Jena 07743, Germany

A B S T R A C T

Due to the absence of microstructure and virtually infinite compositional versatility, glassy materials are perfect candidates for rational methods to predict structure-property relationships. In the most common models of glass structure, a three-dimensional network backbone and less-localized interstitial species are taken as the fundamental constituents of a glass. Such theoretical frameworks break down for the wide range of glass-forming compositions in which covalent bonding does not percolate and, therefore, there is no network in the classical sense. Most prominently, this applies to the class of ionic glasses, presently emerging as a separate class of amorphous materials, which includes classical invert glasses, geometrically frustrated compounds, mixtures of simple salts and ionic liquids, and organic–inorganic hybrids.

Here, we will critically review ionic glasses as a distinct group of materials in which structural predictions are complicated by the dominance of long-range bonding interactions which lack directionality. First, we will reassess Zachariasen's and Hägg's rules for conventional glass formation in an attempt to broaden the current understanding of glass formation to include ionic glasses. By reviewing the compositions and properties of many simple and complex glass-forming salts, we discern the commonalities shared by ionic glasses and their unique characteristics as compared to other classes of glass. Specifically, we discuss the importance of non-percolating topology to dynamic properties such as mechanical behavior, ionic conductivity, relaxation processes and crystallization. We also demonstrate the relevance of the adjacent fields of ionic liquids and amorphous metal organic frameworks, and highlight possibilities for fruitful interdisciplinary exchange. The structure of ionic glasses, being dominated by Coulombic interactions, lends itself to high compositional flexibility and therefore, creates significant opportunities in the search for optical materials with enhanced transmission windows, fast ion conductors, phase-change materials, or, *e.g.*, adapted solders, hermetic seals and dielectric insulators.

1. Introduction

At times it is tempting to classify glass as the fourth state of matter; it has the disordered structure of a liquid, yet its physical properties resemble those of a solid [1]. However, unlike the other states, glass is not a thermodynamically stable phase; it is a metastable state only produced by kinetic means. For example, it is largely admitted that if a sufficiently fast cooling rate can be achieved, any molten mixture can be undercooled so as to solidify into a glass.

Between inorganic network glasses and the sometimes more prolific vitreous metals, ionic glasses are often disregarded as a separate family of materials. This is unjustified in many ways. Since the early 1900s, glasses from simple salts, salt mixtures and mixtures of ionic minerals have been among the most prominent examples for studying the physics of glass formation. Ionic glasses have not only served as a proxy for the properties of molten salts, for example, in a geophysical context, but also produced some of the most promising ion conductors. In a wider sense, they are indeed applied in numerous every-day products: sensor packages, control units, antenna devices and other electronic components in cars and mobile phones would probably not function without low-temperature co-fired ceramics involving ionic glass compositions. Yet they are fundamentally different from either of the classical network-forming glasses or vitreous metals.

In this article, we review the field of ionic glasses: Starting from a

Abbreviations and Symbols: *a*-MOFs, amorphous metal organic framework; *a*-ZIF, amorphous zinc imidazolate framework; *B*, brittleness (μm^{-1}); BO, bridging oxygen; CR, crack resistance (N); CRN, continuous random network; CRR, cooperatively rearranging regions; CTE, coefficient of thermal expansion; EXAFS, extended X-ray absorption fine structure; *G*, shear modulus (GPa); GF, glass former; HDA, high density amorphous phase; HMFG, heavy-metal fluoride glasses; IL, ionic liquid; *K*, bulk modulus (GPa); LDA, low density amorphous phase; *m*, fragility index; MME, mixed-modifier effect; MOF, metal-organic framework; MRN, modified random network; *n*, index of refraction; NBO, non-bridging oxygen; NMR, nuclear magnetic resonance; PCM, phase change material; PDF, pair distribution function; Q^n, nomenclature for structural glass species, n = number of bridging (oxide) bonds between polyhedra; RCP model, random close-packing model; RDF, radial distribution function; T_1, spin-lattice relaxation time (ms or s); T_g, glass transition temperature (°C or K); T_l, liquidus temperature (°C or K); T_{rg}, reduced glass transition temperature; ZIF, zinc imidazolate framework; β, non-exponential parameter 'stretch exponential'; η, viscosity; ν, Poisson's ratio; ρ, density (g/cm); σ, conductivity (S/cm)

* Corresponding author.
E-mail address: lothar.wondraczek@uni-jena.de (L. Wondraczek).

https://doi.org/10.1016/j.nocx.2020.100054
Received 17 February 2020; Received in revised form 16 June 2020; Accepted 17 June 2020
Available online 25 June 2020
2590-1591/ © 2020 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

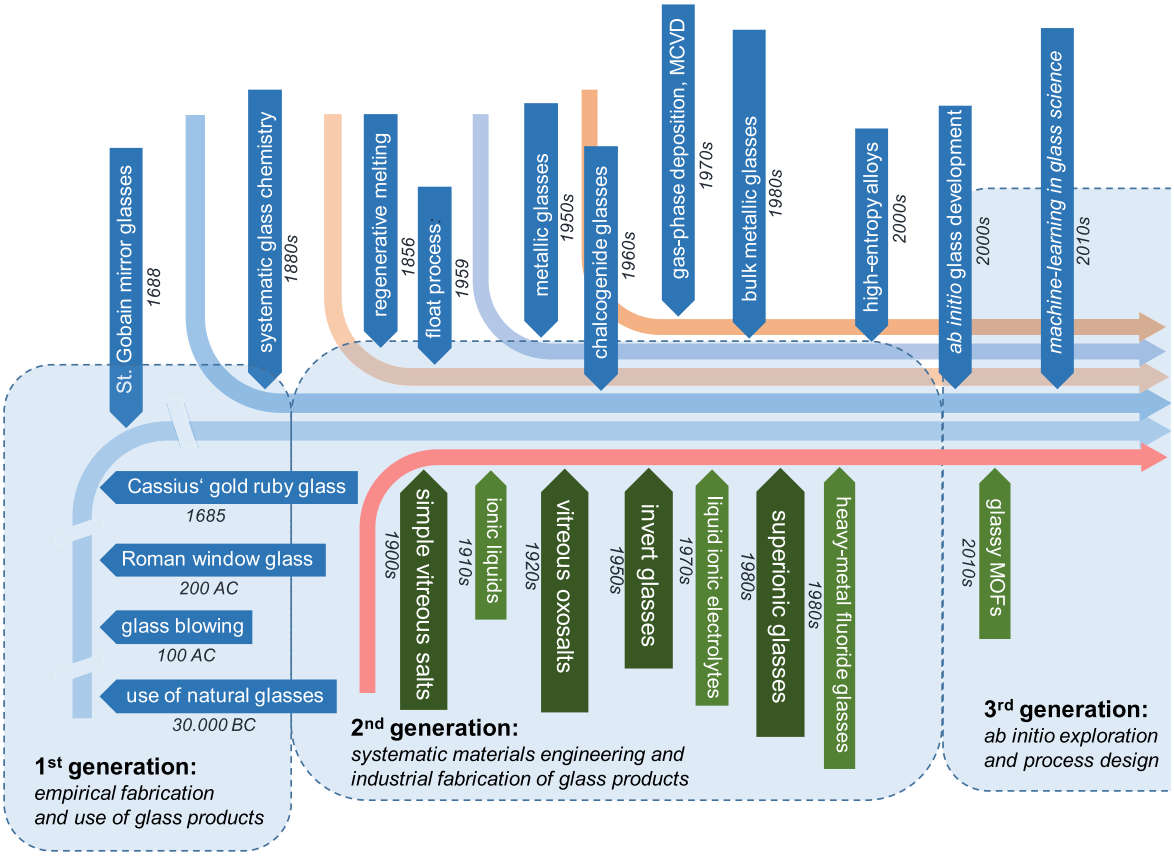


Fig. 1. Timeline of glass chemical development and glass engineering. Ionic glasses and related materials are highlighted in green. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

short summary as to how they do not follow the widely-spread models of network glass structure, we will introduce the various groups of ionic glasses from invert glasses and simple salts to oxoanionic materials and more complex multi-anionic mixtures. We will then look at the various pieces of information which have been obtained on their structure and topology, trying for a classification between different degrees of superstructural polymerization. On this basis, we will summarize some of the properties and applications which make use of the specific structure and chemistry of ionic glasses. Finally, we will highlight some noteworthy relations to the hybrid analogs of ionic liquids and amorphous metal-organic frameworks.

Modern research into the physics and chemistry of ionic glasses started in parallel to the earliest attempts of systematic glass formulation (see Fig. 1), however, both scientific communities initially evolved in relatively disconnected ways, with the latter being driven by rapid progress in glass engineering and applications (*e.g.*, O. Schott, introducing methodologies which initiated industrial glass development [2]), while the former was motivated by fundamental physical questions, and engaged in by crystallographers and geophysicists (*e.g.*, V. M. Goldschmidt and, later, his student W.H. Zachariasen formulating fundamental principles of crystal chemistry which helped and still help predict glass-forming chemistries and corresponding glass structures [3]). Yet, strong similarities and joint interest were found when the group of classical silicate glasses evolved to now include ionic invert glass chemistries, and glass formation was discovered not only in simple salt melts, but also in the more complex oxosalts and polyionic mixtures. Nowadays, fluoride glasses are well-established as materials for active and passive IR optics (*see* references in [4]), superionic glass formulations have entered the field as promising candidate materials for application in next-generation energy storage devices, and fully new material classes are being discovered in the area of organic/inorganic hybrid ionic glasses.

With the advent of model-based property prediction [5] and the broader-scale introduction of machine-learning tools in glass formulation [6], well-informed basis sets are required which contain sufficiently detailed information on the underlying glass structures and structure-property correlations.

The most common models for describing the structure of non-metallic inorganic glasses are based on Zachariasen's network hypothesis: the backbone of a glass lattice is built from interconnected (corner-sharing) metal-oxide polyhedra [7]. Depending on the number of constraints per polyhedron, networks can be one-, two- or three-dimensional [8]. The degree of polymerization can be quantified by the average number of directional bonds per polyhedron. It is determined by the presence of secondary cation species (denoted as network modifiers), which do not participate directly in network formation. In order to compensate for the positive charge of the network modifiers, non-bridging anion species are formed which disrupt the covalent network. The differentiation between network formers and network modifiers is mostly empirical and often ambiguous; it is frequently made on the basis of oxide bond strength or bond covalence [1].

According to this continuous random network (CRN) theory, the creation of a bulk glass network requires a minimum degree of polymerization [9]. However, for many years, there have been glass compositions used in technical applications which fall below this limit; in particular, certain solders, enamels and glass seals with tailored thermomechanical properties are very highly depolymerized. Since in those cases, the classical categorization of glass components into network formers, modifiers and intermediates could not readily be applied, these compositions have been denoted as *invert* glasses, *i.e.*, glasses in which

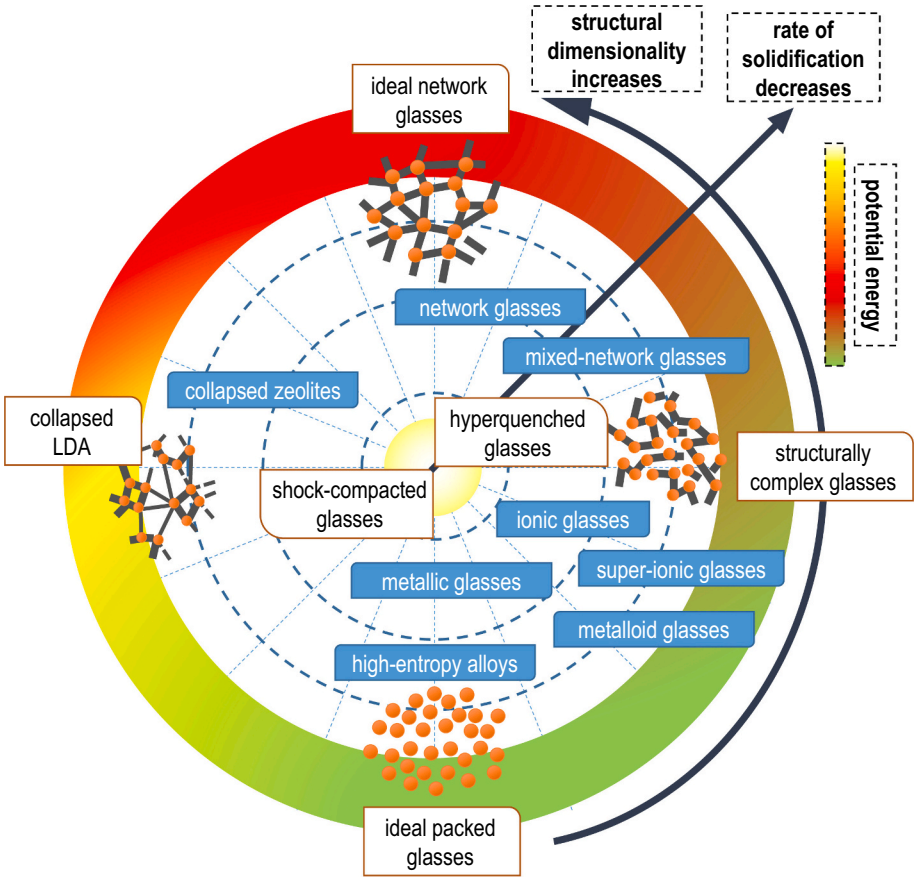


Fig. 2. Properties of known glass families: structural dimensionality, potential energy and rate of solidification. Low density amorphous (LDA) phases pertain to amorphous zeolites, metal organic frameworks (MOFs) and other open framework structures.

the roles of network former and modifier species become inverted. In even sharper contrast to the classical network hypotheses, glasses formed from simple salt mixtures without any classical polymerization have been studied since at least the1960s [10]. Originally primarily for fundamental reasons and for insight into the phenomenology and properties of salt melts, research activity in this field has been growing in recent years, for example in the quest for solid-state ion conductors or phase change materials [11–17].

In the following, we will consider the chemistry of inorganic, non-metallic glasses in which the structure does not percolate in the sense of a CRN, *i.e.*, through directional, covalent bonds. Since in these glasses, ionic crosslinking is a dominant structural feature, we will adopt the term ionic glasses, but also extend its use beyond systems in which bonding is purely ionic [10]. In Fig. 2, we place ionic glasses (glassy salts, superionic glasses, and mixed-network compositions, *i.e.*, invert glasses, and the high-density amorphous [HDA] phase of glass-forming metal-organic frameworks [MOFs]) in reference to other known glass families using three qualitative criteria: structural dimensionality, potential energy and rate of solidification. As will be demonstrated herein, ionic glasses often possess locally ordered polyhedra, giving them higher dimensionality than ideal packed glasses (metallic or metalloid) while still lacking the long-range connectivity of ideal network glasses. Due to the reduction of the degree of polymerization and consequent relaxation of topological constraints, ionic glasses crystallize more readily than network glasses, revealing that their structures and topologies are more similar to those of the corresponding crystals, *i.e.*, they have lower potential energies than either network glasses or collapsed zeolitic structures. The rate of solidification which is needed to form an ionic glass often lies between those of network and metallic glasses. For the following discussion, it is worth noting that ionic liquids as well as MOF-derived glasses possess properties similar to those found in ionic glasses.

This review is laid out into the following sections: first, we will reconsider the classical concept of ionic field strength and its use in glass chemistry, insofar as it can be employed to develop a more definitive classification for ionic glasses. Subsequently in Section 2, we introduce several compositional families of glasses known to possess predominantly ionic bonding: invert glasses, oxoanionic glasses (nitrates, sulfates, carbonates and hydrates), halides, haloborates and halophosphates and then, finally, superionic glasses (this being a property-based grouping, rather than a composition-based one). Section 3 discusses different classification criteria that allow for empirical identification of likely ionic glasses. Building on the glass-forming compositions catalogued in Section 2, the commonalities of ionic glasses' formation, structure and topology will be examined in Sections 4 and 5. In Section 6, we will use properties common to ionic glasses, yet unique compared to other types of glass, to further distinguish ionic glasses from other glass types. From the characteristic properties established in Section 6, Section 7 then highlights existing and potential applications for ionic glasses. Finally, Sections 8 and 9 demonstrate how the emerging connections between ionic glasses and the adjacent fields of ionic liquids and MOF-based glasses can stimulate the ongoing research in each individual field.

1.1. Ionic field strength in classical network hypotheses

When attempting to synthesize and understand new glass compositions, Dietzel's [18] simplistic categorization of metal oxides based on field strength (F) has provided immeasurable guidance to glass scientists: To satisfy Zachariasen's hypothesis of glass formation [7], start

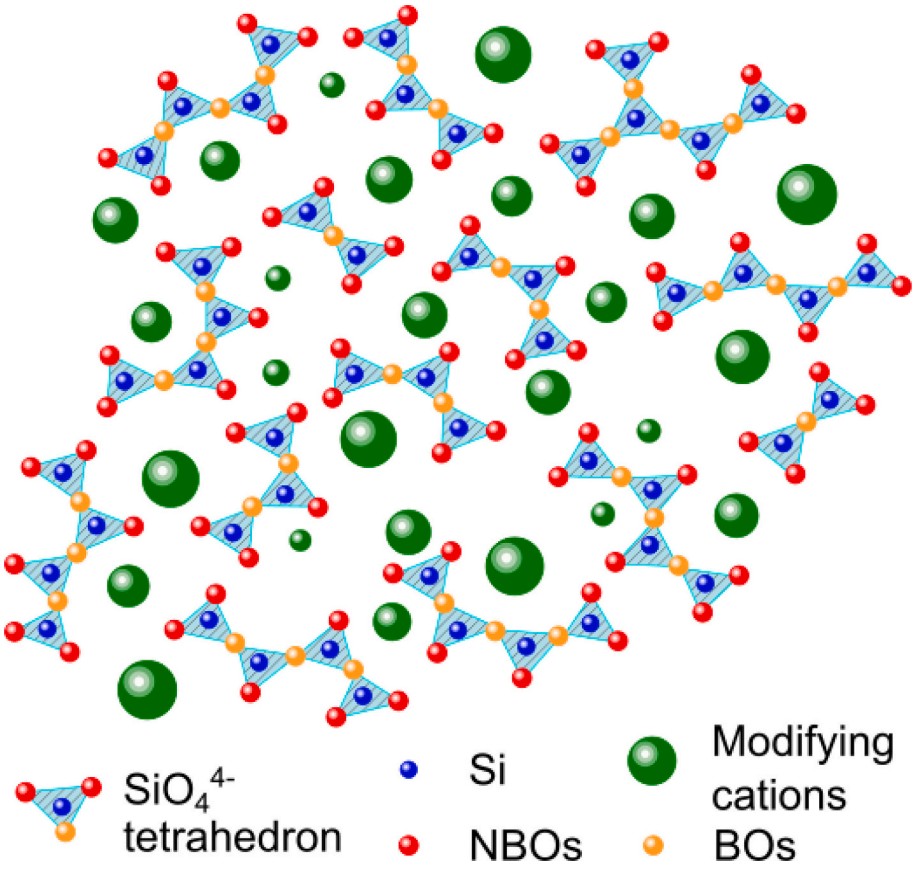


Fig. 3. Two-dimensional schematic of the structure of an invert glass. Reproduced with permission from Trap and Stevels [28]. © Deutsche Glastechnische Gesellschaft (DGG).

with a conventional glass former oxide, add some modifier oxide and, perhaps, a smaller amount of intermediate oxide. In this approach, the modifier is normally less abundant than the glass former and is often considered to play only a secondary role of depolymerizing the network, *i.e.*, it creates non-bridging oxygens (NBOs) by replacing a covalent bond with an ionic bond. According to Dietzel, glass modifiers are defined as having low field strengths, $F < 0.4$ Å^{-2} ($F = Z/a^2$, where Z is the ionic charge and a is the ionic radius), while formers have high field strengths ($F \geq 1.5$ Å^{-2}) and intermediates are found with $0.5 \leq F \leq 1.0$ Å^{-2} [18].

Field strength is related to the ionicity of the metal oxide bonds, as the bonds involving components with low field strength possess high amounts of ionic character and high field strength bonds have more covalency or directionality to their bonding environment. Here, we employ the somewhat simplistic notion of bond directionality in order to classify bonding according to the topological nature of the localized bonding functions [19]. Dietzel's classification criteria was modelled on oxide glasses, and rather than an absolute field strength parameter, Poulain [20] found a ratio between cation and anion field strength (in this case, $F = Z/a$) to work well for determining glass forming fluoride and chloride compounds, while extending the concept of network former to low-field strength cations which reduce ordering through decreased intercationic and interanionic repulsion.

Many glass modifiers lie on the cusp of this former-modifier division; for example, Pb^{2+} and Mg^{2+} transform from a glass modifier to an intermediate depending on their local environment [21,22]. An increase in the cationic content of the glass should increase the ionic conductivity, but that is not the case for xPbO–(100 − x)P_2O_5, where a threshold exists: at > 50 mol% PbO acts more like a former and additional Pb^{2+} increases the activation energy [21]. In MgO–B_2O_3 glasses, the high field strength of Mg^{2+} strongly inhibits the formation of BO_4-units and encourages boroxol ring formation [23]. Generally, addition of modifying cations degrades mechanical properties, however, in the case of Li_2O, the cation surprisingly settles in the interstices and increases Young's modulus [24]. Modifiers can adopt varying coordination numbers based on composition, such as changing from 6- to 4-fold coordination, however, the oxidation state remains the same and the bonding character is still largely ionic. For example, when increasing the coordination number from Mg[4] to Mg[6] the field strength increases from 0.38 to 0.61 Å^{-2} (*i.e.*, in Dietzel's terms, from ionic to intermediate, but much lower than that found for a typical glass former). Yet these differences are sufficient to significantly affect trends in properties, thus, exemplifying the importance of modifier bonding and ionic interactions in all glass properties [25].

To understand structure–property relationships in oxide glasses, most structural investigations focus on quantifying bridging oxygens (BOs) and non-bridging oxygens (NBOs), by separating glass components into clear modifier *vs*. former categories. However, this perspective excludes the interactions between the modifier and the NBOs, which clearly play a defining role in the glass properties. One of today's most prevalent topological models of glass, the modified random network (MRN) model of Greaves [26], highlights the importance of cation–cation and cation–network interactions. Segregated modifier channels and cation interactions are used to explain the well-studied mixed-modifier effect (MME): a deviation from linearity in the physical properties as a function of mixing when two or more unlike modifier cations are present in the glass. The deviation can be small, as in the case of static properties such as density or molar volume, or it can be quite large (orders of magnitude) for properties which involve structural dynamics like ionic conductivity or chemical durability. The magnitude of the MME appears to be correlated with the differences between the mixing cations *and* anions. For example, doping of halide

salts (F^-, Cl^- and Br^-) causes the conductivity to change by orders of magnitude in Li, Na borates [27]: again very clearly, ionic interactions play a significant role in the evolution of glass structure and properties.

Ionic interactions become essential when the glass composition contains more modifier than former species; now rather than the typical, highly connected 3-dimensional covalent network that is the basis of CRN theory, the glass structure is dominated by ionic interactions between cations and small oxoanions. *Invert glasses* are one of the first reported *ionic glasses* and will be discussed further in the next section; for now, they offer a starting definition for *ionic glasses* in terms of chemical composition, in which the fraction of modifier species exceeds that of the network formers. Fig. 3 (from Trap and Stevels [28]) displays a proposed invert glass structure: a glass containing more network modifier, alkali and alkaline earth cations (Na, K, Ca, Sr, Ba), than network former (SiO_2).

It is clear from Fig. 3 that the behaviour of the modifier cations is equally important as that of the former; compared to typical covalent glass formers, the role of the modifier is likely more deterministic in ionic glasses, where the modifier is the major compositional component and ionic, rather than covalent, cross-linking is the dominant structural motif [29]. This is clearly evident in the behaviour of invert glasses, where the trend in many properties reverses (or *inverts*) direction after a majority of the former–former interactions are mediated through ionic NBOs, rather than covalent bonds (as discussed further in Section 2.1). Structures dominated by ionic interactions cannot only be found in invert glasses (silicates, phosphates and borates with more modifier than former), but also in other types of ionic glasses: oxoanionic (nitrates, sulfates, carbonates, and hydrates), halides (fluoride, chloride, bromide, and iodide), super-ionic (mixtures of ions, defined on the basis of having “liquid-like” electric conductivity and charge-carrier activation energy [30], *e.g.* silver phosphates, lithium sulfides [31–34]) and combinations of these glass families such as sulfophosphates, sulfoborates, haloborates, halophosphates, and chalcohalides. The common structural features and properties that are unique to these classes of glass, as well as current and potential applications, will be discussed in this review.

Since ionic glasses contain only a minority of glass former and have structures similar to the one represented in Fig. 3, Zachariasen’s rule of glass formation is no longer valid, *i.e.*, that a glass must be a, “*three-dimensional network* where *at least three corners of each oxygen polyhedron must be shared* [quoted from [7]],” (or, in modern words and acknowledging more complex chemistries which allow for higher coordination numbers on part of the cation and the anion species, *where the percolation threshold must be reached in three-dimensional space*). Instead, we demonstrate that the ionic glasses can be formed when a sufficiently complex mixture of chemical components provokes *geometrical frustration* (*see*Section 4). Moreover, the categories created by Dietzel [18] do not apply; when there are only dimers or isolated polyhedra of conventional glass-forming species among cross-linking cations, the categories of network former, intermediate and network modifier lose relevancy. Instead, the glass-forming ability and structural roles of chemical components will be discussed in terms of a common glass stability parameter, the ratio of the radii of the metal (cation) and the anion (r_M/r_X) [35].

Compared with covalent glasses, ionic glasses display unique trends in their physical properties. Most prominently, this concerns their dynamic behaviour: mechanical, viscous, diffusive, electrical, and thermal transport properties. For example, ionic glasses have reduced viscosities leading to a propensity to crystallize and the highest ionic conductivity of any glass family. Trends in mechanical properties demonstrate that ionic glasses are more akin to metallic glasses with higher Poisson’s ratios, higher structural compactness and lower dimensionality, as compared to the network-forming oxide glasses. Relaxation behaviours in ionic glasses are generally non-Arrhenius, yet, relaxation processes of different chemical components are often decoupled, leading to often lower entropies of melting. All such observations can be related to the reduced bond directionality of ionic glasses. In the extreme case of glasses obtained from simple salts, a less constrained coordination shell results in higher configurational entropy, yet at the same time the simple chemistry necessitates lower structural and dynamic heterogeneity. On the other hand, glasses which contain more complex cations, mixtures of multiple cation or anion species, or even larger covalent species such as chain segments, have the potential for higher structural and dynamic heterogeneity. This in turn greatly affects dynamic properties; in addition to the mere type of bonding (*i.e.*, ionic or covalent) the variation between types of bonding within the same glass becomes a determining factor.

2. Types of ionic glasses

We will begin by identifying different anions that can form ionic glasses (Fig. 4). The following differentiation is based on the occurrence of these anion species, starting with the invert glass polyhedra.

2.1. Invert glasses

Glass is not normally considered an important material for the electronic age; yet, the glass technology required to produce airtight

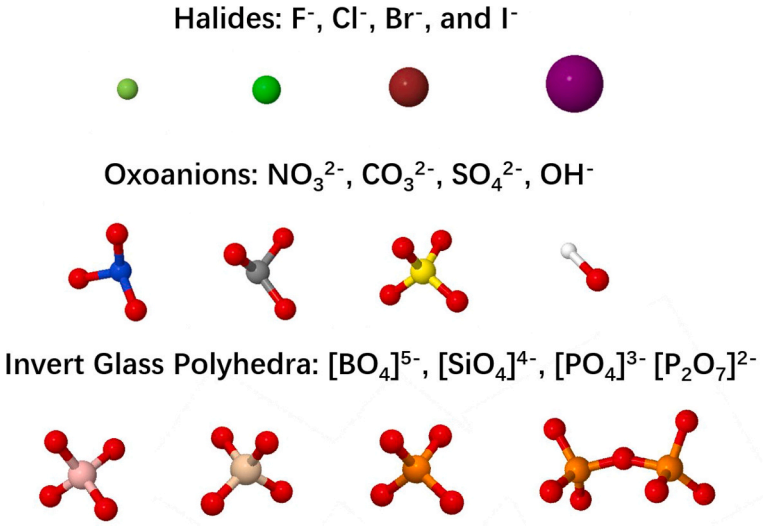


Fig. 4. Variety of anions that form ionic glasses. Generated using the Jmol software package [36].

envelopes for light bulbs was used in the first vacuum tubes produced in the turn of the 20th century. Today, glass solders or sealing glasses are widely employed to create connective layers between a variety of materials including other glasses, ceramics, metals or semi-conductors [37–44]. In addition to being an insulator, a glass solder can hermetically encapsulate components on rough and uneven surfaces. The most common glass solders are used to create glass-to-metal bonds in a variety of applications: construction of vacuum tubes or insulations [45,46], electric discharge tubes, incandescent light bulbs [37], glass-encapsulated semiconductor diodes, reed switches, pressure-tight glass windows in metal cases, and metal or ceramic packages of electronic components [47,48]. More recently, glass-ceramic–metal bonds have been used where stronger mechanical bonding and/or higher refractory properties are required such as in solid oxide fuel cells and cathode ray tubes [49,50].

The applicability of a glass solder requires two essential properties: good wetting and matched coefficients of thermal expansion between the two (or more) involved materials [41,44]. Practically speaking, the solder must also have low viscosity at relatively low temperatures (with a softening temperature which is typically less than 550 °C, for example, to prevent oxidation or reaction with a metal part [44,50]) in order to flow onto and wet the surface of the other material without damaging either component. At the same time, for many applications, it must enable geometric stability at high temperature as well [38], which is often achieved by partial crystallization.

Given the importance of a low softening temperature and the fact that the addition of modifier species lowers the viscosity, high-modifier glasses – invert glasses – were found to be the ideal candidates for glass solders. Typically, these glasses are composed of some SiO_2, B_2O_3 and/or P_2O_5 (since they do not reduce at high temperatures) and some intermediate oxides (such as Al_2O_3 or PbO), while the majority of the composition is classical modifier species such as alkali or alkaline earth oxides [37–40,44]. In some cases, only alkaline earth cations are used; alkalis tend to promote long-term degradation due to high diffusion rates (for the same reason, they may compromise the dielectric performance, *e.g.*, in applications of electronic packaging [39]) or, in the case of Cs^+, high polarizability. Many commercial compositions used to contain high amounts of lead; however, for lower toxicity, since the late 1990s these are being replaced by alternative systems with similar properties (albeit chemically more complex), such as Li_2O–Na_2O–K_2O–CaO–SiO_2–B_2O_3–ZnO–TiO_2–BaO–Al_2O_3 or Al_2O_3–ZrO–BaO–SrO–CaO–MgO–Y_2O_3–La_2O_3–B_2O_3–SiO_2 [51]. At high modifier loading, most of the simpler glass formulations are prone to phase separation [28]. By combining at least three modifying cations of different sizes (Na_2O, K_2O, CaO, SrO, BaO), Trap and Stevels [28] extended the miscibility region and avoided the common problem of crystallization in silicates and phosphates. Although not mentioned explicitly in their original study, this was probably the first application of geometric frustration in order to widen the glass forming region so as to reach otherwise inaccessible compositions. A similar approach was later taken by Inoue for stabilizing amorphous alloys at slow cooling rates [52], opening the field of bulk metallic glasses (BMGs).

The term *invert* glass was coined by Trap and Stevels [28] on the basis of two observations: most obviously, the inverted ratio of modifier to former, but also regarding a point of inversion (or opposite trends as compared to conventional glasses) found in properties involving a dynamic component. They introduced a structure parameter, Y, defined as the average number of bridging oxygens (BOs) per tetrahedron, which is related to the number of NBOs, X, by $X + Y = 4$. If R is the ratio of oxygens to former atoms, [O]/[GF], then $X + 1/2Y = R$. Therefore, $Y = 8 - 2R$ for tetrahedrally coordinated glass formers or $Y = 6 - 200/p$ for silicate glasses, where p is the mol% of SiO_2. The authors found a significant change in the relaxation behaviour of the invert glasses at $Y = 2$, as evidenced by trends in dielectric and mechanical loss, as well as viscosity (Fig. 5): at $Y = 2$, there appears to be a change in bonding character that cannot be explained simply by the reduction of BOs. On the other hand, this value of Y is much less important for average, or static properties, such as density, refractive index and low-frequency dielectric permittivity.

The composition at $Y = 2$ has structural relevance since as Y decreases towards two, the silicate network changes from fully connected tetrahedra in a 3d network to a non-percolating lattice of chain segments. As more modifier is added, Y decreases further and at $Y = 1$, the silicate network is (on average) composed of $Si_2O_7^{6-}$ tetrahedra pairs, while at $Y = 0$, only isolated SiO_4^{4-} tetrahedra exist. In fact, in the Na_2O–K_2O–CaO–BaO series, an average structure of half of $Si_2O_7^{6-}$ tetrahedra pairs and half isolated SiO_4^{4-} tetrahedra ($Y = 0.5$) was achieved; this composition is undeniably dominated by ionic interactions between short silicate chains and cations. As Y decreases past 2, it is clear that the short silicate chains contribute less and less to relaxation behaviour, while the interactions between the modifiers and NBOs quickly become the dominant determinants of dynamic behaviour. This conclusion is not only of historical interest, but it gives us a starting position to further define ionic glasses: in addition to containing more modifier than former, the contribution of percolating former to dynamic properties is weak at $Y \leq 2$.

Dietzel's theory of glass formation [18] has faced some criticisms since its inception in 1933; a major critique being that it requires the classification of intermediate species: ions that are neither former nor modifier. Similarly, the Y classification only works if one can neatly divide anions and cations into former and modifier, or in other words all oxygens must be either BO or NBO. This is clearly not the case for many elements: growing evidence indicates that a sharp distinction between ions is arbitrary and that bond directionality can vary significantly (as demonstrated by the many possible subclasses of shared and closed-shell bonding types [19]). In fact, several invert compositions with intermediate-like cations, such as BaO, CdO, MgO and ZnO have been shown to extend the glass-forming region to a lower-than-expected Y and cause an inversion in properties well-below the common threshold of $Y = 2$ [53]. As a result, Trap and Stevels [28] modified Y such that there is a Y calculated from the composition (Y_c) and a phenomenological (real) Y (Y_r). If we assume that the inversion always happens at $Y = 2$ and otherwise continue on the traditional viewpoint of network formers and network modifiers, then some conventional modifying ions such as MgO must be behaving as glass forming ions (denoted as M_f) and contribute to the dynamic properties as directional BOs; the difference between Y_c and Y_r can be used to calculate M_f.

Trap and Stevels' 1959 [28] determination of the number of intermediate cations undergoing a transition to glass former is exemplary; this was before NMR spectroscopy was routinely used to quantify Q^n-units[1] or coordination spheres. Yet, they found a method to estimate the structural changes occurring in a glass, very similar to how modern scientists quantify the coordination change of Al from tetrahedral- (former) to octahedral- (modifier) coordination. If we extend Trap and Stevel's measure of NBOs, Y, to compositions whose structures have been fully analyzed by ^{31}P and ^{27}Al NMR, such as the sodium aluminophosphate system [54,55], we find that if $[Na_2O] \geq [P_2O_5,]$ then as Al_2O_3 increases, Y reaches a minimum of 1.2 – 1.3 and then rebounds as Al[6] converts into Al[4]. Interestingly, the beginnings of this coordination change and calculated Y minima strongly correlate with the non-linear trends and maxima found in both static and dynamic properties (density, molar volume, refractive index, CTE and T_g [54,55]). If we do not consider Al[4] to be a former, then Y decreases monotonically with decreasing $[P_2O_5]$ (increasing $[Al_2O_3]$) and the inversion behaviour occurs at $Y = 1.1$ in both series in which it occurs.

[1] The n in Q^n-units refers to the number of bridging oxide bonds (BOs) per cationic polyhedra; for example, pure silica would be expected to be composed of fully connected SiO_4^{4-} tetrahedra or Q^4-units, while $NaPO_3$ consists of phosphate tetrahedra chains or Q^2-units with two BOs and two non-bridging oxygens (NBOs).

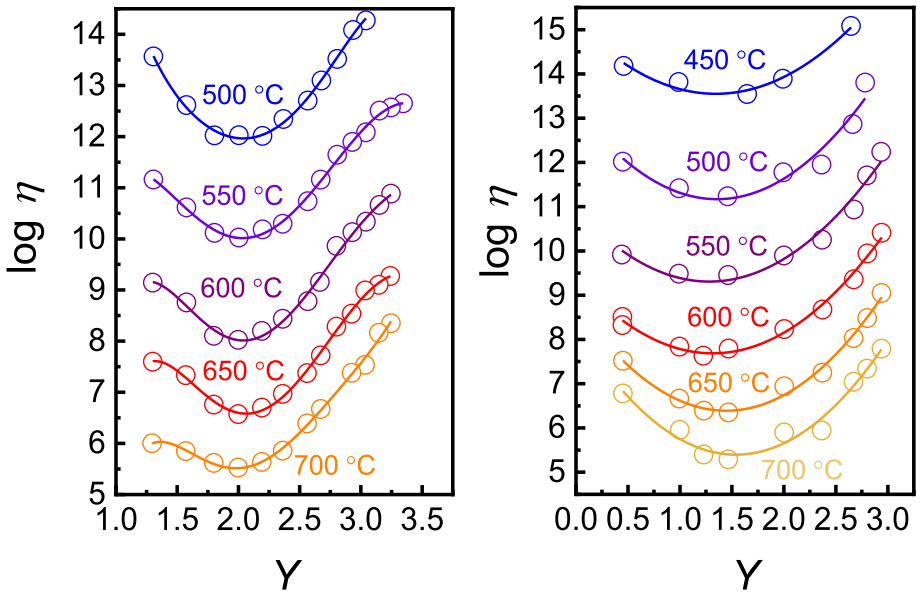


Fig. 5. Viscosity isotherms of two invert glass systems, Na_2O–K_2O–CaO–SrO–BaO–SiO_2 (left) and Na_2O–K_2O–MgO–CdO–ZnO–SiO_2 (right). Reproduced with permission from Trap and Stevels [28]. © Deutsche Glastechnische Gesellschaft (DGG).

A more recent study of calcium silicate glasses with large amounts of intermediate oxides, $(2.5 - x)CaO{\cdot}x/3M_2O_3{\cdot}2SiO_2$, where M = Y, La, In, Al, Ga, also found the typical inversion as more CaO was replaced with an intermediate oxide (Fig. 6) [56]. FTIR corroborated the idea that Al_2O_3 and Ga_2O_3 behaved more like glass formers in this system, while La_2O_3, Y_2O_3 and In_2O_3 acted more like modifiers. A multi-spectroscopic study of a very broad range of PbO–SiO_2, up to 83 mol% PbO, indicated that PbO was incorporated into the silicate network as normal until 60 mol% PbO (increase in NBOs, shorter silicate chains) and then thereafter a separate PbO network with mostly orthosilicate (Q^0) is formed [57]. Even typical glass formers without large amounts of intermediate oxides, *e.g.*, metaphosphate ($NaPO_3$), could be viewed as invert glass formers: the Na^+ cations largely interact electrostatically with the negatively charged, covalently bonded phosphate chains ($^{\delta-}$O—P(OP)$_2$—O$^{\delta-}$). As another example, fluorophosphate glasses are known to contain P(O, F)$_4$ tetrahedra with terminal P–F$^-$ bonds ionically bonded to cations [58], in addition to the typical ionic bonding expected in alkaline earth or heavy metal fluoride glasses. This is similar to the B(O,F)$_3$ units in fluoroborates being described as a 1d spaghetti-like glass [59].

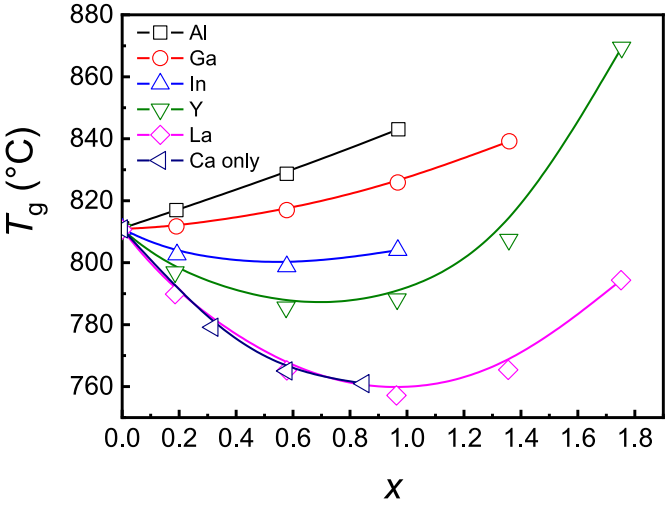


Fig. 6. Glass transformation temperature T_g (± 2 °C) vs. composition of some silicate glasses, expressed as the x values of the formulas of mixed calcium-trivalent cation series: $(2.5 - x)CaO{\cdot}x/3M_2O_3{\cdot}2SiO_2$, where M = Y, La, In, Al, Ga, and a simple calcium-only binary series: $(2.5 - x)CaO{\cdot}(2 + 0.5x)SiO_2$. Reproduced with permission from Branda et al. [56]. © Elsevier.

In comparison to SiO_2, B_2O_3 and P_2O_5 form meta-, pryo- or ortho-glasses much more easily [60–66]. Non-percolating short chains of SiO_4-tetrahedra require more extreme glass melting methods, such as the geometrical frustration employed in the model silicates studied by Trap and Stevels [28], or containerless levitation laser melting in the case of Mg_2SiO_4 [66]. On the other hand, completely amorphous binary alkali borates with more than 70 mol% of each alkali oxide were synthesized without any need for a specialized cooling method [67]. An invert, *i.e.*, greater than 50 mol%, magnesium borate glass can also be produced normally; yet, that composition has the lowest BO_4-unit concentration compared to other alkaline earths. Borate's propensity to form ionic glasses is further confirmed by orthoborates being the anionic components of high-melting and viscous ionic liquids (ILs); the charge dispersion across the ion was compared to that of a common IL anion, bis(trifluoromethane)sulfonimide ($(CF_3SO_2)_2N^-$ or TFSI$^-$) [68].

Polyphosphate is the category of all phosphate compositions in a $xM_yO_z{:}(1 - x)P_2O_5$ series with phosphate as the minor component; it includes metaphosphate, pyrophosphate and orthophosphate. Polyphosphates have been studied extensively, *see* [69] and the references within. Uniquely, pyrophosphate compositions with metal oxides having less than 6-fold coordination can usually be produced using conventional melt-quenching techniques. In contrast, other glass formers, (silicates or aluminates), usually require more unconventional methods such as laser heating, containerless processing, rapid quenching or sol-gel methods in order to avoid crystallization. In the pyrophosphate region of the CaO–Na_2O–MgO–P_2O_5 glass system Q^0 and Q^2 were found to be favoured over Q^1-units, while in the orthophosphate region, the Q^n-speciation departed significantly from the chemical order model (perhaps similar to silicate invert glasses), which was attributed to unreacted MgO [62]. Although distinct from ILs with orthoborate as a major component, several different potassium phosphate salts have been successfully dissolved in imidazolium-based ILs [70].

2.1.1. Relation to mineral glass

We do not aim here to summarize the wide breadth of literature on mineral melts and their glasses, but instead to demonstrate that a

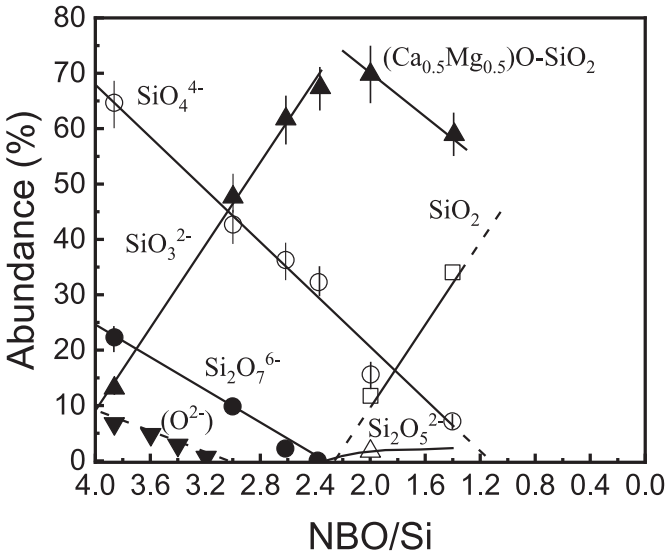


Fig. 7. Relative abundance of anionic structural units as a function of bulk melt NBO/Si and type of metal cations for binary melts (data from Mysen et al. [71]). Reproduced with permission from Mysen [72]. © Annual Reviews Inc.

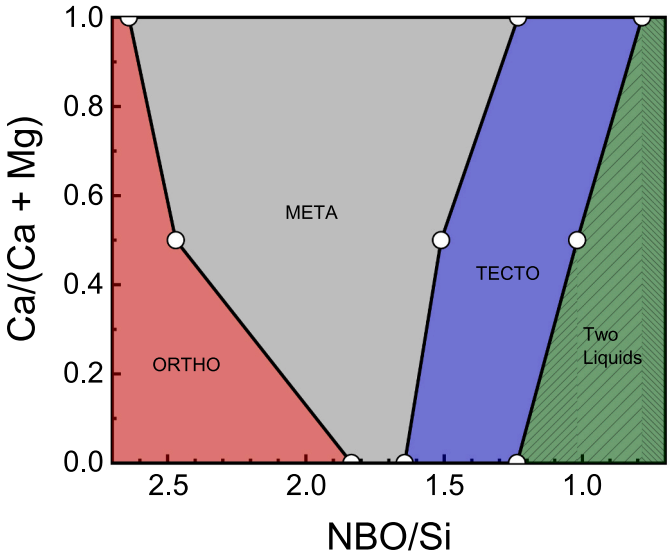


Fig. 8. Relations between orthosilicate/metasilicate, metasilicate/tectosilicate, and liquid immiscibility in the system CaO-MgO-SiO_2 as a function of Ca/(Ca + Mg). Reproduced with permission from Mysen [72]. © Annual Reviews Inc.

geologist would be familiar with invert glasses and therefore, ionic glass compositions, as they are apt to form in the extreme conditions provided by the earth. A rich and motivating source of glass-chemical understanding and successful glass 'recipes' has been the field of geological, or *mineral* melts. Their compositions can be found in magmas and lavas, which crystallize to form common rocks with similar stoichiometric ratios. Many principles used to discuss glass structure, such as the common Q^n-unit nomenclature, adopt the understanding of the anionic building blocks found in crystals. Studies of *mineral* glasses demonstrate that several Q^n species can coexist (Fig. 7) and that their relative numbers are dependent on the identity of the modifier (Fig. 8).

In addition to 3d network silicates (tectosilicates), crystallographers are very well aware that crystals and, therefore, glasses, can be composed of isolated SiO_4^{4-} orthosilicates (nesosilicates), dimer $Si_2O_7^{6-}$ pyrosilicates, and 1d chain metasilicates $Si_nO_{3n}^{2n-}$ (inosilicates). Thus, from a crystallographic and/or geologic perspective, ionic melts and glasses can be defined as being composed of mostly orthosilicates/pyrosilicates with some inosilicates (metasilicates). As discussed in Section 2.1, for *invert* glass this should occur for Y of less than 2, however, as Fig. 8 demonstrates, the type of modifier also determines the threshold composition between inosilicate (metasilicate) and orthosilicate.

2.2. Oxoanionic glasses

An oxoanion is an anion containing one or more oxygen atoms bonded to another element; thus, it is possible to call the anionic species of the ortho-composition of any covalent glass former, *e.g.* SiO_4^{4-}, PO_4^{3-}, BO_4^{5-} *etc.*, an oxoanion. The generic formulae for an oxoanion monomer and a condensed oxoanion are $A_xO_y^{z-}$ and AO_n (*e.g.*, SiO_4^{4-} and SiO_2); the condensed form is comprised of polyhedral units of corner- or edge-sharing polyhedra, as are chains of conventional glass network formers. This results in a very broad variety of possible (poly-) oxoanionic glasses, some of which we will discuss in the following paragraphs. Nonetheless, we wish to limit our use of the term *oxoanionic* glasses to glass formers in their isolated mono- or dinuclear forms, *e.g.*, PO_4^{3-} and $P_2O_7^{4-}$, and *not* in long chains. On this note, the following sections about nitrate, sulfate, carbonate and hydrate glasses regard glass formers for which no or only little evidence has been reported for polymerization, *i.e.*, *only* mononuclear oxoanions. However, some of these compounds do show evidence of forming BOs in combination with typical network glass formers.

2.2.1. Nitrates

Nitrate glasses represent a well-studied group of inorganic ionic glasses; mixtures of metal nitrates readily form glasses ($NaNO_3$, KNO_3, $RbNO_3$, $MgNO_3$, $CaNO_3$, $CdNO_3$), *see*Fig. 9. While closely related, they still differ from the simplest glassy salts (*e.g.*, halide glasses, *see*Section 2.3), as single-component glasses (*i.e.*, end members of the compositional series) do not form glasses as easily [10]. Nitrate glasses are a model system for oxoanionic glasses, with structural investigations indicating the presence of two distinct, relatively ordered, types of nitrate site [73–78]; *see*Section 4. Similarly, multiple investigations of the dynamic relaxation processes near T_g and corresponding fragilities having been reported [79–82]; *see*Section 6.4. On the limiting side, they are extremely vulnerable to hydrolysis, and so appropriate protocols for synthesis and storage must be employed. Syntheses typically involve extensive batch dehydration and melting in sealed inert or vacuum atmosphere.

The best-studied nitrate glass system, KNO_3–$Ca(NO_3)_2$, has a T_g of approximately 60 °C [81]. In nitrate glasses, the T_g was found to decrease when a monovalent or divalent cation was replaced with a more

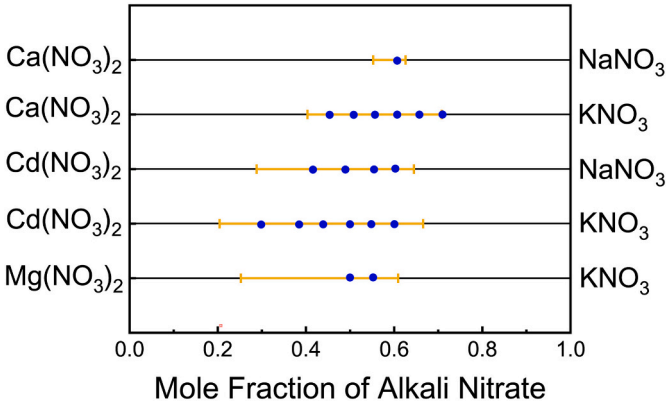


Fig. 9. Glass formation in divalent ion-alkali-nitrates. Gold lines represent the range of the glass-forming regions according to Thilo et al. Blue circles are compositions prepared and analyzed by Raman spectroscopy. Reproduced with permission from Furukawa et al. [74]. © American Institute of Physics.

polarizable cation of the same charge. In contrast, the coefficient of thermal expansion (CTE) and heat capacity at T_g seem to not change when Ca^{2+} is replaced by Cd^{2+} [83]. Nitrates have poor electric conductivity when compared to other ionic glasses; at T_g, $3KNO_3$–$2Ca(NO_3)_2$ was reported to have a conductivity of less than 10^{-10} S/cm [84].

The shear modulus of nitrate glasses is very low, only 3 – 5 GPa; in this sense they tend towards liquids, which have a shear modulus of zero [85]. Additionally, the relationship between viscosity and temperature returns to Arrhenius behaviour near T_g, which has been noted in many liquids [85]. As such, nitrate glasses have often been studied as a proxy for the properties of their melts. This is of interest commercially since molten nitrate salts are used as electrolytes [86], phase change materials (metal nitrate hexahydrates [17]) or, *e.g.*, as low-melting salts in chemical strengthening of glasses [87,88]. As another example, ethylammonium nitrate is considered one of the earliest examples of ionic liquids (reported in 1914), and nitrate salts are still relevant today as their solubility in water and other ILs makes them versatile compounds in chemical synthesis [89].

2.2.2. Sulfates

Similarly to nitrates, it is possible to form oxoanionic glasses with sulfate as the only anion, but single-component sulfate glasses are not known to vitrify easily [10]. The K_2SO_4–$ZnSO_4$ system is the best studied, with glasses of 30 – 80 $ZnSO_4$ mol% having been synthesized through conventional melting with a modified water-cooled splat-quench procedure [90], and, in some cases, dry nitrogen bubbled into the melt [91]. The density (2.7 – 3.3 g/cm^3) and molar volume of the K_2SO_4–$ZnSO_4$ system (Fig. 10), as well as the refractive index (~ 1.5) and molar refractivity were found to vary linearly with component substitution, indicating an ideal mixture without the introduction of new bonding types [90]. Several ternary $ZnSO_4$-based glasses, *see*Fig. 11, have been produced as well, however, instead of forming clear glasses [91], some were reported to have brownish coloration [92].

The T_g values of sulfate glasses (180 – 300 °C) tend to be low in comparison to other inorganic glasses, depending on the measurement method and composition. In the case of K_2SO_4–$ZnSO_4$, there appeared to be no correlation between T_g and the amount of each cation [90]. On

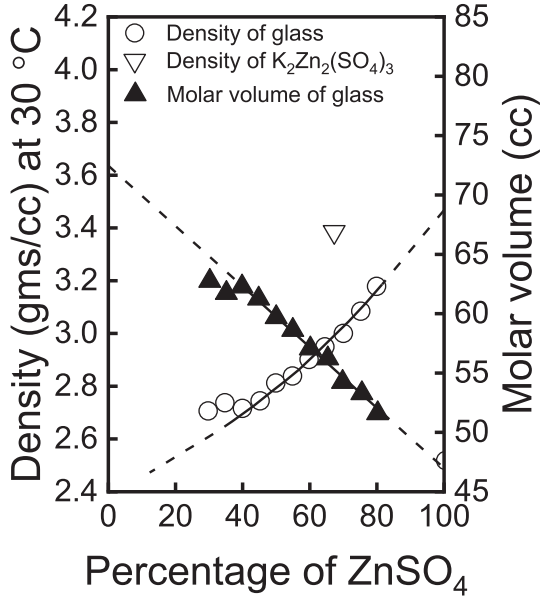


Fig. 10. Densities and molar volumes of crystallized glasses in the K_2SO_4–$ZnSO_4$ system. Reproduced with permission from Narasimham and Rao [90]. © Elsevier.

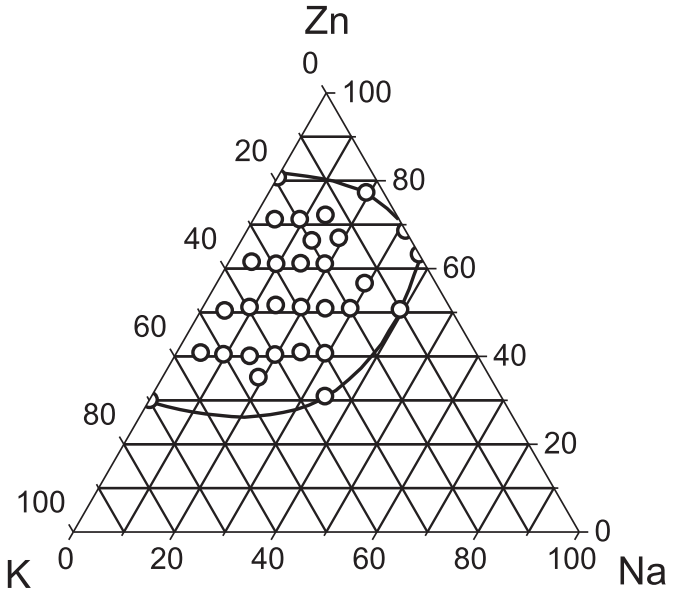


Fig. 11. Glass formation region in the ternary sulfate system K_2SO_4–Na_2SO_4–$ZnSO_4$. Reproduced with permission from Sundar and Rao [91]. © Royal Society of Chemistry (United Kingdom).

the other hand, despite the reduction in density (and therefore looser packing), the T_g of the ternary system K_2SO_4–Na_2SO_4–$ZnSO_4$ surprisingly increased when Na_2SO_4 was replaced by K_2SO_4 (with [$ZnSO_4$] held constant) [91]. The authors explained this odd trend as being the result of K^+ having a higher coordination number than Na^+ and its addition increasing the minimum number of ions required for co-operative rearrangement (relaxation processes are discussed further in Section 6.4). A different trend was found again in K_2SO_4–$ZnSO_4$–$Fe_2(SO_4)_3$, where T_g decreased with the addition of K_2SO_4. Mossbauer spectroscopy indicated a loosening of the structure, *i.e.*, an increase in intermolecular and intramolecular distances (excluding intratetrahedral sulfate distances) [92]. As with many of the less conventional types of ionic glasses, data interpretation is often complicated due to stoichiometric uncertainty, such as from potential water uptake and volatilization of certain components.

As might be expected from glasses having only weak ionic bonds and a strongly polarizable glass former, the hardness and elastic moduli of sulfate glasses are relatively low compared to those of nearly any covalent glass former: Vickers hardnesses of 1.5 – 2 GPa, Young's moduli of 4 – 6 GPa and bulk moduli of 9 – 17 GPa [90,91]. A striking feature of the K_2SO_4–$ZnSO_4$ system is that the reported bulk moduli are 2 – 3 times larger than Young's moduli. Correspondingly, the Poisson's ratios are also unusually high (0.39 – 0.44 [90]); they lie in the rage of typical metallic glasses and approach that of an incompressible liquid. As for thermal expansion, it has been suggested that due to similar ionic interactions, most oxoanionic glasses, namely nitrates and sulfates, have similar CTEs of ~10^4/°C [83,90].

Although studies of pure sulfate glasses are limited in number, there are many mixed-former compositions with high sulfate content. There have been several contradictory structural studies on mixed-former sulfophosphate and sulfoborate compositions, including Li_2O–xSO_3–$(50 - x)P_2O_5$ [93], (Pb,Na)-borosulfophosphate [94], (Na,Zn)-sulfophosphate glass [95], sodium fluorosulfophosphate [96], xNa_2SO_4–$(100 - x)P_2O_5(–H_2O)$ [97], lithium sulfophosphate [98,99], silver sulfophosphate [100], sodium sulfoborate [101], sodium-iron sulfoborate [102] and sulfotelluride [103] glasses. Furthermore, it has been shown that a substantial amount of SO_3 can be added to silicate glasses; both xNa_2O–$(1 - x)SO_3$–PbO–SiO_2 and barium sulfoborosilicate show evidence of Si—O—S bonds [104,105]. Several structural investigations indicate that the sulfate does *not* add to the network former backbone

and remains as an isolated oxoanion, however, glasses containing small ions such as lithium, sodium or water appear to be exceptions (*see*Section 5 for further discussion).

Addition of SO_4^{2-} to $ZnO–Na_2O–SO_3–P_2O_5$ and $Li_2SO_4–LiPO_3$ leads to a decrease in T_g [53,98]. Yet, for this glass system, addition of SO_4^{2-} also increased chemical durability and dynamic fragility [106], despite concomitant reduction of the amount of glass former and network connectivity [53,107]. On the other hand, addition of sulfate to $Li_2SO_4–LiPO_3$ decreased the molar volume, and correspondingly increased the Young's modulus (Y), shear modulus (G) and Poisson's ratio (μ) [98]. There is evidence of phase separation and crystallization when a high amount of sulfate is added in sulfophosphates [100,108], perhaps around the same composition at which sulfate begins to interact directly with the phosphate backbone. Similarly, lithium sulfoborates also showed evidence of two-phase liquid phase separation [11], and phase separation and crystallization were observed at high sulfate content in $xNa_2O–(1-x)SO_3–PbO–SiO_2$ and barium sulfoborosilicate [104,105].

Seemingly the largest problem with sulfate glass synthesis is volatization; in fact, due to its production of SO_2 gas, sulfate is used in many glass melts as a fining agent to promote melt homogeneity [101]. Additionally, the use of pure sulfate glasses in any real-world application is strongly compromised by their high solubility in water [91,109].

2.2.3. Carbonates

Since nitrates and sulfates both form glasses on their own, it should come as no surprise that stable, pure carbonate glasses can also be produced. The equilibrium of carbonate-containing melts, $CO_2{}^{mol.}{\cdot\cdot}O \leftrightharpoons CO_3$, indicates that pressure and temperature can be used to force CO_3 to remain in the melt. Indeed, there are numerous studies on dissolved molecular CO_2 and ionic CO_3 in stable minerals and melts under pressure, as well as on geologically relevant CO_3^{2-}containing glasses [110–113]. Of the three oxoanionic glass families discussed to this point, carbonates are the least stable, volatizing easily during melting and requiring high pressure synthesis procedures [114,115]. Indeed, it is common for glass scientists to use metal carbonates as reagents to form glass melts, rightly assuming that all carbonate will be lost as CO_2 (but also that intermediate carbonatic melts will facilitate batch dissolution). Compared to sulfates, under ambient pressure carbonates have a much smaller window of glass formation, possibly due to the carbonate ions' relatively high symmetry and inability to be easily distorted [114] (although not stated explicitly, this may apply to nitrates as well). In addition, weak Van der Waals forces between the carbonate ions promote crystallization [116]. However, with a high field strength cation, such as Mg^{2+} or Li^+, and geometrical frustration (structural environments which inhibit crystal formation), these factors can be overcome [114]. As early as in 1929, Eitel and Skaliks proved that $K_2CO_3–MgCO_3$ can form a disordered solid [109,117]. A much later (over 30 years) follow-up study found $K_2CO_3–MgCO_3$ glasses to have lower refractive indices than its corresponding crystals, yet found no changes in the carbonate anion itself [114]. It is also possible to produce limited ternary mixtures by adding small amounts ($\leq$ 10 mol %) of $MnCO_3$, $PbCO_3$ and $CaCO_3$ [114].

In view of their low stability, carbonate glasses do not seem to be of interest for practical use. Like sulfates and nitrates, water can also present a practical problem in carbonates, where it was observed to initiate devitrification [114]. However, as with the nitrate compounds, they may again act as model systems for studying the structure of liquid carbonates, *e.g.*, in a geophysical context or for high-pressure/high-temperature electrolyte applications [118,119]. Indeed, mixtures of molten alkali carbonate salts have found utility in applications such as the low-cost, molten carbonate fuel cell [120]. Additionally, the use of complex carbonate ions as both solvents and reagents has proven to be a cost-effective, more environmentally friendly alternative for the production of many ILs *via* carbonate-based ionic liquid synthesis [121,122]. On the same line, the mutual solubility between ILs and complex carbonate ions has led to the testing of IL-carbonate mixtures for improving the safety of electrolytes [123].

2.2.4. Hydrates

We may easily assume that a large fraction of the published data on (poly-)oxoanionic glasses is compromised by unknown amounts of residual water which remained in the glass after melting or were incorporated during measurements or storage. On the other hand, the propensity of these compounds to water sorption can be exploited in the purposeful preparation of hydrate glasses. For example, the poly-oxoanionic $CaF_2–Ca(OH)_2–CaCO_3–BaSO_4$ system contains water as hydroxyl groups and demonstrates the extreme compositional flexibility of ionic glasses [114]. As with the carbonates, water may also enter the glass matrix in several forms, namely $H_2O_{(l)}$ and bound hydroxyl groups [114,124]. Although water is regularly discussed as an impurity or detriment in silicate glasses, its many forms can be found at high concentrations in ionic glasses, such as the $6.5Ba-SO_4–37.1CaCO_3–24.3CaF_2–32.1Ca(OH)_2$, $KHSO_4$, and $Ca(NO_3)_2{\cdot}4H_2O$ compositions [81,109,117]. Oftentimes, some water (up to 5 wt%) can be added to ionic glasses without much effect on the structure [125]. Based on the lack of changes in the pryo- and orthophosphate regions of the FTIR spectra, it has been noted that water is likely present as isolated H_2O molecules which are "*hydrogen-bonded to the BO*" [125]; this is in contrast to other compositions where small amounts of water add as hydroxyl groups [126]. There also exists literature on hydrate glasses, such as ammonia, sulfuric acid and nitric acid hydrates [124,127]. While glass-forming hydrates are a broad subject on their own, we would not go into further detail here.

As a final remark regarding the application of Dietzel's traditional classifications [18] to oxoanionic glasses, we note that, with the exception of the hydrates (where H^+ has the field strength of an intermediate), all of the oxoanions (*i.e.*, SO_4^{2-}, NO_3^- and CO_3^{2-}) contain a cation which has sufficiently high field strength to be firmly categorized as a glass former [128] (as do conventional glass forming units like SiO_4^{4-} and PO_4^{3-}).

2.3. Halide glasses

According to Goldschmidt's theorem of isomorphism [3] certain tetrahedral compounds such as $ZnCl_2$ and BeF_2 provide structural analogues to silica (SiO_2), exhibiting an almost identical ratio of anionic and cationic radii. Aside from this structural similarity, the weaker interionic bonding in these compounds was predicted to result in reduced thermal stability. $ZnCl_2$ has since been known as an easy glass former [129]. In the case of vitreous BeF_2, experimental verification of the principle of isomorphism was obtained in the 1970s [130]. Compared to vitreous silica with a T_g of $\sim$ 1200 °C [131], the T_g of BeF_2 is $\sim$ 250 °C [132,133], and that of $ZnCl_2$ is $\sim$ 105 °C [134].

A rich variety of metal halide compounds are now known to form glasses; several reviews, books, and article collections have been published on the subject [135–151]. Initially, fluorides of Mg, Ca and Al were used to improve the chemical and mechanical stability of BeF_2 [152]. The most prominent of today's halide glasses, however, are based on the binary $ZrF_4–BaF_2$ (ZB) system discovered by Poulain and Lucas [153,154]. While it is rumored that this discovery was made accidentally, it has also been assumed that glass formation was facilitated by the intermixing of a variety of superstructural fluorozirconate units. Fluoride glasses based on the ZB binary are now exceptionally well-studied due to their suitability for optical applications that require low mean dispersion and wide transmission windows. Currently, there are three major families of commercial fluoride glasses: fluoroaluminates (AlF_3), fluorozirconates (ZrF_4, HfF_4) and fluoroindates (InF_3, GaF_3). A few highly specialized glass manufacturers produce these in relatively large quantities. Other fluoride families are still under development and currently of scientific interest only.

One important factor, in particular, for optical applications, is purity

of the starting reagents, where significant research efforts have been devoted to reagent purification, minimization of impurities and alternative ways of glass manufacture [155–161]. Another continuing experimental problem with halide glasses is their tendency to crystallize [162]; due to their low viscosity, they often require rapid cooling (*e.g.*, ~ 1 K/s for ZrF_4 glasses) which can hinder the production of crystal-free, bulk samples (~3 – 4 mm thick is typical). However, when appropriate melting protocols are employed, large fluoride samples can be produced, as in the case of a 50 kg fluoride window prepared for the NRL (*see* Fig 3.2 p.85 in: Infrared Fiber Optics, CRC press, 1998). Furthermore, the contribution of microcrystals to optical losses in fluoride glass fibres can be reduced to not exceed 0.5 dB/km – proving that crystal-free fluoride glasses can be consistently fabricated.

In general, chlorides and bromide melts have higher viscosities, which may lead to a larger glass-forming region than fluorides, but this comes along with a trade-off of higher hygroscopicity. Despite these issues, the high densities of the heavy-metal based compounds make them attractive for optical applications. The densities of fluorides are significantly higher than those of other glass systems, *i.e.*, within 4 – 6 g/cm^3 is common [163], while 6.5 g/cm^3 (ThF_4–ZnF_2–NaF_2–YF_3) is achievable [164]. Using chloride anions, the density is somewhat lower, for example, around 3.5 g/cm^3 for 30NaCl–30KCl–40$ThCl_4$ [165].

2.3.1. Glass-forming compositions

In contrast to oxide glass-forming rules, many stable fluorine-based compositions, *i.e.*, ones with good glass forming ability, contain only *conventional* intermediate (Zr, Al, Th, La, Y, Fe, Cr, V, Ga, Sc, Pb, Mn, Sc, Cd, and Zn) and modifier cations (Li, K, Rb, Cs and Ba), without any *conventional* glass former [166]. Poulain et al. argued that "*the physical existence of a vitreous network based on the coordination polyhedra of the smaller cations in the complex structure of the fluoride glasses is far from being obvious,"* and that *"insofar as they may also be described as a random ionic packing, the contribution of large cations to the vitrification process cannot be neglected*" [quoted from [145]]. Fluoroberyllates represent the major exception, where BeF_2 (see above) is known for having relatively covalent bonds with a structure similar to SiO_2 [166]. Most fluoride glasses are heavy metal based (HMFG), with ZrF_4 by far being the most common tetravalent component or "glass progenitor;" Fig. 12 demonstrates the effect cooling rates have on the glass-forming region in a typical ZrF_4-based glass.

Due to the requirement for ZrF_4, AlF_3, VF_3, *etc*. to be present in the melt, it is common to call the tetravalent (or trivalent) cation fluoride compound the "network-former" (although "glass progenitor" may avoid confusion when making comparisons with other glass families) [162]. Still, as with invert glasses, it is possible to form fluoride glasses composed of more modifier than former: in fact, ThF_4–LiF with up to 70 mol% LiF, as well as ZnF_2–MF glasses with 98 mol% CsF, and 95 mol% KF or RbF, vitrify readily, indicating how little "network-former" is required. The strongly ionic character of fluorine bonds has been said to "*form strong cation-anion-cation bridges, leading to the sluggish relaxation behavior (increase in viscosity) near the freezing point necessary to favor glass formation*" [quoted from [167]]. Poulain highlighted this significant difference: "*This implies that their glass formation follows rules different from those for oxide glasses. Zachariasen's rules, in particular, cannot be applied here in their original form, neither for structural descriptions, nor for the prediction of new vitreous systems*" [quoted from [164]]. The exact ways in which the formation and structure of halides, and other ionic glasses are different from those of the more conventional covalent glasses are elaborated further upon in Section 4 and 5.

As already noted, similarly to BeF_2, single component chloride glasses from $ZnCl_2$ have been studied extensively [162,168]. Generally, $ZnCl_2$-based mixtures [169], $BiCl_3$-based mixtures [162,170], $CdCl_2$-based mixtures [164] and $ThCl_4$-based mixtures [165] with other metal chlorides and halides easily vitrify. On the other hand, like ZrF_4, pure $BiCl_3$ is not known to readily form a glass. Interestingly, glasses based

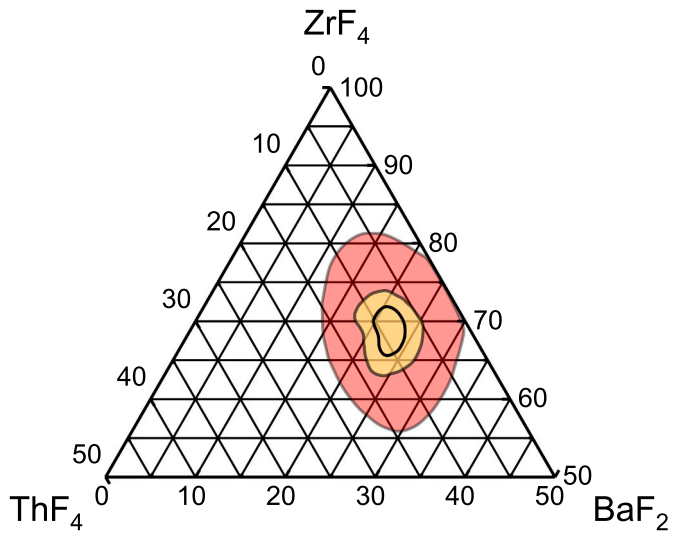


Fig. 12. Limit of glass formation in the ZrF_4–ThF_4–BaF_2 system for different cooling rates: quenched glasses (red); cast into mold at 20 °C (orange); cast into mold at 250 °C (black). Reproduced with permission from Poulain [166]. © Academic Press, Inc.

on pure $TeCl_2$, Te_3Cl_2 and $TeBr_2$, with small amounts of sulfur needed to form TeI-based glasses (so-called TeX glasses) have also been explored for their unique optical properties [171,172].

In contrast to oxides *and* other ionic glasses (fluoride and oxoanionic glasses), simply adding several different metal cations *does not* necessarily increase the glass forming ability of chloride glasses. Mixed-cation chlorides do not easily form glasses (for example CuCl–$PbCl_2$–$RbCl_2$ [173] and LiCl–KCl–CsCl–$BaCl_2$ [174] have markedly small glass forming regions). Additions of KCl surprisingly reduce the glass-forming ability of $ZnCl_2$ [162], and the same is true for $ThCl_4$ (*see* list of typical glass-forming systems of non-fluoride halides [175]). Additions of KCl to $ZnCl_2$ of more than 60 mol% rapidly increase the liquidus temperature [165]. In contrast to cation mixing, mixed-halide compositions appear to vitrify much more readily (as low as 1 at% is sufficient to prevent crystallization) [160,176–178], as evidenced by the comparison in Fig. 13, where the glass-forming region increases substantially with the introduction of fluoride to a chloride system. For example, up to 20, 40 and 60 mol%, respectively of $CdCl_2$, $BaCl_2$, and $PbCl_2$ can be added to fluorides [160,179–181], and more than 30 mol % of alkali chloride (LiCl, NaCl or KCl) was successfully added to the ZrF_4–SrF_2–MCl system [182]. On the other hand, $BiCl_3$ acts like a "normal" glass-former, where mixtures with alkali halides and other salts, like TlCl, increase its glass forming ability [162].

As with $ZnCl_2$, almost pure glasses can also be obtained from $ZnBr_2$, with only 0.5 – 1.0 mol% of As_2O_3 required for stabilization [183,184]. Here, studies of Kadono et al. (*see*Fig. 14) and Kawamoto et al. provide some unique examples of non-fluoride halide glass compositions, such as $ZnBr_2$–KBr–MBr_2–AgBr where M = Ba, Pb, and LiBr–KBr–CsBr–-$BaBr_2$. However, the chemical durability of these bromide glasses is poor [174,183,185,186]. Similar to BaF_2, $BaBr_2$ was also found to prevent devitrification; KBr appeared to be a necessary modifier (where AgBr had no stabilizing effect), while $ZnBr_2$ was the glass progenitor. Bromide appears to be similar to chloride where the best glass-forming bromides are $CdBr_2$ and $PbBr_2$ [180] On the other hand, $BiBr_3$ appears to be analogous to ZrF_4 in that it cannot form glasses alone, but forms a variety glasses when mixed with other components [162]. Not much literature is available on pure iodide glasses, some exceptions are LiI–KI–CsI–BaI_2 [174] and pure PbI_2 [187]. It is possible to add iodine salts to chlorides and bromides [173,187]. *E.g.,* 50 mol% KI can be added in the $ZnCl_2$–KI system [169] and 40 mol% KI in the $CdCl_2$–KI–CdF_2 system [164]

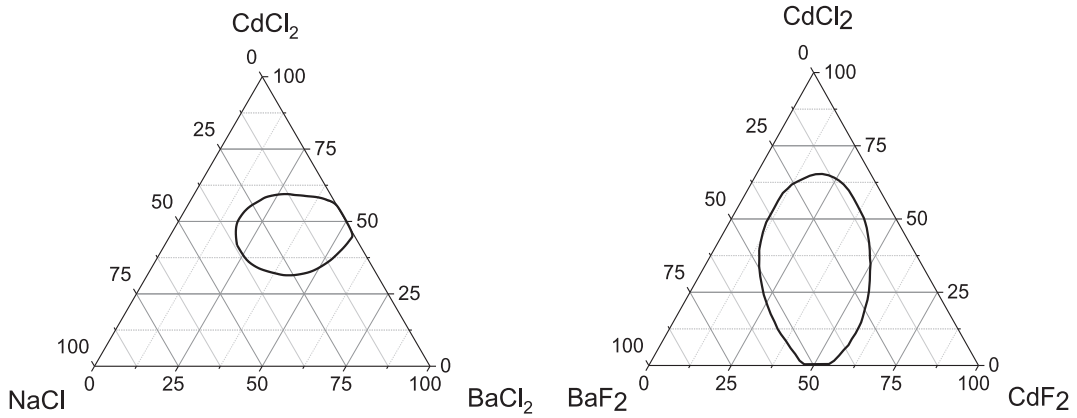


Fig. 13. Glass-forming regions in the $CdCl_2$–$BaCl_2$–NaCl and $CdCl_2$–CdF_2–BaF_2 systems. Reproduced with permission from Poulain [164]. © Elsevier.

It has been theorized that halosulfate mixtures such as K_2SO_4–$CuCl_2$ and K_2SO_4–$CoCl_2$ should form glasses as well [10]. Indeed, there have been halosulfate compositions reported, such as K_2SO_4–$ZnCl_2$ and K_2SO_4–$ZnSO_4$–KCl, which were expected to form better glasses due to the liquidus temperature decreasing more than the T_g [10]. $ZnSO_4$–NaF mixtures with 35 – 70 mol% $ZnSO_4$ poured onto cooled copper plates were reported to have T_g values of only 200 – 350 °C and linear trends in T_g, density and molar volume [188]. Such low melting temperatures and T_g values for the inorganic component enable hybrid salt mixtures such as $ZnCl_2$–C_5H_5N·HCl, demonstrating the strong link between ionic glasses and ionic liquids (*see*Section 8) [189].

Due to their low melting temperatures and consequent ease of volatization, care must be taken to melt halide glasses at the lowest temperature possible. Usually, fluoride (and other halide) glasses are melted directly from the salt components, but they can also be made from oxide reagents, fluoride-exchangers like ammonium bifluoride or reactive atmospheres [164,190]. In order to avoid oxidation at high temperatures, appropriate crucible materials must be selected. Furthermore, melting is usually performed in inert, dry atmosphere, often using sealed capsules or ampoules, *e.g.*, in order to avoid reactive evaporation in the form of hydrogen halides.

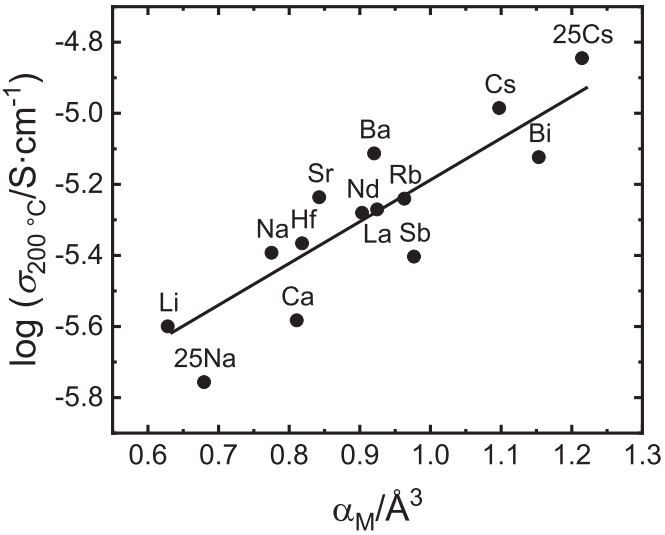


Fig. 15. Dependence of conductivity (at 200 °C) on average polarizability, α_M, of constituent cations in ZrF_4-BaF_2-MF_n glasses. Reproduced with permission from Kawamoto et al. [219]. © Elsevier.

2.3.2. Chemical durability

It is not surprising that (in comparison to covalent oxide glasses) halide glasses are more vulnerable to water attack. In some cases,

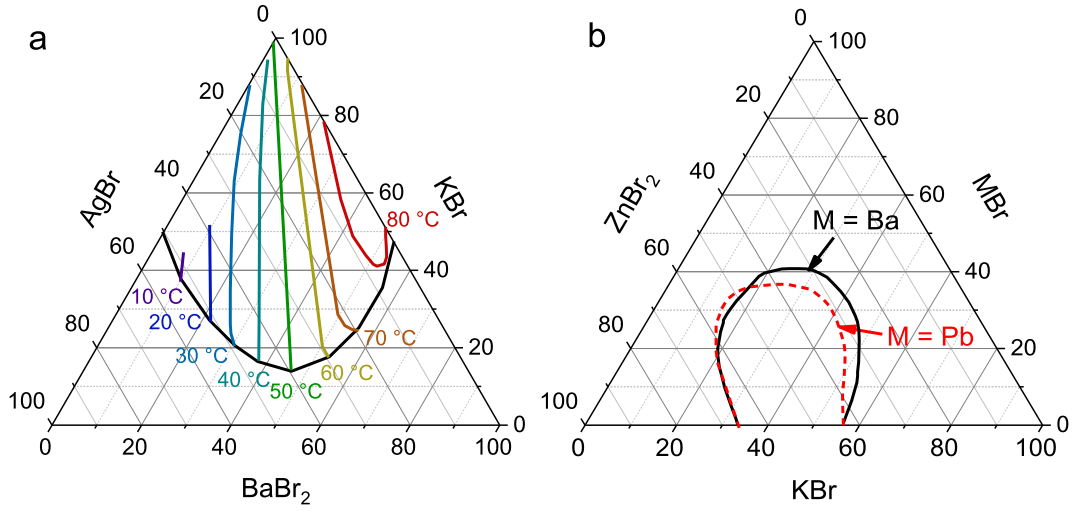


Fig. 14. (a) Glass-forming region and isothermal curves of the glass transition point of the pseudo-ternary system with equal parts KBr–$BaBr_2$–AgBr (50 mol%) and $ZnBr_2$ (50 mol%). (b) Glass-forming regions in the $ZnBr_2$–KBr–MBr_2 system, M = Ba, Pb (mol%). Reproduced with permission from Kadono et al. [185]. © Trans Tech Publications Ltd.

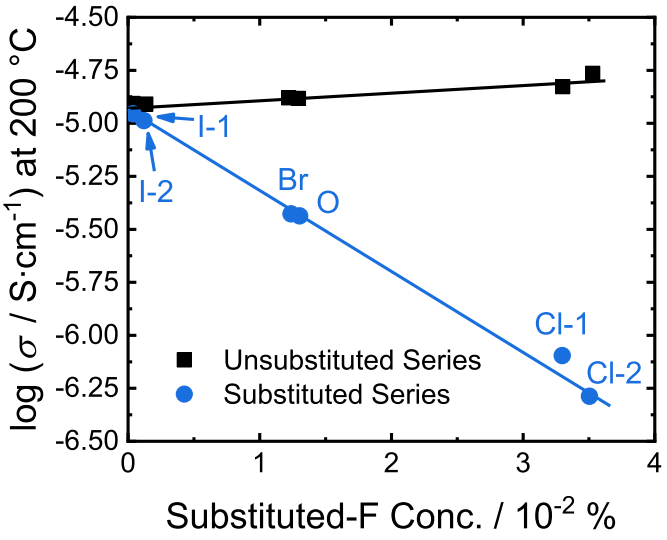


Fig. 16. Variation of conductivity at 200 °C with the substituted fluoride-ion concentration in ZrF_4-BaF_2-CsF glasses partially substituted by Cl^-, Br^-, I^- and O^{2-} ions. Reproduced with permission from Kawamoto et al. [205]. © Elsevier.

especially in those compositions containing alkali salts, they can react with humidity in the air alone, resulting in visible surface degradation, gelation or even bulk flow. However, most fluoride glasses are stable under normal atmospheric conditions, but will form a hydroxide layer in liquid water [190]. Generally, it has been found that components leach congruently, although particularly high ZrF_4 content may trigger inhomogeneous dissolution, whereby ZrF_4 is leached at a higher rate compared to the metal components [191]. Nonetheless, some fluoride glasses, such as the AlF_3–ScF_3-based glasses or the ZnF_2–BaF_2–ThF_4–LnF_3 system, are very stable even when immersed directly in running water [164,166].

Chloride glasses are less stable than fluorides, where after a few minutes of exposure to air, a white oxidation layer forms [162], and $ZnCl_2$ glasses have been called water soluble [168] and would require a protective coating when used in real-world applications [162]. Their dissolution process has been likened to that of chloride crystals: a 'packed' body 'which 'completely collapses' in the presence of the polar water molecules [192,193]. Naturally, hygroscopicity increases with increasing anion polarizability, with bromides being even more hygroscopic [194,195]. Only in some cases can fluoride admixture be used to enhance the water resistance of the other halide glasses [196]. Additionally, components with low solubility in water, such as copper chloride [195] and lead halides, $PbCl_2$, $PbBr_2$ and PbI_2,were found to noticeably increase the durability of non-fluoride halides, compared to thorium [192,193], zinc or cadmium halides [197]. It has been discussed that the addition of only small amounts of oxides to halide glasses can increase their chemical durability without major changes in other physical properties [198]. Other studies state that glasses containing AlF_3 did not have the expected decrease in chemical durability upon chloride addition [199].

Besides visible degradation, water uptake also interferes with optical transparency, particularly in the NIR–MIR spectral range. For this reason, special care must be taken even during glass melting, including careful drying of all starting reagents; nonetheless, most halide glasses can otherwise be synthesized normally [165]. An exception are $BiCl_3$-based glasses, which were reported to require a chlorine atmosphere in order to prevent oxidation and/or clustering [194]. Although hygroscopicity is primarily an experimental problem, it is still interesting to study fundamentally. Such studies can aid in the understanding of the structural role of water and its relevance in, *e.g.*, optical properties. In turn, such insight would be beneficial in cases where glass dissolution is desired, *e.g.* for biomedical applications [200–202].

2.3.3. Thermal and mechanical properties

Typically, halide glasses have low T_g values in comparison to conventional oxide glasses. From the viewpoint of applications, this is both a disadvantage and an advantage, as will be discussed in Sections 7 and 8. Fluoride glasses typically have T_g of 200 – 350 °C [163,203], yet, additions of ZrF_4 or HfF_4 can enhance T_g to as high as 530 °C [204]. Glasses containing other halides in addition to fluorine can have T_g around 200 °C [164,205]. The experimental melting temperatures are correspondingly low (700 °C at most) except for ZrF_4 and AlF_3, where T_m can be as high as 800 °C, especially for the purpose of melt fining (generally, low alkali and/or alkali earth content or higher valence cations have higher T_g and T_m [204]). In a comprehensive study of liquidus temperatures, it was found that T_g/T_m and T_g/T_l were roughly 0.7 and 0.6 for all HMFGs, which corroborates the 2/3 rule of thumb used for conventional glass formers [164,204,206,207]. There are still some clear exceptions with $40BaF_2$–$40ScF_3$–YF_3 and $28ThF_4$–$28ZnF_2$–$15BaF_2$–$28YbF_3$ having a T_g/T_l of 0.5 and 0.51, respectively [164]; low T_g and comparably high T_l indicate extended supercooling regimes. Based on studies of other ionic glasses, fluoride mixtures might be expected to be ideal and have monotonically varying T_g as a function of composition, however both thermal measurements [166] and several structural studies of the BaF_2-ZrF_4 system demonstrated that the structures and bonding of the components in the mixture were different as compared to those of the end members (*see* [35] for an in-depth summary of these structural studies (Sections 7 and 8)).

In mixed chloride glasses, T_g is reduced to 30 – 60 °C, with $11PbCl_2$-40KCl-$49BiCl_3$ and $55BiCl_3$-45KCl having values of 57 and 45 °C, respectively [162,194]. These low values of T_g appear to be common to all chloride glasses, as the addition of high-valence chlorides, *e.g.*, $ThCl_4$, $BiCl_3$, $PbCl_2$, or even of metal fluorides, has little effect on the T_g. Besides pure $ZnCl_2$, another known exception is 30NaCl-30KCl-$40ThCl_4$, which has a T_g of 130 °C [165], likely due to the high T_m of $ThCl_4$ (770 °C [208]). It has been noted that $ZnCl_2$ loses its silica-like polymeric structure upon the addition of a second component, and has a large freezing point depression [10]. Interestingly, $PbCl_2$ appears to cause the largest increase in T_g, while for alkali chlorides T_g surprisingly decreases with increasing alkali radius (contrary to the often-obsered "*regular*" relationship). Bromides have even lower T_g (0 – 40 °C) [194]. However, care must be taken due to the large spread of published values, potentially resulting from contamination of batch materials and glasses with water, oxygen or other impurities.

As with other ionic glasses, halides and particularly heavy metal fluoride glasses exhibit Poisson's ratios of around or above 0.3. This signifies these glasses' intermediate structural role between three-dimensional networks and fully packing-determined ensembles with non-directional bonding [209,210]. Also bulk and shear moduli vary only as far as variations occur in the halogen bond strength, in particular, as most glasses are based on ZrF_4, AlF_3 and BaF_2 [136,211]. Due to their denser packing, weaker and less directional bonding, and lower phonon energy, the thermal expansion of halides is somewhat higher when compared to classical alkali silicate glasses, *e.g.*, 10 – 30 ppm/K [204], with heavier halide glasses having higher CTEs [164]. Mechanical properties undergo pressure-induced variations which are mostly in accordance with the congruent compaction of a closely packed structure, rather than expressing pronounced anomalies, *e.g.*, those found for the open structures of silicates and borates [211,212]. Published data of the mechanical properties of chloride glasses are still sparse, perhaps because of the difficulties in producing samples of larger size.

2.3.4. Electrical properties

As shown in Fig. 17, fluoride glasses have relatively high electrical conductivities, on the order of 10^{-4} – 10^{-6} at 200 °C for common ZrF_4-based glasses [205,213]. InF_3-based glasses can have 10^{-2} S/cm (with E_a values of 0.5 – 0.7 eV). For example, $35InF_3$-$30SnF_2$-$25PbF_2$ has a conductivity of 6.3×10^{-4} S/cm at 150 °C [205]. Other fluoride glasses, such as BaF_2-UF_4-MnF_2-AlF_3, have much lower conductivities of 10^{-6} –

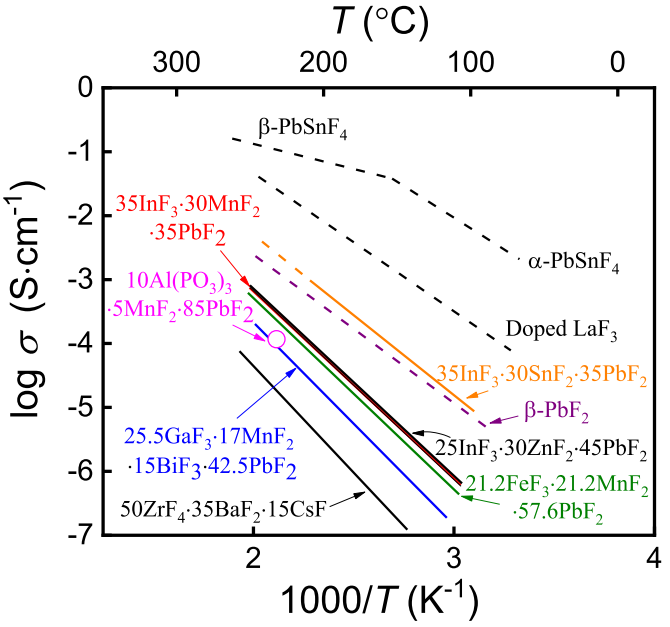


Fig. 17. Electric conductivity of some fluoride glasses. Crystalline fluorides (dashed lines) and an oxyfluoride glass (circle) are reported for reference. Reproduced with permission from Kawamoto et al. [205]. © Elsevier.

10^{-7} S/cm at 200 °C [203]. Since the conductivity is higher than in some fluoride crystals, it has been argued that the glassy state facilitates fluorine mobility. The conductivity was found to improve when the average F coordination number and cation polarizability increase (*see*Fig. 15), as well as when the M—F bond strength and fluoride packing density decrease, such as through increased disorder [214–218].

Rather than being dependent on the number of network modifiers, conductivity is determined by the number of non-bridging fluorines and the packing arrangement of the MF_x polyhedra: a large distribution of polyhedral shapes and sizes increases the number of relatively mobile terminal $F_{,}^{-}$ which can hop between neighbouring cation polyhedra [215,220,221]. Similarly at odds with conventional glass formers, substitution of glass 'stabilizers' (ZrF_4, ThF_4) by alkali cations such as NaF and LiF generally increases the resistivity by electrostatically trapping fluoride anions; moreover, this interaction is much more evident for immobile Na^+ compared to mobile Li^+ [220–224]. Indeed, in the case of mobile species as Li^+, F^- and Li^+ have both been found to be responsible for the ionic conductivity at low Li/F ratio, while at higher Li/F ratio, only Li^+ was responsible [216,222,225–228].

Since many studies indicate that alkalis are not major contributors to ion conduction in fluoride glasses, the appearance of a significant mixed-alkali effect in $56ZrF_4$-$14BaF_2$-$6LaF_3$-$4AlF_3$-xLiF-$(20 - x)$NaF glasses was initially surprising, with the authors reporting that between ⅔ and ¾ of the current may be carried by mobile alkali cations [220]. However, the observed minimum conductivity in mixed Li-Na fluoride glasses resembles the mixed-alkali effect only phenomenologically. It differs mechanistically and also depends on the total mol% of alkali: at 10 mol% alkali, only F^- is mobile and the different types of sites created by the cations results in the σ minimum, yet at 30 mol% alkali there is a transition from anionic to cationic conduction (at low LiF (or high immobile NaF), only anionic F^- conduction occurs until sufficient Li^+ is added to be the major conducting species) [229].

Substitution of fluoride by other halides results in a reduction of conductivity, regardless of the type of cations present (*see*Fig. 16) [205,230,231]. Also replacement of fluorine by oxygen was reported to reduce the electrical conductivity [205]. This is apparently contrary to the many published studies which indicate that the mixed-former (or mixed-anion) effect should generally increase the electric conductivity [232]. It can be understood when realizing that halides, arguably the glass former, are the mobile species; in ZrF_4-based glasses, F^- has been found to be the major mobile species [214,220]. Consequently, a phenomenon analogous to the mixed-modifier effect occurs, wherein chloride ions block the fluoride channels and behave as "traps" due to their large size and polarizability [230]. Tatsumisago et al. [233] compare the mixed-cation and mixed-anion effect in fluorides: the presence of more than one type of cation or anion reduces conductivity of the mixed species, while the mobility of the non-mixed species (cations in the mixed-anion composition, for example) increases. For some discussion about different models of ionic conduction in fluoride glasses, *see* Ravaine et al. [215].

2.3.5. Optical properties

Due their unique optical properties, fluoride glasses are the subject of numerous (ongoing) scientific studies, and have entered a broad range of commercial products [132,135]. They tend to have refractive indices comparable to those of normal window glass ($n \approx 1.5$), but lower optical dispersion (higher Abbe number), a wider optical transmission window and lower intrinsic attenuation or losses [234]; all of these characteristics are due to their lower anion polarizability (low atomic number, high electronegativity) and the comparable weakness of the strongest cation–anion vibration [162] (the latter making halide glasses some of the most interesting host matrices for optical activator species, for which the probability of non-radiative relaxation of excited states is reduced by increasingly complex multi-phonon processes [235–237]).

Since fluoride glasses are common and successful components of modern optical (most prominently, fiber) devices, we wish to turn our attention to the optical properties of the lesser known non-halide glasses. Chloride, bromide and iodide glasses have high dispersion (low Abbe numbers, $\nu_d \approx 13$) when compared to fluoride glasses (similar to that of chalcogenides [238]), which can sometimes be desirable for corrective optics [194]. This is demonstrated in Fig. 18 where the blue line (Bi-based chlorides and bromides) is outside of the lightly shaded region of other optical glasses. The Abbe number is correlated with the polarizability of the atom species' environment, which in turn can be accurately described using the optical basicity scale, as described further in Section 3.3.

Special interest in non-fluoride halide-containing glasses arose from their extended transparency in the IR spectral region, *e.g.*, 1 – 20 μm depending on the halide species [158,159,172,175,179,195,239–241], with some studies reporting good transmission down to 25 μm [175,195] or even 33 μm [180]. In addition to active optics this, at least in theory, enables optical information processing at higher wavelengths and, therefore, lower Rayleigh scattering. Fibres of TeX glasses, where X = Cl, Br and I, where demonstrated at 5 dB/m at 10.6 μm [239], compromised by material purification. It was argued for $ZnCl_2$ that the optical loss may reach as low as $\leq 10^{-3}$ dB/km in the 3.5 – 5 μm region [168,242]. Such figures, however, have never been realized thus far due to the unavailability of ultra-clean material processing logistics, in particular for avoiding oxidation and for reducing the water content of these highly hygroscopic materials to the extreme minimum [168,243]. Several ongoing studies therefore discuss the deleterious effects of impurities (such as oxides and water), not only on the optical properties of halide glasses [158–161], but also on their chemical stability. In cases where the addition of small amounts of oxides did not modify the glass' low IR cut-off significantly, unfortunately, only the thermal stability improved, not the chemical durability [183].

Higher refractive indices can be achieved by increasing the average atomic number, *e.g.* by replacing fluorine with chlorine and bromine [103,182], however, there is an antagonist effect, where single-charged ions are weakly bound and lead to lower refractive indices as compared to oxides [194]. Still, the refractive indices of chloride and bromide glasses are significantly higher than those of fluoride glasses, *i.e.*, 1.90 – 3.04, and also exceed those of oxide glasses [103,172,194,238]. It is

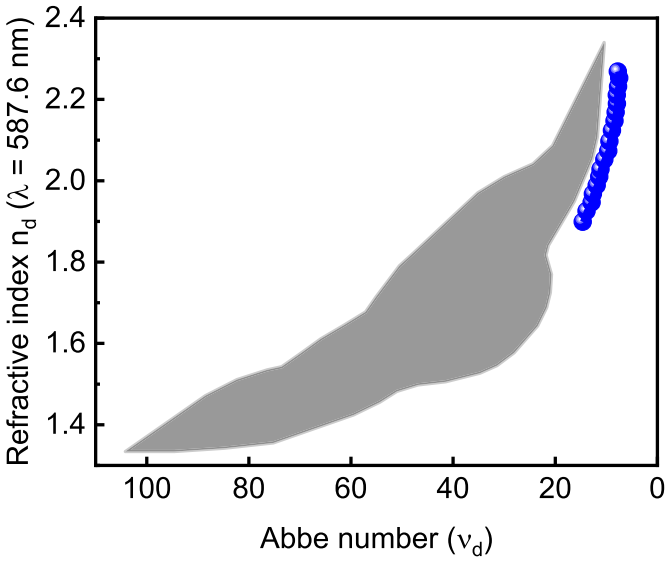


Fig. 18. Index of dispersion island for optical glasses (in gray, after Polukhin [244]) and the combined results of the chloride and bromide glasses (bright blue strip to right of diagram). Reproduced with permission from Ziegler and Angell [194]. © OSA Publishing.

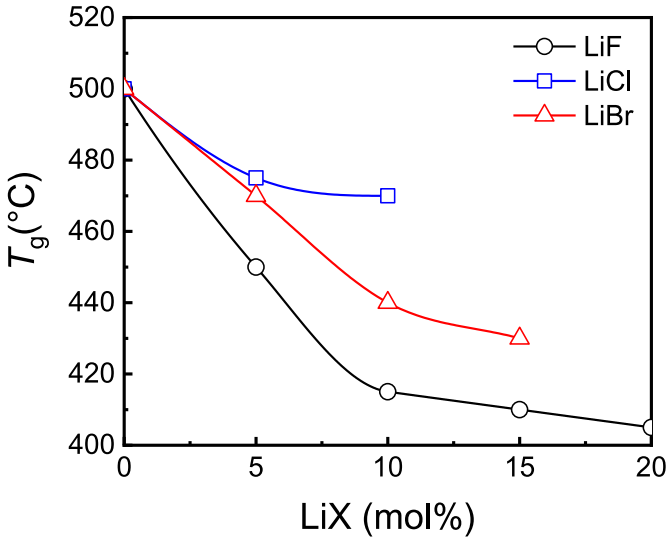


Fig. 19. Glass transition temperature T_g as a function of lithium halide content in (70 – y)B_2O_3-30BaF_2-yLiX for X = F, Cl, Br. Reproduced with permission from El-Hofy and Hager [245]. © Wiley-VCH.

believed that n could be increased further for ionic glasses by increasing the interaction strength between ions: "*only the fact that the single-charged anions are bound somewhat weakly in comparison with comparable oxides prevents quite exceptional refractive indices from being generated*" [quoted from [194]], however, this would potentially reduce transparency in the visible region. Since adding heavier (larger) halide species weakens bonding between the cations and anions (and lowers the energy of the conduction band), the transmission window of chloride and bromide glasses shifts to longer wavelengths (*i.e.*, the frequencies of the UV and IR edges decrease [165]). Some non-fluoride halide glasses have been reported with optical transmission only at > 1 μm [172,197]): in contrast to transparent oxide glasses which have low atomic number anions and strong bonding, transparent halide glasses have high atomic numbers and relatively weak bonding [194].

2.4. Haloborates, halophosphates and others

Phosphates and borates fall comfortably in the glass former category, yet both glass families can accommodate typical modifier compounds (alkali and alkaline earth) well past the 50:50 meta-composition into pryo- and ortho- compositions. Perhaps because of the relatively low stability of their short chains and isolated oxoanionic tetrahedra, the phosphate and borate network can be disrupted by ionic halide salts to form halophosphates or haloborates. The existence of these unique glass systems validates the idea that invert glasses and halide glasses both fall under the category of ionic glasses; despite one family being oxoanionic, they both interact through mostly ionic bonding and are chemically compatible.

A study of the thermal properties of (70 − y)B_2O_3-30BaF_2-yLiX, where X = F, Cl, Br, indicated that initially ($y \leq 10$ mol%) the halide adds to the network forming B—X—Li bonds with smaller-radii halides being more likely to add to the network, and then after $y \geq 10$ mol%, halide atoms add interstitially (*see*Fig. 19) [245]. An interstitial halide salt, instead of being bound to the network (the degree of which can be quantified as $[BO_{4-n}X_n]/[BO_3]$), increases the overall conductivity and lowered the activation energy for charge transport. Surprisingly, for the same glass series, all lithium halide salts increased in density and elastic moduli (Fig. 20) [246]. Even more surprising was the observation that chlorine and bromine increased the elastic moduli more than fluorine, which was explained by differences in bond length, the stretching force constant between the bonds and the average cross-linking density. Conversely, a comparison of lead haloborates revealed that when oxygen was exchanged by fluoride, the density increased, while bromide and chloride exchange decreased the density despite their higher molecular weight (explained as expansion and weakening of the network) [247]. Finally, several papers investigating sulfoborate and sulfophosphate compositions have focused on luminescence properties [248–251].

The data shown in Fig. 21 demonstrate that it is possible to add up to 40 mol% $MgCl_2$ and 10% $AlCl_3$ to the base barium metaphosphate glass in the $Ba(PO_3)_2$-MCl_2-$AlCl_3$ series, where M = Mg, Ca, Sr, Ba [252]. Similar systems $NaPO_3$-BaF_2-REF_3 (RE = rare earth) and $NaPO_3$-$Al(PO_3)_3$-MF_n were explored by Matecki et al. [253] and Rigout et al. [254], respectively. Up to 42 $ZnCl_2$ mol% can be introduced in the $NaPO_3$-$ZnCl_2$ system [251], while up to 30 mol% P_2O_5 can be introduced into the $PbCl_2$-$PbBr_2$-P_2O_5 and PbF_2-$PbBr_2$-P_2O_5 systems (again with lead halide enhancing the chemical durability) [255].

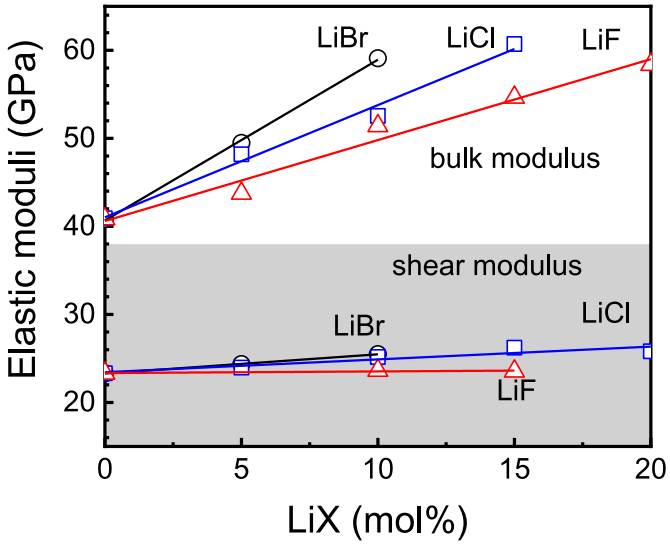


Fig. 20. Variation of the glass transition temperature, T_g with lithium halide content, in (70 – y)B_2O_3-30BaF_2-yLiX, where X = F, Cl, Br. Reproduced with permission from Hager and El-Hofy [246]. © Wiley-VCH.

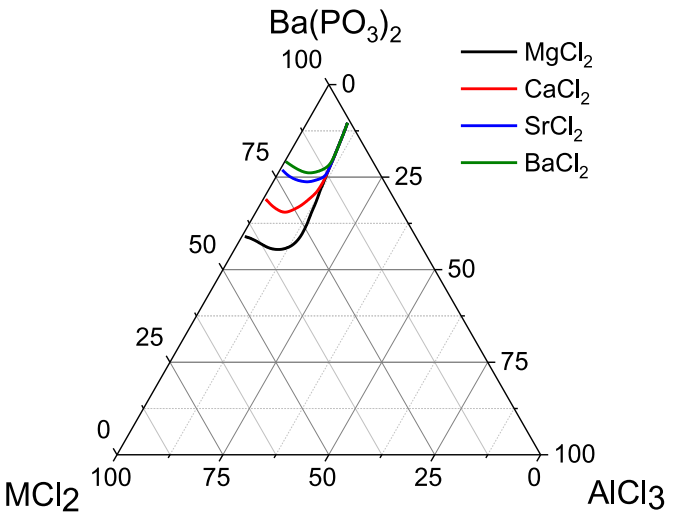


Fig. 21. Glass-forming regions in the $Ba(PO_3)_2$-MCl_2-$AlCl_3$ systems for M = Mg, Ca, Sr and Ba (data in mol%). Reproduced with permission from Wasylak and Czerwosz [252]. © Elsevier.

Compared to silicates, where only small amounts of chlorine (1 – 2 mol %) can be added, phosphates and borates are much more miscible with metal chlorides. The $Ba(PO_3)_2$-MCl_2-$AlCl_3$ series showed interesting properties [252]. As expected from the pure halide glasses, chloride addition lowered the viscosity of the melt and was found to increase the tendency to crystallize. At the same time, T_g increased with chlorine content suggesting that the strength of the bonding between the phosphate tetrahedra increases as chloride is added. Density displayed non-linear trends, similar to those observed for invert glass: despite the large molar weight of $BaCl_2$ or $SrCl_2$, sufficient MCl_2 salt begins to decrease the density (and molar refraction) indicating a loosening of the structure (depolymerization of the phosphate network). A clear inversion was found also in the thermal expansion data; upon the introduction of some MCl_2 rigidity increases (as the CTE decreases) and then sharply decreases (as the CTE increases) thereafter [252].

The complexity of the underlying dynamic properties was demonstrated on the system of $NaPO_3$-Na_2SO_4-NaCl by linking ion mobility and local plastic flow events [256] (Fig. 22). Starting from the pure sodium metaphosphate, additions of either sulfate or chloride salt transform the glass into an ionic material and moderately enhance ionic conductivity. The distinct structural roles of sulfate or chloride anions became visible when comparing the activation energies of charge carrier formation and shear transformation. For halophosphates, a positive linear trend between both quantities was found, signifying that local plastic deformation is strongly linked to charge carrier formation. Yet, in the sulfophosphate glass, a more narrow negative relationship indicated more homogeneous flow, potentially linked to the higher polarizability of the sulfate anion [256].

Although less studied than haloborates and halophosphates, it is possible to add large amounts of halides to germanate [257], telluride [103,258–261] and antimonate glasses [262–266] as well. Kozhukharov et al. [103] reported several glasses in the TeO_2-R_nH_m and TeO_2-R_nO_m-R_nH_m systems, where R_nO_m is a metal oxide and R_nH_m is a non-transition metal halide. Increased halide content was found to dehydrate the glass and increase transmission by reduction of absorbing -OH groups as a result of reactive evaporation. Replacement of TeO_2 by PbF_2 and $PbBr_2$ increases density, while addition of $PbCl_2$ has the opposite effect in the binary TeO_2-$PbCl_2$ system. Dubois et al. [263] report many glass-forming compositions based on $(1 - x)Sb_2O_3$-xR_mX_n, where $0.05 < x < 0.60$, R = alkali, alkaline-earth or transition metal and X = F, Cl, Br or I. These glasses are generally resistant to atmospheric water, and exhibit refractive indices greater than 2. Unlike pure halide glasses, however, they are not highly transparent in the UV nor IR regions (with some being yellow or orange, depending on the halide species). Binary CuI-Sb_2O_3 glasses require rapid quenching, but with the addition of a third component can be cast normally. Due to their large thermal stability range they are more suitable for fibre-drawing than other halide glasses [264]. Nonetheless, bulk pieces of halide antimonates without crystals [265], dark inclusions, bubbles or other inhomogeneities are challenging to synthesize [264].

On a finally note, the introduction of nitrogen into oxide glasses greatly improves the physical properties: the replacement of a divalent O^{2-} by N^{3-} results in a higher bond density within the glass [267]. When high pressures and a glass forming nitride are used, it is possible to form stable ionic glasses with only N^{3-} as the glass former. Grande et al.[267,268] demonstrated several glasses in the Li_3N-Ca_3N_2-P_3N_5 system, which are air-stable, $n \approx 2$, and have hardnesses exceeding that of silica.

2.5. Superionic gasses

Unlike the other categories of ionic glasses in this review, superionic materials are not a distinct chemical class of materials. Instead, they are defined roughly on the basis of their ion conduction performance: they have exceptionally high ion conductivity, typically $\sigma > 10^{-4}$ S/cm at room temperature, and a low activation energy for ion transport, < 20 kJ/mol [12,219]. The term *fast-ion conductor* is also sometimes used for these materials; for a full review of the many possible compositions, *see* Minami [219] and Tuller et al. [269]. Some of the most prominent superionic conductors are typical ionic glass formers, *e.g.*, halides, borates, phosphates, sulfates, and/or glass-forming sulfides (GeS_2, P_2S_5, SiS2, B_2S_3) to which large amounts of silver, lithium, copper or sodium ions are added. Since they all seem to follow similar fundamental principles of chemical design, here, we treat them jointly as a distinct group of ionic glasses, although there is some clear overlap with the previous chapters.

Silver-cesium mixed-halide glassy salts were found to have conductivities of 10^{-4}– 10^{-3} S/cm at RT with crystalline inclusions improving the σ significantly (10^{-2}– 10^{-1} S/cm); the chloride-bromide glasses were extremely unstable to crystallization [270]. Kadono et al. [271] reported $CuCl_xBr_{1-x}$ glass to have σ of 10^{-3} at RT, while LiI based glasses were much lower, only 10^{-6} at RT; Li conductivity improved with increasing size of the anion while Cu^+ conduction appeared indifferent to halide identity. The addition of halides to alkali- and silver-containing borate glasses improves the ionic conductivity by several orders of magnitude. Even in the lithium diborate system, 10^{-3} S/cm at 200 °C can be reached at only 35 mol% Cl replacement [272]. The effect can be even more pronounced in phosphate glasses, where σ can be improved (E_a decreased) by introducing more mobile species or using a larger halide anion [273], for example, when adding AgI to a silver metaphosphate matrix [31,32]. However, there exists a limit to improvement, with σ saturation being explained by NBOs no longer being formed and/or liquid-liquid immiscibility with subsequent halide crystal formation [272,273].

Occasionally iodide is added to a chloride or bromide melt, but reported data are sparse given the increased hygroscopicity imparted by the iodide compound. Yet, there exists a wholly unique family of glasses which have large amounts of iodide, the chalcohalides [274–278]: GeSI, GeS_2-GaI [279], $Te_2Se_nIAs_m$ [238,280], GeTeAsSeI [159], AsSI-SbSI [281,282], and $Ag_7I_4AsO_4$ [84]. Similarly, one can find iodide with other glass formers, such as LiI-Li_2S-P_2S_5 [283], CuI-CuO-P_2O_5 [284] or CuI-Sb_2O_3 [264] (for a more complete list, *see* [219]). The structure of these surprising glass compositions indicates they are indeed ionic glasses (see Section 5). However, stronger arguments for the classification of this family of compositions as being *ionic* are found when observing their optical properties, dynamic behaviour near T_g (Section 6.4) and ionic conductivity. Ingram et al. have called $Ag_7I_4AsO_4$ glass an optimized conductor as its conductivity approaches that of the best-performing solid electrolytes, such as crystalline $RbAg_4I_5$. Although the

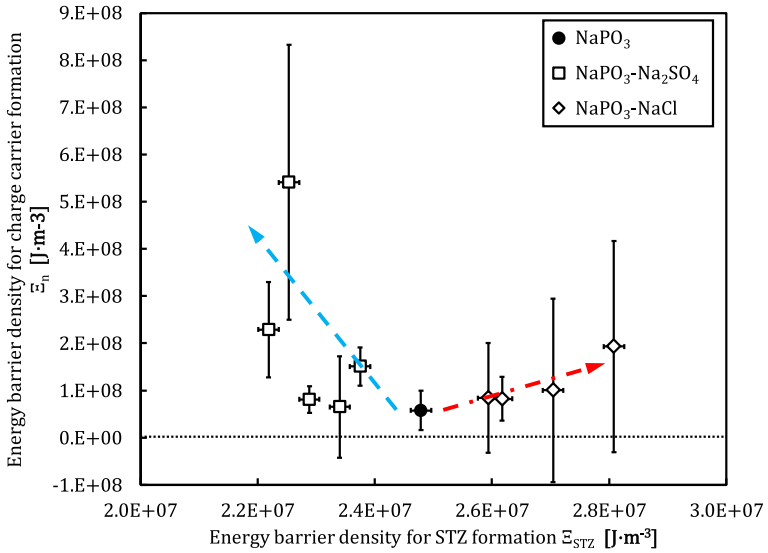


Fig. 22. Energy barrier density for the formation of charge carriers and shear transformation zones, respectively, in the $NaPO_3$-NaCl-Na_2SO_4 glass system. Reproduced with permission from Poletto Rodrigues et al. 2019 [256]. © The Authors.

iodide itself is immobile, the high polarizability and low charge of the iodide creates a loose environment, enhancing the mobility of Ag^+ (most Ag is found in the iodide region), and conductivities of 10^{-2} – 10^{-3} S/cm^{-1} at room temperature are obtained [84]. Moreover, like other halides, the GeS_2-GaI system was found to have light transmission extending deep into the IR region, down to ~ 20 μm [279].

In addition to the many superionic compositions containing silver and iodide, sulfide-based and oxysulfide-based glasses have shown promise for solid-state electrolyte applications, with conductivities up to 5×10^{-3} S/cm [13]. Elemental sulfur has a high theoretical capacity, but shows difficulties for practical use due to the formation of polysulfides [13], making oxysulfides the more attractive option. Oxysulfide glasses with large amounts of conventional glass formers, Li_3BO_3 (25 mol%), Li_2SiO_4 (20 mol%), Li_3PO_4 (40 mol%), Li_4GeO_4 (15 mol%), added to the $0.6Li_2S$-$0.4SiS_2$ system were reported with σ in the superionic range, and possibly contain Si_2S_6O units (with one BO) which prevent crystallization [33]. Using NMR to study lithium-containing oxysulfide glasses, evidence was found for structural units with both bridging oxygens and non-bridging sulfurs, which may impart higher resistance to crystallization and Li^+ mobility, respectively [34]. Taking full advantage of the compositional flexibility of glassy materials, various compositional adjustments can be used to further improve σ, for example, small additions of other components such as Li_4SiO_4, replacement of P_2O_5 by P_2S_5, or of GeO_2 by GeS_2 (utilizing the mixed-former effect) [33,34].

In a more general perspective, ionic conductivity studies of superionic glasses have allowed scientists to probe open questions about ionic conduction in all glass types, such as the relationship between σ_{dc} and carrier concentration [285], mismatch effects in mixed cation systems [286,287], the ion site memory effect [288], the role of free volume and/or *connective tissue* [289–292], elastic strain energy [292], and relaxation via transport processes [12]. For example, the migration mechanism of silver in RI-$AgPO_3$ and RI-Ag_2O-B_2O_3 compositions (R = Na^+, K^+, Ag^+, Pb^{2+} or Hg^{2+}) has been explored by many authors; studies discuss the formation of α-AgI-like clusters [31,293,294], simple free volume expansion [289,291], random distribution of iodide without clustering [295–297], or jumps of Ag^+ between mixed oxoanion sites [298].

2.6. Summary

Several basic properties of the glass compositions reviewed in Section 2 and whose structures and properties will be discussed further in Sections 5 and 6 are tabulated in Table 1. Given the chemical complexity of some of the glass compositions, for example the fluoride systems, some of the compositions are simply described by the names of their chemical components without information about stoichiometry. Additionally, one can find short preparation details, which we have limited to crucible material, atmosphere and melting temperature. In the case of the other properties, only a range of values is given, thus the readers are invited to refer to the original study for more information.

3. Classification criteria

A purpose of this review is to both define and easily *identify* ionic glasses. In Sections 2, 5 and 6, we discuss the compositions, structure and properties, respectively, of ionic glasses, which can help identify ionic glasses on a phenomenological basis. Nonetheless, there is still a further need for a clear-cut definition which allows new glass compositions to be classified as ionic, or at least, which can describe trends in ionicity within a given glass series. Several different criteria exist to categorize inorganic glasses; three common, relevant scales are average ionicity, polarizability and optical basicity [310,311]. Ionicity versus covalency is a complex and perhaps overused scale, yet it remains popular due to its relative ease of use. Optical basicity and polarizability can be related directly to the experimental refractive index of a glass and thus can be empirically determined. Ionic glasses appear well-suited for these types of quantification: to a first approximation their refractive indices, molar refractivities, polarizabilities, and non-linear optics are well-correlated with compositional parameters [91,311,312]. In spite of this, most studies have concerned trends in optical basicity and polarizability only in oxide glasses, indicating that more work should be done to extend these decidedly practical approaches to other types of glasses [310,313]. As a starting point, many halide glasses have had their optical properties characterized [162,194,235–237].

3.1. Ionicity

It is pertinent to first discuss the definition of *ionicity* before considering ionic glasses further. Catlow and Stoneham [314] have reviewed the many scales of ionicity; they caution using the terms ionicity and covalency indiscriminately, however, they also convincingly demonstrate the usefulness of the terms when applied appropriately. The authors evaluated the applicability of ionicity scales and find them to be

Table 1
Basic properties of important types of ionic glasses (superionic glasses are spread throughout).

Composition[c]	Preparation details[b]: Crucible material – atmosphere – temperature	T_g (°C)	Density (g/cm^3)	Young's Modulus (GPa)	Bulk Modulus (GPa)	CTE (10^{-6}·K^{-1})	σ (S/m)[a]	Ref.
Invert glasses – highly modified conventional glass formers								
K_2O-CaO-SrO-SiO_2	Pt – air – 6 hrs at 'suitable		2.72 – 3.40					[28]
Na_2O-K_2O-CaO-BaO-SiO_2	temperatures'		3.11 – 3.52					
PbO-K_2O-CaO-SrO-SiO_2			3.96 – 6.54					
TiO_2-K_2O-CaO-SrO-SiO_2	Quartz – air – 3 hrs at 'suitable		3.12 – 3.41					[299]
PbO-TiO_2-K_2O-CaO-SrO-BaO-SiO_2	temperatures' between 1460 and 1550 °C		3.16 – 3.76					
$(2.5-x)$CaO-$x/3M_2O_3$-$2SiO_2$ (M = Y, La, In, Al, Ga)	Pt – air –1550 °C	758 – 870						[56]
xPbO-$(1-x)SiO_2$		307 – 466						[57]
MgO-SiO_2 series	tungsten – air – 2327 °C	717 – 762						[300]
MgO-Al_2O_3-SiO_2 series								
xPbO-$(1-x)P_2O_5$	porcelain – air – 800 to 1000 °C		3.48 – 5.3				10^{-10} –10^{-5} (25 – 500 °C)	[21]
xZnO-$(1-x)P_2O_5$		424 – 461	3.14 – 3.77					[29]
$(34-x/4)$CaO-$(26-x/4)Na_2O$-xMgO-$(40-x/2)P_2O_5$	silica – air – 1350 °C		2.674 – 2.808					[62]
$50Na_2O$-xAl_2O_3-$(100-x)P_2O_5$	Pt – air – 800°C to 1500 °C	305 – 395	2.537 – 2.635					[55]
$(50-x)$CaO-xMgO-$50P_2O_5$	SiO_2 – 1100 to 1250 °C 2 h	535 – 564	2.441 – 2.64	53.9 – 61.4				[301]
$20Na_2O$-44ZnO-$(30-x)P_2O_5$		585 – 602	3.089 – 3.152		47.57 – 51.36			[212]
Oxoanions (nitrates, sulfates, carbonates, hydrates) + conventional glass formers								
xCa$(NO_3)_2$-$(1-x)KNO_3$	vacuum – 400 °C	62 – 81						[78]
xZnSO_4-$(1-x)K_2SO_4$	Pt – N_2 – 600 °C	184 – 232	2.84 – 3.1	5.423 – 5.635			$< 10^{-4}$	[90]
ZnSO_4-K_2SO_4 mixtures	Pt – N_2 – 607 °C	451 – 508	2.732 – 3.080		8.608 – 10.2383			[91]
$(D_2O)_{0.67}(NH_3)_{0.33}$	NH_3 gas – T_m < 170 K - quenched w/ liquid N_2							[124]
$50Li_2O$-xSO_3-$(50-x)P_2O_5$	porcelain – air – 800 °C	256 – 300						[93]
42ZnO-$20Na_2O$-$(38-x)P_2O_5$-xSO_3	Pt – air – 800°C		3.11 – 3.22				1×10^{-8} – 1.58×10^{-7}	[95]
42.2ZnO-19.79Na_2O-xSO_3-$(100-x)P_2O_5$	Pt – air – 700 °C	329 – 302						[53]
Li_2SO_4-Li_2O-P_2O_5 series	porcelain – air – 850 °C	267 – 318	2.33 – 2.36		29.4 – 46.5			[99]
xAg$_2SO_4$-$(100-x)AgPO_3$	silica – vacuum – 700 °C						1.99×10^{-5} – 3.31×10^{-4}	[100]
Magnesium carbonate–phosphate glass	sealed Pt tube – 1400 to 1550 °C for 3 – 6h							[302]
Halides								
Distilled BeF_2	under vaccum – 850 °C		1.968					[130]
Distilled BeF_2	HF-H_2 gas mixture at 850 °C	250	1.982					[133]
Undistilled BeF_2	HF-H_2 gas mixture at 850 °C	285	1.995			6.8	$6.7*10^{-17}$ at 150 °C	[133]
ZrF_4, BaF_2, LaF_3, ThF_4, AlF_3, NaF, HfF_4, YF_3, LiF, ZnF_2, CaF_2, BaF_2, SrF_2, CdF_2, YbF_3		256 – 450	4.0 6.43			10.9 – 20.5		[145]
$Zr_{0.60}Ba_{0.33}Th_{0.07}F_{3.34}$			4.8			4.3(57°C)		[190]
$Hf_{0.60}Ba_{0.33}Th_{0.07}F_{3.34}$			6.2			11.38 (227 °C)		
ZrF_4, BaF_2, LaF_3, AlF_3, NaF, LiF, HfF_4, ZnF_2, YbF_3, ThF_4, PbF_2, MnF_2, GaF_3, YF_3		515 – 623						[206]
ZrF_4-BaF_2-LaF_3-NaF-AlF_3				57.41 – 59.65				[211]
ZrF_4-BaF_2-CsF series	Pt – N_2 – 950 °C for 20 min	228 –313	4.16 – 4.64		76 – 100	16.1 – 17.8		[214]
ZrF_4, HfF_4, ScF_3, ZnF_2, GaF_3, FeF_3, InF_3 (glass formers); many metal fluoride additives	Pt – N_2 (with NH_4F·HF added)– 600–950 °C for 10–30 min – plate quenched	155 – 360					$10^{-2.6}$ – $10^{-5.74}$ at 200 °C	[205]
$ZnCl_2$ glassy film	rapid cooling in a dry air stream within dry chamber	102						[129]
$ZnCl_2$	HCl gas followed by dry N_2 flush-quench in dry liquid N_2	122.3				6.6	1.3×10^{-12} at 150 °C	[134]

(*continued on next page*)

Table 1 (*continued*)

Composition[c]	Preparation details[b]: Crucible material – atmosphere – temperature	T_g (°C)	Density (g/cm^3)	Young's Modulus (GPa)	Bulk Modulus (GPa)	CTE (10^{-6}·K^{-1})	σ (S/m)[a]	Ref.
(45-x)KCl-xNaCl-55BiCl	chlorine gas atmosphere	29 – 45						[162]
30NaCl-30KCl-40$ThCl_4$	Pt – N_2 – 650 °C for 15 min	130	3.52					[165]
LiX-KX-CsX-BaX_2 (X = only single-halide Cl, Br, I)	silica crucible – N_2 – 450 to 550 °C for 15 mins	40 – 74					5.3×10^{-9} (Cl) – 5.8×10^{-7} (I)	[174]
xKCl-(1 − x)$BiCl_3$	sealed Pyrex tube – Cl_2	25 – 45						[194]
$ThCl_4$-$PbCl_2$-RCl, R = Na, Ag, Cu	N_2, carbon and red light	110 – 143	3.72 – 4.05					[192]
$ZnBr_2$	Pt crucible – N_2 – 500 °C for 15 min, plate quenched	122	3.77					[303]
$ZnBr_2$ with 0.5 – 1 mol% other oxides, sulfides and nitrides	Pyrex ampoule – N_2 – 450 °C, cooled in air	126 – 130						[183]
CdF_2, 50 BaF2, 20ZnF_2, ThF_4, $CdCl_2$, $BaCl_2$, 10KCl, 10NaCl	open Pt crucible; long Pt tube – N_2 or air – casting	170 – 354	4.0			21.5		[160,196]
ZrF_4-BaF_2, with $BaCl_2$		261 – 300	4.312 –4.662				$10^{-4.7}$ – $10^{-5.2}$ (200 °C)	[230]
ZrF_4-SrF_2-MCl, M = Li, Na, K, Rb, Cs	Pt crucible – air – plate quenched	218 – 286	4.00 – 4.35			15.3 – 18.2		[182]
PbX_2 (X Cl, Br, I), including pure vitreous PbI_2, and AgI, KCl, KBr	Covered vitreous silica crucible – dry, flowing N_2 – 450 to 650 °C for 20 mins, plate quenching	48.1 – 83.1						[187]
$PbBr_2$-PbI_2-Cu(Br)Cl-KBr	alumina crucible – N_2 – slowly heat up to 500 °C for few mins, quenched in air	51 – 72						[197]
CsAg mixed-halide (Cl, Br, I)	Rapid cooling of molten mixtures	40 – 65					1.0×10^{-4} – 2.8×10^{-1}	[270]
$TeCl_2$, Te_3Cl_2, $TeBr_2$ and TeI_2 with small amounts of sulfur	sealed Pyrex tube – vacuum – 300 °C for several hours, air cooled	75 – 83				31 – 74		[171]
$ZnBr_2$-KBr-MBr_2 and $ZnBr_2$-KBr-MBr_2-AgBr, M = Ba, Pb	Pt – nitrogen – > 450 °C for 20 min, vigorous stirring – plate quenching	30 – 85						[185]
Haloborates, halophosphates, halosulfates, halosulfophosphates and others								
90.1 AgI-9.9Ag_3BO_3	sealed silica tube – 500 to 600 °C for 30 min					19.0 – 24.5		[304]
B_2O_3-BaF_2, with LiF LiCl, or LiBr	porcelain – 950 °C for 40 min	405 – 500	3.402 – 3.562	58.32 – 67.54			1.19×10^{-10} –7.09×10^{-13}	[245] [246]
82AgI-13.5Ag_2O-4.5B_2O_3	silica - air – 600 °C	80					10^{-2} – 10^{-1}	[14], [15]
(100 − x)$NaPO_3$-$xAlF_3$	silica – 800 °C for 1h Pt – 850 °C for 30 min	260 – 405	2.52 – 2.71					[305]
$NaPO_3$-BaF_2-REF_3 (RE = rare earth)	open Pt crucible – air and N_2 – 300 °C, up to 850–1000 °C, down to 750 °C for 30 min, preheated molds	254 – 302	3.33 – 4-06			19.0 – 23.2		[253]
(100 − x)(38.1SiO_2:55.5CaO:6.3P_2O_5)-xCaF_2, or -xSrF_2	Pt/Rh – 1 h at 1550 °C –quench in cold water	558 – 790 562 – 783						[201]
(100 − x)$NaPO_3$-$xZnCl_2$	glazed porcelain – 650 to 850 °C		2.525 – 2.720		40.93 – 59.27		1.303×10^{-11} –2.842×10^{-7} (393 °C)	[251]
(100 − x)$AgPO_3$-xAgI		353 – 459	4.35 – 5.41				10^{-4} – 10^{0}	[31] [32]
(100 − x)AgI-xAg_3PO_4	quartz – 900 °C for 15 min	54.0 – 94.6	6.06 – 6.45		10.42 – 16.08			[295]
(100 − x)$AgPO_3$-xAgI	porcelain – 600 to 700°C 2h	74.85 – 162.85						[297]
x(2LiCl)·y(Li_2O)·(100 − x − y)P_2O_5 and xAl_2O_3·y(2LiCl)·(100 − x − y)P_2O_5	flowing, dry N_2 – 650–700 °C – plate quenched in air	2.1 – 2.62					$1\times10^{-3.58}$ (120 °C)	[292]
PbF_2-$PbBr_2$-P_2O_5 or $PbCl_2$-$PbBr_2$-P_2O_5	porcelain – air – 600 ° for 15 mins or 450–470 °C for 20 mins	210 – 281; 146 – 190	5.28 – 5.81; 4.75 – 5.46					[255]

(*continued on next page*)

Table 1 (*continued*)

Composition[c]	Preparation details[b]: Crucible material – atmosphere – temperature	T_g (°C)	Density (g/cm^3)	Young's Modulus (GPa)	Bulk Modulus (GPa)	CTE (10^{-6}·K^{-1})	σ (S/m)[a]	Ref.
$NaPO_3$-$Al(PO_3)_3$-MF_n, M = Li, Na, Ca, Ba, Ga	Pt crucible – air – brass mold near T_g	285 – 354				18.2 – 24.3		[254]
$(100-x)$NaF-xZnSO_4	sealed glassy carbon – air – 873 to 973 K for 30–40 min	212 – 247	3.188 – 3.446			17.1		[188]
$NaPO_3$-AlF_3-Na_2SO_4	Pt – 850 to 1000 °C	289 – 412	2.5 – 2.753	36.9 –70.5				[96]
0.8TeO_2-0.2$TeCl_4$; 0.8TeO_2-0.2WCl_6	covered alumina crucible – air – 300 °C for 10 mins, 700 °C for 10 mins, RT steel molds	330 – 350	5.267 – 5.870					[259]
TeO_2-PbO-$PbCl_2$	covered Pt crucible – air – 700 to 800 °C for 10 min	189 – 273	5.581 – 6.680	29.04 – 38.74	24.77 – 28.52			[258]
TeO_2-R_nH_m and TeO_2-R_nO_m-R_nH_m; R_nO_m = metal oxide and R_nH_m = non-transition metal halide	Au or Pt crucible – dry air flow – 873 K – poured unto copper plate	225 – 300	5.5 – 6.35			13.0 – 22.0		[103]
$(1-x)Sb_2O_3$-xR_mX_n; R = alkali, alkaline-earth or transition; X = F, Cl, Br or I	covered alumina or glass carbon – air – 1000–1200 °C for 5 min – quenching and casting	165 – 330	5.1 ($PbCl_2$)			18.8 – 22.7; 2–4 ($PbCl_2$)		[263,264]
Li_3N-Ca_3N_2-P_3N_5	High pressure, sealed Pt ampoules – vacuum – 1000 for 30 °C min, 1400 °C for 20 min – rapid quenching	> 700						[267,268]
Chalcohalides or halochalcogenides								
$(1-x)Ga_2S_3$CsCl-xGa_2S_3·GeS_2	quartz ampoule – vacuum – 1100 °C for 5 hours – air or water quenched	30 – 120				34 – 44		[275]
GeS_2-Ga_2S_3-$MnCl_2$	silica ampoules – vacuum – 950–970 °C for 3 hours – quenched in cold water	305 – 423						[306]
$Ag_7I_4AsO_4$	die - 75 MPa to 375 MPa – RT	65					10^{-1} (65 °C)	[84]
GeS_2-Ga-I series		163 – 400	3.14 – 4.2				2.40×10^{-7} – 5.13×10^{-3}	[279]
AsS-SbI series	silica ampoules – vacuum of 10^{-3} mbar – 750 °C for 20 h		3.654 – 3.983	20.82 – 12.83				[282]
As_2Se_3-GeTe-AgI		149 – 240	4.661 – 5.234			14.906 – 21.788		[158]
GeTeAsSeI	Sealed distillation ampoule – vacuum – 120 °C for 4 h, distilled 300–800 °C	182 – 203	3.95 – 4.17					[159]
Amorphous metal organic frameworks								
a-ZIF-4, *a*-Zn(Im)$_2$ (GIS), *a*-ZIF-62, *a*-ZIF-76	Ar – 437 to 590 °C	287 – 343	1.625(2) – 1.671(1)	6.4(0.4) – 8.5(0.2)			10^{-3}	[307]
a-ZIF-4, *a*-ZIF-62, *a*-ZIF-76, *a*-Zn(Im)$_{1.33}$(mbIm)$_{0.67}$	Ar – 437 to 590 °C	292 – 318		6.12– 6.89				[308]
a-ZIF-62	Ar – 500°C	322	1.566					[309]

[a] at room temperature, if not otherwise stated.
[b] if a complex melting profile was used, only maximum dwell temperature is stated.
[c] composition in the same form as originally stated in the reference.

useful tools for understanding trends in physical properties within classes of materials, such as molten salts. At the most basic level, ionicity is the difference in electronegativity between a cation an and anion (or a metal centre and a ligand) as defined by Pauling's thermochemical electronegativity scale [315]. There are several different scales of electronegativity; including Phillips' spectroscopic electronegativity scale and some other, less-known ionicity scales based on the effective charges of ions [314,316,317]. Alternatively, a comprehensive list of electronegativities, including those of many polyanions, was calculated by assuming that "*the attractions of the two nuclei for the valence electrons become equalized*" [quoted from [318]].

Overall, as one might expect, based on either Pauling's [315] or Phillips' [316] scale crystalline alkali halide salts are the most ionic, followed by metal oxides, with chalcogenide compounds being the least ionic. Tetrahedrally coordinated F^- and O^{2-} anions are particularly close in size (131 and 138 pm, respectively [319,320]); furthermore, both have high electronic affinities and would be expected to form ionic glasses. Interestingly, the electron affinity of oxygen for the first electron is also comparable to that of fluorine, while the acceptance of the 2nd e^- is much more destabilizing for O^{2-}, perhaps highlighting how electron density on oxygen determines ionicity: while O^- makes relatively covalent bonds, O^{2-} forms purely ionic bonds. Other studies find ionicity to increase with the amount of modifier in silicate crystals [314]; thus, average ionicity appears to be a good first approximation for categorization of ionic crystals and, accordingly, ionic glasses.

In the field of ionic liquids (Section 8), ionicity has been defined "*as the effective fraction of ions available to participate in conduction*" [quoted from [321]], and quantified by the adherence of the conductivity to the Nernst–Einstein equation, with molten salts having only a 10% deviation from this equation (high ionicity) and other ionic liquids with longer organic cations having 20 – 50% deviations (more ion pairing, lower ionicity) [321,322].

3.2. Electronic polarizability

In the limit of low electric field strength (linear elasticity), polarizability is simply the proportionality coefficient between an external electric field and the dipole moment in the atom or molecule it induces [323]. Given that a dipole moment depends on the charge and separation distance of said charges, it is clear that the electronic polarizability depends on the formal charge and the ease of distortion of the electron function (which is related to the separation distance). For some elements, there also exists highly polarizable electron pairs found off the bond axis that must also be taken into account when discussing polarizability; specifically, the s-orbital lone pairs of heavy metal cations such as Sn^{2+}, Sb^{3+}, Te^{3+}, Bi^{3+} and Pb^{2+} as well as s-orbital electron pairs in the transition metals Ti^{4+} and Zr^{4+} can substantially affect the polarizability (and refractive indices) of oxide bonds involving those elements. In the same electric field, electron clouds of species with higher polarizabilities deform more easily compared to those with lower polarizabilities. We assume that for a given element bonds with higher polarizability are *less* directional (more distortable) and therefore *more* ionic. Measuring the average linear (and, at high electric field, such as lasers, also non-linear) polarizability of a glass is important technologically, in particular, for optical applications, where this quantity can be used to predict and tailor static properties [313,324,325]. For example, the average electronic polarizability of the O^{2-} anion in oxides very strongly correlates with refractive index and molar volume [326].

3.3. Optical basicity

In conventional oxide glasses, the oxygen anion typically donates an electron pair to a neighbouring metal centre, making it a Lewis base [327]. The oxide–metal bond is a donor–acceptor (or dative) pair, which generally has some covalent character. Duffy and Ingram [317] were the first to quantify this behavior by extending the theory of optical basicity (Λ) to glasses. They used changes in the electronic structure of colour centre probe ions, such as Th^+, Pb^{2+}, and Bi^{3+}, to quantitatively compare (on the basis of UV-Vis spectra) the electron-donating properties of different glass media [317,328]. The resulting optical basicity scale correlates the electron-donating character of the oxide to the frequency shift between the glass-incorporated ion and a free ion. The frequency shift is normalized over the *possible* frequency range of the *spectroscopic indicator* (between zero basicity and maximum basicity, *i.e.*, zero and maximum electron-sharing), which allows Λ to be compared between different types of *spectroscopic indicators* and, thus, enables extension of the approach to other glass systems, including non-oxides. The optical basicity concept has been applied to glassy oxides as well as fluorides and sulfides, allowing it to be calculated for many different types of glass compositions [329–331]. Although Duffy [329] cautions against comparing Λ between different anions on the same scale (such as oxides compared with fluoride and chloride) he was able to include fluorine in his Λ calculations after the use of appropriate *basicity moderating parameters*. Overall, optical basicity is an easily accessible parameter for most glass compositions; however, its relation to ionicity must be used with caution. In particular, this applies for cases in which chemical selectivity is known or expected to cause non-random ion distributions and, thus, deviations from the underlying mean-field approximation. In such cases, the local site basicity may very strongly differ from the expected average basicity.

There is an established positive correlation between polarizability (as measured from refractive index) and optical basicity, especially for oxides [312,324,326,330]. This has been exploited in the case of germanate glasses, where the typical spectroscopic probe cannot be used [332]. Generally, for sp-block metal-oxide glasses, optical basicity and polarizability are positively correlated with the electron density on the oxygen, *i.e.*, its ionicity. The drawback to this approach is that it is possible to have a highly ionic bond without high polarizability, *e.g.* NaF; PbO provides an example of a highly polarizable, non-ionic crystal. Nonetheless, several papers have been written regarding the empirical correlations of *average* electronegativity with *average* polarizability and optical basicity [310,324,333], proving that within certain limits optical basicity can be a reasonable measure of ionicity. Indeed, when transition metals are excluded, there is a strong inverse correlation between the optical basicity of binary oxides ($MO_{z/2}$) and the electronegativity of M [330].

Within a given compositional glass series (holding the identity of the anion constant), it is possible to use the trends in Λ as a measure of ionicity. For example, Λ (and therefore polarizability and ionicity) was found to correlate positively with the amount of modifier in conventional glass-formers, and also generated the expected ionicity trend in alkali borates ($K^+ > Na^+ > Li^+$). It was found that sodium metasulfate and metaphosphate had lower optical basicity compared to sodium metaborates, with sodium metasilicates being the most optically basic [317,331]. This indicates that the oxide bond is the most polarizable, *i.e.*, most ionic, in silicates, which agrees with the ionicity calculated from Pauling's electronegativity. This trend may also help explain the much higher solubility of sulfate in heavily-modified phosphate and borate compositions compared with low sulfate miscibility in silicate glasses (Section 2.2.2).

3.4. Summary

Based on oxide polarizability and optical basicity, Dimitrov and Komatsu [310] separated glasses into three categories that look largely similar to Sun's [334] or Dietzel's [18] classic categories: semi-covalent acidic oxides (conventional glass-formers), ionic or basic oxides (intermediate oxides) and very ionic or very basic oxides (modifier

oxides). Of course, there are some stark exceptions with SrO, BaO, CdO and PbO having the highest polarizability and optical basicity values, indicating incorrectly that they are also the most ionic or basic oxides (however, these oxides also have significant d-shell character). Nonetheless, the fact that either ion polarizability or optical basicity can nearly recreate the typical modifier and former distinctions for oxides, makes them worth considering when attempting to classify glass types. As a result, we highly encourage scientists investigating ionic glasses to measure the experimental polarizability and calculate the theoretical optical basicity of their glasses as a starting point in determining ionicity. Furthermore, these parameters can be used to explain trends in optical and electrical properties [335–338].

4. Glass formation in ionic glasses

As pointed-out in Section 2, with the exception of simple glassy salts, most ionic glasses have complex compositions; few single-component ionic glasses are known, and for those which are, *i.e.*, BeF_2 and $ZnCl_2$, their structures have been determined to be more polymeric in nature. The formation of typical ionic glasses has been discussed on "*the basis of (a) differential field strengths of cations, (b) distortion of anions and (c) the low liquidus temperatures*" [quoted from [90]].

The first criterion for forming ionic glasses, (*a*), requiring cations of different field strengths, appears to apply for all types of ionic glasses, in which both addition of more components and increased chemical difference between components decrease the immiscibility gap (*i.e.*, increases glass forming ability). The formation of invert glasses has been attributed to geometric frustration (as found in metallic glasses [339]) following the 'principle of confusion', wherein competition between several accessible crystal structures 'frustrates' crystal growth. In contrast to a case where only one crystal structure is preferred and crystallizes easily out of the melt, inefficient packing of different and distorted polyhedra requires more time to achieve the required space-filling geometries for crystallization.

Mysen [340] points out that glasses with large compositional variety have large entropies of mixing (sometimes called configurational entropy), and therefore will lose entropy from the melt more slowly, thereby increasing glass forming ability. The liquid will retain sufficient entropy when cooled through the empirical glass transition temperature (this assumes all materials have a glass transition if cooled sufficiently fast). Mysen [340] uses the example of forsterite (Mg_2SiO_4) which is notoriously difficult to vitrify; its configurational entropy is near zero at the same temperature where the viscosity is sufficiently high to prevent further rearrangement (*i.e.* T_g). Only by rapid cooling (with a predicted rate of ~ 700 °C/min) could vitreous Mg_2SiO_4 be formed from a liquid at a sufficiently high fictive temperature, creating a structure with high configurational entropy. Using containerless levitation melting combined with laser heating Kalampounias et al. [341,342] managed to form glass along the $MgSiO_3$ and Mg_2SiO_4 compositional join; with the additional stability provided by Mg–O–Mg bonds, a transition from a more covalent $MgSiO_3$ glass (lower entropy of fusion) to a more ionic Mg_2SiO_4 glass (higher entropy of fusion) was discovered from vibrational spectra analysis. It appears that the configurational entropy conferred by the additional oxygen species (anionic oxygen, O^{2-}, located between Mg^{2+}) increases the glass forming ability. Similar ideas have led to the assumption that glasses which form from simple salts with low configurational entropy (rather than from conventional glass formers) might be good examples of ideal glass formers (and for the same reasons *bad* glass formers) [10].

The second criterion, (*b*), which requires distortable anions, appears to be particularly valid for oxoanionic glasses: carbonates are the most symmetric and least distortable oxoanion, and also have the smallest range of glass formation, while the glass-forming range of sulfates is significantly larger. On the other hand, much more research has been conducted on fluorides as compared to chlorides and bromides (possibly due to their higher hygroscopicity), so it is more difficult to judge whether distortability is important for glass formation in halides. Intuitively, one might readily expect that the ability to accommodate structural distortion facilitates structural flexibility and, thus, reduces the constraints of structural ordering (hence, increasing the number of possible configurational states).

Finally, the third requirement, (*c*), of a low liquidus temperature (T_l), has been used to compare the glass forming ability of different glass compositions by the calculation of the reduced glass transition temperature, $T_{rg} = T_g/T_l$ [343]. A higher T_{rg} (or lower T_l) implies higher viscosity at the nose of the cooling curves, making it easier to kinetically suppress crystallization. The T_g/T_l values of HMFGs and other polycomponent halides are often lower than those of conventional oxide glasses (e.g., 1/2 vs. 2/3) [164,206], so T_{rg} often predicts that ionic glasses are less stable against crystallization. There have been several different metrics for glass-forming ability involving melting, glass transition, kinetic fragility and crystallization temperatures, which could be of interest when looking to understand glass formation in ionic glasses [343,344].

5. Structure and topology of ionic glasses

In order to structurally define ionic glasses, we begin by briefly discussing the conventional glass structure theory, the continuous random network (CRN), as laid out by Zachariasen in 1932 [7] and already introduced in the first section of this review. Early glass theorists asserted that the local bonding found in crystals and glasses is the same and that the glass forming ability of any metal oxide can be determined from the comparison between the two structures, *i.e.*, less difference equals better glass-forming ability. This is stated in one of Zachariasen's conditions for glass formation: that "*the substance can form extended three-dimensional networks lacking periodicity with an energy content comparable with that of the corresponding crystal network*" [quoted from [7]]. His theory of the vitreous state required three-dimensional linking of glass-forming polyhedra: "*at least three corners of each oxygen polyhedron must be shared*" [quoted from [7]].

A contemporary, Hägg, pointed out in 1935 [345] the restrictiveness of Zachariasen's requirement of a 3-dimensional network between the oxygen polyhedra with the counterexample of orthophosphoric acid, a one-dimensional glass. Zachariasen later clarified that in H_3PO_4 the PO_4-tetrahedra are joined together through a hydrogen midpoint between the two tetrahedra [346]; thus, Zachariasen may have unknowingly included ionic glasses in his early definition; glass formers connected through modifiers could form 3d networks. Hägg [345] insisted on evaluating the propensity to glass formation by examining the melt; if the melt had strong interatomic forces and large or sufficiently irregular groups, then the construction of a crystal lattice would become less energetically favourable. Ionic crystals, where only small, mobile ions were present, would not form a glass easily; he turned out to be prescient about the necessity of large organic molecules as components of ionic liquids (Section 8).

Building upon the CRN model proposed by Zachariasen, several authors used X-ray scattering to obtain further information on glass structure, particularly when network modifiers had been added. Namely, Warren and Biscoe [347] found that Na^+ created NBOs and sat in the interstices of the silica network surrounded by, on average, six oxygen atoms. Greaves [26] went one step further and defined the modified random network (MRN, Fig. 23), in which the modifier cations form distinct ion channels. The MRN model is commonly used today to explain glass properties and behaviour, especially in regards to dynamic processes, *e.g.*, the mixed-alkali effect. For example, the lowered viscosity of modified glasses is attributed to shear forces which are assumed to occur along the interfaces between the ionic channels and covalent network regions [348,349]. The MRN requires there to be clear-cut distinctions between glass-former (covalent bonding) and glass-modifier (ionic bonding), such as by Dietzel's field strength parameter discussed in Section 2.

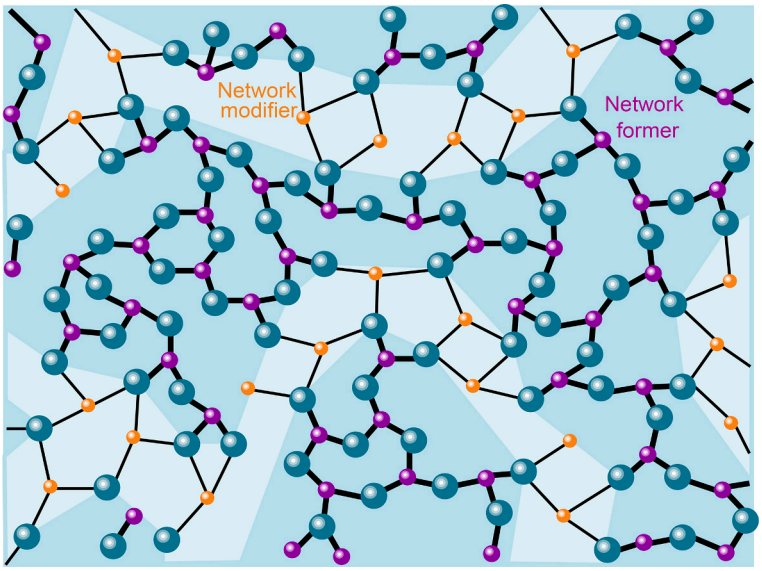


Fig. 23. Modified Random Network (MRN) for silicate glasses according to Greaves. Network forming cations are shown in purple, oxygens in teal and modifying cations in orange. Modifier channels are shown in light teal and the network component in dark teal. Reproduced with permission from Greaves [350]. © Mineralogical Society of Great Britain and Ireland.

As an alternative approach to understanding glass structure, the topological constraint model has been developed since the 1970s. It strictly applies to glasses with only covalent bonding and was originally applied to chalcogenide glasses [351–354]; it proposes that many physical properties of glass, such as thermal and some mechanical properties, can be accurately explained from network topology and rigidity alone. Using classical percolation theory, Phillips [351] and Thorpe [352] delineated a rigidity percolation threshold which differentiates rigid and floppy glass networks and, thereby takes coordination number into account. The constraint theory has since been successfully applied to systems with some ionic character, such as for predicting T_g, fragility and hardness in alkali borates [355,356] and alkali phosphates [357], however, it is known to be less successful for glasses with less directional, *i.e.*, ionic, bonds [25,358,359].

Other statistical models, *e.g.*, the Bernoulli statistical model [360,361] and the Boltzmann-weighted statistical mechanical model [362], are no exception: "*certain assumptions need to be fulfilled, including charge stability in the glass and that all modifier ions associate with the network, likely preventing the model from being applicable on invert glasses*" [quoted from [362]]. Simulating ionic glasses with molecular dynamics (MD) has a similar problem: an ionic system is less constrained and has more parameter variation leading to flatter energy minima. It is presently unclear which structural analysis method will lead to accurate models of *ionic topologies*, *i.e.*, which will show local connectivity among the ionic regions, and either large modifications of current paradigm or entirely different approaches may be required. It is possible that perspectives which emphasize the long-ranging character of ionic interactions, chemical selectivity, packing rules and/or structural heterogeneity will be helpful in this. Some of the further difficulty of modelling ionic glass systems in detail comes from the experimental challenges, as will be discussed in the next few paragraphs.

Connectivity in glass formers is usually quantified through the concentration of BOs and NBOs as measured by some direct or indirect spectroscopic technique: most frequently, SSNMR (*e.g.*, ^{31}P, ^{29}Si, ^{11}B and/or ^{27}Al) or vibrational spectroscopy are employed for this. In the case of ionic glasses, there are few conventional BOs, instead there are mostly anions cross-linked ionically by cations, or BOs in polynuclear anion species such as $P_2O_7^{4-}$. In addition, the local environments of typical modifiers are frequently difficult to study. Vibrational spectroscopy may provide access to cage-rattling modes of cation species [301]. The signal from NMR spectroscopy is either narrow (Li, Na) making it difficult to confidently separate distinct cationic sites, or very broad (K, Rb, Mg, Sr, Ba) thereby preventing accurate quantification as it is not possible to fully excite all sites in one scan. For example, ^{6}Li and ^{23}Na have been studied in a glass, but the chemical shift range for any Li or Na chemical environment is fairly narrow (roughly 2 and 30 ppm maximum, respectively [363]) making small changes in the spectra difficult to deconvolute.

Since common glass analysis techniques are not always suitable for ionic glasses, other methods must be employed to quantify connectivity and glass structure. Generally, diffraction techniques are equally sensitive for both covalent and ionic glasses (and ionic liquids), however, both covalency of bonds and polarizability of electron clouds can introduce difficulties with interpretation. Radial distribution function analysis (RDF), today called total scattering or pair distribution function analysis (PDF) from X-ray or neutron scattering yields the most structural information regarding long-range order, while extended X-ray absorption fine structure (EXAFS) spectroscopy only provides information about short (interatomic) distances [167]. However, the long-range disorder and macroscopic isotropicity result in only a one-dimensional correlation function, which makes the accuracy of the underlying fitting model very important. The shape and density of the electron cloud must be modelled accurately for the bonding type, otherwise the obtained coordination numbers and distributions of interatomic distances may be erroneous [167].

In addition to requiring varying and accurate theories to model ionic bonds, ionic bonding is simply less locally ordered than its covalent counterpart. Ionic bonds are more difficult to define — reported coordination spheres for many modifiers vary (for example K^+ in glasses has been found to have 5 – 9 neighbours [18,116,334,348,364]). Ionic bonds are no longer strictly directional, making many of our current structure-property models inapplicable. Among those is the concept of connectivity for network glasses — instead of network bonds, we must consider cross-linking cations as the linkers of the anions. On first view, this presents a large shift from the current perspective, where normally the structural role of the cations is considered secondary to that of the network former. New models of the structures of ionic glasses need to be explored and correlated with their macroscopic properties. Aside from some of the already mentioned principles, another simple starting point can be found in the application of packing rules such as those used for ionic crystals [3,365,366].

When discussing the different types of ionic glasses, it becomes clear that their structures are composed of ions (from the previous paragraphs, it is noteworthy that these ions can be composed of only one single atom or multiple atoms, such as the oxoanions SO_4^{2-} or CO_3^{2-}), and that they can be mononuclear (*e.g.*, PO_4^{3-}) or polynuclear ($P_2O_7^{4-}$).

Following the definitions of *invert* and *mineral* glass in Sections 2.1 and 2.1.1, respectively, ionic glasses must have a major fraction of Q^n-units with $n \leq 1$, such that the glass lacks any percolating network structure in the sense of CRN theory. These compositions are found in the immiscibility gap for alkali silicates [367] and can only be synthesized using a mixture of several cations (invert glass) or using intermediate cations such as ZnO or MgO, *e.g.*, 67MgO-33SiO_2 (Mg_2SiO_4) [341].

On the other hand, as laid out in Sections 2.4 and 2.5 phosphates and borates can easily form pyro- and ortho-compositions. As stated by Brow, polyphosphate compositions are comparable to silicate invert glasses, since their structures are composed of "*isolated tetrahedra and small molecular fragments linked through their non-bridging oxygens by larger, smaller valence ('non-glass forming') cations*" [quoted from [69]]. As the orthoborate composition is approached in alkali borates, Li^+ and Na^+ disrupts the borate network, converts BO_4 units into BO_3, and forms mostly pyroborate ($B_2O_5^{4-}$) with some loose BO_4 tetrahedra, metaborate (chains) and orthoborate (BO_3^{3-}). At high Rb^+ and Cs^+ content BO_4 units were retained, but with two NBOs per tetrahedron [67,368]. With a high cationic field strength MgO-B_2O_3 possesses "*boroxol rings and highly charged pyroborate triangular dimers*" [quoted from [369]]. This wide variety of possible structural units highlights borate's ability to easily form many different, highly charged, oxoanionic species.

In phosphate matrices, the addition of halide salts, such as AgX (where X = Cl, Br, I) does not appear to modify pyrophosphate ($Ag_4P_2O_7$) dimers, instead the presence of AgI_4 units was confirmed by PDF analysis and Raman spectroscopy [370]. The molar volumes of several different families of invert glasses and crystalline AgI show linear trends with respect to AgI concentration (Fig. 24). It has been argued that since the slopes of the silver invert glasses are similar to α-AgI, AgI is simply sitting in the interstices of the glass network; this seems to definitely be the case for pyrophosphate and pyroborate ($Ag_4P_2O_7$ and $Ag_2B_4O_7$), but less true for metaphosphate and metaborate compositions ($AgPO_3$ and Ag_3BO_3). It is likely an indication that adding silver halides, or any halide salt, continues to disrupt the network, *i.e.*, convert BOs into NBOS, at the meta-composition. Yet by the pyro-composition, ionic halide salts can only add to the growing halide salt ion channels, which then in turn come to dominate the structure with only a few short, isolated conventional glass-forming oxoanions remaining (as demonstrated by Trap and Stevel's schematic of invert glass shown in Fig. 3).

Similar observations were made on lithium di- and metaborate

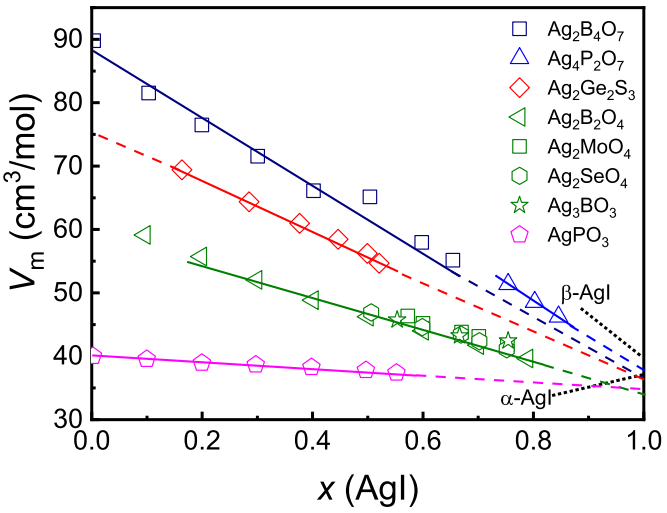


Fig. 24. Molar volume as a function of AgI content (mol%) in various glass systems. The ideal molar volumes of AgI are indicated by the dashed lines; for β- and α-AgI they are respectively 41.3 and 39.2 cm^3/mol. Reproduced with permission from Mercier et al. [370]. © Elsevier.

compositions, where additions of Cl were found to reside only in the interstices. Here, they were reported to significantly affect the number and bonding environment of the Li^+ sites (by Cl^- association and/or lowering of density) as well as the effective Li^+ 'jump' distance. Chlorine addition decreased E_a, T_g and density, and increased σ into the superionic region, therefore it has been argued that substantial modification of the network is occurring, yet at the same time, Cl was not found to enter the borate network [272,371]. Similarly, spectroscopic analyses of these glasses did not find evidence of strong interactions between the borate network and chloride anions; only changes which were expected from the decrease of the [O]/[B] ratio were observed [372]. Also, Cl^- was said to destroy the P–O–P network in LiCl–Li_2O–P_2O_5 glasses, but evidence of P–Cl bonding was never discussed [292].This is interesting as it indicates the large *indirect* influence halide addition can have on glass structure and properties. Although chlorine and bromine are thought to only add interstitially, fluorine appears to be different; many structural studies disagree about whether fluorine may add to the network and create non-bridging sites such BO_2F, BO_2F_2, BOF_3, and BO_3F [246], or whether it occupies interstitial sites only as with the other halides [245,373]. A study of lithium haloborates (F, Cl, Br) indicated that the halide adds to the network for up to about 10 mol% of halide addition. For higher amounts, it is incorporated interstitially, with fluorine being more likely to add to the network [246].

Chalcohalides [274–278], such as As_2S_3-$BiCl_3$, As_2S_3-AsI and As_2S_3-SbI_3, have structures analogous to those of halophosphate and haloborate glasses which include Cl, Br or I. Halide addition creates terminal bonding and reduces the connectivity of chalcogenide glasses from 3d corner-sharing to chain-like edge-sharing polyhedral structures [276,374]. Interestingly, As_2S_3-$BiCl_3$ glasses were described as being prone to phase separation and very inhomogeneous having covalent As_2S_3 and ionic $BiCl_3$ regions (with elemental S_8 inclusions) [276]. From XPS and Raman spectroscopy, As_2S_3-AsI and As_2S_3-SbI_3 structures were described as AsI_3 or SbI_3 molecules dissolved in the As_2S_3 polymeric network (Fig. 25) [281]. It is very difficult to distinguish former vs. modifier in compositions such as As_2S_3-AsI and As_2S_3-SbI_3; yet it can be said that AsI_3 or SbI_3 behave like nitrate or sulfate, *i.e.*, as oxoanions, since they do not participate in the As_2S_3 network and bond *via* S···I interactions only. Given the ion-complex structure of AsI_3 or SbI_3, and how this structure can incorporate roughly equal amounts of chalcogenide, we infer that chalcohalides have structures similar to other ionic glasses (such as sulfoborate/sulfophosphate (Section 2.2.2) or haloborate/halophosphate (Section 2.4) compositions where the sulfate or halide ions, respectively, do not disrupt the conventional glass network). Although iodide increases the density of As_2S_3-AsI and As_2S_3-SbI_3, it reduces the hardness and elastic moduli, likely due to the formation of weaker As-I or Sb-I bonds [282].

An early report on $KHSO_4$ glass discusses a requirement for glass formation: distortion of the structure and prevention of crystallization by adding two very different cations, such as K^+ and H^+; overall, this appears to be a general principle for many ionic glasses and corroborates our assertions in Section 4. Coordination numbers for ionic glasses may be higher than those found in conventional glass-formers: In sulfate glasses, the cations K^+, Na^+ and Zn^{2+} had twelve, eight and six neighbours respectively [92], while the commonly accepted values for the same cations in oxide glasses are 8/9, 6 and 4, respectively [1,18,334]. When K_2SO_4-$ZnSO_4$ glasses were determined to have nearly the same theoretical and experimental molar volume, the authors could accurately describe the structure using the random close packing (RCP) model of quasispherical ions: the cations, zinc (octahedrally coordinated) and potassium (12-fold coordinated), were found to fill the interstices between well-packed sulfate anions all cross-linked through ionic bonding [90,92]. The RCP model of equally sized spheres creates holes with distorted tetrahedral shapes [91]. Therefore to create an octahedral hole, Zn^{2+} must rearrange and distort the SO_4^{2-} "spheres;" these steps are shown in Fig. 26. This is an interesting example of a case

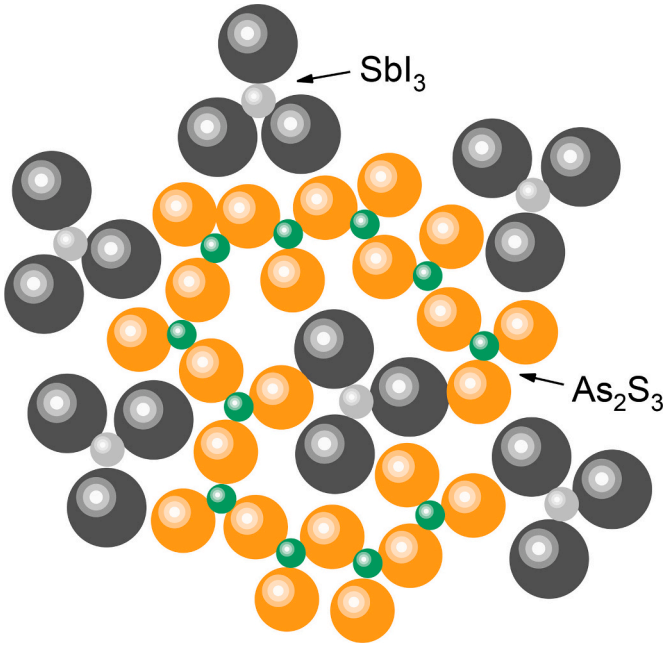


Fig. 25. Schematic representation of the structure of vitreous As_2S_3-SbI_3. Reproduced with permission from Kurushkin et al. [281]. © American Ceramic Society.

where the polarizability of the ions comes into play, *i.e.*, one cannot simply treat ionic glasses as random hard spheres when predicting their preferred structure and topology.

A large range of compositions in the sodium sulfoborate family have been synthesized; SO_4^{2-} addition does not introduce large structural changes (when the borates were already BO_4 tetrahedra, *e.g.* in $25Na_2O$-$75B_2O_3$), until concentrated sulfate regions are formed and then sulfate begins to modify the glass-former network, replacing BO_4 units [101]. A Raman spectroscopic study of $x Na_2SO_4$-$(1 - x)(0.4Fe_2O_3$-$0.6P_2O_5)$ with $0 \leq x \leq 0.45$ revealed that Na_2SO_4 additions resulted in the depolymerization of the phosphate network ($Q^1 \rightarrow Q^0$) [102]. In $x Na_2O$-$(1 - x)SO_3$-PbO-SiO_2, the authors observed that when Na_2O was added, the structure changed from isolated SO_4^{2-} orthosulfates to one where sulfate actively participated in the silicate backbone, forming Si—O—S bonds [104]. In a different study of barium sulfoborosilicate, ^{29}Si and ^{11}B NMR revealed SO_4^{2-} to modify the silica network until sulfate began to preferentially bind Ba^{2+} and form $BaSO_4$ crystals [105]. This appears to be the opposite trend of haloborates, where halides add to the borate network only at low halide concentrations, while sulfates only add when large concentrations of sulfate or modifier (sodium or barium) are already present.

The ways in which sulfate could be incorporated into the phosphate network *via* S—O—P bonding have been hypothesized at length [375]. Nepomiluev et al. [97] used ^{31}P NMR to observe different structures for the $x Na_2SO_4$-$(100 - x)P_2O_5$(-H_2O) system, which when $x > 0.5$ formed a mixed sulfate-phosphate network with S—O—P bonds. Lithium sulfophosphate glass at high sulfate content was the only ionic system found to contain S—P bonds (dithiophosphate, $Li_3PS_2O_2$), indicating an entirely different connectivity and, possibly, topology [99]. On the other hand, conductivity values were the same for the $AgPO_3$-Ag_2O and Ag_2SO_4-$AgPO_3$ series, signifying that σ is dependent only on Ag^+ concentration and no major structural changes are occurring, *i.e.*, no sulfate is incorporated into the phosphate network [100]. Other structural studies of sulfophoshate glasses have found similar results. In Li_2O-xSO_3-$(50 - x)P_2O_5$ [93], (Pb,Na)-borosulfophosphate [94] and (Na,Zn)-sulfophosphate glass [95], no evidence was found for SO_4^{2-} or sulfophosphate polymerization (S—O—S or S—O—P bonds, respectively), indicating that sulfate remained as isolated ions. Yet, sulfate addition has been said to contribute to the depolymerization of the phosphate network until the glass is composed of only highly charged ions [95]. Given the indirect nature of ^{31}P NMR spectroscopy, it is uncertain whether sulfophosphate glasses contain only P—O—P, M^+··O—P, and M^+··O—S bonds, or whether S—O—P and S—P are also possible. These discrepancies demonstrate the experimental difficulties inherent to elucidating a glass structure composed of few BOs, *such as in ionic glasses*.

Aluminum sulfofluorophosphate glasses were found to contain only Q^1 and Q^2 phosphate units cross-linked by aluminum octahedra, yet no evidence of sulfate cross-links has been found [58,96,376]. From ^{31}P NMR spectroscopy, the authors proposed the structure shown in Fig. 27 where PO_4^{3-} and SO_4^{2-} are connected through cations only (SP-type structure in Fig. 27). Furthermore, from ^{19}F NMR, the authors found that AlF_3 forms *only* Al[6], thereby forming an ionically bonded FP-type structure [58].

Structural studies of chloride-containing tellurites show the formation of mixed-anion Te polyhedra: Te(O, Cl)$_3$ and Te(O,Cl)$_4$ units [258–260]; although the Te—Cl bond energy is in the far-IR range and often cannot be detected directly, the positions the tellurium-related vibrations in the IR spectra suggest weaker, *i.e.*, chloride, bonding. In general, greater chlorine content increases the number of trigonal pyramidal Te(O, Cl)$_3$ units. A study by Dubois et al. [262] uses the crystallization products of a $PbCl_2$-Sb_2O_3 glass to identify likely local bonding, discovering mixed-anion Pb(O, Cl)$_x$ polyhedra and a

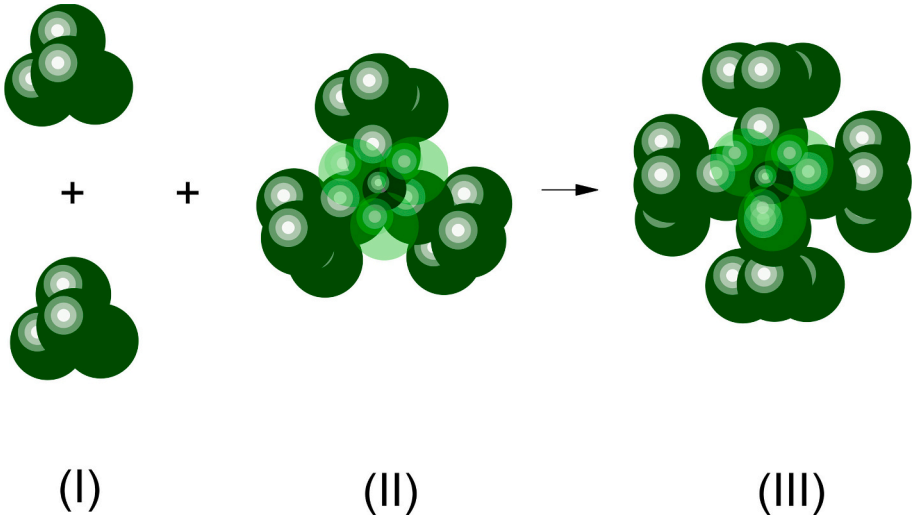


Fig. 26. Schematic representation of the stages through which a zinc cation (black) attains octahedral coordination with oxygen anions from sulfate groups (shown in dark green and light green when in front of zinc centre): Zn ions start from tetrahedral sites (holes, II), then, as the Zn^{2+} attracts two more sulfate ions (I) the Zn^{2+} site must open up and (III) rearrange into a stable octahedral coordination. Reproduced with permission from Narasimham and Rao [90]. © Elsevier.

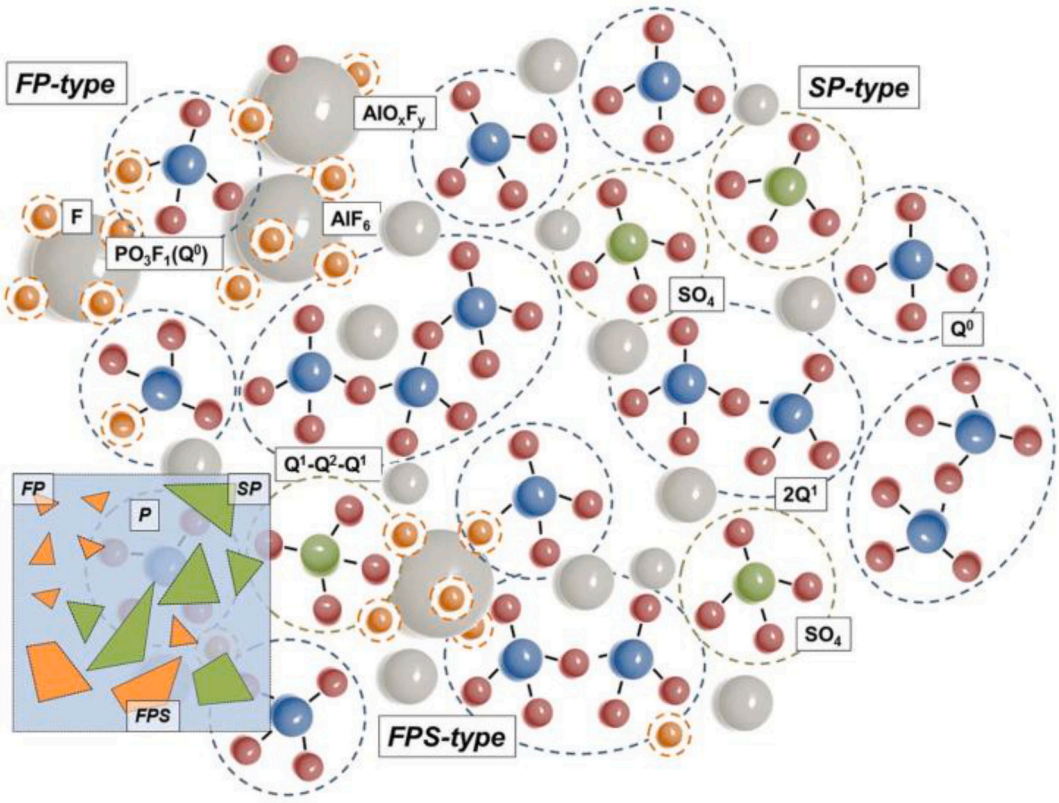


Fig. 27. Schematic of a possible structure of fluoro-sulfo-phosphate glasses. The inset schematically depicts chemical heterogeneity in the three types of glass. The gray balls depict cation species. Reproduced with permission from Le et al. [96]. © The Authors.

preference of SbO_4 over SbO_3 as $[PbCl_2]$ increases.

In a modern investigation, glass compositions were prepared along the $Mg_3P_2O_8$-$MgCO_3$ join; as expected from studies of other ionic glasses, the resulting structures were found to be an ionic "*network of isolated carbonate (CO_3^{2-}) and orthophosphate (PO_4^{3-}) interconnected by Mg^{2+} cations*" [quoted from [125]], *i.e.*, Mg–CO_3–Mg–PO_4–Mg–CO_3–Mg–PO_4 linkages. Interestingly, the authors found that the same composition would not form a glass when Mg^{2+} was substituted for Ca^{2+}, highlighting the importance of the cation in determining glass-forming ability and structure in ionic glasses.

Many spectroscopic and structural analyses of nitrate glasses have indicated that they also exhibit characteristic topological features. Raman [73,74], ^{15}N NMR [75] spectroscopy, neutron scattering [76], EXAFS [77], as well as PDF analysis from X-ray scattering [78] have been used to examine the structures of these nitrate mixtures. Depending on the chosen technique, there is some debate over the existence of distinct nitrate sites and order within nitrate glasses. Following Raman spectroscopic analysis, Leong et al. [73] and Furukawa et al. [74] could not comment confidently on the nitrate sites in Ca$(NO_3)_2$:KNO_3. Leong et al. observed two distinct Raman bands upon addition of $Ca(NO_3)_2$ to KNO_3, but could not assign it to sites dominated by Ca^{2+} *vs.* a mixture of Ca^{2+} and K^+ sites. Furukawa et al. also concluded that different orientations and perturbing forces could result in the same Raman spectrum, thus exact structural assignments are difficult to make. Nonetheless, both groups of authors felt that the addition of $MgNO_3$ created more distinct nitrate sites in $Mg(NO_3)_2$:KNO_3, indicating that the ordering in these types of glasses is strongly dependent on the electric field strength of the cations. While several authors used EXAFS and PDF analysis to study $Ca(NO_3)_2$:KNO_3, Wang et al. [78] used PDF analysis to generate one of today's most complete pictures, shown in Fig. 28. Here, the divalent cation is tetrahedrally coordinated by four nitrates, whereas the monovalent cation is closely sandwiched between two nitrate ions. A much later study of $2Ca(NO_3)_2$-$3RbNO_3$ found a very similar arrangement with the Ca^{2+} located inside of nitrate cages and Rb^+ playing the same role as K^+ [377]. These two cation sites are likely related to the slow and fast NO_3 populations observed by ^{15}N NMR near T_g (Fig. 29) [75]. The contrasting information which is obtained between vibrational spectroscopic methods, X-ray scattering and NMR again highlights the experimental challenges associated with determining the atomic structure of ionic glasses.

We will now turn the discussion to the structures of glasses with

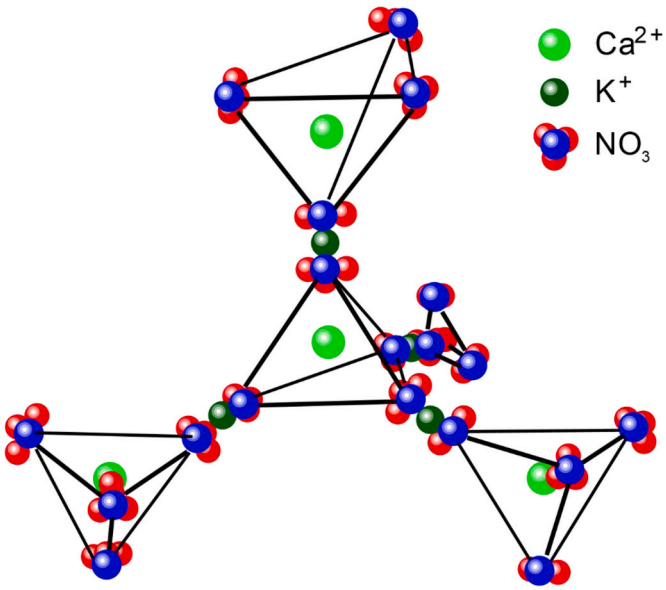


Fig. 28. Schematic of the structure of $Ca_{0.33}K_{0.67}(NO_3)_{1.33}$ glass based on PDF analysis. Reproduced with permission from Wang et al. [78]. © Elsevier.

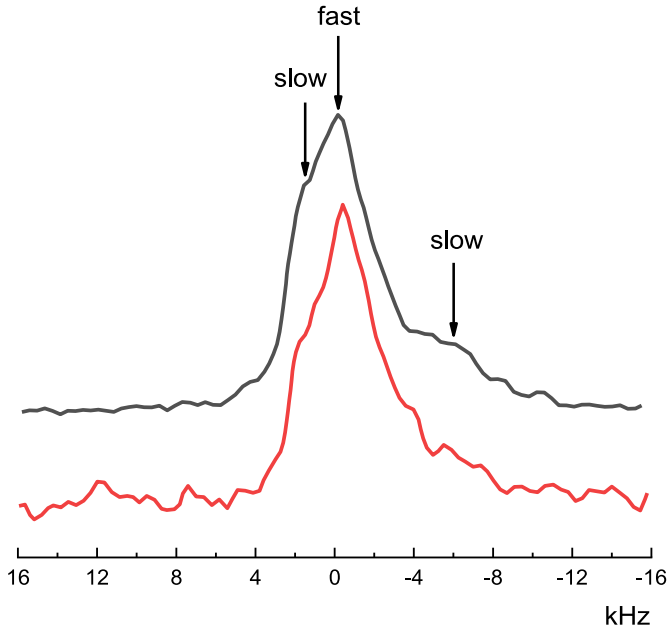


Fig. 29. Experimental ^{15}N static NMR spectra of $Ca(NO_3)_2$-KNO_3 obtained at T = 361 K as a function of variable waiting times after complete saturation of the magnetization. The top and bottom spectra correspond to waiting times of 500 and 10s, respectively. The slow- and fast- relaxing components are indicated. Reproduced with permission from Sen and Stebbins [75]. © American Physical Society.

simple (mono-atomic) anion species, like halides. Since there are smaller electronegativity differences between the different types of cations, the bonding is similar between ion pairs and there is no need for modifier and former distinctions in many fluoride glasses. Without the directional bonding inherent to Zachariasen's CRN model, the RCP model used for sulfate glasses can also be employed here. The interatomic potentials are spherically symmetric and the atoms are modelled as appropriately polarizable balls [378]. This is true for most HMFGs, where the bonds have some degree of ionicity and the closed shell configuration of F^- can be treated as spherical and undeformed [35]. As a result, most analyses of fluoride glass structures start from random-packing or random-network assumptions, depending on the preferred local topology. Unfortunately, the introduction of many types of components leads to significant inaccuracies in both approaches [35]. Modern techniques therefore involve a combination of modelling and experimental approaches [35]. Given the theory of glass formation in HMFGs, and its possible application to many other ionic glasses, we can say that using a variety of components helps to ensure glass formation and homogeneity. Unfortunately, this makes structural analysis much more difficult: in a pair correlation function there may be upward of 15 independent components. Their separation is presently nearly impossible without isotopic or isomorphous substitution.

Compared to other halide glasses, the structures of fluoride glasses are relatively well-studied. At the same time, structural features of fluoride glasses are much less well-understood compared to those of conventional glass-formers. The first main issue is the lack of a consistent structural unit, analogous to the SiO_4^{4-} tetrahedron in silicate glasses. Instead, metal fluorides have a large range of coordination polyhedra, for example Zr^{4+} has 6, 7, or 8 neighbours in crystalline compounds [35]. For the simple binary fluoride glass ZrF_4-BaF_2, diffraction and EXAFS studies determined that coordination numbers vary from 7 to 8.4 for Zr^{4+} [379–381], and a huge variance of 8 to 15 was found for Ba^{2+}; this demonstrates the substantial difficulties inherent to probing large, polarizable cations [35]. In agreement, many MD simulations found the coordination of Zr to be 7.7F to 8F [382–384]. Most recently, a ^{19}F NMR study confirmed that Zr has 8F neighbours (7.8F ± 0.2F) in barium fluorozirconates [385].

In disagreement with the above, many Raman spectroscopic studies [167,386,387] have been interpreted in that Zr^{4+} consistently has octahedral coordination environments, rather than the typical CN of 8. The archetypical fluoride, ZrF_4 likely has mostly corner-sharing polyhedra, but up to 22% edge-sharing polyhedra have also been reported (*see* [35] for a comparison of structures in several fluoride compositions); however, no edge-sharing fluorides were detected by ^{19}F NMR in ZrF_4-BaF_2 (only corner-sharing bridging fluoride and two types of non-bridging fluorides) [385]. Once again, differences in results between structure analysis techniques emphasize the difficulty of accurately observing ionic glasses. A study of $ThCl_4$ glass suggested it to have the same structure as ZrF_4, forming a network of distorted polyhedra with high coordination numbers (4 shorter bonds and 4 longer bonds) [165], with alkali cations going into the interstices, increasing the number of non-bridging chlorines and average thorium coordination [192]. On the other hand, the introduction of $PbCl_2$ to $ThCl_4$ has little effect on the structure and may simply add as separate $PbCl_2$-rich chains [192]. In the case of mixed-halide glasses, the preferential formation of Y—Cl bonding modified the number of bridging and non-bridging fluorines through the glass structure [388]. At this point, we note again that one of the main challenges to studying halide glasses (other than fluorides) is water sorption, particularly the formation of -OH bonds [165].

In contrast to HMFGs, vitreous BeF_2, $ZnCl_2$, and $ZnBr_2$ have a high degree of regularity to their structure, *i.e.*, corner-sharing regular tetrahedra, due to more covalent bonding necessitating exact intratetrahdral bonding angles (110° and 131° for $ZnCl_2$ and $ZnBr_2$, respectively [184]); all three compounds have been called structural analogs to silica [3,130,389,390]. However, moving away from the vitreous SiO_2 network, the structure of $ZnCl_2$ glass has been described as a random close-packed array of Cl^- spheres with Zn^{2+} in corner-sharing tetrahedral "holes", where the reduction of Zn—Cl—Zn bonds and random rotation of angles only approximate a SiO_2-like structure [389,391]. In pure liquid $ZnCl_2$, there is believed to be a small fraction of unbonded or terminal Zn—Cl units [392], which may be similar to the terminal bonds observed in molten ZnI_2 [390], and a small number of higher order rings [391]. Interestingly, the network-like structure of $ZnCl_2$ is reported to disappear in the presence of a second component, such as LiCl, KCl, CsCl, or KNO_3, with a sharp decrease in T_m and cation-stabilized $ZnCl_4^{2-}$ tetrahedral complexes forming [10,392,393]. A study of $ZnCl_2$-KI and $ZnBr_2$-KI found KI to behave like a modifier: it increases the glass-forming ability of $ZnCl_2$ and $ZnBr_2$ by breaking up the SiO_2-type network to form chains akin to metasilicate [387]. In the $ZnCl_2$-$AlCl_3$ series, the addition of an intermediate ($AlCl_3$) decreases fragility as $AlCl_4^-$ tetrahedra increase overall connectivity by adding corner-sharing Zn—Cl—Al bonds [393]. These observations very strongly corroborate earlier ideas about isomorphism, however, the dominant bonding type changes when moving from a covalent silica network, or a modified silicate glass, to salt-based glasses with clearly ionic bonding.

5.1. Stability parameter classification

Throughout the decades, many different parameters, both thermodynamic and kinetic, have been used to predict the stability of (halide) glasses. One of the earliest examples simply tracks trends in the periodic table to predict glass formation of ionic glasses, and correctly identifies NO_3^-, CO_3^{2-} and SO_4^{2-} as glass formers [394]. Similar to Sun's bond strength parameter for oxide glasses, Rawson and Portier both used the bond strength (and T_g and field strength, respectively) to categorize halide glasses [395]. Using the field strength [395], or cationic and anion field strength ratio (F_a/F_c) is also a common method [20,146]. On the other side, there are completely empirical approaches based on the crystallization and glass transition temperatures [396], or on the cooling rates which are necessary for glass formation [397]. However, for any chosen model, there are always counterexamples [395–398].

This being true already within the sub-class of halides, finding a common stability parameter for all types of ionic glasses can only result in a practical compromise.

Grande et al. [399] found thermodynamics to determine good glass formation in BeF_2- and ZrF_4-based glasses: *near* deep eutectics was necessary, but too close resulted in crystallization. Overall, ionic glasses are less constrained by directional bonding, thus, viscosity will rarely be the glass-forming limitation. Thus, glass formation in ionic glasses appears to be a thin line between atomic configurations that are so stable that crystallization is unavoidable, and those which are so unstable that no appreciable mixing occurs at all.

We find it interesting that Lucas [400] said that fluoride glasses must have a highly polarizing cation, such that it can delocalize the electron on F^-, to create edge- and corner-sharing polyhedra, but also a second large cation with low field strength to introduce non-bridging fluorines and disorder. Gan [192] stated something similar, although more cryptically, a year before: that fluoride glasses should have strong *intra*polyhedral and weaker *inter*polyhedral bonding; only with two different cations can one create this situation. A comparative IR study found that although BeF_2 had distinguishable bending and stretching vibrations, HMFG had only two very broad and undecipherable bands, populated by any number of bonding vibrations [401]. We believe these two broad bands could be the strong *intra*polyhedral and weak *inter*polyhedral bonding discussed by Lucas [400] and Gan [192]. The presence of these two types of bonding is not unlike those within an oxoanion (covalent bonding) and the ionic bonding between the cation and the oxoanion.

From its application to other glass types [3,35], the stability of a component in ionic glasses, MX_n (where M = metal and X = anion), can be described by the radius ratio of the two components, r_M/r_X. This is of course an extension of Pauling's rules of packing for complex ionic crystals [365], where the cation at the centre of a polyhedron must be sufficiently large to (formally) have contact with all of the coordinating anions, *i.e.*, r_M/r_X cannot be too small. If the radius ratio is above a critical value, another anion will be added to the coordination sphere, as is the case with borate (BO_4 units) and germanate (GeO_5 units) glass systems. Smaller, more electronegative cations result in low r_M/r_X (coordination, n = 3 or 4 [25]), which in turn leads to well-defined structural units. Larger, more polarizable cations have larger n, with higher variance and distortion in their polyhedra. At large enough r_M/r_X, edge or even face sharing may occur without a significant strain energy [167].

Although simple, the radius ratio can be used to identify ionic glasses without directly considering bonding character; the atomic radii should reflect electron-sharing rather than closed-shell interactions. For example, $n \leq 4$ is indicative of glasses with *isolated* local order, *i.e.*, polyhedra with directional bonds, while $n > 4$ are large singular ions with more local disorder, *i.e.*, variations of coordination and angles. Such a categorization appears to work well in oxide glasses and many halides: BeF_2 and $ZnCl_2$ have a coordination number of four and can form single-component glasses (like conventional glass formers), while other metal fluorides have larger, more variable coordination spheres (like conventional modifier species).

This categorization is, of course, very similar to Dietzel's categories of glass former and modifier. However, the criterion can be extended to all non-metallic inorganic glass compositions as it considers the *radius ratio between cation and anion*. For example, if we consider BeF_2 and $ZnCl_2$ according to Dietzel's criterion we find field strengths of 0.84 (a = 1.54 Å) and 0.37 (a = 2.31 Å), respectively, classifying these glass-formers firmly as intermediate and modifier, respectively [402,403]. Thus, it appears that the classical field strength calculation is not valid for halides, and perhaps for all highly ionic glasses.

In Table 2, we consider r_M/r_X of many different cations and anions (oxides and halides) and $r_{M\text{-}O}/r_X$ mononuclear complex anions (for example $2K^+{\cdot}SO_4^{2-}$ is $r_{S\text{-}O}/r_K$). However, we know from the above discussion that multicomponent halides also have structures akin to oxoanions, so we could write ZrF_4-BaF_2 as $2Ba^{2+}{\cdot}ZrF_8^{4-}$ and $BiCl_3$-KCl as $5K^+{\cdot}BiCl_8^{5-}$. From this, we identify three categories of relevance to ionic glasses: oxoanionic glass-formers (some with the potential to form longer chains, *i.e.*, conventional glass formers), *strong* ionic glass formers (this label will be discussed in Section 6.4.1), and highly ionic glass components (typical modifying oxides, but also halide and multi-atomic anion compounds that cannot form glasses alone).

Table 2
Tabulation of the stability parameter r_M/r_X, using Shannon-Prewitt ionic radii [319,320] or $d_{M\text{-}O}/d_X$ from crystal bond lengths [404–413].

Ion/Former	r_M (pm)	r_X (pm)	r_M/r_X or $r_{M\text{-}O}/r_X$
OH^-	−38	138	−0.28
H_2O	−18	138	−0.13
NO_3^-	−10.4	138	−0.08
BO_3^{3-}	1	138	0.01
BO_4^{5-}	11	138	0.08
SO_4^{2-}	12	138	0.09
CO_3^{2-}	15	138	0.11
PO_4^{3-}	17	138	0.12
SiO_4^{4-}	26	138	0.19
BeF_2	27	133	0.20
AsI_3	58	220	0.26
GeO_2	39	138	0.28
As_2S_3	58	184	0.32
SbI_3	76	220	0.35
$ZnCl_2$	74	181	0.41
TeO_2	66	138	0.48
$ThCl_4$	94	181	0.52
MgO	72	138	0.52
ZrF_4	72	133	0.54
MgF_2	72	133	0.54
KI	151	220	0.69
KCl	151	181	0.83
K_2O	151	138	1.09
KF	151	133	1.14
$2Mg^{2+}{\cdot}SiO_4^{4-}$	207	164	1.26
$3Mg^{2+}{\cdot}(PO_4^{3-})_2$	203	153	1.33
$Zn^{2+}{\cdot}SO_4^{2-}$	195	145	1.34
$2Ba^{2+}{\cdot}ZrF_8^{4-}$	205	152	1.35
$3K^+{\cdot}PO_4^{3-}$	203	148	1.37
$5Mg^{2+}{\cdot}(BO_4^{5-})_2$	212	141	1.50
$Mg^{2+}{\cdot}CO_3^{2-}$	210	128	1.64
$4K^{2+}{\cdot}SiO_4^{4-}$	283	164	1.73
$Ca^{2+}{\cdot}CO_3^{2-}$	234	128	1.83
$Ca^{2+}{\cdot}(NO_3^-)_2$	250	122	2.05
$5K^+{\cdot}BO_4^{5-}$	281	137	2.05
$2K^+{\cdot}SO_4^{2-}$	306	147	2.08
$5K^+{\cdot}BiCl_8^{5-}$	298	138	2.16
$K^+{\cdot}NO_3^-$	287	127	2.26

We observe that all oxoanionic (nitrates, sulfates, carbonates and hydrates) and conventional glass formers that form ortho-compositions (borates, phosphates and silicates) are found below 0.19 (negative r_M and resulting r_M/r_X indicate a significant amount of covalent bonding). SiO_2 has r_M/r_X of 0.19 and it can be forced to form glass past the meta-composition, but only through rapid quenching or in the form of multi-component invert glasses [28,341]. Many non-percolating mono- and dinuclear oxoanions condense into conventional covalent glasses (SiO_2, P_2O_5, B_2O_3) or do not form glass alone (SO_4^{2-}, NO_3^-, CO_3^{2-}), thus, oxoanions are only ionic glasses when large amounts of conventional glass modifiers (K^+, Na^+, Mg^+) are present. Consequently, this first category of ionic glasses only serves to indicate the types of covalently bonded complex ions which can form isolated oxoanions in glasses.

Next we have *strong* ionic glasses; like polymeric covalent glasses, they can form single component glasses [20,130,134,414], and they can form isolated (mononuclear or dinuclear) anions, *i.e.*, ortho- or pyro-compositions, when combined with components that form non-bridging anions, such as other alkali oxides or halides [10,392,393]. For example, as noted in the previous section, pure $ZnCl_2$ and BeF_2 both form a weakly bonded, more locally disordered, silica-type network [3,130,389]. The addition of modifying chlorides easily produces ionic

$ZnCl_4^{2-}$ and $Zn_2Cl_7^{3-}$ [392]. On the other hand, due to their ability to have more than four neighbours, GeO_2 and TeO_2 likely have more ionic bonding character than do those glass formers limited to $n \leq 4$ [415,416].

Finally, we have highly ionic glass components which are normally unable to form glasses on their own, but require many components (geometric frustration) or extreme synthesis methods to produce glasses. In this category, from KF upwards, we consider the packing of two polyhedra (rather than atomic radii) and calculate the ratio of metal–oxide to former–oxide bond lengths, $d_{M\text{-}O}/d_{X\text{-}O}$ (as determined from their corresponding crystal structures [404–413]). For example, K_4SiO_4 and Mg_2SiO_2 are found in this third category, as invert glasses rely on configurational entropy to form and generally must include many components to vitrify easily. The one clear exception is K_3BO_3; alkali borates can form glass with 70 mol% M_2O, but this is likely because the many forms of borate (boroxyl rings, BO_4^{5-} vs. BO_3^{3-}) create geometric frustration [417]. There is evidence that the reason for the high glass forming ability of borate glass are the "*major differences in atomic structures of the parent liquids and their isochemical crystal phases are cause for poor nucleation*" [quoted from [60]].

One may notice that the categories of ionic glasses do not separate the families of glass on glass-forming ability alone, the main differences originate from the varying type of bonding, coordination number and the ability to easily form ions, *i.e.*, to stabilize the extra charge (as determined by electronegativity, polarizability, resonance structures and/or electron-withdrawing groups). As discussed earlier the type of bonding parallels the r_M/r_X ratio: in general, the lower the ratio, the more covalent the interactions, which should also roughly correlate with coordination number. Furthermore, based on packing alone, the ideal or most stable coordination number as a function of radius ratio can been determined [167]. Fig. 30 compares the ideal relationship (as indicated by the bands of colour and structural models) with our calculated radius ratios (on the r_M/r_X-axis) and reported coordination numbers (on the CN-axis). The radii and coordination number (CN) values for crystalline oxide and fluoride compounds are readily available from Shannon and Prewitt [128]. However, only a few coordination numbers of alkali or alkaline earth cations are known in oxoanionic glasses, such as for K_2SO_4, Na_2SO_4 and $ZnSO_4$ [90–92], nonetheless, we expect them to be larger than in covalently bonded glasses or crystals. Furthermore, their ionic radii can only be estimated from the corresponding CN determined from oxide or fluoride crystals [128]. As such, we expect these values to be the least accurate.

As expected following Goldschmidt [3], the points corresponding to SiO_2 and BeF_2 fall on top of each other in Fig. 30. $ZnCl_2$ has a larger r_M/r_X, about halfway between the ideal radius ratio for a CN of four and six. It is important to remember that glass components with radius ratios above the critical radius ratio for that CN (as indicated by the red dots in Fig. 30) are considered structurally stable, while glass compositions with radius ratios below the ideal are unstable [365]. Therefore, tetrahedral symmetry is a stable configuration for $ZnCl_2$ and given that the bond between Zn^{2+} and Cl^- is significantly more covalent than BeF_2 (Δelectronegativity of 1.51 vs. 2.41), it may be that the additional bond directionally enforces and stabilizes tetrahedral symmetry. Indeed, Desa et al. [391]found that a significant amount of covalency was required to calculate the correct Zn–Zn correlation function in MD simulations of $ZnCl_2$. Nonetheless, it may not just be the stability of the existing glass coordination, but also the instability of other possible coordinations; for example, $ZnCl_6^{4-}$ octahedra are higher in energy than tetrahedra due to the d^{10} zinc electron configuration.

Since even SiO_2 is slightly below the respective critical r_M/r_X, the type of bonding must play a role. Similarly, ZrF_4 and $ThCl_4$ are found below their 'ideal coordination', indicating that according to their r_M/r_X, they would prefer to form octahedra, yet they are found in 8-fold or 9-fold coordination in mixtures of metal halides [167]. The clear exceptions appear to be borates and phosphates, which despite having r_M/r_X below the corresponding critical r_M/r_X, form glasses easily.

In addition to BeF_2, other ionic glasses which can be formed from a single compound and fall near the appropriate critical r_M/r_X are Mg_2SiO_4, $Mg_3(PO_4)_2$, and $Mg_3(BO_3)$. One exception is $MgCO_3$, which is near its 'ideal' radius ratio, but is not known to form glass on its own; however, it was shown to form a glassy mixture with phosphate (while $CaCO_3$ did not) [125]. For clearly ionic components, there appears to be a lower acceptable difference between its r_M/r_X and the corresponding ideal one; although MgF_2 and NaCl are in the 'correct' stability band, *i.e.*, have matching r_M/r_X, range and CN, they still do not form glasses easily (unlike Sb_2O_3 or Bi_2O_3 which are much further from the ideal radius ratio and still are considered glass formers). Indeed, the difference threshold appears even smaller for the CN = 8 (orange) band, where despite Na_2SO_4 being almost on top of the 'ideal configuration' for octahedra, it still does not form glass alone. Once again, glass-forming, highly modified borates and phosphates (K_3PO_4 and K_3BO_3) appear to be exceptions.

Using the simple metric of Table 2 and Fig. 30, similarities between apparently different glass-forming species can be identified easily, for example, the complete miscibility in the tellurium chloride system of $xTeO_2$:$(1-x)ZnCl_2$, with $0 \leq x \leq 100$ [418]. Here, Raman spectra

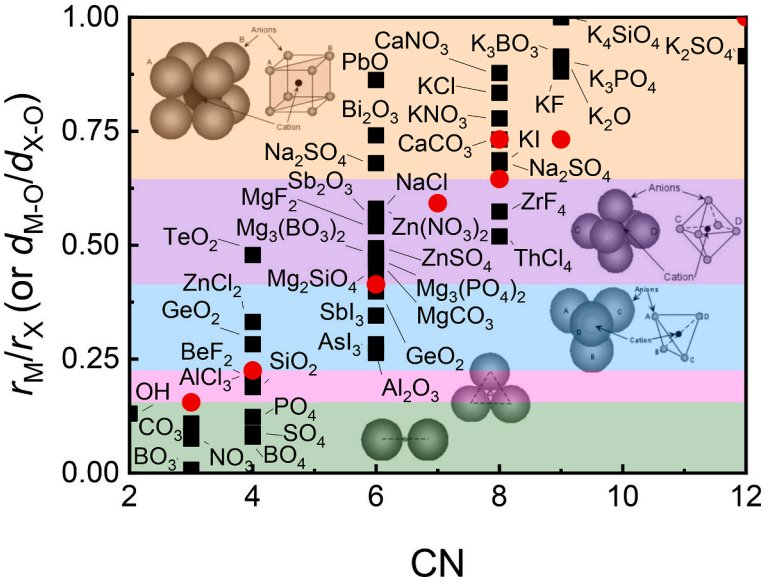


Fig. 30. Correlation between coordination number (CN) and radius ratio for various types of ionic glasses. Instead of the typical r_M/r_X calculation, for glasses with oxoanions, the ratio of metal–oxide to former-oxide bond–lengths, $d_{M\text{-}O}/d_{X\text{-}O}$ is used (as determined from their corresponding crystal structures [404–413]). The red dots represent the critical radius ratio for each CN (as depicted by the polyhedra) according to Pauling [365] with the different colored bands indicating where each CN is stable.

indicated that $ZnCl_2$ broke Te—O—Te bridges and formed bridging chlorines bonded to Zn^{2+} (Te—Cl—Zn) [419]. As further examples, AsI_3 and SbI_3 in the solid-state have been known to form highly anisotropic octahedra with three short and three longer bonds [420], yet these components appear to form neutral molecules when added to As_2S_3 glass [281]. This highlights the subtle difference between molecular and ionic glasses; research is required into these types of systems: do ionic AsI_x^{y-} or SbI_x^{y-} form?

In invert glasses, the oxide bonds between the modifiers become more important than the BOs in determining dynamic properties. The interactions within the *ion channels* of Greaves' MRN model become as important as those between conventional modifier and former: the anionic oxygen, O^{2-}, in a Mg—O—Mg bond in a Mg_2SiO_4 glass, the LiCl channels that form in a $(70 - y)B_2O_3$-30BaF_2-yLiCl glass, or a telluride tetrahedron being replaced by a zinc chloride in $x TeO_2$-$(1 - x)$ $ZnCl_2$ (recall that $ZnCl_2$ is traditionally categorized as a modifier, $F = 0.37$ Å^{-2}). Given that large amounts of halide, such as chloride, can be added to the point where halide reaches conventional former content [252,418], we have to assume that the strong ionic interactions within halide salts (cation-anion-cation) are sufficiently similar to those between network and modifier in pyro- or ortho-compositions of conventional glass formers. For example, highly modified, conventional glass formers (like K_4SiO_4, K_3PO_4, and K_3BO_3) show evidence of metal halides, sulfates, and hydrates bonding directly to the covalent network backbone (instead of just acting as cross-linking cations in ion-channels).

Since the classical definition of glass-former and modifier in ionic glasses becomes blurred, we must look to the relative differences between components (the stability parameter r_M/r_X) in order to judge to ionic glass formability and potential structure. As Simmons et al. discuss, "*simple radius ratio considerations mean that, in the melt, even the network modifying cations will tend to restructure their local environments in order to achieve their optimum first coordination shell*" [quoted from [35]]. Using Table 2 and Fig. 30 helps to determine whether a chemical composition forms an ionic glass, and which type of ionic glass.

5.2. Further considerations of topology

The many different glass structures discussed in this Section demonstrate how the concept of a conventional glass former can be transformed into a nearly ionic glass structure interacting through only Coulombic forces. These ionic structures are composed of modifying cations, halides and/or more complex anion species. Thus, the nature of bonding beyond the first coordination shell becomes a matter of further interest. For example, whether having *isolated* local order such as in the oxoanionic glasses creates large differences compared to glasses with mostly local disorder (e.g., in halides) is a question which has yet to be answered.

We expect that at length-scales beyond the short-range order around the anions (so-called medium range order), the Coulombic interactions can be modelled similarly for oxoanionic and halide glasses, however, as the chain lengths increase the topologies will be increasingly different. Given the compositions and methods of glass formation used to produce ionic glasses, we expect them to have more chemical heterogeneity than covalent glasses. Nonetheless, network topology and structural heterogeneity must be considered separately. We make the assumption that highly ionic glasses would tend to have smaller electron density fluctuations; as we see in halide glasses many different bonding environments are comparable in energy, and therefore, ionic glasses seem to exhibit lower structural heterogeneity [167].

On the other hand, highly modified invert or oxyanionic glasses, or glasses which contain multiple anion (and cation) species with strongly varying sizes, electronegativities and polarizabilities [96], may tend towards local chemical selectivity and form regions with variable bond directionality and/or ion channels; therefore, fluctuations in bonding and electron density may become even more pronounced than they are in conventional network-forming glasses [58,95]. Oxoanionic glasses tend to form isolated polyhedra and therefore have restricted connectivity, thus, at some threshold, the connectivity of the anions will create similar structural heterogeneity as in modified covalent glasses. We know that having larger structural units creates a sluggish kinetic environment for glass to form in ionic liquids, so there must be a threshold where sufficiently long chains of anions dominate the properties, such as the $Y = 2$ criterion for invert glasses.

6. Properties common to ionic glasses

6.1. Optical properties

Although halides have interesting optical properties (*see* Poulain [421] for optical applications of mixed-anion ionic glasses), most of them are not unique to this glass family and therefore, are not unique to ionic glasses. In fact, it is possible to find decidedly non-ionic glasses with similar refractive indices and Abbe numbers to halides. Furthermore, even within the halides, chlorides and bromides have much higher dispersion than fluorides. This is because of the large variation of polarizability, atomic number, and bond strength across the set of possible glass components. Nonetheless, we still infer that most conventional 3d network glasses will usually be less polarizable than invert glasses made with high amounts of conventional modifying cations, or than ionic glasses composed of less-conventional oxoanions discussed in the previous sections. The same is true for bonding: the strongest cation–anion interaction can be much weaker in ionic glasses than in covalent glasses. Unfortunately, polarizability (and atomic weight) and bond strength act opposingly on the refractive index. This effect may be lessened regarding the optical dispersion or Abbe number (ν_d) however, there is still no distinguishing trend for ionic glasses.

Ionic glasses provide the potential for enhanced *tunability* of optical properties, *i.e.*, the ease of substituting one type of cation or anion for a different type (in contrast to conventional silicate glasses, which always remain within the stringent limits of having oxygen anions and silicate tetrahedra). Haloborate compositions provide an extreme example, where it is possible to switch-out an isolated BO_4 unit for Cl^- without other major topological changes occurring. Such flexibility can be used to adapt a range of different properties for applications of potential interest, *e.g.*, phonon energy, refractive index, non-linear polarization, interionic coupling and energy transfer processes. For example, the refractive index of fluoride glass has been said to be able to be "*tailored by varying compositions which allows HMF glasses to access new regions of index/dispersion space*" [quoted from [136]]. In chloride glasses, the Abbe number has been described as being "*manipulated in a predictable manner*" [quoted from [194]]: compared to conventional glass formers, ion coordination and polarizability are significantly more tunable in ionic glasses (applying the classifications of Section 5) [194]. For this tunability, their spectroscopic (*e.g.*, fluorescence, emission/absorption and transmission) properties can be easily modified by compositional changes. In general, weaker bonding leads to wider transmission windows, and increased polarizability makes chloride and bromide glasses transparent further into the IR region, but also red-shifts the UV edge as compared to fluoride compositions [194]. Similarly, the UV absorption edge in haloborate glasses was found to shift to lower energies as the concentration of LiX increased [311]. This weak bonding can also be beneficial for observing emission wavelengths not seen in more covalently bonded glasses, *i.e.*, the 1.3 μm emission of Pr^{3+} present in Ga_2S_3-La_2S_3 glass, but not seen in silica; by having lower phonon energy, some phonon-assisted non-radiative transfers are absent in ionic glasses.

The tunability of ionic glasses is most evident in fluorescence. It is possible to predict the fluorescence of halide mixtures using the theory of hard acids and bases; one expects colour centres to interact with their like (either hard or soft) counterpart [251]. The peak cross-section is determined by the line strength and linewidth of the transition as well as by the refractive index of the host [250]: "*The rules for tailoring large*

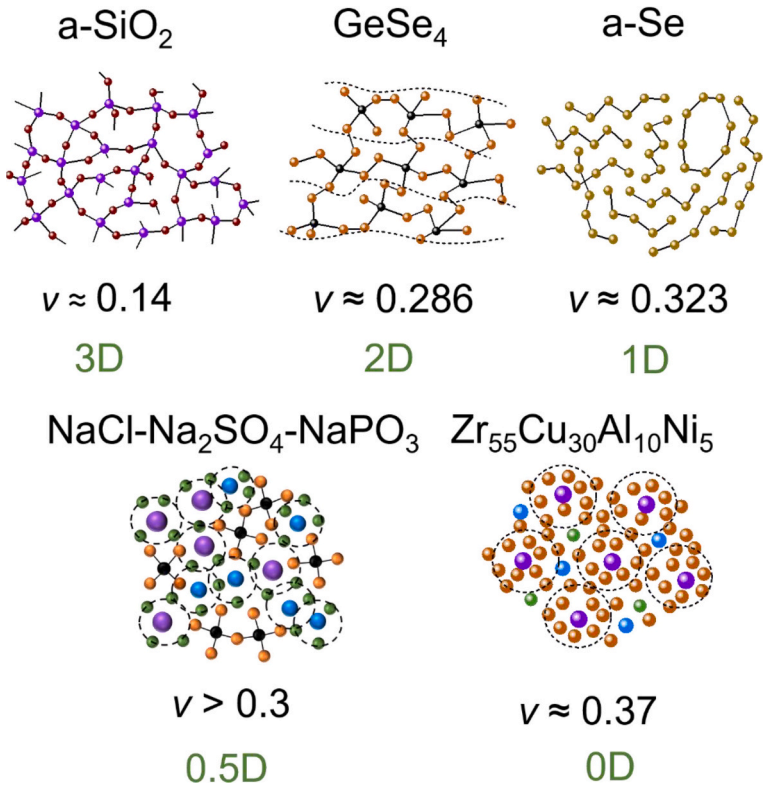


Fig. 31. Schematic drawings of glass atomic networks with increasing Poisson's ratio and decreasing structural dimensionality. Reproduced with permission from Rouxel [8]. © American Ceramic Society. The typical ionic glass of NaCl-Na_2SO_2-$NaPO_3$ was added for reference.

cross sections include the selection of glass compositions having large, polarizable anions to increase the line strength and refractive index and cations with small charge:radius ratios to decrease the effective emission linewidth" [quoted from [250]]. Larger stimulated emission cross-sections result in higher gain coefficients, for example in laser applications; cross sections in halides (except fluorides) are higher than those achievable in silicates, phosphates, tellurites and fluorophosphates [250]. For example, $ZnCl_2$ has a cross-section of $\sim 5.4 \times 10^{-20}$ cm^2 [251].

Additionally, since a narrower linewidth is achieved for an ion in a more isotropic chemical environment *i.e.*, a low electric field gradient, it is beneficial if the fluorescent ion can gain ligands easily, as is the case in a weakly coordinating glass former. This is evident from the narrow linewidth produced by replacing divalent oxide anions with monovalent halides (Cl^-, Br^-, I^-); not only are they less strongly bound, but they are also more polarizable [251]. Nonetheless, the bandwidth of absorption and emission lines of a disordered glass will always intrinsically be wider than the one of the corresponding ordered crystal. Additionally, a larger refractive index can also widen the emission cross-section by retarding the phase velocity of the excitation beam, leading to enhanced interactions between the excitation light and optically active colour-centres.

Generally, it is thought that the optical properties of coloured metal centres can be used to probe the ionicity or covalency of all glass formers [422]. Coloured metal centres have been used to probe the ionicity of the environment in chlorosulfate and pure sulfate glasses; the blue shift of the UV band signified increased ionicity in sulfate glasses compared to when chloride was present [422].

Similarly, variations in optical emission spectra have been used to study relaxation processes of nitrate glasses, another approach which may be applicable to all ionic glasses [79]. If changes in the size, shape or surrounding chemical environment of a coloured probe molecule occur, optical spectroscopy can be used to determine the length scale of the relaxations near T_g. Time-dependent spectrophotometry [79,82] appears well-suited for the study of ionic glasses, wherein small amounts of 'probing molecules', *e.g.*, a visible metal centre, such as Co^{2+}, or a halide salt, such as KCl, can be easily dissolved into the molten mixture. Co^{2+} (like other transition metal ions) is optically sensitive to its coordination sphere: dodecohedric Co^{2+} surrounded by four nitrates (523 nm) transitions to tetrahedral Co^{2+} with chloride neighbours (623, 660 and 690 nm), indicating a structural rearrangement at higher temperatures. The lag time between the change in temperature of the experimental setup and the chemical equilibrium between Co^{2+}_{tet} and Co^{2+}_{dod} can be minimized until the 'matching' temperature-cycling frequency is found and assumed to be on the order of the relaxation processes. In a molten environment, experiments demonstrated nitrate and Cl to be interchanging very rapidly [79,82]. In other optical studies of nitrate glasses authors have noticed an anomalous shift in the UV absorption around T_g [423], which can also be used to probe relaxation times. Many further studies have taken advantage of unpaired spins and performed electron paramagnetic resonance (EPR) or electron spin resonance (ESR) to confirm that the bonding in sulfate glasses is ionic [424], monitor crystallization processes [425] or observe inhomogeneity [101,104].

6.2. Mechanical properties

In conventional glasses, the connectivity of the glass network is intuitively related to the mechanical properties of the macroscopic sample. For example, the ratio of BO_3 and BO_4 units has been used to explain the anomalous maximum found in many properties for sodium borates [356]. The differing connectivity of the structural units (3- vs. 4-fold) explains the significant mismatch between the mechanical properties of borate glasses (containing large amounts of BO_3 units) and crystals (when composed of BO_4 units) [426]. Ionic glasses have atypical structures: rather than strings of monomers, they are ideally composed of simple ionic "balls" of different polarizability, size and coordination sphere (*see*Fig. 27) [96]. In that sense, ionic glasses bridge the gap between the more covalent glasses, like a-Se (1d chains) and

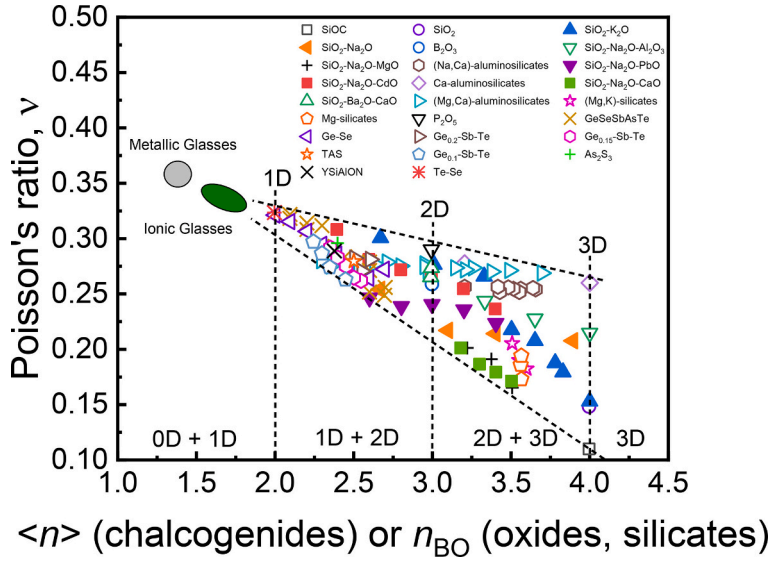


Fig. 32. Relations between Poisson's ratio and the average coordination number $< n >$ or the number of bridging oxygen per glass-forming cation n_{BO}. Reproduced with permission from Rouxel [8]. © American Ceramic Society. Reported and expected values for ionic glasses are indicated by the green ellipse.

disordered metals (Fig. 31). A schematic of NaCl-Na_2SO_2-$NaPO_3$, has been included in Fig. 31; it possesses both the locally ordered tetrahedra of α-SiO_2 and the randomly packed spheres of metallic glasses (the dimensionality and Poisson's ratio of NaCl-Na_2SO_2-$NaPO_3$ are strictly estimates).

The fact that ionic glass series can be made that progress from 1d chains to mostly isolated ions is of great interest because of the resulting mechanical properties: as shown in Fig. 32, there is a correlation between structural dimensionality and Poisson's ratio (ν) [8], where ionic glasses have $\nu > 0.3$ [90,427]. However, little is presently known experimentally about the mechanical properties of non-metallic glasses with n_{BO} between 0 and 1 (dimensionality between 0 and 1d). A similar correlation between number of BOs (connectivity) and maximum densification can be observed in Fig. 33, where we expect ionic glasses to fall in between $GeSe_4$ and metallic glasses, and therefore having a maximum densification of 2 – 4%. For example, TeO_2-PbO-$PbCl_2$ glasses were found to have Poisson's ratios 0.27 – 0.33 and molar volumes similar to HMFG [258]. Interestingly, a Brillouin study of vitreous $ZnCl_2$ found ν_{trans}/ν_{long} to be 0.56, translating to a Poisson's ratio of 0.27 [428], perhaps indicating that the connectivity of $ZnCl_2$ is between 1D and 2D.

Fig. 34 shows the compactness and Poisson's ratios of ionic sulfophosphate glasses. In agreement with Fig. 33, these values are between those of common covalent formers and volume-conserving (*i.e.*, high Poisson's ratio) metallic glasses [427]: the compactness of the structures of ionic glasses was found to increase with ionicity [95]. Finally, some ionic glasses have been found to be less brittle, but also less crack-resistant than typical glass formers (*see*Fig. 35) [427]; however, due to the limited body of data which is presently available, no general conclusions can be drawn at this point. Future work could include synthesizing ionic glasses with mechanical properties closer to those of metallic glasses, as one open question is whether ionic glasses can approach some of the exemplary mechanical properties of disordered metals, *e.g.*, high Poisson's ratio, compactness, hardness and fracture toughness [427,430]. Although mechanical properties are difficult to measure for hygroscopic oxyanionic glasses, one study reported K_2SO_4-$ZnSO_4$ to have a Poisson's ratio between 0.39 – 0.44 with each composition's bulk modulus being 2 – 3 times higher than the Young's modulus [90]. Such mechanical properties place these sulfates in the same region as typical metallic glasses in Fig. 33 and Fig. 34. Additional data should be generated along this line in order to determine general trends in the mechanical behaviour of ionic glasses.

Given that metallic glasses can form both inhomogeneous (fragile) and homogenous (strong) lattices (Fig. 36) with different fractions of weakly (high ν, low G) and strongly bonded regions (low ν, high G), a further question is how fragility and structural heterogeneity affect the mechanical properties of ionic glasses [431] and, in particular, whether this can be used to reach the high ductility found in some metallic glasses. For example, by introducing structural heterogeneity, Das et al. [430] created a metallic glass with strain reaching ~ 18%. A similar approach, which utilizes the relationships between structural heterogeneity and mechanical properties, could potentially be adapted to ionic glasses.

6.3. Ionic conductivity

Generally, ionic glasses have higher conductivity and lower E_a in comparison to conventional network glasses. It could be argued that they simply have higher concentrations of mobile charge carriers, however, there are various additional factors, including the lower entropy of melting and higher polarizability of the individual components. Indeed, trends in other properties, such as T_g, in glasses composed of conventional formers and halides or oxyanions demonstrate that the bonding of the covalent backbone changes despite an absence of direct interaction between the halides or oxyanions and the network (*see*Sections 2.4, 2.5 and 5 for more information). From a classical viewpoint "*halogen (fluorine, chlorine, bromine, iodine) ions and different inorganic salts with large-sized anions (S^{2-}, SO_3^{2-}, SO_4^{2-}, etc.) are usually used as modifiers. Upon introduction of these anions, the structure of the initial glass is loosened, the energy of bonds between charge carrier ions and the glass matrix decreases, and the electrical conductivity, as a rule, increases*" [quoted from [98]].

As noted in Section 2.3.4, in comparison with oxide glasses with the same cations (Zr^{4+}, Al^{3+}, *etc.*) fluoride glasses have much higher conductivities at 200 °C: 10^{-2} – 10^{-7} S/cm [203,205]. This is due to the mobile charge carrier being fluoride instead of a cation like Li^+ or Ag^+ [214]. As such, all of the common trends, such as the "mixed-alkali" and "mixed-former" effects, are reversed [205,230]. Although this reversal is most pronounced for fluorides, in other ionic compositions the anion can also be relatively mobile (in comparison to covalent glasses); this extra mobile component can affect the frozen-in structure and the energetics of the conductivity pathways, as is discussed below.

Given that glass is a kinetic solid, the structural arrangement is dictated by the competition between melt viscosity and

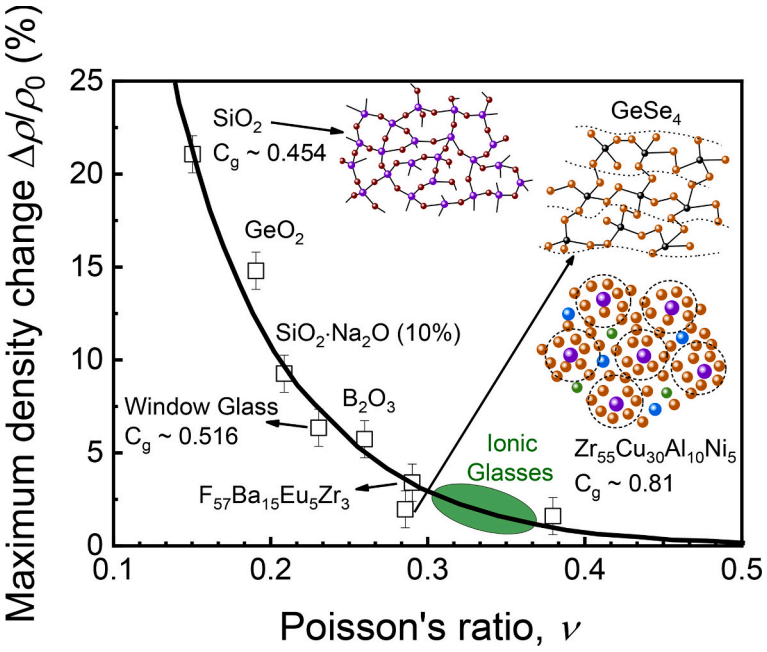


Fig. 33. Relation between the maximum of structural densification during high-pressure treatments at room temperature and Poisson's ratio of various groups of glasses. Reproduced with permission from Rouxel [429]. © American Physical Society. Reported and expected values for ionic glasses are indicated by the green ellipse.

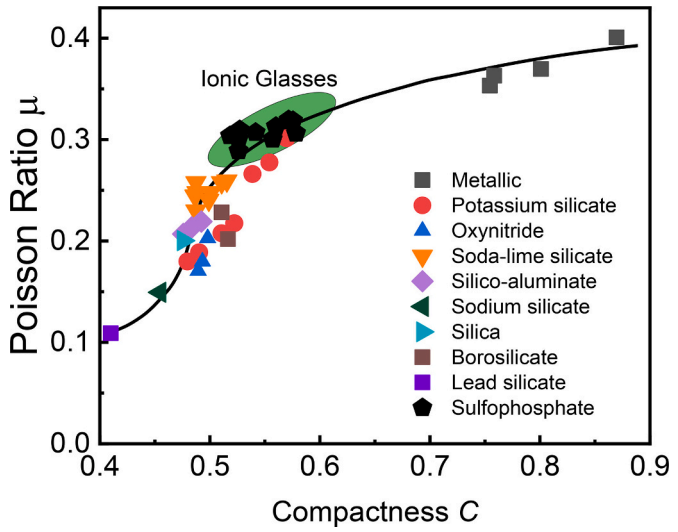


Fig. 34. Poisson's ratio as a function of structural compactness for various types of glasses. Ionic glasses (highlighted in green circle) from the sulfophosphate family are placed in between low-dimensional network glasses and metallic glasses (data from Striepe et al. [427]). Other data compiled from [8]. Reproduced with permission from Striepe et al. [427]. © Elsevier. Reported and expected values for ionic glasses are indicated by the green ellipse. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

thermodynamics, and the resulting structural heterogeneity will affect the ionic conductivity. Differences in viscosity create differing spatial regions of low and high mobility, described by Ingram et al. [290,433] as "clusters" and "connective tissue", respectively (Fig. 37). In Fig. 37, the foreign cations are cations which should prefer to be in the clusters, yet are found in the connective tissue. The *cluster bypass model* is used to explain the sharp drop in conductivity when even a small amount of a different cation is introduced. It purports that the concentration of the mobile ions in the connective tissue dictates the ion conductivity: "*In other circumstances, the accumulation of 'foreign material' in the connective tissue might be the cause of conductivity enhancement, as for example in the 'mixed anion' or 'network' effects as well as in the effects of the addition of 'dopant' salts such as LiCl*" [quoted from[290]]. Indeed, small amounts of Cl^- or Br^- (not F^-) have increased σ in sodium borate glasses [27], and SO_4^{2-} does the same in sulfophosphate glasses [11,98]; this could be due to increased structural heterogeneity and maximum viscosity difference, per the *cluster bypass model*. At the same time, a behaviour analogous to the mixed-alkali effect has been observed in mixed-halide glasses, where the F^- conductivity is lowered by the presence of Cl^- or Br^-; this represents an opportunity to extend the many models of MAE-type behaviour to an entirely different class of glasses.

In a covalently bonded glass, the viscosity of the network former is normally much higher than that of the other components. In the case of highly ionic glasses, all species may have similar diffusivity and therefore the structural and dynamic heterogeneity are equally low: distinct *connective tissue* and ion channels do not exist. In a highly ionic system, it is possible that all of one type of species is completely unbound in the glass and thereby high conductivities are achieved. This phenomenon is sometimes denoted *half-melting* and manifests in particularly low entropy of melting. Without the added complication of directional bonding, polarizability variations may govern which cation remains in the connective tissue and is mobile, and which cation is sequestered in the clusters. Thus, ionic glasses are an exciting opportunity for glass scientists to observe more thermodynamically driven glass structures and use classical electrostatic interactions to tailor electrical properties.

The electrical conductivities (~ 10^{-6} S/cm at RT) of silver and lithium sulfophosphate indicate that all of the Ag^+ or Li^+ is essentially unbound and can move freely through the glass structure, therefore resulting in pure cationic conductivity [98,100]. Increased sulfate content also leads to significantly increased electrical conductivity (up to 100 times [11,98]), which may result from increased ionicity and polarizability of the sulfate group, lower network connectivity and/or the mixed-anion effect [93,100]. One study argues that the mechanism of ion migration changes upon addition of sulfate: rather than lithium moving through channels of the phosphate network, the high-sulfate region contains vacancies into which the lithium can easily hop (since in crystalline Li_2SO_4 only half the accessible positions are filled) [98]. In their investigation of sulfoborate glass, Smedley et al. considered that SO_4^{2-} may increase the salt-like character and allow the mobile charge carrier to have a packing efficiency closer to that of an "ideal glass" [11]. The addition of sulfate to a conventional glass former demonstrates both cases described in the above paragraph: increased ionicity leads to similar mobilities of the cation and anions, and therefore, less

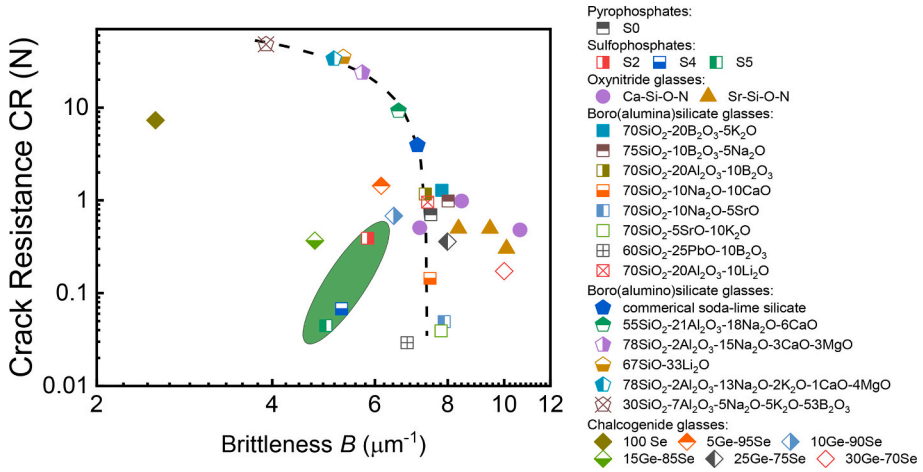


Fig. 35. Relations between crack resistance and brittleness of glasses, with ionic sulfophosphate glasses highlighted in green. Reproduced with permission from Striepe et al. [427]. © Elsevier. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

distinction between ion channels and former (Li^+ in SO_4-rich regions), yet at the same time, the polarizability of the anion (SO_4^{2-} vs. PO_4^{3-}) also determines the regions in which the mobile ion forms clusters.

In Fig. 38, a comparison is performed between superionic glasses (those with $\sigma \geq 10^{-3}$ S/cm at operating temperature) and a conventional glass former, $Na_2O{:}3SiO_2$ [12]. In essence, the data demonstrate the similarity in high-temperature conductivity between molten KNO_3 and superionic AgI-$AgBO_2$ glass; however the superionic glass has a lower activation energy. Compared to other glasses, the structure is such that in superionic glasses, the energy mismatch between jumping ions and their target sites is small, resulting in easily accessible channels for ion conduction. Since some of the energy cost of the ion "hop" is caused by the ion site being forced to rearrange to accommodate the ion leaving or entering, the lower activation energy for the ionic glass follows from its decreased local order [434].

6.4. Relaxation processes

Glasses have been called the, "*ideal objects for the study of relaxation processes* [quoted from [435]]", as they exhibit continuous changes over a large temperature range [435]. Relaxation processes are defined as the, "*return of a macroscopic system towards the thermodynamic equilibrium after the equilibrium of the system has been disturbed* [quoted from [435]]"; they are generally local, reversible movements of atoms, with electrical, mechanical and structural relaxation processes being the most commonly studied. Different measurement methods offer different advantages and probe different relaxation timescales, *see* [79]. For example, dielectric measurements can probe much larger frequency ranges than other measurement techniques, allowing separation of relaxation processes over a wide range of temperatures, while NMR can assign relaxation processes to structural units in the material. All of these measurements share a common feature: they inject energy into the system, usually at a constant rate, and then measure changes in the system as a function of response time. Superionic glasses in particular lend themselves to kinetic studies; the highly decoupled nuclear-spin lattice relaxation and d.c. conductivity 'jump' frequencies, *i.e.*, the different E_a for different species, allow for better understanding of unique relaxation modes in glasses [436,437].

6.4.1. Fragility

$Ca(NO_3)_2{:}KNO_3$ has been used as a model ionic material for studying the relaxation processes of disordered solids and their melts [79–82]. A plot of log(η) as a function of inverse temperature (*Angell* plot [438]) of this material and several other types of glass forming

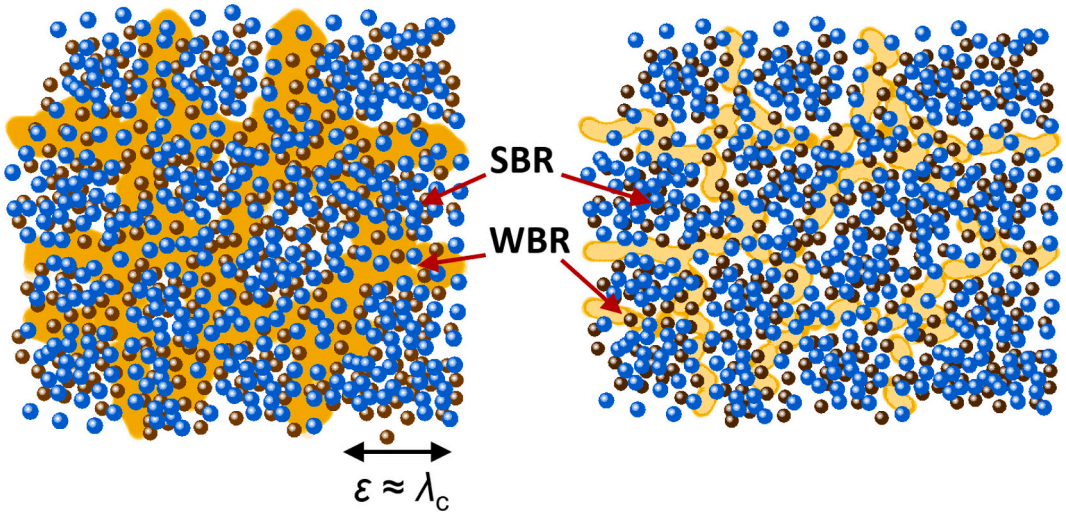


Fig. 36. Representation of a static heterogeneous structure of inhomogeneous Pd- and Pt-based (left) and more homogeneous Zr-based (right) metallic glasses. Heterogeneity manifests in strongly bonded regions (SBR) and weakly bonded regions (WBR), which are thought to facilitate plastic deformation. The correlation length, ε, is approximately equal to the size of the elastic heterogeneity (cross-over wavelength), λ_C. Reproduced with permission from Ichitsubo et al. [432]. © Elsevier.

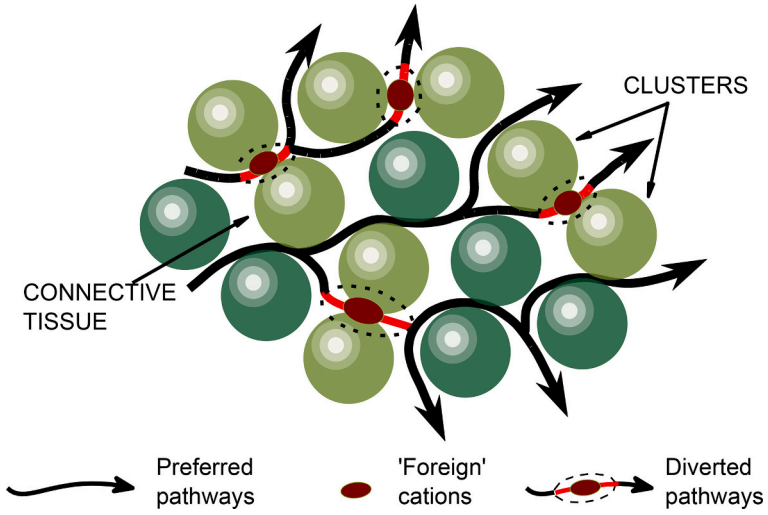


Fig. 37. Schematic representation of the cluster bypass model. The arrows show the locations of the pathways for ionic migration within the '*connective tissue*' and '*diversions*' caused by the presence of foreign cations. Reproduced with permission from Ingram et al. [433]. © Elsevier.

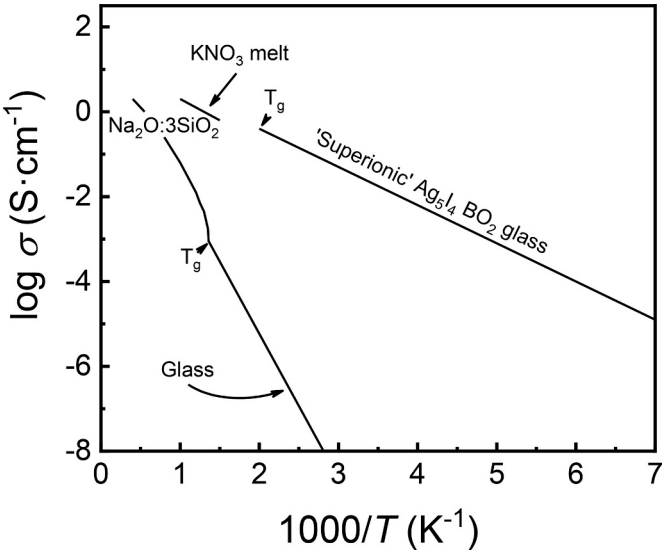


Fig. 38. Comparison of the temperature dependence of ionic conductivity in molten and vitreous $Na_2O{:}3SiO_2$, molten KNO_3 and vitreous $Ag_5I_4BO_2$. Reproduced with permission from Ingram [12]. © Elsevier.

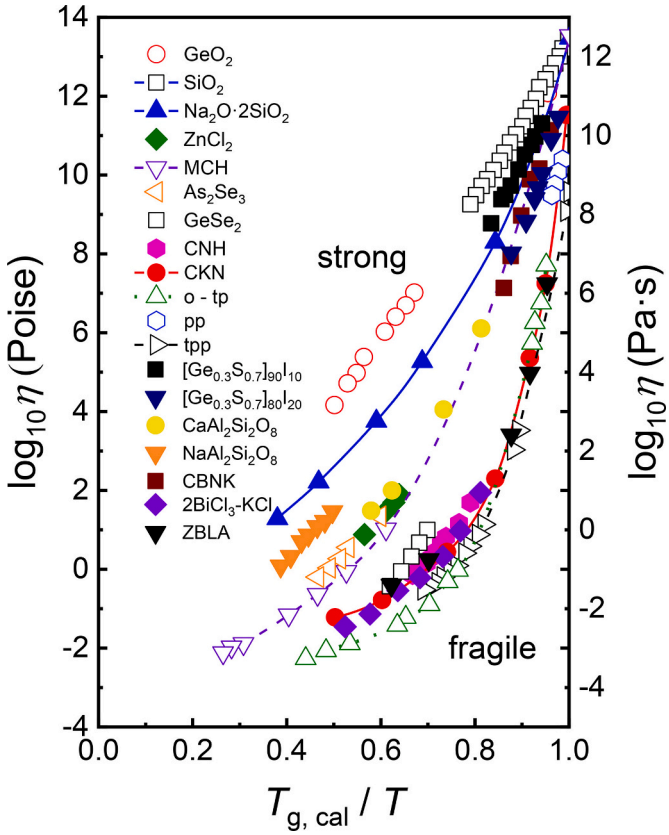


Fig. 39. Angell plot of the viscosity of a variety of supercooled liquids. The inverse temperature scale is normalized with respect to the calorimetric glass transition temperature T_g. The following acronyms have been used: MCH: methylcyclohexane; *o*-tp: *ortho*-terphenyl; pp: phenolphthalein; tpp: triphenylphosphite; CKN: $2Ca(NO_3)_2{\cdot}3KNO_3$; CNH: $Ca(NO_3)_2{\cdot}4H_2O$; CBNK: $Cd_{0.5}Ba_{0.13}N_{0.34}K_{0.03}Cl_{0.66}F_{0.97}$; ZBLA; ZrF_4-BaF_2-LaF_3-AlF_3. *Lines* are drawn to guide the eye. Data taken from Böhmer and Angell [79], Seddon et al.[445] and Matecki et al. [444].

liquids is provided in Fig. 39, where the inverse temperature is normalized to T_g so that the plot shows a range between 0 (high T) and 1 (low T) [79]. It can be seen that ionic glasses, CKN: $2Ca(NO_3)_2{\cdot}3KNO_3$ and CNH: $Ca(NO_3)_2{\cdot}4H_2O$ both have strikingly non-Arrhenian behavior; they are *fragile* as compared to *strong*, conventional glass-formers such as like SiO_2 and GeO_2. Ionic glasses are often kinetically *fragile* glass-formers (with some prominent exceptions such as BeF_2 and $ZnCl_2$): their dynamic properties, such as viscosity, dielectric constant, sound wave velocities and specific heat capacity are keenly temperature-dependent and depart significantly from Arrhenian behaviour. Over the years, many different models have been used to reconcile this departure from Arrhenian behaviour in molten salts close to and above T_g; for a more complete history we suggest reading [30,79,84]. In fact, the only glasses in Fig. 39 which are more fragile are molecular glasses (*o*-tp: *ortho*-terphenyl; pp: phenolphthalein; tpp: triphenylphosphite). It has been proposed that the nature of the relaxation processes can be fundamentally different between ionic and molecular glass formers as well as between strong and fragile liquids [75].

As shown in Fig. 39, fluoride glasses are also notably fragile (ZBLA and CBNK), especially ZrF_4-based glasses [439–442], which have high configurational entropies due to their range of coordination numbers and polyhedral connectivity (corner- vs. edge-sharing), and fluoride's ability to be bridging or terminal [443]. Other types of halide glasses are similarly fragile, for example $2BiCl_3$-KCl [444]. Generally, large

amounts of terminal, dangling bonds are required for high fragility; Ge-S-I becomes fragile when I = 20 mol%, due to sufficient terminal Ge–I bonds, reducing connectivity and surpassing the rigidity threshold [274,445]. Moreover, halide glasses with large amounts of conventional glass formers were found to be less fragile [265,446]. This is confirmed for conventional network formers as well: $NaAl_2Si_2O_8$ vs. $CaAl_2SiO_8$ in Fig. 39.

Among the counterexamples, $ZnCl_2$ has a fragility comparable to that of the chalcogenide As_2Se_3. From the discussion in Section 4, we know that $ZnCl_2$ forms a structure analogous to that of SiO_2, composed of tetrahedra with restricted bond-angles. It has been argued that $ZnCl_2$ is a strong glass former due to its relatively slow rate of bond breaking and small increase in entropy with increasing T (*see* discussion below) [134,447]. The same "strong" behavior is found for BeF_2 [448] and was explained by its relatively ordered local structure, again akin to SiO_2 and GeO_2. Therefore, it is not surprising that the 3d network-like structure of $ZnCl_2$ exhibits *strong* behavior; it is more interesting that $ZnCl_2$ is still more fragile than Na_2O-$2SiO_2$, demonstrating that its overall bonding topology is less restrictive than that of an alkali disilicate composition.

Typically the types of materials which exhibit the largest deviations from linearity in their viscosities, *i.e.*, are the most fragile, are molten salts which lack directional bonding between and within chemical species [449]. Angell et al. [79] and Vilgis et al. [450] argue that the directionality of the bonding and size of coordination sphere in glass is intimately related to the kinetic fragility: fragile glasses have non-directional bonds with flexible coordination spheres allowing many different degenerate structural states to be accessible at the same temperature. As a result, when the temperature approaches T_g there is a large jump in the number of accessible states, and therefore the configurational entropy (so-called *structural fragility*). On the other hand, conventional glass formers with directional bonding are limited in the number of energetically similar and accessible structural states and therefore there are no large changes in the local structure and the configurational entropy increases exponentially throughout the glass transition. For metallic glasses, such relations between kinetic fragility and structural fragility have been demonstrated in detail [451]; it will be interesting to see whether a similar general picture also exists for ionic glasses.

Alternatively, the non-linear temperature dependence of kinetic processes and the jump in configurational entropy can be explained by the cooperativity of the glass' constituents. As a liquid cools, clusters (*i.e.*, *cooperatively rearranging regions*, or CRR) combine to form larger regions, thereby decreasing in number and losing configurational entropy. Conversely, at higher temperatures the thermally induced structural changes increase the configurational entropy and decrease the activation energy for kinetic processes [80]. We expect ionic glasses to contain various bonds of a similar energy and therefore, to a first approximation, whose properties would depend on temperature similarly. Thus, the large jump in activation energy upon cooling is likely caused by the sudden activation of many similarly sized clusters. Angell's work suggests the existence of two distinctive classes of glassy materials: polymers/network glasses that couple to form subunits, and molecular glasses such as aromatic alcohols which can have different cooperativities but the same fragility [79]. Since the effective activation energy is dependent on the size of the cluster and the rearrangement of *a single molecule in a cluster* [79], we presume that the above explanations are only valid for highly ionic glasses. When the ionic glass is more heterogeneous and has some components with covalent bonding, we expect that the 2nd-nearest-neighbour interactions will be much more important.

Although it is practically impossible to readily distinguish between types of bonding in different liquids with the same kinetic fragility (m) at T_g, the mechanical properties can help separate glass types further. A comparison of m and mechanical deformability (high frequency bulk and shear moduli, B_∞/G_∞) is shown in Fig. 40. This plot demonstrates

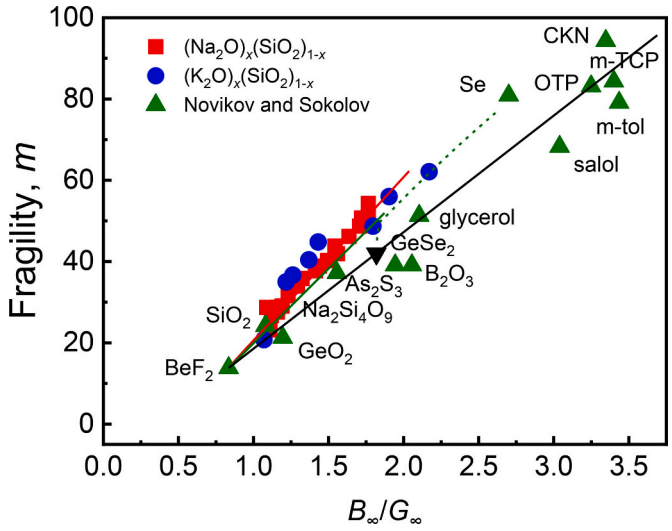


Fig. 40. Correlation between the ratio of the high frequency bulk and shear moduli, B_∞/G_∞, and m for a range of single-component glasses (green triangle, from [452]), together with values for $(Na_2O)_x(SiO_2)_{1-x}$ (red square, up to 40 Na_2O mol% [453]) and for $(K_2O)_x(SiO_2)_{1-x}$ (blue circle, up to 40 K_2O mol% [454]). Reproduced with permission from Greaves and Sen [106]. © Taylor and Francis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that highly ionic glasses and invert glasses are fragile and have volume-conserving behaviour (as discussed in Section 6.2), while polymeric ionic glasses are stronger and more shape-conserving. Indeed, the halide family represents an extraordinary case for study because "*molten halides therefore span the whole spectrum of fragility in a way that no other glass-forming system does*" [quoted from [106]]. Like ionic glasses, molecular glasses can also be fragile and be easily mechanically deformed. Consequently, when attempting to identify an ionic glass, it is always important to consider the type of bonding, *i.e.*, the stability of the ionic components.

There has been at least one attempt to relate ionicity (defined as the square of the difference in electronegativity normalized by bond distance) to fragility in glasses, with higher ionicity being roughly correlated with higher fragility [455]. Many studies focus on relating the fragility (non-Arrhenius behaviour) of many kinetic processes to the structural heterogeneity or level of disorder in a glass [30,80,106,456]; it is assumed that additional degenerate states (configurational entropy) are responsible for the non-linear relaxation. Sen summarizes: "*In terms of energy landscapes, where strong liquids like silica are characterized by broad minima, those of fragile modified silicate melts are distinguished by numerous sharp minima that reflect the increased variety of configurations resulting from the depolymerization of the network by the modifying oxide*" [quoted from [106]].

We know that monoanionic glass melts, *i.e.*, halide glass melts, have higher configurational entropy and fragility than conventional glass formers (despite oxygen being an anion we do not classify oxide glasses as monoanionic glasses since their bonding character generally has a high covalent component). Yet, it is clear from the structure of many ionic glasses that there can be regions of high ionicity (like isolated AgI or Na_2SO_4 clusters) interspersed between regions of lower ionicity (like chains of $AgPO_3$ or $NaPO_3$), similar to those shown in the cluster bypass model (Fig. 37). We would expect ionic glasses with mixed bonding regions, such as oxyanionic–monoanionic glasses, to have the most structural heterogeneity, higher than in conventional or monoanionic glasses. Therefore, these glasses are expected to have higher structural and kinetic fragility. However, these mixed bonding-type ionic glasses are not necessarily the most fragile; in fact, nitrates appear to be the most fragile. Greaves and Sen claim that if the medium-range inhomogeneity caused by network former and ion channels could be

excluded, a "*simple **inverse** relationship between melt fragility and glass inhomogeneity should prevail*" [quoted from [106]]. Thus, without the heterogeneity introduced by mixed bonding types, more fragile melts would be more homogeneous on the long-range scale. This appears to indicate that highly ionic systems, where the bonding environment and mobility of the anion and cation are nearly indistinguishable, should be kinetically fragile. Yet, we observe that the fragilities of ionic systems such as $ZnCl_2$ and BeF_2 (in Fig. 39 and Fig. 40, respectively) are low, indicating a more complex relationship between structural and kinetic fragility than that observed in metallic glasses [451].

6.4.2. Electrical relaxation

Since $Ca(NO_3)_2$:KNO_3 displays significant fragility, *i.e.*, lower-than-expected viscosity well-above T_g, its electrical relaxation processes as a function of T are also of interest. At temperatures below T_g, the electrical conductivity is dominated by individual ions diffusing in a solid network, however this 'solid-like' ionic conduction can still be found well above T_g [80]. Howell et al. ascribed the non-exponential electrical relaxation behaviour to the effects of microscopic heterogeneity [80]. However, eventually the conduction mechanism gradually changes to a 'liquid-like' conductivity, where the movement of the ions and rearrangement of the network are on the same timescale (these rearrangements are more important than increased diffusivity).

Like nitrate glasses, also $Ag_7I_4AsO_4$ glass displays non-exponential conductivity similar. Rather than having a sudden change in σ from a phase transition, it gradually falls off in the 'linear' Arrhenius range ($10^{-4} - 10^{-9}$ S/cm). The conductivity of molten salts generally falls off somewhat above T_g [457], while, uniquely, the conductivity of $Ag_7I_4AsO_4$ becomes nonlinear below T_g (Fig. 41, T_g = 65 °C [84]). Grant et al. pointed out that both nitrate and iodioarsenate glasses depart from Arrhenius behaviour at ~ 10^{-6} S/cm; the authors attributed this to the same physical cause found in $Ca(NO_3)_2$:KNO_3 glass: an eventual complete "unfreezing" of the network, leading to an increase in the dynamical behavior of the anions [84]. On the other hand, sufficiently below T_g, the structure remains frozen and the cations are forced to migrate through a static anionic lattice. The non-linear conductivity of $Ag_7I_4AsO_4$ glass is in accordance with the reduced plasticity found below T_g, as measured from shear relaxation [84]. It appears that dynamic electrical and mechanical processes have a similar underlying mechanism in these ionic glasses.

Further information can be obtained from comparing the temperature-dependent dielectric and mechanical response spectra. The relaxation processes (primary and secondary) of a ZrF_4-based glass were described as "*remarkably similar to those in molecular glasses*" [quoted from [458]], and unlike those of typical inorganic glasses, *e.g.*, silicates [458]. The measured electric and mechanical responses were very similar: they followed the same rate and had maxima at the same frequency, indicating that the motion of the same structural units was responsible for the relaxation seen in both sets of experiments. This is contrast to covalent glasses, where the dielectric and mechanical loss relaxation rates can be different by a factor of 30 [459], and may indicate a common trend for ionic glasses: electric and mechanical response functions could be caused by the same process. However, Angell and Böhmer caution that, generally, the width and shape of the resonance peak (decoupling β-relaxation) can vary significantly between both types of experiments [30,79].

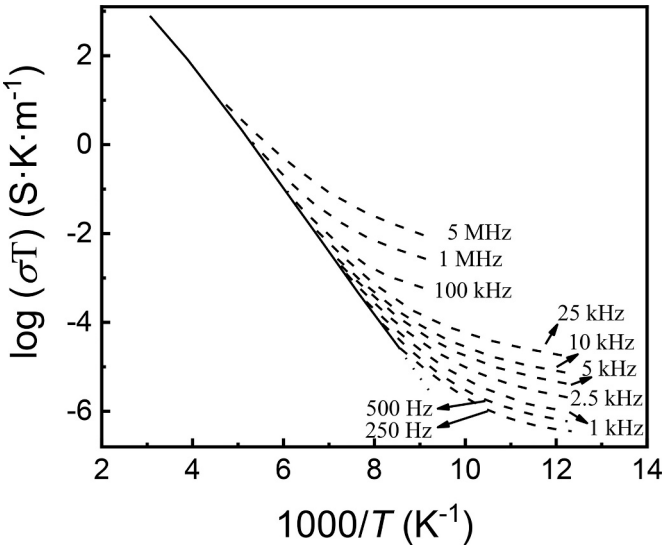


Fig. 41. Temperature-dependence of the d.c. conductivity of $Ag_7I_4AsO_4$ glass at fixed frequency. Reproduced with permission from Grant et al. [84]. © American Chemical Society.

6.4.3. Spin-lattice relaxation

NMR spectroscopy is very different from viscosity or impedance measurements, probing the relaxation of magnetic moments *via* the spin-lattice relaxation time, T_1. The spin-lattice relaxation time is a measure of the mobility of the lattice: higher energy vibrational and rotational states are more likely to provide the fluctuating fields required to capture energy from an excited state [460]. The main difference is that for impedance and some types of mechanical measurements the frequency and temperature are easy to vary, while NMR measurements are usually limited to temperature variation only. Indeed, it has been proven that decay of T_1 shares the same temperature dependence as electrical and mechanical response spectra [30,461,462]. Since NMR spectroscopy can assign relaxation processes to corresponding structural units in a material, it has the potential to identify the structural groups that correspond to each relaxation mode, such as in the case of two distinct groups in nitrate glasses [75]. As such, it has the power to provide clear answers about whether differences of mobility can be ascribed to different chemical regions, like those predicted by Ingram's cluster bypass model [433].

6.4.4. Decoupling of relaxation processes

Fig. 42 highlights the different relaxation processes which can happen in a non-polymeric, fragile glass. The slowest and most fundamental mode is called the α-relaxation, but other processes, such as β-relaxation (common secondary hops and rotations of chemical species) or ionic conductivity can decouple from the primary α-relaxation. When a glass transforms into a liquid, there are several different relaxation processes happening in parallel, which cannot easily be studied individually because the processes are strongly interacting; for example, localized diffusion of ions may or may not be coupled to the long-range motions which govern viscous flow (although densification will eventually affect ionic motion) [458].

The relaxation behavior of liquids is often fit with a stretched exponential function, with the non-exponential parameter β found between 0 and 1 [30]. At T_g, nitrates have been reported to have dielectric loss β parameters of 0.74 [30], and HMFGs have 0.61 [458]. Thus, the impedance spectra of ionic glasses are more likely to be Lorentzian in shape and dominated by only one relaxation process (at a given temperature; over a broad range of temperatures the appearance of additional relaxation processes is expected). Consequently, relaxation modes inseparable in covalent glasses are observable in ionic glasses; for example, shear viscosity relaxation modes are much more easily observed in fragile glass formers [79]. Presumably these relaxation processes happen in conventional glass formers as well, but at a lower proportion, making the study of fragile, ionic glasses important for all glass types [81]. Angell highlights that neither the kind of relaxation process which dominates primary relaxation, nor the relative strength of each mode relaxation, is known [30]. In particular, relaxation parameters that are correlated with entropy or free volume would be of interest in understanding the glass transition.

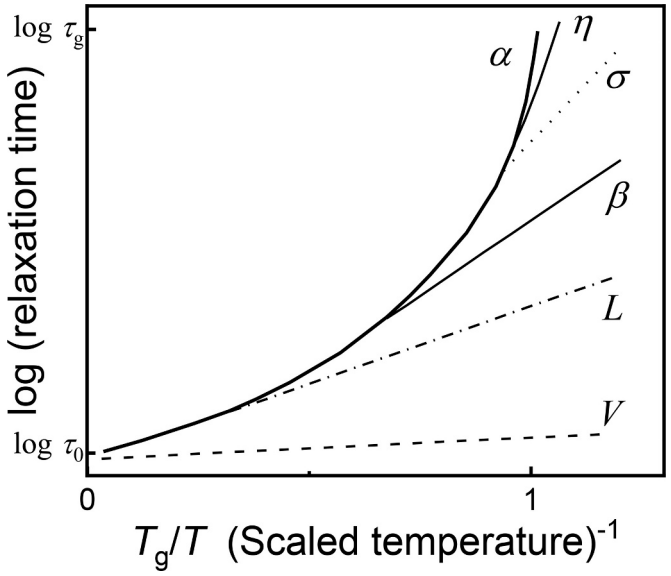


Fig. 42. Schematic relaxation map for a non-polymeric glass-forming material. Apart from the primary or α-relaxation, vibrational (V) and local modes (L) can exist. β-relaxations are almost universally observed, and conductivity (σ) decoupling is a major factor in ion conducting materials. Shear viscosity (η) decoupling is particularly apparent in fragile glass-formers. Reproduced with permission from Böhmer and Angell [79]. © Springer-Verlag.

6.5. Crystallization

Halides are ideal for the study of crystallization from a melt, as many compositions likely undergo homogeneous nucleation [463]. Furthermore, they can possess complex and useful crystallization behaviour: in HMFGs, several different crystalline phases exist which, if metastable, can interconvert below T_L [206]. The $ZnCl_2$-KI system is also unique in that it undergoes stable amorphous–amorphous macrophase separation close to the liquidus curve; with its low viscosity and low T_m, $ZnCl_2$-KI "*is a model system for studies on physicochemical properties of super critical melts*" [quoted from [169]]. Additionally, a thermodynamic study revealed amorphous and crystalline $ZnCl_2$ to have nearly the same heat capacity, making it easier to separate kinetic and thermodynamic effects [134].

As shown in Fig. 43, crystallization generally decreases σ (due to reduction of free volume), however there are exceptions. For example, in their review Tatsumisago and Hayashi demonstrated that the metastability of glass can lead to inclusions of high-temperature, superionic, crystalline phases, which have larger volumes and are metastable at room temperature (RT) [304]. These phases are not achievable by regular solid-state reactions, and yield glass-ceramics with conductivities of 10^{-1} S/cm at RT. The most common metastable phases studied thus far are α-AgI (by suppression of the α ➔ β transformation) in $82AgI\text{-}13.5Ag_2O\text{-}4.5B_2O_3$ [14,15] and $Li_7P_3S_{11}$ in $70LiS\text{-}30P_2S_5$ glass [13]. Orthoborate appears to be a conventional glass-former which can easily form the basis of many superionic glass compositions, however superionic glasses composed of only silver halide salts, with only small deviations from Arrhenius behaviour, have been studied as well [464].

As with other ionic glasses, the authors found the sulfate glasses to have high configurational entropy in the multi-component glass-forming compositional region, and lower configurational entropy near the endpoint members [90,91]. A thermodynamic study of sulfophosphate melts found that compositions with only one type of cation have a single eutectic and constant activation energy for viscous flow, while mixed-cation compositions formed intermediate compounds [465]. The same behavior has been found in mixed-cation sulfate glasses, where the mixed system forms an intermediate solid compound and the mixture has much higher E_a for viscous flow than the end members [465].

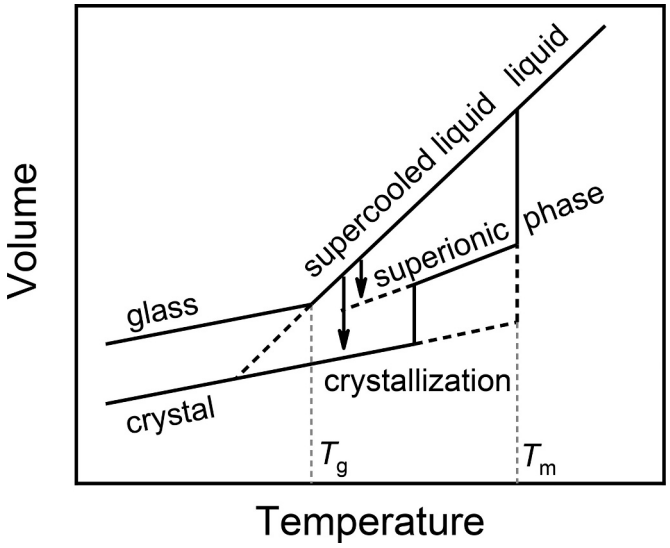


Fig. 43. Volume changes from glass to crystal with increasing temperature. Reproduced with permission from Tatsumisago and Hayashi [13]. © Elsevier.

Since sulfate self-segregates when added to a typical glass former, addition of sulfate appears to encourage formation of sulfate-containing crystals, which may be used to design sulfate glass-ceramics [11,100,104,105]. Crystallization studies of K_2SO_4-$ZnSO_4$ reported sharp crystallization peaks and the presence of strongly second-order kinetics as evidenced by the non-linearity of the rate of crystallization with temperature (Fig. 44) [90]. For many sulfates, often only one crystallization peak was observed, yet the crystallization energies were much higher than would be expected from a simple reduction in volume, signifying that the structural groupings are much different in the glass than in the corresponding crystals [91].

In addition to ionic conductivity, crystallization strongly influences various other macroscopic properties, in particular, optical and mechanical ones. In order to design specific property profiles there has been much investigation into understanding the mechanisms of crystallization in oxide glasses [466]; the composition, structure and nanoscale heterogeneity of a glass network play a large role in its crystallization behaviour. We expect the degree of heterogeneity

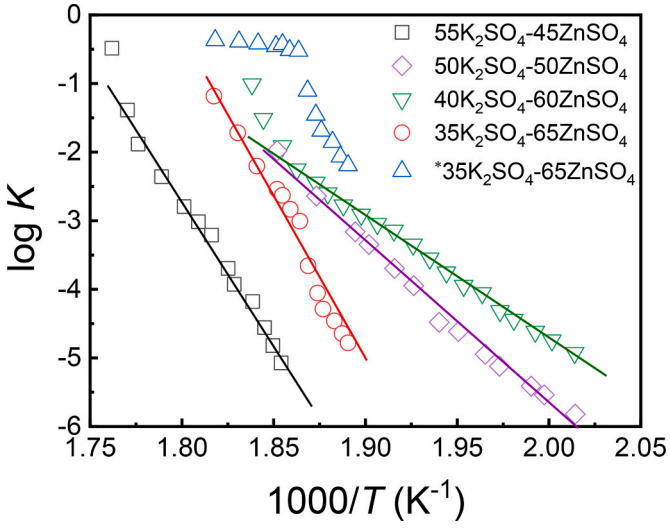


Fig. 44. Rate of crystallization (K) as a function of inverse temperature in K_2SO_4-$ZnSO_4$ glasses. Data were calculated assuming second-order kinetics, with the exception of *$35K_2SO_465ZnSO_4$. Reproduced with permission from Narasimham and Rao [90]. © Elsevier.

(configurational, structural and dynamic) of the starting glass to affect crystallization kinetics, phase of crystals, phase purity, number of defects and crystal size. Structural and dynamic heterogeneity play large roles in the viscous slowdown near the glass transition and further observation of this kinetic transition in ionic glasses is needed [75]. Such studies of ionic glass-formers enable investigations of the glass transition without interference from the directional bonding found in conventional glasses. If the fundamental relationships between the glass transition, crystallization and different types of heterogeneity are properly understood, such knowledge could allow for controlling crystallization in improved ways.

7. Applications

Although we find ionic glasses to be of significant fundamental interest, the properties common to most ionic glasses make them of practical use as well. Their main attributes which can be exploited for applications are their low T_g and T_m values, low or high chemical durability, high ionic conductivity, high Poisson's ratio (leading to less brittleness and more plasticity) and their often unusual optical properties. Furthermore, ionic glass' compositional flexibility, in particular that of the anionic species, makes them a truly tailorable glass family, allowing for optimization in several scientific and engineering fields.

Low T_g and low viscosity above T_g make them ideal candidates for composite devices, whether it is as glass solder in established electronics applications or to provide structural reinforcement in a glass-polymer composite. In the case of the very low-melting ionic glasses, the inorganic glass phase could be added to a polymer composite material by co-forming. Ziegler and Angell have pointed out that some of these glasses are stable enough to anneal 10 °C below their T_g for almost a year without any crystallization [194].

Ionic glass' compositional tunability could be taken advantage of in all of its applications, with the greatest opportunities in optics and photonics. As we discuss above in Section 6.1, by modulating the identity of the anion it is possible to widen or shift the transmission window into the deep IR or UV, and to modify the refractive index, Abbe number, transmission losses and non-linear optical properties. Furthermore, the optical basicity can greatly affect the frequency and shape of the emission or absorption spectra of colour centres, which can be important for laser applications and filter glasses. Large stimulated emission cross-sections are of interest for low-threshold, high gain laser applications [168,250]. Finally, the long lifetimes of fluorescent rare-earth elements in ZBLAN glasses has been attributed to low non-radiative decay rates, especially below 3200 cm^{-1}; this is a consequence of much lower multiphonon process rates compared to other typical host materials, such as SiO_2 or YAG [235–237].

High solubility of many types of ions makes them ideal candidates for fluorophore hosts; ionic glasses with lowest T_g even let envision doping with organic (aromatic) molecules. The main issue facing many ionic glasses in laser applications would be water contamination; for example the theoretical quantum efficiency has yet to be realized in Nd-doped $ZnCl_2$ glasses [250]. The effect of atmospheric moisture has been measured in optical fibres as well: water undergoes a limited surface reaction that does not change optical transmission, but dramatically decreases the tensile strength; however, a controlled atmosphere inhibits hydrolysis at the surface [166].

Generally, hygroscopicity is regarded as a disadvantage, however, there are situations in which glass dissolution is advantageous. The main example is bioactive glass [200–202]; we know highly modified conventional glass formers, such as potassium silicates, sulfates and halides, dissolve readily in distilled water or even gelate if left overnight in normal atmosphere [91,109,162,168,190,194]. Such dissolution behavior can be used when designing resorbable glass compositions with tailored, or even personalized ion release, notably involving not only the cation species but also the anions. Furthermore, there has been research into dissolving glass homogenously for applications that require solution feedstocks [467], such as in the production of solution-based coatings [468].

Since sulfur is often a limiting factor in the waste vitrification [469], the relatively high solubility of sulfur (as sulfate) in ionic glasses could be useful also in that field. Many studies about sulfate solubility in heavily modified silicates and borosilicates have been published, as these are glass compositions commonly used for waste storage [105,470–472]. Nonetheless, we believe that a reduction in silicate content would increase the sulfate solubility immensely, for example, by moving towards an ionic sulfoborate composition. It remains to be explored whether the chemical stability of such glasses could be tailored in a way to allow for long-term immobilization.

There are some applications in which crystallization is desirable, such as in phase change materials (PCMs) for information or energy storage [473]. Although optical storage media is a mature field, there has been renewed interest in using PCMs for non-volatile electronic storage purposes [474]. In addition to utilizing differences in optical properties between glass and crystal, there are also electrical differences between the two phases which can be treated as Boolean switches [475]. One of the main hurdles for this application is finding a glass which crystallizes homogeneously at a sufficiently low voltage [474]. Here, ionic glasses may be a possible option due to their non-volatility and the low voltage required for crystallization.

Several ionic glass compositions have already been tested for their use as solid electrolyte materials. As Minami [219] points out, compared to crystals, glasses have no grain boundaries, are isotropic and can easily be shaped into various forms, making them more readily adaptable to a given application. In general, glasses are a desirable material for electrolytes because only the cation is mobile, removing the need for a plastic separator which can break during usage and result in a potentially dangerous shorting of the electrical cell. Single-cation conduction is also extremely beneficial in other ways; in addition to easier optimization of σ, it also results in "*fewer side reactions and very wide electrochemical windows*" [quoted from [13]]. Due to their high σ, superionic glasses have been tested in thin film and coin-type batteries. Only 10^{-6} S/cm is required for thin films [12], while coin cells using the Li_3PO_4-Li_2S-SiS_2 system have conductivities of 10^{-3} S/cm [16]. As shown in Fig. 45, ionic sulfide-based glasses or glass-ceramics have better σ than many crystalline electrolytes and, most importantly, their conductivities approach the conductivity of liquid electrolytes at room temperature. These sulfide-based glasses or glass-ceramics also display properties important for implementation; small volume change, little capacity degradation and high charge-discharge efficiency after many cycles [13].

8. Relationship to ionic liquids

Most theories of glass formation distinguish sharply between glass formers and modifiers; thereby declaring that covalent, directional bonding is vital for successful glass formation. However, there is an entire field of study devoted to materials which interact mainly through Coulombic charge–charge attractions, yet easily form glasses: ionic liquids. Their propensity to form glasses is aptly described by Wilkes et al.: "*the melting temperature of many ionic liquids can be problematic, since they are notorious glass-forming materials.*" [quoted from [89]].

There is no fundamental difference between molten salts and ionic liquids, other than a shift in working temperature. Much like many ionic glasses above their T_g, ionic liquids (ILs) are simply molten salts, or compounds dominated by ionic interactions, yet with melting points below 100 °C [89]. They can be composed of any combination of different types of anion and cation, although generally the cation is an organic species in order to lower the melting temperature [89]. Of course, not all ILs can form a glass [80], but many compositions, such as halide salts, are already studied as ionic glasses. Given the many similarities between ionic glasses and ILs, much of the theoretical and practical understanding about ionic liquids is likely also applicable to

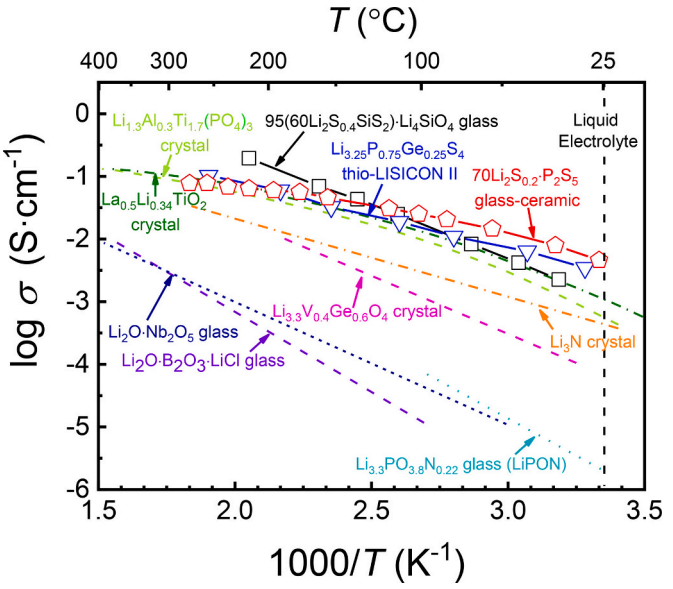


Fig. 45. Temperature dependence of conductivity for typical inorganic solid electrolytes. Reproduced with permission from Tatsumisago and Hayashi [13]. © Elsevier.

ionic glasses.

Ionic liquids have low liquidus temperatures (− 100 – 400 °C [476]) and broad melting transitions, which are often extended below the liquidus line by the glass transition [477]. Although ionic liquids are more viscous than common solvents, these materials are still liquids with viscosities ≈ 0.01 – 5 P at RT [478], with a few exceptional ILs having viscosities of up to 10^3 P at 10 °C [479]. Some ionic glasses, such as nitrates, also have low viscosities at temperatures only just above T_g, *e.g.*, 10 P when $T = 1.25T_g$ (they are highly *fragile liquids*, *see*Fig. 39); therefore the distinction between ionic glasses and ILs *really* is only a matter of melting temperature (caused by the weaker bonding of the organic component). The Coulombic interactions in ILs are relatively weak, with energies on the order of intermolecular interactions in molecular liquids; therefore, other types of bonds such as van der Waals and hydrogen bonding are also important [478]. In addition to ion size, the strength of all these forces must be considered when examining changes in viscosity. Interestingly, a small addition of a *cosolvent*, *i.e.*, a molecular liquid such as acetonitrile, can significantly reduce viscosity [478]. The addition of a molecular glass into an ionic one has not, to our knowledge, been explored, and could provide an interesting topic of further study. From our perspective, molecular glasses and ionic glasses share some topological similarities as they both lack percolating networks and yet have different types of bonding (presumably these contrasts also could be seen in their physical properties). We can imagine that adding NaCl to a phosphate network might be an example of the reverse process: adding an ionic glass to a molecular one (if one can imagine a 3-d polyphosphate as one giant molecule), and we would expect to see a substantial reduction in viscosity.

ILs also have complex thermal behaviour, with solidification being a kinetic transition and, in some cases, evidence of solid–solid transitions [480] akin to those observed in some ionic glasses (Section 6.5). The melting temperature is related directly to the Coulombic attraction; to decrease T_m the charge density should be minimized (ionic charge should be low, ion size large) [89]. To prevent crystallization, one must decrease the lattice energy of the crystal, for example by use of flexible, asymmetric ions that increase charge separation (packing inefficiency), disrupt lattice isotropy and generate more rotational and vibrational modes. This is analogous to increasing chemical, structural and/or dynamic heterogeneity, which we know greatly increases glass-forming ability in ionic glasses (whether by geometrical frustration or by mixing a conventional glass former with an ionic glass, *e.g.*, in haloborates, *see*Section 4).

There are striking similarities between the guiding structural principles we have described for common ionic glasses and those which apply to IL-derived glasses. For example, strong correlations between the local structure of the solid and liquid have been found for ILs [481]. Similarly, like mixed glass formers, the ionic liquids LiF-NaF-KF [481] and $ZnCl_2$-RbCl [414] form new liquid structures with fully interacting species, while retaining the local structure of their components. Another striking example is $AlCl_3$: the liquid was found to be composed of polymeric, corner-sharing tetrahedra, but when an alkali halide was added it broke the Al—Cl—Al linkages and promoted the formation of $AlCl_4^-$ species [482]. This process is analogous to the conversion of BO_3 to BO_4 units in alkali borate glasses.

Many structure analysis techniques used to study ionic glasses, especially those used for halide glasses, are also applicable to ILs, including neutron diffraction, X-ray scattering and EXAFS [481]. In ILs there is a correlation between organic chain length and the range of order: longer chains result in intermediate ordering and pronounced diffraction peaks at longer scattering vectors. The organic chain length could act similarly to the chain length of a conventional glass former (*e.g.* metaphosphate) in an ionic glass, where the added directionality of covalent bonding limits possible topologies. On the other hand, computational modelling of ILs faces many of the same problems as does that of glasses: "*liquid phase* (amorphous structure)*, a high viscosity, the size of the constituent ions, the type of interactions that occur, and the electronic structure*" [quoted from [483]]; these challenges prevent in-depth insights into the constructive principles of IL-derived glasses. Nonetheless, many of these challenges might be overcome in the near future, as quantum chemical, classical MD and Monte Carlo, and *ab initio* MD studies are starting to yield insightful structural and dynamical information likely applicable to ionic glasses as well [483].

Given that impurities and water are often present in ILs and can affect their properties (*e.g.* viscosity and density [122,484], the effects of such could be a point of interest for ionic glasses as well. As discussed in the Sections 2.2.1 (nitrates), 2.2.2 (sulfates), 2.2.4 (hydrates), and 2.3 (halides), sorption of water represents a significant challenge to the applicability of ionic glasses. Water has been shown to normally be present in ILs (even hydrophobic ones have > 1 wt.% of H_2O [485]). Even very small amounts of impurities, e.g., 0.001 to 0.09 wt%, can drastically affect macroscopic properties (*e.g.*, ionic conductivity [478,486]). Thus, as discussed above, it is very worthwhile to take the same care and attention when synthesizing both ionic liquids *and* glasses. Understanding of the structural role of water is critical to the production of accurate chemical analyses.

Much initial research interest in ILs was related to their use as electrolytes in battery and capacitor applications [487]. Therefore, like the field of ionic glasses, optimization of ionic conductivity is of import in IL research as well. Some ILs possess lower-than-expected σ ($\sim 10^{-3}$ S/cm); this is rationalized by ion pairing and/or ion aggregation reducing unbound charge carriers as well as the hinderance effected by the larger organic ion species. Others ILs have very high σ of 10^{-1} S/cm at RT, however, presently only weak correlations between σ and the size and type of cations/anions are known. A better guide for improvement has been the empirical Walden rule (suggested by P. Walden [488]), according to which higher conductivity is correlated to higher viscosity. The complex thermal behaviour of ILs near T_g manifests in the conductivity as a significant departure from Arrhenian behaviour. Given the extensive literature regarding dynamics in glass-forming liquids, namely fragility and cooperativity (*see*Section 6.4.1), the temperature dependence and interdependence of conductivity and viscosity may be an area in which the field of ILs has something to gain from glass science: "*Are the dynamics of ionic liquids similar to that of normal molecular liquids, or do the strong Coulombic forces present in these systems give rise to different dynamical behavior*?" [quoted from [483]].

Until now much of the utility of ILs has originated from their

versatility as solvents of a variety of inorganic, organic or organometallic species for reaction syntheses or electrochemical processes [487]. Like ILs, ionic glasses also have the "*potential for tailoring size, shape and functionality offer opportunities for control in reactivity unobtainable with molecular solids. Changing ion type, substitution and composition produces new ionic liquid systems, each with a unique set of properties that can be explored and applied to the issues*" [quoted from [489]]. This potential for optimization is most evident in the optical properties of halide glasses (*see*Section 6.1) and demonstrates the chemical versatility that is a hallmark of ionic glasses. Many applications for ILs take advantage of their tailorable chemical functionality, porosity and reactivity, such as their ability to store gases and catalysts, which is similar to that of amorphous MOFs [489,490].

9. Relationship to amorphous metal organic frameworks

Another class of materials related to ionic glasses is that of amorphous metal organic frameworks (*a*-MOFs). Like ionic glasses and ionic liquids, *a*-MOFs are glasses composed of interacting ions, which display many of the characteristic properties of glassy materials lacking a percolating covalent network (as will be illustrated in this Section). Over 60 000 different crystalline metal organic framework (MOF) structures have been reported to date [491], underscoring that MOFs possess the same chemical flexibility which makes ionic glasses a highly tunable class of materials. MOFs are of scientific and commercial interest for a wide range of applications, as their tailorable porosity and chemical functionality make them ideal for gas sorption and separation, magnetism, catalysts, and photoluminescence applications [492,493]. In the last decade, amorphous MOFs have emerged as a new class of glasses. The most commonly studied *a*-MOFs are zeolitic imidazolate frameworks (ZIFs), which possess topologies analogous to zeolites: tetrahedrally coordinated metal centres (Zn^{2+}, Co^{2+}) bridged by anionic imidazolate organic linkers [493]. When heated, ZIFs undergo solvent loss and structural collapse into an amorphous phase (these processes may or may not be concomitant). Depending on the type of ZIF, the amorphization process is followed by either melting from the amorphous phase or recrystallization into a high-density, stable "zni" phase (which then melts at a higher temperature) [492–494]. Glasses can then be formed by quenching these melts. Similarly, many amorphous ZIFs (*a*-ZIFs) can be produced by pressure or ball-milling [494]. Neutron and X-ray scattering experiments have demonstrated there to be no structural difference between amorphous phases produced from simple collapse upon heating, quenching from the melt, and ball-milling [494]. The formation of *a*-ZIFs involves a true reconstructive phase transition, rather than the displacive transition generally found for other open network structures [492]. This unique behaviour has been attributed to the relative weakness and lability of the Zn—Im—Zn linkage, resulting in its propensity to break and reform and extreme mechanical flexibility [309,495–499].

Upon amorphization, the crystalline and amorphous pair distribution functions (PDFs) of *a*-ZIFs are nearly identical up to 6 Å (approx. equal to $d_{Zn\cdots Zn}$ for Zn—Im—Zn), indicating the persistence of the first coordination shell, *i.e.*, two Zn^{2+} ions bridged by an imidazolate linker, following amorphization [492,493,497,500]. Yet, at longer correlation distances (> 10 Å), the amorphous peaks broaden and are eventually overwhelmed by the background (> 25 Å), demonstrating large differences between the crystal and amorphous structures over medium- and long-range length scales [492,493]. The topologies of crystalline ZIFs are often compared to that of fused silica: the organic linker and tetrahedrally coordinated Zn^{2+} replace oxygen and Si^{4+}, respectively [309,492,500]. In fact, both Zn—Im—Zn and Si—O—Si linkages have bond angles of 145°; the major geometric difference is the longer intertetrahedral distances, *i.e.* $d_{Zn\cdots Zn}$ is larger than $d_{Si\cdots Si}$. These similarities often lead *a*-ZIFs to be grouped together with fused silica [309,493,494,501], specifically as an example of Zachariasen's CRN [7]. On the other hand, there are several structural arguments for *a*-ZIFs to be categorized as ionic glasses. In crystalline ZIFs, it is possible for the organic linkers and metal centres to be substituted entirely without substantial modification of the framework topology through post-synthetic ligand or metal ion exchange (solid/liquid phase processing methods akin to ion-exchange strengthening of inorganic glass [495,502]). In silica, boria or P_2O_5 crystals, and their corresponding pure covalent glasses, it is not possible to perform cation exchange in the solid state. Zeolite synthesis offers a good example of where Si or Al *can* migrate freely, but only as fully coordinated, geometrically restricted SiO_4^{4-} and AlO_4^{5-} polyhedra; generally in water as $Si(OH)_4$ and $Al(OH)_4^-$ [503,504]. Even in conventional as-melted quenched silicate and aluminosilicate glasses, there are always geometric constraints enforced by the 109° intrapolyhedral angles (O—Si—O or O—Al—O) and interpolyhedral angles (Si—O—Si or Al—O—Al), which can only vary 135° to 160° [505].

It is important to note that *a*-ZIFs are denser than their crystalline parents [307,492], while the opposite is generally true for covalent network glasses (*c.f.* fused silica and α- or β-quartz [1]); consequently, it is impossible for the topology of the *a*-ZIFs to be conserved during amorphization, as a displacive transition upon heating would necessarily increase the volume. In fact, a proposed melting mechanism shows an Im group leaving, before another Im group approaches to recreate tetrahedrally coordinated Zn^{2+}, yet the new Zn—N bond suggested by Gaillac et al. [496] would adopt a position almost 90° different from the original bond (highlighted by gold boxes in Fig. 46). As discussed above, this ability to rearrange on a larger scale is a result of the weakness and high lability/flexibility of the Zn—N bond (or variability in ∠Zn—Im—Zn) compared to that of the Si—O bond (or variability in ∠Si—O—Si) [309,495–499]. In an reverse Monte Carlo study, this structural freedom was found to come from the Im ring being able to twist out of its original plane relative to the zinc centre [499]. This propensity for rearrangement is also strongly evident in the transition of ZIF-4 from the low density amorphous (LDA) phase to the high density amorphous (HDA) phase. In amorphous zeolites, the LDA phase has been determined to have an open structure that requires no bond breaking to be formed from the crystalline phase; it retains the crystalline topology, *i.e.*, bonding and connectivity, yet has a broader and more acute distribution of average Si–O–Si bond angles (centred around 125° compared with crystalline 145 – 150°) [506]. On the other hand, the HDA phase requires significant topological reconstruction compared to the crystal (or the LDA phase) [506]; we believe this HDA phase is only accessible due to the ionicity of the Zn^{2+} and imidazolate, which allow easy reconfiguration and produce several different, stable configurations in the HDA phase.

The bonding differences between silica and *a*-ZIFs are also manifested as partial ligand decoordination (and, presumably, undercoordinated zinc centres) well below T_g [307,507]. Although in pure molten silica, the formation of ionic species through the breaking of the Si—O—Si bonds must occur [508], in the glassy state 'dangling' terminal oxygen atoms are unheard of (^{29}Si NMR shows all silicon to be Q^4 species, *i.e.*, connected to four neighbouring SiO_4^{4-} tetrahedra [505]). As such, it appears the relative stabilities of undercoordinated Zn^{2+} cations and imidazolate anions are higher than those of the corresponding ionic species in pure silica glass.

Since the weaker and more labile Zn—N bond results in a wider breath of possible glassy topologies than does Si–O [506] (including fully ionic species [495,502] and bond decoordination [307,507]), the coordination sphere surrounding each ion in an *a*-ZIF is more easily distorted and reconfigured. Similarly, as has been shown in this review, ionic glasses also allow for much more local disorder around each cation. As a consequence, we classify *a*-ZIFs as ionic glasses based on their topology, especially the HDA phase. We suggest that the zinc-centered tetrahedra experience similar geometric constraints to a silica tetrahedron, but rather than possessing neighbouring topologically constrained oxygen bridges, instead there are *always* neighbouring labile, easily distorted Zn—Im—Zn linkages [309,495–499] (much like in

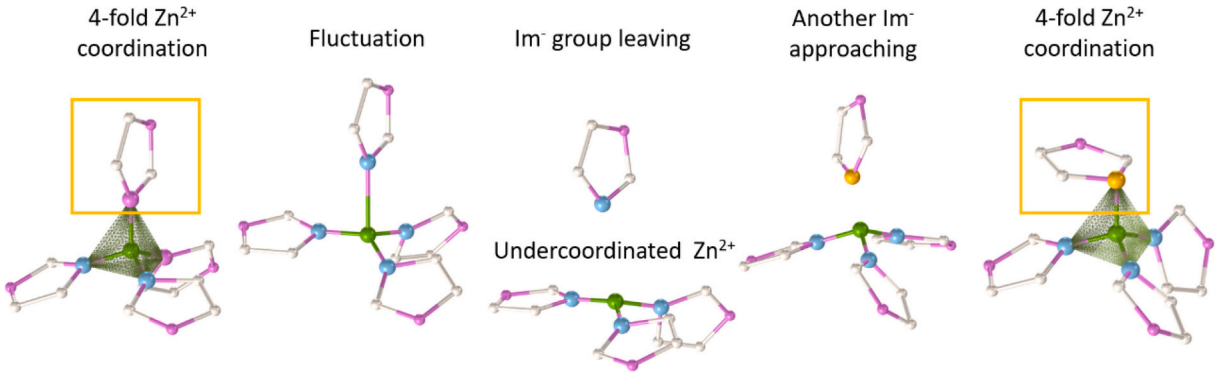


Fig. 46. Molecular mechanism of ZIF-4 melting. Visualization of a representative imidazolate exchange event. Zn, green: N (initially coordinated), pink: N, light blue; N (coordinated after exchange), gold; C, white. Reproduced with permission from Gaillac et al. [496]. © Nature Publishing Group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

silicate invert glasses that consistently have cations with distortable coordination spheres coordinated to silica polyhedra). If we put *a*-ZIFs into the context of other ionic glasses surveyed in this review we can compute r_M/r_X: in this case the ratio $d_{Zn–N}/d_{Im}$ is 1.965 Å/2.25 Å, which equals 0.87 and puts *a*-ZIFs into the 'highly ionic' glass category. This category generally contains components from which single compound glasses cannot be produced and geometrical frustration must be employed for successful glass formation. One exception to this categorization is potassium borates, which form glasses despite being a simple binary composition of highly modified ionic species. It has been suggested that *a*-ZIFs, like glassy borates, may simply possess many energetically equivalent, yet stable states leading to a structure too significantly different from their crystalline counterparts to have effective nucleating sites [60,417]. However, the high viscosities of ZIFs just above T_g has also been reported to suppress crystallization in *a*-ZIFs [308,309].

Following this hypothesis that *a*-ZIFs are ionic glasses, we can consider several properties which should help distinguish themselves from other glasses, specifically in contrast with covalent network glasses such as silica. The mechanical properties of *a*-ZIFs offer one example, since they do not exhibit ductile fracture during spherical scratch testing [308]. Furthermore, the high Poisson's ratio (volume-conserving) and ductility found for an *a*-ZIF [309] agree strongly with their classification as ionic glasses following the discussion in Section 6.2: in Fig. 32 and Fig. 33 we show that ionic glasses have high Poisson's ratio, somewhere between that of metallic and chalcogenide glasses. Specifically, in glasses high Poisson's ratio is correlated with the number of constraints (Fig. 32), a low maximum density change (Fig. 33), high compactness (Fig. 34) and high crack resistance (assuming high ν is inversely related to brittleness) (Fig. 35). We argue that, like ionic glasses, *a*-ZIFs have relatively few constraints (due to their labile Zn–N bonds) as well as low maximum densification, or high compactness. The fact that *a*-ZIFs generally are more dense than their (metastable) crystalline counterparts, including the dense ZIF-zni phase indicates that they have a high packing density, as would be expected for a packing of ionic components [307]. Similarly, *a*-ZIFs are less prone to cracking, less brittle and more ductile than typical covalent network glass formers, proving that they cannot be considered analogous to amorphous silica [308,309]. It is believed that introducing structural heterogeneity into metallic and ionic glasses could improve strength and ductility; it is therefore interesting that the ductility of *a*-ZIFs is already high [308,309].

Another indication that *a*-ZIFs are ionic glasses comes from scratch testing and nano-indentation stress-strain curves: the overlap of lateral deformation and normal indentation behaviour suggests that the material deformation modes are independent of direction [308]. We would not expect that to be the case for covalent network glasses, where the deformation modes are affected by the loading constraints (specifically, an order of magnitude less lateral stress is required to produce the same deformation found for normal stress [509]). Direction-independence implies a lack of bonding directionality that would be expected for glasses with a low number of constraints, *i.e.* ionic and metallic glasses. The experimentally derived strain-rate sensitivity of the hardness of several differently prepared *a*-ZIFs was found to be similar to glassy polymers and Se-rich GeSe chalcogenide glasses [308].

Although high ionic conductivity is generally a property common to ionic glasses, the size of the ion and field strength play a role, and in the case of *a*-ZIFs, neither Zn^{2+} nor imidazole contribute significantly to conductivity. However, it is possible to achieve high ionic conduction by incorporating ILs into crystalline and amorphous ZIFs [510]. Since some *a*-ZIFs exhibit an amorphous–amorphous phase transition [511], it would be interesting to investigate whether the higher ionicity of the HDA phase affects the ionic conductivity.

The final argument for *a*-ZIFs being classified as ionic glasses comes from their relaxation processes. The collapsed HDA phase of *a*-ZIF-4 is considered an intermediately fragile glass ($m = 41$) [511]; this value is on par with ionic glasses ($m = 50 – 100$ [512]) as well as chalcogenide and metallic glasses (*see*Fig. 39). An extra entropic contribution in the HDA phase can be easily rationalized by a structural motif common to ionic glasses: both linker and metal centre ions have the easily distorted coordination spheres required for dense packing.

Other indications of fragility are large structural and volume changes between the crystal and melt [106]. The understanding of amorphous ZIFs is often complicated by the fact that the initial crystalline phase is a metastable, open framework structure (leading to the large negative $\frac{\Delta\rho}{\rho_{glass}}$ reported in [309]), and a compositionally equivalent, stable, crystalline phase is not available for comparison. Although we agree with the conclusion of the thermodynamic analysis of ZIF-4 performed by Bennett et al. [511] (that the HDA phase has more configurational entropy than the LDA phase), the HDA phase being *both* more dense *and* more disordered is very unusual in the context of glass science. We believe this apparent paradox to be reconciled when the fact that LDA and HDA phases have different stable, crystalline reference states is considered. This difference would be strange in an inorganic glass, yet the metastability of both crystalline and amorphous ZIFs allows them to access dramatic structural transitions at relatively moderate temperatures and pressures.

When comparing ZIFs to silica, it is important to note that SiO_2 has a pertinent high-pressure crystalline polymorph, stishovite, wherein the Si^{4+} are octahedrally coordinated, while the oxygen atoms have three nearest-neighbours. Increased pressure has been found to correlate with electron density localization, *i.e.*, ionicity, in stishovite [513]. Similarly, ZIF-4 has several polymorphs including the ambient pressure α-zni phase and the high-pressure, less compliant, β-zni phase (formed at 0.5 – 0.8 GPa) [514]. Consequently, the HDA phase of ZIF-4 may have more

ionic bonding, as found in its high-pressure polymorphs (which are analogous to stishovite for silica). Unlike in the case of SiO_2, where topological constraints are sufficiently strong to prevent the formation of a dense amorphous structure with bonding akin to stishovite, the high porosity and lability of a ZIF allows many topologically different crystalline and amorphous states to be accessed. This is an interesting conclusion; it suggests that pressure-induced amorphization necessitates a more ionic structure that incorporates some of the topology of the next most stable high-pressure polymorph.

In contrast to most ionic glasses, ZIF-62 is known to have a high viscosity near T_m ($\eta = 10^5$ Pa·s) and high glass-forming ability, which has been attributed to the high steric hindrance created by its large organic linkers and the close-packing of the glass structure [308,309]. Indeed, using a bulkier linker has been proven to increase the glass-forming ability of *a*-ZIFs [309]. Consequently, we believe *a*-ZIFs to have an 'artificially deflated' dynamic fragility, where the 'frustrated' network dynamics raise the activation energy for relaxation in the liquid phase. Based on the linear correlations between dynamic fragility and T_g Qin and McKenna [512] found for hydrogen-bonded organic glasses (and arguably for organic and inorganic ionic glasses as well), the T_g range for most *a*-ZIFs ($T_g = 300 - 350$ °C) corresponds to a fragility index of 115 – 220. As discussed in Section 6.4.1 there is a difference between dynamic and thermodynamic fragility: dynamic fragility is correlated to the number of accessible relaxation modes as determined by the E_a at T_g, while thermodynamic fragility, *i.e.*, configurational entropy as determined by heat capacity measurements, is related to the number of accessible configurational states. It is clear that *a*-ZIFs have high amounts of configurational states afforded them by the lability of the Zn—N bond; similarly, we expect them to relax heterogeneously as the bonding within the linker is directional and different than the Zn—Im—Zn bonding. The fragilities of *a*-ZIFs as measured by SAXS (assumed to be a measurement of glass dynamics) and thermodynamic methods are roughly equal [309,511], however, it should be noted that any correlation between these two values is controversial and depends greatly on the glass type [512]. All ionic glasses are fragile thermodynamically; they possess high configurational entropy (whether by having many components, many bonding types, or easily distortable coordination shells), although highly ionic glasses such as halides can have less dynamic fragility or heterogeneity. Since invert or oxyanionic glasses possess both thermodynamic and dynamic heterogeneity, we assign *a*-ZIFs specifically to this category of ionic glasses.

10. Conclusions and outlook

By reviewing information about several different categories of ionic glasses (invert, oxoanionic, halide, and superionic glasses as well as mixtures thereof), we have made conclusions about their shared compositional trends, glass-forming abilities, local structures, topologies and properties:

- the compositional ratio at which there is more modifier than conventional former (based on Dietzel's classifications) corresponds to the *inversion* in trends in dynamic properties, indicating that consideration of non-percolating ionic species is crucial for understanding glass properties
- many ionic liquids cannot readily form single-component glasses, instead, geometric frustration is required (except for 'strong' ionic glass formers, *see*Section 5.1)
- ionic glasses are usually hygroscopic; water is a regular contaminant and likely to significantly affect properties (like ionic liquids)
- ionicity in a solid is ambiguously defined, but for a glass there are empirical optical measurements that have been related to polarizability, Pauling's ionicity measure and optical basicity
- highly ionic glasses exhibit low structural dimensionality (*i.e.* they lack a percolating network) and can often be accurately described by random close packing models; ionic glasses with some covalent bonding character or covalently bonded ions (oxoanions) are more difficult to model and represent challenges for common structural analysis techniques, such as MD or PDF
- a simple radius ratio, r_X/r_A, can be used to determine the degree of stability of the glassy state; highly ionic glasses are unstable glass formers unless they are near an 'ideal' radius ratio, as is the case for BeF_2 or Mg_2SiO_4 (and the tolerance appears to diminish as the coordination number increases)
- oxoanionic and mononuclear ionic glasses can have the same topology; the importance of local order imposed by the polyhedra of the oxoanions remains to be evaluated; thus far, percolation appears to be the principal determinant of dynamic properties
- the dominating Coulombic interactions result in high Poisson's ratio, volume-conservative flow, and often high ionic conductivity in the glassy state, and low viscosity in the liquid state
- higher local disorder, high configurational entropy and high fragility are pertinent to the structural dynamics of ionic glasses
- like other glasses, ionic glasses provide extraordinary chemical tunability, akin to the solvation properties of ionic liquids and chemical diversity of MOFs

Covalency strongly influences all dynamic processes, but is *not* necessary for glass formation. Zachariasen's glass-forming rules are not applicable to ionic glasses; metals in halides have large coordination spheres and their coordination polyhedra can be edge-sharing. Furthermore, it is not necessary for three corners to be directionally bonded (shared); ionic cross-linking is sufficient for glass formation. Dietzel's field strength parameter does not classify many ionic glasses as glass formers; instead, we found the radius ratio, r_M/r_X, (modified to $d_{M\text{-}O}/d_X$ for oxyanionic glasses) to predict stability of ionic glass formers and give some guidance to those compositions likely to be highly ionic glasses. To facilitate identification of ionic glasses (over other groups of glasses) we encourage the implementation of some simplistic classification criteria: ionicity, polarizability and optical basicity.

Highly ionic glasses act as prototypes for the ionic glasses: they are the most unstable kinetically and require glass formation *via* geometrical frustration. When the glass structure is composed of only mononuclear oxyanions or halides, random close packing of quasispherical particles interacting through Coulombic forces satisfactorily describes glass structure, however, in many glass compositions, bridging anions may also form, introducing directional bonding. 2nd-nearest-neighbour interactions that result in network percolation are especially critical in ionic glasses, where the topology, and therefore structural and dynamic heterogeneity, can vary significantly. Accordingly, current topological models and MD simulation protocols must be modified when intended for application to ionic glass formers.

Features characteristic to ionic bonding, *e.g.* compositional flexibility and a lack of directional bonding, produce trends common to most ionic glasses in optical and mechanical properties as well as in the dynamic processes of ionic conduction, relaxation and crystallization. Heterogeneity (configurational, structural and dynamic) is a central motif; ionic glasses are typified by their high Poisson's ratio, high ionic conductivity, and high fragility, as well as highly decoupled relaxation modes and low viscosity in the liquid state.

Comparison of ionic glasses and liquids could result in a fruitful exchange of knowledge for both disciplines. For example, the Walden rule (viscosity $\propto$ conductivity), which demonstrates that many interactions, such as strong ion association and relaxation effects, are not the main determinant of conductivity in ionic liquids, may also be true for ionic glasses [321,322]. Similarly, amorphous MOFs are another external field of study with significant fundamental overlap with ionic glasses – notably they share compositional flexibility, bonding lability, topology (*i.e.,* 'dangling' bonds), high Poisson's ratio and high fragility. Since the incorporation of ionic liquids into the pores of amorphous MOFs creates a stabilizing effect on the Zn—N bond, it may be possible to combine *a*-MOFs with more prototypical ionic glasses.

The low T_g, T_m, viscosity, and chemical durability in combination with high ionic conductivity, Poisson's ratio (less brittleness, more plasticity), and compositional flexibility of ionic glasses are advantageous for applications in the fields of optics, electronics, memory storage, energy storage, and biomedical engineering. Scientists should expand the current definition of glass formers to include many of the oxoanionic and polyanionic compositions discussed herein; a more thorough fundamental understanding of glass formation and a broader range of available glass properties will be obtained. Ionic glasses present exciting opportunities to understand and exploit heterogeneity by combining bonds of variable degrees of localization.

Declaration of Competing Interest

The authors declare no competing interests.

Acknowledgements

This project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (ERC grant UTOPES, grant agreement no. 681652). CC is grateful to the Natural Sciences and Engineering Research Council of Canada (NSERC) for financial support. The authors wish to thank their present and former colleagues, students and collaborators for many years of fruitful discussion.

References

[1] A.K. Varshneya, Fundamentals of the Glassy State, in: Fundam. Inorg. Glas, 1st ed, Academic Press, Inc, San Diego, 1994, pp. 13–25, https://doi.org/10.1016/B978-0-08-057150-8.50025-2.
[2] O. Schott, Beitraege zur Theorie und Praxis der Glasfabrikation, Friedrich-Schiller University of Jena, Jena, Germany, 1875.
[3] V.M. Goldschmidt, Die gesetze der krystallochemie, Naturwissenschaften. 21 (1926) 477–485.
[4] X. Zhu, N. Peyghambarian, High-Power ZBLAN glass fiber lasers: review and prospect, Adv. Optoelectron. 2010 (2010) 1–23, https://doi.org/10.1155/2010/501956.
[5] J.C. Mauro, A. Tandia, K.D. Vargheese, Y.Z. Mauro, M.M. Smedskjaer, Accelerating the Design of Functional Glasses through Modeling, Chem. Mater. 28 (2016) 4267–4277, https://doi.org/10.1021/acs.chemmater.6b01054.
[6] H. Liu, Z. Fu, K. Yang, X. Xu, M. Bauchy, Machine learning for glass science and engineering: A review, J. Non. Cryst. Solids. 119419 (2019), https://doi.org/10.1016/j.jnoncrysol.2019.04.039.
[7] W.H. Zachariasen, The atomic arrangement in glass, J. Am. Chem. Soc. 54 (1932) 3841–3851, https://doi.org/10.1021/ja01349a006.
[8] T. Rouxel, Elastic properties and short-to medium-range order in glasses, J. Am. Ceram. Soc. 90 (2007) 3019–3039, https://doi.org/10.1111/j.1551-2916.2007.01945.x.
[9] T.Y. Yoo, J. Tran, S.P. Stahlheber, C.E. Kaainoa, K. Djepang, A.R. Small, Site percolation on lattices with low average coordination numbers, J. Stat. Mech. Theory Exp. 2014 (2014) P06014, , https://doi.org/10.1088/1742-5468/2014/06/P06014.
[10] C.A. Angell, Sulfate and sulfate-chloride glasses, J. Am. Ceram. Soc. 540 (1965).
[11] S.I. Smedley, C.A. Angell, Highly conducting Li^+-rich inorganic glasses, Solid Sate Commun. 27 (1978) 21–23.
[12] M.D. Ingram, Superionic glasses: Theories and applications, Curr. Opin. Solid State Mater. Sci. 2 (1997) 399–404, https://doi.org/10.1016/S1359-0286(97)80079-4.
[13] M. Tatsumisago, A. Hayashi, Superionic glasses and glass-ceramics in the Li_2S-P_2S_5 system for all-solid-state lithium secondary batteries, Solid State Ionics. 225 (2012) 342–345, https://doi.org/10.1016/j.ssi.2012.03.013.
[14] T. Saito, M. Tatsumisago, T. Minami, Phase transformation and lattice strain of α-AgI stabilized in superionic glass, J. Electrochem. Soc. 143 (1996) 687–691, https://doi.org/10.1149/1.1836501.
[15] M. Tatsumisago, T. Saito, T. Minami, M. Hanaya, M. Oguni, X-ray diffraction and electric conductivities at low temperatures of α-AgI stabilized in a rapidly quenched orthoborate glass matrix, J. Phys. Chem. 98 (1994) 2005–2007, https://doi.org/10.1021/j100059a003.
[16] K. Takada, N. Aotani, K. Iwamoto, S. Kondo, Solid state lithium battery with oxysulfide glass, Solid State Ionics. 86–88 (1996) 877–882, https://doi.org/10.1016/0167-2738(96)00199-3.
[17] A. Sharma, V.V. Tyagi, C.R. Chen, D. Buddhi, Review on thermal energy storage with phase change materials and applications, Renew. Sustain. Energy Rev. 13 (2009) 318–345, https://doi.org/10.1016/j.rser.2007.10.005.
[18] A. Dietzel, Die kationenfeldstärken und ihre beziehungen zu entglasungsvorgängen, zur verbindungsbildung und schmelzpunkten von silikaten, Z. Elektrochem. Angew. Phys. Chem. 48 (1942) 9–23, https://doi.org/10.1002/bbpc.19420480104.
[19] B. Silvi, A. Savin, Classification of chemical bonds based on topological analysis of electron localization functions, Nature. 371 (1994) 683–686, https://doi.org/10.1038/371683a0.
[20] M. Poulain, Glass formation in ionic systems, Nature. 293 (1981) 279–280, https://doi.org/10.1038/293279a0.
[21] K. El-Egili, H. Doweidar, Y.M. Moustafa, I. Abbas, Structure and some physical properties of PbO-P_2O_5 glasses, Phys. B Condens. Matter. 339 (2003) 237–245, https://doi.org/10.1016/j.physb.2003.07.005.
[22] J.R. Allwardt, J.F. Stebbins, Ca-Mg and K-Mg mixing around non-bridging O atoms in silicate glasses: An invesigation using ^{17}O and 3QMAS NMR, Am. Miner. 5 (2004) 777–784.
[23] E.I. Kamitsos, M.A. Karakassides, G.D. Chryssikos, Vibrational spectra of magnesium-sodium-borate glasses. 2. Raman and mid-infrared investigation of the network structure, J. Phys. Chem. 9 (1987) 1073–1079, https://doi.org/10.1021/j100289a014.
[24] M.R. DeGuire, S.D. Brown, Dependence of Young's modulus on volume and structure in alkali silicate and alkali aluminosilicate glasses, J. Am. Ceram. Soc. 67 (1984) 270–273, https://doi.org/10.1111/j.1151-2916.1984.tb18845.x.
[25] B.P. Rodrigues, L. Wondraczek, Cationic constraint effects in metaphosphate glasses, J. Chem. Phys. 140 (2014), https://doi.org/10.1063/1.4879559.
[26] G.N. Greaves, EXAFS and the strucure of glass, J. Non-Cryst. Solids. 71 (1985) 203–217.
[27] H. Jain, H.L. Downing, N.L. Peterson, The mixed alkali effect in lithium-sodium borate glasses, J. Non. Cryst. Solids. 64 (1984) 335–349, https://doi.org/10.1016/0022-3093(84)90187-X.
[28] H.J.L. Trap, J.M. Stevels, Physical properties of invert glasses, Glastech. Ber. 5 (1959) 32–52.
[29] Y. Onodera, S. Kohara, H. Masai, A. Koreeda, S. Okamura, T. Ohkubo, Formation of metallic cation-oxygen network for anomalous thermal expansion coefficients in binary phosphate glass, Nat. Comm. 8 (2017) 1–8, https://doi.org/10.1038/ncomms15449.
[30] C.A. Angell, Dynamic processes in ionic glasses, Chem. Rev. 90 (1990) 523–542, https://doi.org/10.1021/cr00101a006.
[31] B.P. Rodrigues, H. Ebendorff-Heidepriem, L. Wondraczek, Decoupling mobility and charge carrier concentration in AgR-$AgPO_3$ glasses (R = Cl, Br, I), Solid State Ionics. 334 (2019) 99–104, https://doi.org/10.1016/j.ssi.2019.02.009.
[32] A.C.M. Rodrigues, M.L.F. Nascimento, C.B. Bragatto, J.L. Souquet, Charge carrier mobility and concentration as a function of composition in $AgPO_3$-AgI glasses, J. Chem. Phys. 135 (2011) 1–7, https://doi.org/10.1063/1.3666835.
[33] M. Tatsumisago, K. Hirai, T. Hirata, M. Takahashi, T. Minami, Structure and properties of lithium ion conducting oxysulfide glasses prepared by rapid quenching, Solid State Ionics. 86–88 (1996) 487–490, https://doi.org/10.1016/0167-2738(96)00179-8.
[34] M. Tatsumisago, T. Minami, M. Tanaka, Glass formation by rapid quenching amounts in lithium silicates containing large amounts of Li_2O, Yogyo Kyokaishi. 93 (1985) 581–584.
[35] J.H. Simmons, C.J. Simmons, R. Ochoa, A.C. Wright, Chapter 2: Fluoride Glass Structure, in: I.D. Aggarwal, G. Lu (Eds.), Fluoride Glas. Fiber Opt, 1991, pp. 37–84, , https://doi.org/10.1016/B978-0-12-044505-9.50008-5.
[36] Jmol: an open-source Java viewer for chemical structures in 3D. http://www.jmol.org/, (n.d.).
[37] W.H. Dumbaugh Jr., Glass envelopes for iodine cycle incandescent lamps, 3,496,401, (1965).
[38] G.L. Thomas, High-temperature glass composition, 4,060,423, (1977).
[39] R.A. Schoenlaub, Glass Composition, 2,334,961, (1943).
[40] R.H. Dalton, Glass-to-Metal Seal, 2,392,314, (1946).
[41] A.W. Hull, E.E. Burger, Glass-to-Metal Seals, Physics (College. Park. Md). 5 (1934) 384–405, https://doi.org/10.1063/1.1745225.
[42] F. Veres, Glass-to-Glass Seal, 2931,142, (1960).
[43] L.H. Dixon Jr., Sealing of electrical semconductor devices, 3,020,454, (1962).
[44] R.G. Frieser, A review of solder glasses, Electrocompon. Sci. Technol. 2 (1975) 163–199.
[45] A. Giostri, M. Binotti, M. Astolfi, P. Silva, E. Macchi, G. Manzolini, Comparison of different solar plants based on parabolic trough technology, Sol. Energy. 86 (2012) 1208–1221, https://doi.org/10.1016/j.solener.2012.01.014.
[46] H.F. Loehrke, Manufacture of glass for electron tube envelopes, 3,121,628, (1964).
[47] M.T. Sebastian, H. Jantunen, Low loss dielectric materials for LTCC applications: A review, Int. Mater. Rev. 53 (2008) 57–90, https://doi.org/10.1179/174328008X277524.
[48] M.R. Gongora-Rubio, P. Espinoza-Vallejos, L. Sola-Laguna, J.J. Santiago-Avilés, Overview of low temperature co-fired ceramics tape technology for meso-system technology (MsST), Sensors Actuators, A Phys. 89 (2001) 222–241, https://doi.org/10.1016/S0924-4247(00)00554-9.
[49] R.K. Datta, Sealing materials for ceramic envelopes, 4,076,991, (1978).
[50] J.F. Ross, Ceramic bonding, 3,281,399, (1966).
[51] A. Flügel, M.D. Dolan, A.K. Varshneya, Y. Zheng, N. Coleman, M. Hall, D. Earl, S.T. Misture, Development of an Improved Devitrifiable Fuel Cell Sealing Glass, J. Electrochem. Soc. 154 (2007) B601, https://doi.org/10.1149/1.2728291.
[52] A. Inoue, T. Zhang, T. Masumoto, Glass-forming ability of alloys, in: 8th Int. Conf. Liq. Amorph. Met. Aug. 31-Sept. 04, Journal of Non-Crystalline Solids, Vienna, Austria (1993) 473–480.
[53] N. Da, S. Krolikowski, K.H. Nielsen, J. Kaschta, L. Wondraczek, Viscosity and softening behavior of alkali zinc sulfophosphate glasses, J. Am. Ceram. Soc. 93 (2010) 2171–2174, https://doi.org/10.1111/j.1551-2916.2010.03707.x.
[54] R.K. Brow, R.J. Kirkpatrick, G.L. Turner, Nature of alumina in phosphate glass: II,

Structure of sodium aluminophosphate glass, J. Am. Ceram. Soc. 76 (1993) 919–928, https://doi.org/10.1111/j.1151-2916.1993.tb05316.x.
[55] R.K. Brow, Nature of Alumina in Phosphate Glass: I, Properties of aluminophosphate glass, J. Am. Ceram. Soc. 76 (1993) 913–918.
[56] F. Branda, F. Arcobello-Varlese, A. Costantini, G. Luciani, T_g and FTIR of $(2.5-x)CaO{\cdot}x/3M_2O_3{\cdot}2SiO_2$ (M = Y, La, In, Al, Ga) glasses, J. Non. Cryst. Solids. 246 (1999) 27–33.
[57] S. Feller, G. Lodden, A. Riley, T. Edwards, J. Croskrey, A. Schue, D. Liss, D. Stentz, S. Blair, M. Kelley, G. Smith, S. Singleton, M. Affatigato, D. Holland, M.E. Smith, E.I. Kamitsos, C.P.E. Varsamis, E. Ioannou, A multispectroscopic structural study of lead silicate glasses over an extended range of compositions, J. Non. Cryst. Solids. 356 (2010) 304–313, https://doi.org/10.1016/j.jnoncrysol.2009.12.003.
[58] C. Calahoo, J. Petrovic, Q.H. Le, U. Werner-zwanziger, J. Zwanziger, L. Wondraczek, Structural Relaxation in Polyanionic Sodium Fluorophosphate Glasses, Front. Mater. 6 (2019) 1–15, https://doi.org/10.3389/fmats.2019.00165.
[59] J. Lucas, Non-conventional spaghetti-type glasses, J. Non. Cryst. Solids. 184 (1995) 21–24, https://doi.org/10.1016/0022-3093(95)00087-9.
[60] E.D. Zanotto, D.R. Cassar, The microscopic origin of the extreme glass-forming ability of Albite and B2O3, Sci. Rep. 7 (2017) 43022, https://doi.org/10.1038/srep43022.
[61] T. Kasuga, Y. Abe, Calcium phosphate invert glasses with soda and titania, J. Non. Cryst. Solids. 243 (1999) 70–74, https://doi.org/10.1016/S0022-3093(98)00820-5.
[62] G. Walter, J. Vogel, U. Hoppe, P. Hartmann, The structure of CaO–Na_2O–MgO–P_2O_5 invert glass, J. Non. Cryst. Solids. 296 (2001) 212–223, https://doi.org/10.1016/S0022-3093(01)00912-7.
[63] L. Montagne, G. Palavit, R. Delaval, Effect of ZnO on the properties of (100 - x)(NaPO3)-xZnO glasses, J. Non. Cryst. Solids. 223 (1998) 43–47, https://doi.org/10.1016/S0022-3093(97)00444-4.
[64] S. Koritala, K. Farooqui, M. Affatigato, S. Feller, S. Kambeyanda, S. Ghosh, E.I. Kamitsos, G.D. Chryssikos, A.P. Patsis, The glass transition temperature of lithium-alkali borates, J. Non. Cryst. Solids. 134 (1991) 277–286, https://doi.org/10.1016/0022-3093(91)90386-K.
[65] B.N. Meera, A.K. Sood, N. Chandrabhas, J. Ramakrishna, Raman study of lead borate glasses, J. Non. Cryst. Solids. 126 (1990) 224–230, https://doi.org/10.1016/0022-3093(90)90823-5.
[66] S. Kohara, K. Suzuya, K. Takeuchi, C.-K. Loong, M. Grimsditch, J.K.R. Weber, J.A. Tangeman, T.S. Key, Glass formation at the limit of insufficient network formers, Science (80-.) 303 (2004) 1649–1652, https://doi.org/10.1126/science.1095047.
[67] E.I. Kamitsos, G.D. Chryssikos, Borate glass structure by Raman and infrared spectroscopies, J. Mol. Struct. 247 (1991) 1–16, https://doi.org/10.1016/0022-2860(91)87058-P.
[68] W. Xu, L.-M. Wang, R.A. Nieman, C.A. Angell, Ionic Liquids of Chelated Orthoborates as Model Ionic Glassformers, J. Phys. Chem. B. 107 (2003) 11749–11756, https://doi.org/10.1021/jp034548e.
[69] R.K. Brow, Review: the structure of simple phosphate glasses, J. Non. Cryst. Solids. 263&264 (2000) 1–28, https://doi.org/10.1016/S0022-3093(99)00620-1.
[70] T. Mourão, A.F.M. Cláudio, I. Boal-Palheiros, M.G. Freire, J.A.P.P. Coutinho, Evaluation of the impact of phosphate salts on the formation of ionic-liquid-based aqueous biphasic systems, J. Chem. Thermodyn. 54 (2012) 398–405, https://doi.org/10.1016/j.jct.2012.05.019.
[71] B.O. Mysen, D. Virgo, A.F. Seifert, The structure of silicate melts: Implications for chemical and physical properties of natural magma, Rev. Geophys. Sp. Phys. 20 (1982) 353–383.
[72] B.O. Mysen, The structure of silicate melts, Annu. Rev. Earth Planet Sci. 11 (1983) 75–97.
[73] W.H. Leong, D.W. James, M.T. Carrick, Structure in nitrate glasses : multiple anion sites or random continuum? J. Raman Spectrosc. 14 (1983) 11–15.
[74] T. Furukawa, S.A. Brawer, W.B. White, Raman spectroscopic study of nitrate glasses, J. Chem. Phys. 69 (1978) 2639–2651, https://doi.org/10.1007/s11460-007-0060-5.
[75] S. Sen, J.F. Stebbins, Heterogeneous NO_3^- Ion Dynamics near the Glass Transition in the Fragile Ionic Glass Former $Ca_{0.4}K_{0.6}(NO_3)_{1.4}$: A ^{15}N NMR study, Phys. Rev. Lett. 78 (1997) 3495–3498.
[76] K. Shibata, T. Kamiyama, K. Suzuki, K. Inoue, Dynamic structure of molten $0.80RbNO_3$-$0.20Sr(NO_3)_2$ mixture of neutron quasielastic scattering, J. Non. Cryst. Solids 117–118 (1990) 120–123.
[77] M. Yao, Y. Kawakita, K. Nagaya, Y. Ohmasa, T. Tsuzuki, M. Inui, H. Endo, EXAFS study of the liquid-glass transition in binary nitrate systems, J. Non. Cryst. Solids. 156–158 (1993) 555–558, https://doi.org/10.1016/0022-3093(93)90020-X.
[78] Y.B. Wang, D.H. Ryan, Z. Altounian, Structural and thermal studies of nitrate glasses, J. Non. Cryst. Solids. 205–207 (1996) 221–224, https://doi.org/10.1016/S0022-3093(96)00229-3.
[79] R. Böhmer, C.A. Angell, Local and global relaxations in glass-forming materials, in: R. Richert, A. Blumen (Eds.), Disord. Eff. Relaxational Process. Springer-Verlag, Berlin, 1994, pp. 11–55, , https://doi.org/10.1007/978-3-642-78576-4.
[80] F.S. Howell, R.A. Bose, P.B. Macedo, C.T. Moynihan, Electrical relaxation in a glass-forming molten salt, J. Phys. Chem. 78 (1974) 639–648, https://doi.org/10.1021/j100599a016.
[81] C.P. Johari, M. Goidstein, Viscous liquids and the glass transition. II. Secondary relaxations in glasses of rigid molecules, J. Chem. Phys. 53 (1970) 2372–2388, https://doi.org/10.1063/1.1674335.
[82] R. Böhmer, E. Sanchez, C.A. Angell, Ac technique for simultaneous study of local and global linear responses near the glass transition: the case of doped $Ca^{2+}/K^+/NO_3$, J. Phys. Chem. 96 (1992) 9089–9092, https://doi.org/10.1149/1.2427699.
[83] K.J. Rao, D.B. Helphrey, C.A. Angell, Thermodynamic properties of M(I)-M(II) mixed nitrate glasses and supercooled liquids, Phys. Chem. Glas. 14 (1973) 26–32.
[84] R.J. Grant, M.D. Ingram, L.D.S. Turner, C.A. Vincent, Optimized ionic conductivity in glass. Vitreous silver arsenate iodide (Ag7I4AsO4) electrolytes, J. Phys. Chem. 82 (1978) 2838–2844, https://doi.org/10.1177/0269881109105679.
[85] H. Tweer, N. Laberge, P.B. Macedo, Inadequacies of viscosity theories for a vitreous KNO_3-$Ca(NO_3)_2$ melt, J. Am. Ceram. Soc. 54 (1971) 121–123.
[86] G.E. McManis, M.H. Miles, A.N. Fletcher, Discharge characteristics of lithium/molten nitrate thermal battery cells using silver salts as solid cathode materials, J. Power Sources. 16 (1985) 243–251, https://doi.org/10.1016/0378-7753(85)80089-6.
[87] A.Y. Sane, A.R. Cooper, Stress Buildup and Relaxation During Ion Exchange Strengthening of Glass, J. Am. Ceram. Soc. 70 (1987) 86–89, https://doi.org/10.1111/j.1151-2916.1987.tb04934.x.
[88] S.S. Kistler, Stresses in Glass Produced by Nonuniform Exchange of Monovalent Ions, J. Am. Ceram. Soc. 45 (1962) 59–68, https://doi.org/10.1111/j.1151-2916.1962.tb11081.x.
[89] J.S. Wilkes, P. Wasserscheid, T. Welton, Introduction, Ion. Liq. Synth. 1 (2008), https://doi.org/10.1055/s-2003-40869.
[90] P.S.L. Narasimham, K.J. Rao, Phase diagram and glass formation in the K_2SO_4-$ZnSO_4$ system, J. Non. Cryst. Solids. 27 (1978) 225–246.
[91] H.G.K. Sundar, K.J. Rao, Formation in the ternary sulphate system, K_2SO_4-Na_2SO_4-$ZnSO_4$, J. C. S. Faraday I. 76 (1980) 1617–1629.
[92] T. Nishida, Y. Takashima, Mössbauer and DTA studies of K_2SO4–$ZnSO_4$–$Fe_2(SO_4)_3$ glasses, Bull. Chem. Soc. Jpn. 59 (1986) 2789–2794.
[93] N. Hémono, F. Muñoz, Dissolution of SO_3 within a lithium phosphate glass network and structure – property relationships, Phys. Chem. Glas. Eur. J. Glas. Sci. Technol. B. 51 (2010) 121–126.
[94] S.V. Stefanovskii, A.I. Aleksandrov, Structure of glasses of the sodium metaphosphate-lead metaborate-sodium sulfate system by the methods of IR and EPR spectrscopy, Zhurnal Prikl. Spectroskopii. 53 (1990) 121–126.
[95] N. Da, O. Grassmé, K.H. Nielsen, G. Peters, L. Wondraczek, Formation and structure of ionic (Na, Zn) sulfophosphate glasses, J. Non. Cryst. Solids. 357 (2011) 2202–2206, https://doi.org/10.1016/j.jnoncrysol.2011.02.037.
[96] Q.H. Le, T. Palenta, O. Benzine, K. Griebenow, R. Limbach, E.I. Kamitsos, L. Wondraczek, Formation, structure and properties of fluoro-sulfo-phosphate poly-anionic glasses, J. Non. Cryst. Solids. 477 (2017) 58–72, https://doi.org/10.1016/j.jnoncrysol.2017.09.043.
[97] A.M. Nepomiluev, R.N. Pletnev, O.B. Lapina, S.G. Kozlova, V.G. Bamburov, Structure of Glasses in the Na_2SO_4– P_2O_5–H_2O System, Glas. Phys. Chem. 28 (2002) 1–4.
[98] I.A. Sokolov, N.A. Valova, Y.P. Tarlakov, A.A. Pronkin, Electrical properties and the structure of glasses in the Li_2SO_4-$LiPO_3$ system, Glas. Phys. Chem. 29 (2003) 548–554, https://doi.org/10.1023/B:GPAC.0000007930.11101.ee.
[99] M. Ganguli, K.J. Rao, Studies of ternary Li_2SO_4-Li_2O-P_2O_5 glasses, J. Non. Cryst. Solids. 243 (1999) 251–267, https://doi.org/10.1016/S0022-3093(98)00832-1.
[100] J.P. Malugani, R. Mercier, B. Fahys, G. Robert, Ionic conductivity of and raman spectroscopy investigation binary oxosalts $(1-x)AgPO_3$-xAg_2SO_4 glasses, J. Solid State Chem. 45 (1982) 309–316.
[101] S.V. Stefanovskii, A.I. Aleksandrov, EPR spectra and structure of sodium sulfate borate glasses, Zhurnal Prikl. Spectroskopii. 54 (1991) 150–154.
[102] Y.M. Lai, X.F. Liang, S.Y. Yang, J.X. Wang, B.T. Zhang, Raman spectra study of iron phosphate glasses with sodium sulfate, J. Mol. Struct. 1013 (2012) 134–137, https://doi.org/10.1016/j.molstruc.2012.01.025.
[103] V. Kozhukharov, H. Bürger, S. Neov, Glassforming, properties and structure of halide tellurite glasses, Mater. Sci. Forum. 67–68 (1991) 143–148, https://doi.org/10.4028/www.scientific.net/MSF.67-68.143.
[104] S.V. Stefanovskii, A.I. Aleksandrov, Glass formation and structure of glasses in the Na_2O-PbO-SiO_2-SO_3 system, J. Appl. Spectrosc. 51 (1990) 56–79.
[105] R.K. Mishra, K.V. Sudarsan, P. Sengupta, R.K. Vatsa, A.K. Tyagi, C.P. Kaushik, D. Das, K. Raj, Role of sulfate in structural modifications of sodium barium borosilicate glasses developed for nuclear waste immobilization, J. Am. Ceram. Soc. 91 (2008) 3903–3907, https://doi.org/10.1111/j.1551-2916.2008.02773.x.
[106] G.N. Greaves, S. Sen, Inorganic glasses, glass-forming liquids and amorphizing solids, Adv. Phys. 56 (2007) 1–166, https://doi.org/10.1080/00018730601147426.
[107] S. Sirotkin, R. Meszaros, L. Wondraczek, Chemical Stability of ZnO-Na_2O-SO_3-P_2O_5 Glasses, Int. J. Appl. Glas. Sci. 3 (2012) 44–52, https://doi.org/10.1111/j.2041-1294.2011.00076.x.
[108] S. Reibstein, N. Da, J.P. Simon, E. Spiecker, L. Wondraczek, Phase separation and crystal precipitation in supercooled sulphophosphate ionic melts, Phys. Chem. Glas. Eur. J. Glas. Sci. Technol. Part B. 53 (2012) 61–67.
[109] W.A. Weyl, T. Förland, Formation of a sulfate glass, J. Am. Ceram. Soc. 33 (1950) 186–188, https://doi.org/10.1111/j.1151-2916.1950.tb12788.x.
[110] H.P. Eugster, G.I. Smith, Mineral equilibria in the searles lake evaporites, california, J. Petrol. 6 (1965) 473–522.
[111] G.R. Osinski, J.G. Spray, Impact-generated carbonate melts: Evidence from the Haughton structure, Canada, Earth Planet. Sci. Lett. 194 (2001) 17–29, https://doi.org/10.1016/S0012-821X(01)00558-1.
[112] A.H. Rankin, M.J. Le Bas, Liquid immiscibility between silicate and carbonate melts in naturally occuring ijolite magma, Nature. 250 (1974) 206–209, https://doi.org/10.1038/252497a0.
[113] Y. Morizet, S.C. Kohn, R.A. Brooker, Annealing experiments on CO_2-bearing jadeite glass: an insight into the true temperature dependence of CO_2 speciation in silicate melts, Mineral. Mag. 65 (2001) 701–707, https://doi.org/10.1180/0026461016560001.

[114] R.K. Datta, D.M. Roy, S.P. Faile, O.F. Tuttle, Glass formation in carbonate systems, J. Am. Ceram. Soc. 47 (1964) 153.
[115] S.E. Ragone, R.K. Datta, D.M. Roy, O.F. Tuttle, The system potassium carbonate-magensium carbonate, J. Pharm. Sci. 70 (1966) 3360–3361.
[116] A.K. Varshneya, Additivity Relationships, Fundam. Inorg. Glas, 1st ed, Academic Press, Inc, San Diego, 1994, pp. 146–148, , https://doi.org/10.1016/B978-0-08-057150-8.50025-2.
[117] W. Eitel, W. Skaliks, Concerning a few double carbonates of alkalis and alkaline earths, Z. Anorg. Allg. Chem. 183 (1929) 263–286.
[118] Y. Kono, C. Kenney-Benson, D. Hummer, H. Ohfuji, C. Park, G. Shen, Y. Wang, A. Kavner, C.E. Manning, Ultralow viscosity of carbonate melts at high pressures, Nat. Commun. 5 (2014) 1–8, https://doi.org/10.1038/ncomms6091.
[119] B.C.H. Steele, A. Heinzel, Materials for fuel-cell technologies, Nature. 414 (2001) 345–352.
[120] J. Collins, D. Qu, G. Chemistry, Molten Carbonate Fuel Cells Learn more about Molten Carbonate Fuel Cells Modern Applications of Green Chem- istry Fuel cells, (2018).
[121] R.S. Kalb, Reaction Chemistry & Engineering Carbonate Based Ionic Liquid Synthesis (CBILS ®): preparation of ultra-pure ionic liquids †, (2017), pp. 432–436, https://doi.org/10.1039/c7re00028f.
[122] C.M. Gordon, M.J. Muldoon, Synthesis of Ionic Liquids, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., Weinheim, WILEY-VCH Verlag GmbH & Co, 2008, pp. 7–23.
[123] N. Dupre, P. Moreau, E. De Vito, L. Quazuguel, M. Boniface, H. Kren, P. Bayle-guillemaud, D. Guyomard, Carbonate and Ionic Liquid Mixes as Electrolytes To Modify Interphases and Improve Cell Safety in Silicon-Based Li-Ion Batteries, Chem. Mater. 29 (2017) 8132–8146, https://doi.org/10.1021/acs.chemmater.7b01963.
[124] M. Frey, H. Didzoleit, C. Gainaru, R. Böhmer, Dynamics in glass forming sulfuric and nitric acid hydrates, J. Phys. Chem. B. 117 (2013) 12164–12174, https://doi.org/10.1021/jp407588j.
[125] M.E. Fleet, X. Liu, X. Wang, Magnesium carbonate-phosphate glass, J. Non. Cryst. Solids. 355 (2009) 2604–2609, https://doi.org/10.1016/j.jnoncrysol.2009.09.006.
[126] J. Deubener, R. Müller, H. Behrens, G. Heide, Water and the glass transition temperature of silicate melts, J. Non. Cryst. Solids. 330 (2003) 268–273, https://doi.org/10.1016/S0022-3093(03)00472-1.
[127] H. Didzoleit, M. Storek, C. Gainaru, B. Geil, R. Bo, R. Böhmer, Dynamics of glass forming ammonia hydrates, J. Phys. Chem. B. 117 (2013) 12157–12163, https://doi.org/10.1021/jp407581q.
[128] R.D. Shannon, C.T. Prewitt, Effective ionic radii in oxides and fluorides, Acta Crystallogr. Sect. B Struct. Crystallogr. Cryst. Chem. 25 (1969) 925–946, https://doi.org/10.1107/S0567740869003220.
[129] C.A. Angell, G.H. Wegdam, J. van der Elsken, FIR spectra of liquid, glass and crystalline states of $ZnCl_2$: order and temperature effects on band shape, Spectrochim. Acta Part A Mol. Spectrosc. 30 (1974) 665–672, https://doi.org/10.1016/0584-8539(74)80189-3.
[130] A.H. Narten, Diffraction pattern and structure of noncrystalline BeF_2 and SiO_2 at 25 °C, J. Chem. Phys. 56 (1972) 1905–1909, https://doi.org/10.1063/1.1677473.
[131] K. Saito, A.J. Ikushima, Reduction of light-scattering loss in silica glass by the structural relaxation of “frozen-in” density fluctuations, Appl. Phys. Lett. 70 (1997) 3504–3506.
[132] C.M. Baldwin, R.M. Almeida, J.D. Mackenzie, Halide glasses, J. Non. Cryst. Solids. 43 (1981) 309–344, https://doi.org/10.1016/0022-3093(86)90380-7.
[133] C.M. Baldwin, J.D. Mackenzie, Preparation and properties of water-free vitreous beryllium fluoride, J. Non. Cryst. Solids. 31 (1979) 441–445.
[134] C.A. Angell, E. Williams, K.J. Rao, J.C. Tucker, Heat capacity and glass transition thermodynamics for zinc chloride. A failure of the first Davies-Jones relation for dTg/dP, J. Phys. Chem. 81 (1977) 238–243, https://doi.org/10.1021/j100518a011.
[135] F. Gan, Optical properties of fluoride glasses: a review, J. Non. Cryst. Solids. 184 (1995) 9–20, https://doi.org/10.1016/0022-3093(94)00592-3.
[136] D.C. Tran, G.H. Sigel, B. Bendow, Heavy metal fluoride glasses and fibers: A review, J. Light. Technol. 2 (1984) 566–586, https://doi.org/10.1109/JLT.1984.1073661.
[137] M. Yamane, C.T. Moynihan (Eds.), Halides Glasses V, Trans Tech Publications, Switzerland-Germany-UK-USA, 1988.
[138] G.H. Frischat, C.T. Moynihan (Eds.), Halides Glasses VI, Trans Tech Publications, Switzerland-Germany-UK-USA, 1991.
[139] D.R. MacFarlane (Ed.), Halide Glasses 7, North-Holland Publishing, Elsevier Science, 1992, , https://doi.org/10.1016/s0022-3093(05)80726-4.
[140] H. Poignant, D. Ronarc'h, D.L. Griscom (Eds.), Halide Glasses 8, North-Holland Publishing, Elsevier Science, 1993, , https://doi.org/10.1016/0022-3093(93)90653-F.
[141] G. Fuxi, D.R. MacFarlane, D.L. Griscom (Eds.), Non-Oxide Glasses 9, North-Holland Publishing, Elsevier Science, 1995, , https://doi.org/10.1016/0022-3093(95)80026-3.
[142] R.F. Bartholomew, A. Bruce, A.G. Clare, P.A. Tick (Eds.), Non-Oxide Glasses 10, 213–2 North-Holland Publishing, Elsevier Science, 1996.
[143] R.M. Almeida (Ed.), Halide Glasses for Infrared Fiberoptics, Martinus Nijhoff Publishers, Dordrecht, The Netherlands, 1987.
[144] J. Lucas, Review: Fluoride glasses, J. Mater. Sci. 24 (1989) 1–13, https://doi.org/10.1007/BF00660926.
[145] M. Poulain, A. Soufiane, Y. Messaddeq, M.A. Aegerter, Fluoride Glasses: Synthesis and Properties, Brazilian J. Phys. 22 (1992) 205–217.
[146] P.P. Fedorov, Fluoroindate glasses, Russ. Chem. Rev. 69 (2000) 705–716, https://doi.org/10.1070/RC2000v069n08ABEH000582.
[147] G. Drexhage, Martin, Heavy-Metal Fluoride Glasses, in: M. Tomozawa, R.H. Doremus (Eds.), Glas. IV Treatise Mater. Sci. Technol, Elsevier B. V, Amsterdam, 1985, pp. 151–243.
[148] J.J. Videau, Portier J., Fluoride glasses, in: P. Hagenmuller (Ed.), Inorg. Solid Fluorides. Chem. Phys. Academic Press, Inc., New York, London, 1985, pp. 309–329, , https://doi.org/10.1007/BF00660926.
[149] I.D. Aggarwal, G. Lu (Eds.), Fluoride Glass Fiber Optics, Academic Press, Inc., San Diego, 1991, , https://doi.org/10.1016/B978-0-12-044505-9.50008-5.
[150] J. Lucas, C.T. Moynihan (Eds.), Halide Glasses I & II, Trans Tech Publications, Switzerland-Germany-UK-USA, 1985.
[151] M.G. Drexhage, C.T. Moynihan, M. Robinson (Eds.), Halide Glasses, Trans Tech Publications, Switzerland-Germany-UK-USA, 1987.
[152] K.H. Sun, T.E. Callear, Fluoride glasses, 2466506, (1949).
[153] M. Poulain, M. Chanthanasinh, J. Lucas, Nouveaux verres fluores, Mater. Res. Bull. 12 (1977) 151–156, https://doi.org/10.1016/0025-5408(77)90157-X.
[154] M. Poulain, M. Poulain, J. Lucas, P. Brun, Verres fluores au tetrafluorure de zirconium propriete optiques d'un verre dope au Nd^{3+}, Mater. Res. Bull. 10 (1975) 243–246, https://doi.org/10.1016/0025-5408(75)90106-3.
[155] Z. Ling, Z. Chengshan, D. Gaoxian, W. Kangkang, ZrOCl2 for fluoride glass preparation, J. Non. Cryst. Solids. 140 (1992) 331–334.
[156] M. Jafary-Zadeh, R. Tavakoli, J.J. Koh, Z.H. Aitken, Y.W. Zhang, Effect of chemical composition and affinity on the short- and medium-range order structures and mechanical properties of Zr-Ni-Al metallic glass, J. Non. Cryst. Solids. 456 (2017) 68–75, https://doi.org/10.1016/j.jnoncrysol.2016.10.042.
[157] M. Poulain, R. Lebullenger, M. Saad, Synthesis of high purity fluorides by wet chemistry, J. Non. Cryst. Solids. 140 (1992) 57–61, https://doi.org/10.1016/S0022-3093(05)80741-0.
[158] J. Cheng, W. Chen, D. Ye, Novel chalcohalide glasses in the As-Ge-Ag-Se-Te-I system, J. Non. Cryst. Solids. 184 (1995) 124–127, https://doi.org/10.1016/0022-3093(94)00601-6.
[159] P. Klaska, X.H. Zhang, J. Lucas, “High Tg” germanium–tellurium–halide glasses, J. Non. Cryst. Solids. 161 (1993) 297–300.
[160] M. Matecki, M. Poulain, Composition adjustments in cadmium fluorochloride glasses, J. Non. Cryst. Solids. 140 (1992) 82–86, https://doi.org/10.1016/S0022-3093(05)80746-X.
[161] J.S. Sanghera, V.Q. Nguyen, R. Miklos, I.D. Aggarwal, Measurement of bulk absorption coefficients of chalcogenide and chalcohalide glasses at 10.6 μm using CO2 laser calorimetry, J. Non. Cryst. Solids. 161 (1993) 320–322, https://doi.org/10.1016/0022-3093(93)90728-G.
[162] C.A. Angell, D.C. Ziegler, Inorganic chloride and mixed halide glasses with low maximum phonon frequencies, Mater. Res. Bull. 16 (1981) 279–283, https://doi.org/10.1016/0025-5408(81)90043-X.
[163] Y. Le Page, G. Fonteneau, J. Lucas, Synthese et etude de verres fluores dans les systemes LnF_3-BaF_3-MnF_3, Mater. Res. Bull. 17 (1982) 647–652.
[164] M. Poulain, Halide Glasses, J. Non. Cryst. Solids. 56 (1983) 1–14.
[165] H. Hu, J.D. Mackenzie, New chloride glasses in the NaCl-KCl-$ThCl_4$ system, J. Non. Cryst. Solids. 51 (1982) 269–272.
[166] M. Poulain, Fluoride Glass Composition and Processing, in: I.D. Aggarwal, G. Lu (Eds.), Fluoride Glas. Fiber Opt. Academic Press, Inc., San Diego, 1991, pp. 1–36.
[167] J.H. Simmons, C.J. Simmons, R. Ochoa, A.C. Wright, Fluoride Glass Structure, in: I.D. Aggarwal, G. Lu (Eds.), Fluoride Glas. Fiber Opt. Academic Press, Inc., San Diego, 1991, pp. 37–84, , https://doi.org/10.1016/B978-0-12-044505-9.50008-5.
[168] L.G. Van Uitert, S.H. Wemple, $ZnCl_2$ glass: A potential ultralow-loss optical fiber material, Appl. Phys. Lett. 33 (1978) 57–59, https://doi.org/10.1063/1.90189.
[169] A.J. Easteal, P.J. O'Rourke, Phase equilibria in the system $ZnCl_2$ + KI, Aust. J. Chem. 27 (1974) 151–159, https://doi.org/10.1016/j.jeurceramsoc.2008.03.011.
[170] L.E. Topol, S.W. Mayer, L.D. Ransom, The heat of fusion of bismuth trichloride—a comparison of calorimetric and cryoscopic determinations, J. Phys. Chem. 64 (1960) 862–865, https://doi.org/10.1021/j100836a009.
[171] X. Zhang, J. Lucas, G. Fonteneau, A new glass family based on tellurium chalcohalides, Mater. Sci. Forum. 19–20 (1987) 67–72, https://doi.org/10.4028/www.scientific.net/MSF.19-20.67.
[172] J. Lucas, X.H. Zhang, The tellurium halide glasses, J. Non. Cryst. Solids. 125 (1990) 1–16, https://doi.org/10.1016/0022-3093(90)90317-F.
[173] C.A. Angell, C. Liu, H.G.K. Sundar, Far IR transmitting halide glasses, Mater. Sci. Forum. 5 (1985) 189–192, https://doi.org/10.4028/www.scientific.net/MSF.5-6.189.
[174] K. Kadono, K. Mitani, K. Kinugawa, H. Tanaka, Novel halide glasses based on systems of LiX (X = Cl, Br, I), J. Non. Cryst. Solids. 122 (1990) 214–215, https://doi.org/10.1016/0022-3093(90)91069-4.
[175] H. Nasu, D.P. Yamato, J. Heo, J.D. Mackenzie, Preparation and properties of non-fluoride halide glasses, Mater. Sci. Forum. 5 (1985) 121–126, https://doi.org/10.4028/www.scientific.net/MSF.5-6.121.
[176] K. Miura, I. Masuda, M. Tokida, T. Yamashita, Chlorine doped fluorozirco-aluminate glass, Mater. Sci. Forum 32–33 (1988) 367–372.
[177] K. Zhang, C. Zhang, Studies of new mixed halide glasses, J. Non. Cryst. Solids. 112 (1989) 173–176, https://doi.org/10.1016/0022-3093(89)90516-4.
[178] M. Poulain, A. Elyamani, Barium and strontium chlorofluorozirconate glasses, Mater. Sci. Forum. 32–33 (1988) 143–147, https://doi.org/10.4028/www.scientific.net/MSF.32-33.143.
[179] J.L. Adam, C. Ricordel, J. Lucas, New compositions of low phonon energy fluoride and chloro-fluoride glasses, J. Non. Cryst. Solids. 213–214 (1997) 30–35, https://doi.org/10.1016/S0022-3093(97)00005-7.
[180] B. Petrová, M. Frumar, E. Cernosková, I. Pavlík, L.N. Blinov, ESR and optical spectra of Co^{2+} DOPED CdX_2-PbX_2-KI GLASSES (X = Cl, Br), Mater. Sci. Forum.

67–68 (1991) 537–542, https://doi.org/10.4028/www.scientific.net/MSF.67-68.537.
[181] G.J. Dai, C.S. Zhang, S.T. Huang, Progress in divalent mixed-halide glasses, Mater. Sci. Forum. 67–68 (1991) 125–128, https://doi.org/10.4028/www.scientific.net/MSF.67-68.125.
[182] A. Elyamani, M. Poulain, Strontium chlorofluorozirconate glasses, Mater. Sci. Forum. 67–68 (1991) 119–124, https://doi.org/10.4028/www.scientific.net/MSF.67-68.119.
[183] L.F. Santos, R.M. Almeida, Zinc halide glasses with improved thermal characteristics, J. Non. Cryst. Solids. 184 (1995) 68–71, https://doi.org/10.1016/0022-3093(94)00640-7.
[184] L.F. Santos, R.M. Almeida, Short and medium range order in zinc halide based glasses, J. Non. Cryst. Solids. 232–234 (1998) 638–643, https://doi.org/10.1016/S0022-3093(98)00528-6.
[185] K. Kadono, H. Nakamichi, M. Nogami, Preparation of glasses in the system ZnBr2 - KBr - MBr2 - AgBr (M / Ba, Pb), Mater. Sci. Forum. 19–20 (1987) 63–66, https://doi.org/10.4028/www.scientific.net/MSF.19-20.63.
[186] Y. Kawamoto, R. Kanno, M. Shojiya, K. Kadono, Structural study of ion-conducting MX-K (or Rb) X-CsX-BaX2 halide glasses (M = Li, Cu or Ag) with single X species (X = Cl, Br or I), J. Non. Cryst. Solids. 184 (1995) 155–159, https://doi.org/10.1016/0022-3093(94)00608-3.
[187] J. Cheng, Z. Jin, New lead-halide-based glass-forming systems, J. Non. Cryst. Solids. 184 (1995) 213–217, https://doi.org/10.1016/0022-3093(94)00614-8.
[188] A.M. Nepomiluev, S.A. Ryzhakov, V.G. Bamburov, Synthesis and properties of glasses in the NaF-ZnSO4 system, Glas. Phys. Chem. 29 (2003) 122–126, https://doi.org/10.1134/S003602360705021X.
[189] L.J. Pollard, M.L. Crowe, W. Strauss, Compressibility measurements of the liquid binary systems KNO_3-$Ca(NO_3)_2$ and $ZnCl_2$-$C_5H_3N_5HCl$, J. Chem. Eng. Data. 16 (1971) 134–136, https://doi.org/10.1021/je60049a003.
[190] M. Robinson, M.G. Drexhage, A phenomenological comparison of some heavy metal fluoride glasses in water environments, Mater. Res. Bull. 18 (1983) 1101–1112, https://doi.org/10.1016/0025-5408(83)90153-8.
[191] C.J. Simmons, H. Sutter, J.H. Simmons, D.C. Tran, Aqueous corrosion studies of a fluorozirconate glass, Mater. Res. Bull. 17 (1982) 1203–1210.
[192] F. Gan, Y. Dai, H. Hu, Preparation, properties and structure of chloride in glasses in systems of ThCl4-PbCl2RCl (R = Na, Cu OR Ag), Mater. Sci. Forum. 5 (1985) 113–120, https://doi.org/10.4028/www.scientific.net/MSF.5-6.113.
[193] Y. Dai, H. Hu, F. Gan, Investigation on water dissolution of THCl4 based glass using dynamic ion selective electrode method, Mater. Sci. Forum. 32–33 (1988) 273–278, https://doi.org/10.4028/www.scientific.net/MSF.32-33.273.
[194] D.C. Ziegler, C.A. Angell, High refractive index, low Abbe number halide glasses, Appl. Opt. 21 (1982) 2096–2098.
[195] M. Fuding, New halides glasses – in the AgBr-$PbBr_2$-CuCl-CsBr SYSTEM, Mater. Sci. Forum. 5–6 (1985) 127–133, https://doi.org/10.4028/www.scientific.net/MSF.5-6.127.
[196] M. Matecki, M. Poulain, de Bea, Cadmium halide glasses, J. Non. Cryst. Solids 56 (1983) 81–86.
[197] K. Zhang, L. Jixiang, Z. Xiujian, L. Hwiweng, Heavy halide glasses based on lead bromide and lead iodide, J. Non. Cryst. Solids. 140 (1992) 225–228.
[198] J. Jiang, G. Zhang, M. Poulain, Effects of oxides in cadmium mixed halide glass, J. Non. Cryst. Solids. 184 (1995) 346–351, https://doi.org/10.1016/0022-3093(94)00654-7.
[199] S. Inoue, D.R. MacFarlane, Glass forming tendency of ZrF_4 - BaF_2 systems containing chlorides, Mater. Sci. Forum. 19–20 (1987) 27–32, https://doi.org/10.4028/www.scientific.net/MSF.19-20.27.
[200] X. Chen, N. Karpukhina, D.S. Brauer, R.G. Hill, Novel Highly Degradable Chloride Containing Bioactive Glasses, Biomed. Glas. 1 (2015) 108–118, https://doi.org/10.1515/bglass-2015-0010.
[201] X. Chen, X. Chen, D.S. Brauer, R.M. Wilson, R.G. Hill, N. Karpukhina, Novel alkali free bioactive fluorapatite glass ceramics, J. Non. Cryst. Solids. 402 (2014) 172–177, https://doi.org/10.1016/j.jnoncrysol.2014.05.025.
[202] D.S. Brauer, N. Karpukhina, V.R. Law, R.G. Hill, R.V. Law, R.G. Hill, Structure of fluoride-containing bioactive glasses, J. Mater. Chem. 19 (2009) 5629–5636, https://doi.org/10.1039/b900956f.
[203] J. Guery, G. Courbion, C. Jacoboni, R. de Pape, Synthese et caracterisation de verres fluores dans in le systeme BaF_2-UF_4-FeF_3-MnF_2, Mater. Res. Bull. 19 (1984) 1437–1442.
[204] A.J. Bruce, The Thermal Properties of Fluoride Glass, Halide Glas. Infrared Fiberoptics, NATO ASI Ser 123 (1987) 149–162, https://doi.org/10.1007/978-94-009-3561-7_9.
[205] Y. Kawamoto, I. Nohara, J. Fujiwara, Y. Umetani, Exploration of fluoride glasses with faster fluoride-ion conduction, Solid State Ionics. 24 (1987) 327–331, https://doi.org/10.1016/0167-2738(87)90139-1.
[206] R. Mossadegh, C.T. Moynihan, N. York, A.J.I. Bruce, M. Hill, M.G. Drexhage, DSC studies of melting behavior of heavy metal fluoride glass compositions, Mater. Res. Bull. 22 (1987) 593–600.
[207] S. Sakka, J.D. Mackenzie, Relation between apparent glass transition temperature and liquidus temperature for inorganic glasses, J. Non. Cryst. Solids. 6 (1971) 145–162.
[208] S. Yoshida, R. Oyamada, K. Kawamura, Raman studies of complex ions in $ThCl_4$-KCl melt using Raman bands shifts, Bull. Chem. Soc. Jpn. 511 (1978) 25–27, https://doi.org/10.1246/bcsj.51.25.
[209] L. Wondraczek, J.C. Mauro, J. Eckert, U. Kühn, J. Horbach, J. Deubener, T. Rouxel, Towards ultrastrong glasses, Adv. Mater. 23 (2011) 4578–4586, https://doi.org/10.1002/adma.201102795.
[210] R. Limbach, B.P. Rodrigues, L. Wondraczek, Strain-rate sensitivity of glasses, J. Non. Cryst. Solids. 404 (2014) 124–134, https://doi.org/10.1016/j.jnoncrysol.2014.08.023.
[211] R. Ota, N. Soga, Elastic properties of fluoride glasses under pressure and temperature, J. Non. Cryst. Solids. 56 (1983) 105–110.
[212] B.P. Rodrigues, S. Sen, L. Wondraczek, Pressure-induced chemical reordering in supercooled polyionic sulfophosphate liquids, J. Non. Cryst. Solids. 499 (2018) 95–100, https://doi.org/10.1016/j.jnoncrysol.2018.07.019.
[213] D. Leroy, J. Lucas, M. Poulain, D. Ravaine, ETUDE DE LA CONDUCTION IONIQUE DE VERRES A BASE DE ZrF_4, Mater. Res. Bull. 13 (1978) 1125–1133.
[214] Y. Kawamoto, I. Nohara, Nohara, Ionic conductivites of ZrF_4-BaF_2-CsF glasses, Solid State Ionics. 22 (1986) 207–212.
[215] D. Ravaine, G. Perera, M. Poulain, Anionic conductivity in fluoride glasses, Solid State Ionics. 9–10 (1983) 631–637, https://doi.org/10.1016/0167-2738(83)90306-5.
[216] J.M. Reau, M. Poulain, Ionic Conductivity In Fluorine-Containing Glasses, Mater. Chem. Phys. 23 (1989) 189–209, https://doi.org/10.1017/CBO9781107415324.004.
[217] V.Y. Kavun, E.I. Voit, S.L. Sinebryukhov, A.B. Podgorbunskii, O.V. Brovkina, E.B. Merkulov, V.K. Goncharuk, Ionic mobility, structure, and conductivity of 45ZrF 4 -35BiF 3 -20CsF glass according to 19 F NMR, IR, raman, and impedance spectroscopy, J. Struct. Chem. 52 (2011) 1102–1105, https://doi.org/10.1134/S0022476611060102.
[218] S.P. Gabuda, R.L. Davidovich, S.G. Kozlova, N.K. Moroz, Phase transitions and ionic mobility in thallium fluorozirconates, J. Struct. Chem. 37 (1996) 340–342, https://doi.org/10.1007/BF02591066.
[219] T. Minami, Recent progress in superionic conducting glasses, J. Non. Cryst. Solids. 95–96 (1987) 107–118.
[220] D. Ravaine, W. Perera, M. Minier, Alkali fluoride containing ZrF_4 - based glasses : electrical properties and NMR investigation, J. Phys. Colloq. 43 (1982) 407–410, https://doi.org/10.1051/jphyscol:1982978ï.
[221] J.M. Bobe, J. Senegas, J.M. Réau, M. Poulain, Ionic conductivity and 19F NMR investigations of some series of ZrF4-based fluoride glasses containing LaF3 or YF3, J. Non. Cryst. Solids. 162 (1993) 169–177, https://doi.org/10.1016/0022-3093(93)90753-K.
[222] J.M. Reau, J. Senegas, J.M. Rojo, M. Poulain, M. Poulain, TRANSPORT PROPERTIES OF ZrF_4-BaF_2-LaF_3-AF (A = Li, Na) GLASSES, J. Non. Cryst. Solids. 116 (1990) 175–178.
[223] J.M. Reau, H. Kahnt, M. Poulain, Ionic transport studies in mixed alkali-fluorine conductor glasses of composition ZrF4BaF2LaF3AF (A = Li, Na) and ZrF4BaF2ThF4LiF, J. Non. Cryst. Solids. 119 (1990) 347–350, https://doi.org/10.1016/0022-3093(90)90308-9.
[224] J.M. Bobe, J.M. Réau, J. Senegas, M. Poulain, F^- ion conductivity and diffusion properties in ZrF_4-based fluoride glasses with various NaF concentrations ($0 \leq x_{NaF} \leq 0.45$), Solid State Ionics 82 (1995) 39–52, https://doi.org/10.1016/0167-2738(95)00192-9.
[225] J.M. Reau, J. Senegas, H. Aomi, P. Hagenmuller, M. Poulain, Alkali fluoride containing fluorozirconate glasses: Electrical properties and NMR investigation, J. Solid State Chem. 60 (1985) 159–164, https://doi.org/10.1016/0022-4596(85)90107-0.
[226] J. Senegas, J.M. Reau, H. Aomi, P. Hagenmuller, M. Poulain, Ionic conductivity and nmr investigation of quaternary glasses in the ZrF4–BaF2–ThF4–LiF SYSTEM, J. Non. Cryst. Solids. 85 (1986) 315–334.
[227] J.M. Réau, X.Y. Jun, J. Senegas, C. Le Deit, M. Poulain, Influence of network modifiers on conductivity and relaxation parameters in some series of fluoride glasses containing LiF, Solid State Ionics. 95 (1997) 191–199, https://doi.org/10.1016/s0167-2738(96)00564-4.
[228] J.M. Bobe, J.M. Reau, J. Senegas, M. Poulain, Ion conductivity and diffusion in ZrF4-based fluoride glasses containing LiF ($0 \leq x_{LiF} \leq 0.60$), J. Non. Cryst. Solids 209 (1997) 122–136.
[229] J.M. Reau, H. Kahnt, M. Poulain, P. Hagenmuller, Cation Effects in some Series of Mixed Alkali Fluoride Glasses, Z. Anorg. Allg. Chem. 598 (599) (1991) 269–277.
[230] R.M. Almeida, J.D. MacKenzie, The electrical conductivity of fluorozirconate and chloro-fluorozirconate glasses, J. Mater. Sci. 17 (1982) 2533–2538, https://doi.org/10.1007/BF00543884.
[231] H. Kahnt, J.M. Réau, A. Elyamani, M. Poulain, Influence of chlorine-fluorine substitution on the electrical properties of some fluoride glasses, J. Non. Cryst. Solids. 160 (1993) 255–260, https://doi.org/10.1016/0022-3093(93)91269-9.
[232] M.D. Ingram, Ion transport in glassy electrolytes, Solid State Ionics. 94 (1997) 49–54, https://doi.org/10.1016/S0167-2738(96)00610-8.
[233] M. Tatsumisago, Y. Akamatsu, T. Minami, Effects of cation-mixing and anion-mixing in halide glasses on ionic conductivity, Mater. Sci. Forum. 32–33 (1988) 617–622, https://doi.org/10.4028/www.scientific.net/MSF.32-33.617.
[234] L.G. Van Uitert, A.J. Bruce, W.H. Grodkiewicz, D.L. Wood, Minimum loss projections for oxide and halide glasses, Mater. Sci. Forum. 6 (1985) 591–606, https://doi.org/10.4028/www.scientific.net/MSF.5-6.591.
[235] L. Wetenkamp, G.F. West, H. Többen, Optical properties of rare earth-doped ZBLAN glasses, J. Non. Cryst. Solids. 140 (1992) 35–40.
[236] Y. Chen, F. Auzel, Investigation of multiphonon assisted side band absorptions in rare-earth-doped halide glasses and crystals, J. Non. Cryst. Solids. 184 (1995) 278–281, https://doi.org/10.1016/0022-3093(94)00625-3.
[237] E.R. Taylor, B.N. Samson, D.W. Hewak, J.A. Medeiros Neto, D.N. Payne, S. Jordery, M. Naftaly, A. Jha, Cadmium mixed halide glass for optical amplification at 1.3 μm, J. Non. Cryst. Solids 184 (1995) 61–67, https://doi.org/10.1016/0022-3093(94)00596-6.
[238] H.L. Ma, X.H. Zhang, J. Lucas, R. Iizuka, T. Yamagishi, T. Yamashita, Refractive index measurement of the TeX glasses, J. Non. Cryst. Solids. 161 (1993) 198–201.

[239] X.H. Zhang, G. Fonteneau, H.L. Ma, J. Lucas, Tellerium halide glass fibers, Mater. Sci. Forum. 67–68 (1991) 371–376, https://doi.org/10.4028/www.scientific.net/MSF.67-68.371.
[240] K. Zhang, Y. Xia, L. Huo, Y. Qihua, The formation and properties of halide glasses in the system CdF2-BaF2-NaX(X = F, Cl, Br), J. Non. Cryst. Solids. 95–96 (1987) 495–500, https://doi.org/10.1016/S0022-3093(87)80149-7.
[241] M. Yamane, H. Kawazoe, S. Inoue, K. Maeda, IR transparency of $48ZnCl_2$·48KBr·4$PbBr_2$ (mol%) glass, Mater. Sci. Forum. 5–6 (1985) 489–493, https://doi.org/10.4028/www.scientific.net/MSF.5-6.489.
[242] L.F. Santos, R.M. Almeida, Long wavelength - transmitting zinc halide based glasses, Mater. Sci. Forum. 32–33 (1988) 155–160, https://doi.org/10.4028/www.scientific.net/MSF.32-33.155.
[243] T. Miyashita, T. Manabe, Infrared Optical Fibers, IEEE J. Quantum Electron. 18 (1982) 1432–1450.
[244] V.N. Polukhin, A review of glass-forming systems used for the synthesis of various types of optical glasses, Sov. J. Glas. Phys. Chem. 6 (1980) 421.
[245] M. El-Hofy, I.Z. Hager, Ionic conductivity in lithium haloborate glasses, Phys. Status Solidi Appl. Res. 199 (2003) 448–456, https://doi.org/10.1002/pssa.200306670.
[246] I.Z. Hager, M. El-Hofy, Investigation of spectral absorption and elastic moduli of lithium haloborate glasses, Phys. Status Solidi Appl. Res. 198 (2003) 7–17, https://doi.org/10.1002/pssa.200305958.
[247] K.C. Sekhar, A. Hameed, G. Ramadevudu, M.N. Chary, M. Shareefuddin, Physical and spectroscopic studies on manganese ions in lead halo borate glasses, Mod. Phys. Lett. B. 31 (2017) 1750180, https://doi.org/10.1142/S0217984917501809.
[248] V.P. Dotsenko, I.V. Berezovskaya, N.P. Efryushina, A.S. Voloshinovskii, P. Dorenbos, C.W.E. Van Eijk, Luminescence of Ce^{3+} ions in strontium haloborates, J. Lumin. 93 (2001) 137–145, https://doi.org/10.1016/S0022-2313(01)00179-X.
[249] A. Meijerink, G. Blasse, Luminescence properties of Eu^{3+}-activated alkaline earth haloborates, J. Lumin. 43 (1989) 283–289.
[250] M.J. Weber, D.C. Ziegler, C.A. Angell, Tailoring stimulated emission cross sections of Nd^{3+} laser glass: Observation of large cross sections for $BiCl_3$ glasses, J. Appl. Phys. 53 (1982) 4344–4350, https://doi.org/10.1063/1.331214.
[251] M.J. Weber, R.M. Almeida, Large stimulated emission cross section of Nd^{3+} in chlorophosphate glass, J. Non. Cryst. Solids. 43 (1981) 99–104.
[252] J. Wasylak, E. Czerwosz, Phosphate-halide glasses, structure and properties, J. Non. Cryst. Solids. 56 (1983) 117–122.
[253] M. Matecki, N. Duhamel, J. Lucas, Sodium fluorometaphosphate glasses with high rare earth ion concentrations, J. Non. Cryst. Solids. 184 (1995) 273–277, https://doi.org/10.1016/0022-3093(94)00624-5.
[254] N. Rigout, J.L. Adam, J. Lucas, Chemical and physical compatibilities of fluoride and fluorophosphate glasses, J. Non. Cryst. Solids. 184 (1995) 319–323, https://doi.org/10.1016/0022-3093(94)00590-7.
[255] W. Zhou, H. Zhao, D. Zhu, Preparation and properties of lead halide phosphate glasses, J. Non. Cryst. Solids. 263 (2000) 277–284, https://doi.org/10.1016/S0022-3093(99)00640-7.
[256] B. Poletto Rodrigues, R. Limbach, G. Buzatto de Souza, H. Ebendorff-Heidepriem, L. Wondraczek, Correlation Between Ionic Mobility and Plastic Flow Events in $NaPO_3$-NaCl-Na_2SO_4 Glasses, Front. Mater. 6 (2019) 1–13, https://doi.org/10.3389/fmats.2019.00128.
[257] H. Morikawa, Y. Shimizugawa, Y. Tabira, F. Marumo, M. Yamane, EXAFS STUDY OF AgBr-Ag2O-GeO2 GLASSES, Mater. Sci. Forum. 32–33 (1988) 391–395, https://doi.org/10.4028/www.scientific.net/MSF.32-33.391.
[258] Y.-H. Wang, A. Osaka, Y. Miura, J. Takada, K. Oda, K. Takahashi, Glass forming range and properties of new oxyhalide glasses in the system TeO_2-PbO-$PbCl_2$, Mater. Sci. Forum. 32–33 (1988) 161–165, https://doi.org/10.4028/www.scientific.net/MSF.32-33.161.
[259] R.A. El-Mallawany, Preparation and characterization of binary oxy-halide tellurite glasses, Mater. Sci. Forum. 67–68 (1991) 149–154, https://doi.org/10.4028/www.scientific.net/MSF.67-68.149.
[260] M.R. Sahar, N. Noordin, Oxychloride glasses based on the TeO2-ZnO-ZnCl2 system, J. Non. Cryst. Solids. 184 (1995) 137–140, https://doi.org/10.1016/0022-3093(94)00606-7.
[261] M.R. Sahar, A.K. Jehbu, M.M. Karim, TeO2-ZnO-ZnCl2 glasses for IR transmission, J. Non. Cryst. Solids. 213–214 (1997) 164–167, https://doi.org/10.1016/S0022-3093(97)00096-3.
[262] B. Dubois, J.J. Videau, M. Couzi, J. Portier, Structural approach of the (xPbCl2 - (1 - x)Sb2O3) glass system, J. Non. Cryst. Solids. 88 (1986) 355–365, https://doi.org/10.1016/S0022-3093(86)80037-0.
[263] B. Dubois, H. Aomi, J.J. Videau, J. Portier, P. Hagenmuller, Ney oxyhalide glasses involving Sb_2O_3, Mater. Res. Bull. 19 (1984) 1317–1323.
[264] G. Poirier, M. Poulain, M. Poulain, Copper and lead halogeno-antimoniate glasses, J. Non. Cryst. Solids. 284 (2001) 117–122, https://doi.org/10.1016/S0022-3093(01)00389-1.
[265] D. Holland, M.R. Sahar, M.M. Ahmed, Thermal behaviour of glasses bases on the Sb2O3-PbCl2-ZnCl2 system, Mater. Sci. Forum. 67–68 (1991) 257–262, https://doi.org/10.4028/www.scientific.net/MSF.67-68.257.
[266] H. Aomi, B. Dubois, J.J. Videau, J. Portier, SUR DE NOUVEAUX VERRES OXYFLUORES A BASE D'ANTIMOINE, J. Fluor. Chem. 31 (1986) 395–403.
[267] T. Grande, J.R. Holloway, P.F. Mcmillan, C.A. Angell, Nitride glasses obtained by high-pressure synthesis, Lett. Nat. 369 (1994) 43–45.
[268] T. Grande, S. Jacob, J.R. Holloway, P.F. McMillan, C.A. Angell, High-pressure synthesis of nitride glasses, J. Non. Cryst. Solids. 184 (1995) 151–154, https://doi.org/10.1016/0022-3093(94)00603-2.
[269] H.L. Tuller, D.P. Button, D.R.R. Uhlmann, Fast ion transport in oxide glasses, J. Non. Cryst. Solids. 40 (1980) 93–118.
[270] N.V. Lichkova, A.L. Despotuli, V.N. Zagorodnev, N.A. Minenkova, Superionic glasses based on silver and caesium monohalides, Mater. Sci. Forum. 67–68 (1991) 601–606, https://doi.org/10.4028/www.scientific.net/MSF.67-68.601.
[271] K. Kadono, K. Mitani, M. Yamashita, H. Tanaka, Y. Kawamoto, K. Ohno, R. Kanno, Ionic conduction in LiX-based and CuX-based glasses (X = Cl, Br, I), J. Non. Cryst. Solids. 140 (1992) 103–106, https://doi.org/10.1016/S0022-3093(05)80750-1.
[272] D.P. Button, R.P. Tandon, H.L. Tuller, D.R. Uhlmann, Fast Li^+ ion conduction in chloro-borate glasses, J. Non. Cryst. Solids. 42 (1980) 297–306, https://doi.org/10.1016/0022-3093(80)90031-9.
[273] A. Levasseur, J.C. Brethous, J.M. Reau, P. Hagenmuller, J.M. Réau, P. Hagenmuller, Etude comparee de la conductivite ionique du lithium dans les halogenoborates vitreux, Mater. Res. Bull. 14 (1979) 921–927, https://doi.org/10.1016/0025-5408(79)90158-2.
[274] A.B. Seddon, Chalcohalides: Glass-forming systems and progress in application of percolation theory, J. Non. Cryst. Solids. 213–214 (1997) 22–29, https://doi.org/10.1016/S0022-3093(97)00060-4.
[275] Y.S. Tver'yanovich, M. Vlček, A. Tverjanovich, Formation of complex structural units and structure of some chalco-halide glasses, J. Non. Cryst. Solids. 333 (2004) 85–89, https://doi.org/10.1016/j.jnoncrysol.2003.09.042.
[276] J. Heo, J.S. Sanghera, H. Nasu, J.D. Mackenzie, The structure and properties of infrared transmitting chalcogenide glasses, Mater. Sci. Forum. 19–20 (1987) 55–62, https://doi.org/10.4028/www.scientific.net/MSF.19-20.55.
[277] F. Gan, Structure, properties and applications of chalcohalide glasses: a review, J. Non. Cryst. Solids. 140 (1992) 184–193.
[278] L. Zan, L. Huang, C. Zhang, New chalcohalide glasses from the Sb2S3-MXn system, J. Non. Cryst. Solids. 184 (1995) 1–4, https://doi.org/10.1016/0022-3093(94)00593-1.
[279] Z.G. Ivanova, V.S. Vassilev, K.G. Vassileva, New IR-transmitting chalcohalide glasses, J. Non. Cryst. Solids. 162 (1993) 123–127, https://doi.org/10.1016/0022-3093(93)90747-L.
[280] H.L. Ma, X.H. Zhang, J. Lucas, C.T. Moynihan, Relaxation near room temperature in tellurium chalcohalide glasses, J. Non. Cryst. Solids. 140 (1992) 209–214.
[281] M.V. Kurushkin, V.A. Markov, A.V. Semencha, M.D. Mikhailov, A.S. Tverjanovich, A.L. Shakhmin, T.V. Larionova, V.D. Andreeva, Determination of AsSI-SbSI glasses short-range structure via raman spectroscopy, XPS and XRD, Int. J. Appl. Glas. Sci. 9 (2018) 85–89, https://doi.org/10.1111/ijag.12279.
[282] A.G. Prabhudessai, S. Balaji, K. Biswas, R. Dasgupta, P. Sarkar, K. Annapurna, Correlation between Raman spectroscopy and mechanical properties of As-Sb-S-I chalcogenide glasses, J. Non. Cryst. Solids (2018), https://doi.org/10.1016/j.jnoncrysol.2018.10.016.
[283] G. Robert, J.P. Malugani, A. Saida, Fast ionic silver and lithium conduction in glasses, Solid State Ionics. 3 (4) (1981) 311–315.
[284] C. Liu, C.A. Angell, Fast Cu+ ion conducting phosphate iodide-glasses, Solid State Ionics. 13 (1984) 105–109, https://doi.org/10.1016/0167-2738(84)90043-2.
[285] J.C. Dyre, P. Maass, B. Roling, D.L. Sidebottom, Fundamental questions relating to ion conduction in disordered solids, Reports Prog. Phys. 72 (2009) 046501, , https://doi.org/10.1088/0034-4885/72/4/046501.
[286] K. Funke, B. Heimann, M. Vering, D. Wilmer, Concept of Mismatch and Relaxation Explains DC and AC Conductivities of Fragile Glass-Forming Ionic Melts, J. Electrochem. Soc. 148 (2001) A395, https://doi.org/10.1149/1.1359193.
[287] K. Funke, R.D. Banhatti, S. Brückner, C. Cramer, D. Wilmer, Dynamics of mobile ions in crystals, glasses and melts, described by the concept of mismatch and relaxation, Solid State Ionics. 154–155 (2002) 65–74, https://doi.org/10.1016/S0167-2738(02)00465-4.
[288] A. Bunde, M.D. Ingram, P. Maass, The dynamic structure model for ion transport in glasses, J. Non. Cryst. Solids. 172–174 (1994) 1222–1236, https://doi.org/10.1016/0022-3093(94)90647-5.
[289] J. Swenson, L. Börjesson, Correlation between free volume and ionic conductivity in fast ion conducting glasses, Phys. Rev. Lett. 77 (1996) 3569–3572, https://doi.org/10.1103/PhysRevLett.77.3569.
[290] M.D. Ingram, Ionic conductivity and glass structure, Philos. Mag. Part B. 60 (1989) 729–740, https://doi.org/10.1080/13642818908209739.
[291] J.D. Wicks, L. Börjesson, G. Bushnell-Wye, W.S. Howells, R.L. McGreevy, Structure and Ionic Conduction in (AgI) (AgPO3) Glasses, Phys. Rev. Lett. 74 (1995) 726–729, https://doi.org/10.1103/PhysRevLett.74.2248.
[292] M. Horiuchi, T. Sei, T. Tsuchiya, Preparation and properties of superionic conducting glasses containing lithium halide, J. Non. Cryst. Solids. 177 (1994) 236–241, https://doi.org/10.1016/0022-3093(94)90536-3.
[293] M. Doreau, J.P. Malugani, G. Robert, Verres mixtes $AgPO_3$-NaI et $AgPO_3$-KI domaine vitreux, conductivite electrique, approche structurale, Electrochim. Acta. 26 (1981) 711–717.
[294] J.P. Malugani, R. Mercier, M. Tachez, Correlation between structural and electrical properties in (1 − x)$AgPO_3$·xMX_2 glasses (M = Pb^{2+}, Hg^{2+}; X = I^-, Br^-, Cl^-) from Raman spectroscopy and ionic conductivity measuremen, Solid State Ionics. 21 (1986) 131–138.
[295] J. Ren, H. Eckert, Anion distribution in superionic Ag_3PO_4 − AgI glasses revealed by dipolar solid-state NMR, J. Phys. Chem. C. 117 (2013) 24746–24751.
[296] P. Mustarelli, L. Linati, V. Tartara, C. Tomasi, A. Magistris, Structure/transport relationships in silver-based oxide glasses: 1-D and 2-D NMR information, Solid State Nucl. Magn. Reson. 27 (2005) 112–121, https://doi.org/10.1016/j.ssnmr.2004.06.007.
[297] E. Kartini, M. Collins, Anomalous temperature dependence of the first diffraction peak in the superionic glass $(AgI)_x(AgPO_3)_{1-x}$, Phys. Rev. B - Condens. Matter Mater. Phys. 61 (2000) 1036–1042, https://doi.org/10.1103/PhysRevB.61.1036.
[298] J. Swenson, S. Adams, Application of the bond valence method to reverse Monte

Carlo produced structural models of superionic glasses, Phys. Rev. B. 64 (2001) 024204, , https://doi.org/10.1103/PhysRevB.64.024204.
[299] H.J.L. Trap, J.M. Stevels, Conventional and invert glasses containing titania: Part 1, Phys. Chem. Glas. 1 (1960) 107–118.
[300] P. Richet, F. Leclerc, L. Benoist, Melting or forsterite and spinel, with implications for the glass transition of Mg_2SiO_4 liquid, Geophys. Res. Lett. 20 (1993) 1675–1678.
[301] K. Griebenow, E.I. Kamitsos, L. Wondraczek, Mixed-modifier effect in (Ca,Mg) metaphosphate glasses, J. Non. Cryst. Solids. 468 (2017) 74–81, https://doi.org/10.1016/j.jnoncrysol.2017.04.036.
[302] M.E. Fleet, X. Liu, X. Wang, Magnesium carbonate – phosphate glass, J. Non. Cryst. Solids. 355 (2009) 2604–2609, https://doi.org/10.1016/j.jnoncrysol.2009.09.006.
[303] H. Hu, F. Ma, J.D. Mackenzie, New halide glasses - bromide glasses based on ZnBr2 I. Vitreous zinc bromide, J. Non. Cryst. Solids. 55 (1983) 169–172, https://doi.org/10.1016/0022-3093(83)90018-2.
[304] M. Tatsumisago, Y. Shinkuma, T. Minami, Stabilization of superionic α-AgI at room temperature in a glass matrix, Nature. 354 (1991) 217–218 https://www.nature.com/articles/201464a0.pdf.
[305] H. Bradtmüller, L. Zhang, C.C. de Araujo, H. Eckert, D. Möncke, D. Ehrt, Structural studies of $NaPO_3$-AlF_3 glasses by high-resolution double-resonance NMR spectroscopy -supporting materials, J. Phys. Chem. C. 122 (2018) 21579–21588, https://doi.org/10.1021/acs.jpcc.8b06162.
[306] B. Petrová, M. Frumar, E. Černošková, V. Černý, New chalcohalide glasses from the system GeS2–Ga2S3–MnCl2, J. Non. Cryst. Solids. 161 (1993) 316–319.
[307] T.D. Bennett, Y. Yue, P. Li, A. Qiao, H. Tao, G.N. Greaves, T. Richards, G.I. Lampronti, S.A.T. Redfern, F. Blanc, O.K. Farha, J.T. Hupp, A.K. Cheetham, D.A. Keen, Melt-Quenched Glasses of Metal-Organic Frameworks, J. Am. Chem. Soc. 138 (2016) 3484–3492, https://doi.org/10.1021/jacs.5b13220.
[308] S. Li, R. Limbach, L. Longley, A.A. Shirzadi, J.C. Walmsley, D.N. Johnstone, P.A. Midgley, L. Wondraczek, T.D. Bennett, Mechanical Properties and Processing Techniques of Bulk Metal-Organic Framework Glasses, J. Am. Chem. Soc. (2019), https://doi.org/10.1021/jacs.8b11357.
[309] A. Qiao, T.D. Bennett, H. Tao, A. Krajnc, G. Mali, C.M. Doherty, A.W. Thornton, J.C. Mauro, G.N. Greaves, Y. Yue, A metal-organic framework with ultrahigh glass-forming ability, Sci. Adv. 4 (2018) 1–7.
[310] V. Dimitrov, T. Komatsu, An interpretation of optical properties of oxides and oxide glasses in terms of the electronic ion polarizability and average single bond strength (review), J. Univ. Chem. Technol. Metall. 45 (2010) 219–250, https://doi.org/10.1016/j.jnoncrysol.2009.11.014.
[311] I.Z. Hager, Optical properties of lithium barium haloborate glasses, J. Phys. Chem. Solids. 70 (2009) 210–217, https://doi.org/10.1016/j.jpcs.2008.10.007.
[312] J.A. Duffy, The refractivity and optical basicity of glass, J. Non. Cryst. Solids. 86 (1986) 149–160.
[313] V. Dimitrov, T. Komatsu, V. Dimitrov, Electronic polarizability, optical basicity and non-linear optical properties of oxide glasses, J. Non. Cryst. Solids. 249 (1999) 160–179, https://doi.org/10.1016/S0022-3093(99)00317-8.
[314] C.R.A. Catlow, A.M. Stoneham, Ionicity in solids, J. Phys. C Solid State Phys. 16 (1983) 4321–4338, https://doi.org/10.1088/0022-3719/16/22/010.
[315] L. Pauling, The nature of the chemical bond. IV. The energy of single bonds and the relative electronegativity of atoms, J. Am. Ceram. Soc. 54 (1932) 3570–3582.
[316] J.C. Phillips, Ionicity of the chemical bond in crystals, Rev. Mod. Phys. 42 (1970) 317–356, https://doi.org/10.1103/RevModPhys.42.317.
[317] J.A. Duffy, M.D. Ingram, Establishment of an optical scale for lewis basicity in inorganic oxyacids, molten salts, and glasses, J. Am. Chem. Soc. 93 (1971) 6448–6454, https://doi.org/10.1021/ja00753a019.
[318] R.T. Sanderson, Electronegativities in inorganic chemistry. III, J. Chem. Educ. 31 (1954) 238, https://doi.org/10.1021/ed031p238.
[319] R.D. Shannon, Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides, Acta Crystallogr. Sect. A. 32 (1976) 751–767, https://doi.org/10.1107/S0567739476001551.
[320] R.D. Shannon, Revised effective ionic radii in halides and chalcogenides, Acta Cryst. 751 (1976), https://doi.org/10.1107/S0567739476001551.
[321] D.R. MacFarlane, M. Forsyth, E.I. Izgorodina, A.P. Abbott, G. Annat, K. Fraser, On the concept of ionicity in ionic liquids, Phys. Chem. Chem. Phys. 11 (2009) 4962–4967, https://doi.org/10.1039/b900201d.
[322] H. Tokuda, K. Hayamizu, K. Ishii, M.A.B.H. Susan, M. Watanabe, Physicochemical Properties and Structures of Room Temperature Ionic Liquids. 1. Variation of Anionic Species, J. Phys. Chem. B. 108 (2004) 15693–16600, https://doi.org/10.1021/jp047480r.
[323] D.J. Griffiths, Electric Fields in Matter, in: A. Reeves (Ed.), Introd. to Electrodyn, 3rd ed., Prentice-Hall Inc., Upper Saddle River, New Jersey, 1999, pp. 160–201.
[324] R.R. Reddy, Y. Nazeer Ahammed, P. Abdul Azeem, K. Rama Gopal, T.V.R.R. Rao, Electronic polarizability and optical basicity properties of oxide glasses through average electronegativity, J. Non. Cryst. Solids. 286 (2001) 169–180, https://doi.org/10.1016/S0022-3093(01)00481-1.
[325] J. Qi, D. Xue, H. Ratajczak, G. Ning, Electronic polarizability of the oxide ion and density of binary silicate, borate and phosphate oxide glasses, Phys. B Condens. Matter. 349 (2004) 265–269, https://doi.org/10.1016/j.physb.2004.03.307.
[326] V. Dimitrov, S. Sakka, Electronic oxide polarizability and optical basicity of simple oxides. I, J. Appl. Phys. 79 (1996) 1736–1740, https://doi.org/10.1063/1.360962.
[327] G.N. Lewis, Acids and Bases, J. Franklin Inst. 226 (1938) 293–313, https://doi.org/10.1016/B978-0-12-811816-0.00008-7.
[328] J.A. Duffy, M.D. Ingram, An interpretation of glass chemistry in terms of the optical basicity concept, J. Non. Cryst. Solids. 21 (1976) 373–410.
[329] J.A. Duffy, A common optical basicity scale for oxide and fluoride glasses, J. Non. Cryst. Solids. 109 (1989) 35–39, https://doi.org/10.1016/0022-3093(89)90438-9.
[330] J.A. Duffy, A review of optical basicity and its applications to oxidic systems, Geochim. Cosmochim. Acta. 51 (1993) 3961–3970, https://doi.org/10.1016/0016-7037(93)90346-X.
[331] J.A. Duffy, Optical basicity: A practical acid-base theory for oxides and oxyanions, J. Chem. Educ. 73 (1996) 1138, https://doi.org/10.1021/ed073p1138.
[332] E.I. Kamitsos, Y.D. Yiannopoulos, J.A. Duffy, Optical basicity and refractivity of germanate glasses, J. Phys. Chem. B. 106 (2002) 8988–8993, https://doi.org/10.1021/jp0207230.
[333] X. Zhao, X. Wang, H. Lin, Z. Wang, A new approach to estimate refractive index, electronic polarizability, and optical basicity of binary oxide glasses, Phys. B Condens. Matter. 403 (2008) 2450–2460, https://doi.org/10.1016/j.physb.2008.01.009.
[334] K.-H. Sun, Fundamental condition of glass formation, J. Am. Ceram. Soc. 30 (1947) 277–281, https://doi.org/10.1111/j.1151-2916.1947.tb19654.x.
[335] D. Möncke, S. Reibstein, D. Schumacher, L. Wondraczek, Irradiation-induced defects in ionic sulfophosphate glasses, J. Non. Cryst. Solids. 383 (2014) 33–37, https://doi.org/10.1016/j.jnoncrysol.2013.04.029.
[336] M.A. Hassan, Effect of halides addition on the ligand field of chromium in alkali borate glasses, J. Alloys Compd. 574 (2013) 391–397, https://doi.org/10.1016/j.jallcom.2013.05.177.
[337] G.D. Chryssikos, E.I. Kamitsos, A.P. Patsis, Effect of Li_2SO4 on the structure of Li_2O-B_2O_3 glasses, J. Non. Cryst. Solids. 202 (1996) 222–232.
[338] J. Swenson, L. Börjesson, Intermediate-range structure and conductivity of fast ion-conducting borate glasses, J. Non. Cryst. Solids. 232–234 (1998) 658–664, https://doi.org/10.1016/S0921-4526(96)01037-X.
[339] A. Hirata, L.J. Kang, T. Fujita, B. Klumov, K. Matsue, M. Kotani, A.R. Yavari, M.W. Chen, Geometric frustration of icosahedron in metallic glasses, Science (80-.) 341 (2013) 376–379, https://doi.org/10.1126/science.1232450.
[340] B.O. Mysen, P. Richet, Configurational Properties, Silic Glas. Melts Prop. Struct, Elsevier, Amsterdam, 2005, pp. 53–68 http://ebooks.cambridge.org/ref/id/CBO9781107415324A009.
[341] A.G. Kalampounias, N.K. Nasikas, G.N. Papatheodorou, Levitation / CO2 Laser Melting and Glass Formation in MgO Rich Mixtures with SiO2: Raman Spectra and Structure, ECS Trans. 16 (2009) 343–355, https://doi.org/10.1149/1.3159339.
[342] A.G. Kalampounias, N.K. Nasikas, G.N. Papatheodorou, Glass formation and structure in the $MgSiO_3$-Mg_2SiO_4 pseudobinary system : From degraded networks to ioniclike glasses, J. Chem. Phys. 131 (2009) 1–8, https://doi.org/10.1063/1.3225431.
[343] Y. Li, S.C. Ng, C.K. Ong, H.H. Hng, T.T. Gob, Glass forming ability of bulk glass forming alloys, 36 (1997), pp. 783–787.
[344] Z.P.P. Lu, Y. Li, S.C.C. Ng, Reduced glass transition temperature and glass forming ability of bulk glass forming alloys, J. Non. Cryst. Solids. 270 (2000) 103–114, https://doi.org/10.1016/S0022-3093(00)00064-8.
[345] G. Hägg, The Vitreous State, J. Chem. Phys. 3 (1935) 363–364, https://doi.org/10.1063/1.1749676.
[346] W.H. Zachariasen, The Vitreous State, J. Chem. Phys. 3 (1935) 162, https://doi.org/10.1063/1.1749676.
[347] B.E. Warren, J. Biscoe, Fourier analysis of X-ray patterns of soda-silica glass, J. Am. Ceram. Soc. 21 (1938) 259–265, https://doi.org/10.1111/j.1151-2916.1938.tb15774.x.
[348] G.N. Greaves, S.J. Gurman, C.R.A.A. Catlow, A.V. Chadwick, S. Houde-Walter, C.M.B.B. Henderson, B.R. Dobson, V.A. Chadwick, S. Houde-Walter, C.M.B.B. Henderson, B.R. Dobson, A structural basis for ionic diffusion in oxide glasses, Philos. Mag. A Phys. Condens. Matter, Struct. Defects Mech. Prop 64 (1991) 1059–1072, https://doi.org/10.1080/01418619108204878.
[349] A. Faivre, F. Despetis, F. Guillaume, P. Solignac, M. Ramonda, Role of mobile cations on microplasticity in alumino-phosphate glasses, J. Am. Ceram. Soc. 93 (2010) 2986–2989, https://doi.org/10.1111/j.1551-2916.2010.03950.x.
[350] G.N. Greaves, Structure and ionic transport in disordered silicates, Mineral. Mag. 64 (2000) 441–446, https://doi.org/10.1196/annals.1326.033.
[351] J.C. Phillips, Topology of covalent non-crystalline solids: Short-range order in chalcogenide alloys, J. Non. Cryst. Solids. 34 (1979) 153–181, https://doi.org/10.1016/0921-4526(95)00390-U.
[352] M.F. Thorpe, Continuous deformations in random networks, J. Non. Cryst. Solids. 57 (1983) 355–370, https://doi.org/10.1016/0022-3093(83)90424-6.
[353] J.C. Phillips, Structure and selectively enhanced Raman spectra of high-silica alkali-silicate glasses, Phys. Rev. B. 32 (1985) 5350–5355, https://doi.org/10.1140/epjc/s10052-012-2186-8.
[354] J.C. Mauro, Topological constraint theory of glass, Am. Ceram. Soc. Bull. 90 (2011) 31–37, https://doi.org/10.1039/c3ee40810h.
[355] J.C. Mauro, P.K. Gupta, R.J. Loucks, Composition dependence of glass transition temperature and fragility. II. A topological model of alkali borate liquids, J. Chem. Phys. 130 (2009), https://doi.org/10.1063/1.3152432.
[356] M.M. Smedskjaer, J.C. Mauro, S. Sen, Y. Yue, Quantitative design of glassy materials using temperature-dependent constraint theory, Chem. Mater. 22 (2010) 5358–5365, https://doi.org/10.1021/cm1016799.
[357] A.I. Fu, J.C. Mauro, Topology of alkali phosphate glass networks, J. Non. Cryst. Solids. 361 (2013) 57–62, https://doi.org/10.1016/j.jnoncrysol.2012.11.001.
[358] B.P. Rodrigues, J. Deubener, L. Wondraczek, Modifier Interaction and Mixed-Alkali Effect in Bond Constraint Theory Applied to Ternary Alkali Metaphosphate Glasses, Front. Mater. 3 (2016) 1700850, https://doi.org/10.3389/fmats.2016.00025.
[359] B.P. Rodrigues, L. Wondraczek, Floppy mode degeneracy and decoupling of

constraint predictions in super-cooled borate and silicate liquids, Front. Mater. 1 (2015) 1–7, https://doi.org/10.3389/fmats.2014.00032.
[360] G. Gao, L. Wondraczek, Heavily Eu3 + -doped boroaluminosilicate glasses for UV/blue-to-red photoconversion with high quantum yield, J. Mater. Chem. C. 2 (2014) 691–695, https://doi.org/10.1039/c3tc31660b.
[361] C.F. Dietrich, Uncertainty, Calibration and Probability: The Statistics of Scientific and Industrial Measurement, 2nd ed., Taylor & Francis, Boca Raton, FL, 1991, https://doi.org/10.1017/CBO9781107415324.004.
[362] M.S. Bødker, J.C. Mauro, S. Goyal, R.E. Youngman, M.M. Smedskjaer, Predicting Q-Speciation in binary phosphate glasses Using statistical mechanics, J. Phys. Chem. B. 122 (2018) 7609–7615, https://doi.org/10.1021/acs.jpcb.8b04604.
[363] J.F. Stebbins, J.V. Oglesby, S.K. Lee, Oxygen sites in silicate glasses: A new view from oxygen-17 NMR, Chem. Geol. 174 (2001) 63–75, https://doi.org/10.1016/S0009-2541(00)00307-7.
[364] A.C. Hannon, B. Vessal, J.M. Parker, The structure of alkali silicate glasses, J. Non. Cryst. Solids. 150 (1992) 97–102, https://doi.org/10.1016/0022-3093(92)90102-P.
[365] L. Pauling, The principles determining the structure of complex ionic crystals, J. Am. Chem. Soc. 51 (1929) 1010–1026, https://doi.org/10.1021/ja01379a006.
[366] H.W. Fairbairn, Packing in ionic minerals, Bull. Geol. Soc. Am. 54 (1943) 1305–1374.
[367] H. Maekawa, T. Maekawa, K. Kawamura, T. Yokokawa, The structural groups of alkali silicate glasses determined from 29Si MAS-NMR, J. Non. Cryst. Solids. 127 (1991) 53–64, https://doi.org/10.1016/0022-3093(91)90400-Z.
[368] P.J. Bray, S.A. Feller, G.E. Jellison, B^{10} NMR studies of the structure of borate glasses, J. Non. Cryst. Solids 38 and 39 (1980) 93–98.
[369] Y.D.D.D. Yiannopoulos, G.D.D. Chryssikos, E.I.I.I. Kamitsos, Structure and properties of alkaline earth borate glasses, Phys. Chem. Glas. 42 (2001) 164–172, https://doi.org/10.1098/rsif.2008.0348.
[370] R. Mercier, M. Tachez, J.P. Malugani, C. Rousselot, Microstructure of silver superionic glasses, Mater. Chem. Phys. 23 (1989) 13–27, https://doi.org/10.1016/0254-0584(89)90014-X.
[371] D.P. Button, R.P. Tandon, H.L. Tuller, D.R. Uhlmann, Fast Li^+ ion conductance in chloroborate glasses II- diborates and metaborates, Solid State Ionics. 5 (1981) 655–658, https://doi.org/10.1016/0167-2738(81)90186-7.
[372] M. Irion, M. Couzi, A. Levasseur, J.M. Reau, J.C. Brethous, An infrared and Raman study of new ionic-conductor lithium glasses, J. Solid State Chem. 31 (1980) 285–294.
[373] Z. Pan, D.O. Henderson, S.H. Morgan, A Raman investigation of lead haloborate glasses, J. Chem. Phys. 101 (1994) 1767–1774, https://doi.org/10.1063/1.467755.
[374] Y.S. Tver'yanovich, M. Vlček, A. Tverjanovich, Formation of complex structural units and structure of some chalco-halide glasses, J. Non. Cryst. Solids. 333 (2004) 85–89, https://doi.org/10.1016/j.jnoncrysol.2003.09.042.
[375] E. Thilo, G. Blumenthal, Zur chemie der kondensierten phosphate und arsenate über sulfatophosphate, Z. Anorg. Allg. Chem. 348 (1966) 77–88.
[376] H. Bradtmüller, L. Zhang, C.C. De Araujo, H. Eckert, D. Möncke, D. Ehrt, Structural Studies of NaPO3-AlF3Glasses by High-Resolution Double-Resonance Nuclear Magnetic Resonance Spectroscopy, J. Phys. Chem. C. 122 (2018) 21579–21588, https://doi.org/10.1021/acs.jpcc.8b06162.
[377] C. Zürn, A. Titze, G. Diezemann, R. Böhmer, ^{87}Rb NMR studies of molten and glassy $2Ca(NO_3)_2$–$RbNO_3$, J. Phys. Chem. A. 103 (1999) 4109–4112.
[378] R. Zallen, Random Close Packing, Phys. Amorph. Solids, WILEY-VCH Verlag GmbH & Co. KGaA, New York, 2004, pp. 49–59.
[379] R. Coupe, D. Louer, J. Lucas, A.J. Leonard, X-Ray Scattering Studies of Glasses in the System ZrF4-BaF2, J. Am. Ceram. Soc. 66 (1983) 523–529, https://doi.org/10.1111/j.1151-2916.1983.tb10595.x.
[380] Y. Kawamoto, PROGRESS IN STRUCTURAL STUDY OF ZrF_4-BASED GLASSES, Mater. Sci. Forum. 6 (1985) 417–426, https://doi.org/10.4028/www.scientific.net/MSF.5-6.417.
[381] Y. Kawamoto, T. Horisaka, Short-range structures of barium, lead, and strontium meta-fluorozirconate glasses, J. Non. Cryst. Solids. 56 (1983) 39–44, https://doi.org/10.1016/0022-3093(83)90443-X.
[382] L.T. Hamill, J.M. Parker, Molecular Dynamics Simulations of Fluoride Glass Melts, Mater. Sci. Forum. 5–6 (1985) 437–440. https://www.scientific.net/MSF.5-6.437 (accessed May 8, 2020).
[383] H. Inoue, H. Hasegawa, I.Y.-P. and chemistry of Glasses, U 1985, A study of the structure of glasses based on ZrF4, Phys. Chem. Glas. 26 (1985) 74. https://inis.iaea.org/search/search.aspx?orig_q=RN:16070740 (accessed May 8, 2020).
[384] J. Lucas, C.A. Angell, S. Tamaddon, Fluoride bridging modes in fluorozirconate glasses by x-ray and computer simulation studies, Mater. Res. Bull. 19 (1984) 945–951, https://doi.org/10.1016/0025-5408(84)90057-6.
[385] R.E. Youngman, S. Sen, A high-resolution 19F NMR spectroscopic study of barium fluorozirconate glasses and related crystals, Solid State Nucl. Magn. Reson. 27 (2005) 77–89, https://doi.org/10.1016/j.ssnmr.2004.06.008.
[386] R.M. Almeida, Short and intermediate range structures in fluoride glasses by vibrational spectroscopy, J. Non. Cryst. Solids. 140 (1992) 92–97, https://doi.org/10.1016/S0022-3093(05)80748-3.
[387] R.M. Almeida, Vibrational spectroscopy studies of halide glass structure, in: R.M. Almeida (Ed.), Halide Glas. Infrared Fiberoptics, Martinus Nijhoff Publishers, Dordrecht, The Netherlands, 1987, pp. 57–72.
[388] T. Yano, J. Mizuno, S. Shibata, M. Yamane, S. Inoue, Y. Onoda, NMR study on glass structure of chlorine-doped AlF3-based glasses with various glass-forming abilities, J. Non. Cryst. Solids. 213–214 (1997) 345–352, https://doi.org/10.1016/S0022-3093(97)00056-2.
[389] R.N. Sinclair, J.A.E. Desa, G. Etherington, V.P.A. Johnson, A.C. Wright, P.A.V. Johnson, A.C. Wright, Neutron diffraction studies of amorphous solids, J. Non. Cryst. Solids. 42 (1980) 107–116.
[390] F. Ma, M. Zhang, Viscosity and structure of molten halide glasses based on chlorides, bromides and iodides, J. Non. Cryst. Solids. 112 (1989) 106–110, https://doi.org/10.1016/0022-3093(89)90503-6.
[391] J.A. Erwin Desa, A.C. Wright, J. Wong, R.N. Sinclair, A neutron diffraction investigation of the structure of vitreous zinc chloride, J. Non. Cryst. Solids. 51 (1982) 57–86, https://doi.org/10.1016/0022-3093(82)90189-2.
[392] G.N. Papatheodorou, O.J. Kleppa, Enthalpies of mixing of charge unsymmetrical binary fused salt systems: ZnX_2-AX (A = Li, Cs, Ag; X = Cl, Br), Z. Anorg. Allg. Chem. 401 (1973) 132–144.
[393] L.E. Pedersen, Viscosity, structure and glass formation in the $AlCl_3$ -$ZnCl_2$ system, (2001).
[394] A. Winter, Glass Formation, J. Am. Ceram. Soc. 40 (1957) 54–58, https://doi.org/10.1111/j.1151-2916.1957.tb12574.x.
[395] J. Portier, Halogenide, chalcogenide and chalcohalogenide glasses: materials, models, applications, J. Non. Cryst. Solids. 112 (1989) 15–22, https://doi.org/10.1016/0022-3093(89)90489-4.
[396] M. Saad, M. Poulain, Glass Forming Ability Criterion, Mater. Sci. Forum. 19–20 (1987) 11–18, https://doi.org/10.4028/www.scientific.net/MSF.19-20.11.
[397] T. Iqbal, M.R. Shahriari, G.H.J. Sigel, A new glass forming ability criterion for multicomponent halide glasses, Mater. Sci. Forum. 67–68 (1991) 225–232, https://doi.org/10.4028/www.scientific.net/MSF.67-68.225.
[398] M.F. Ding, J. Lau, J.D. Mackenzie, Halide glasses based on chlorides, bromides and iodides, J. Non. Cryst. Solids. 80 (1986) 538–542, https://doi.org/10.1016/0022-3093(86)90443-6.
[399] T. Grande, S. Aasland, S. Julsrud, Some comments on the thermodynamic aspects of glass formation in fluoride systems, J. Non. Cryst. Solids. 184 (1995) 114–118, https://doi.org/10.1016/0022-3093(94)00600-8.
[400] J. Lucas, Halide glasses, J. Non. Cryst. Solids. 80 (1986) 83–91, https://doi.org/10.1016/0022-3093(86)90380-7.
[401] A.M. Efimov, The IR spectra of non-oxide glasses of various types: Crucial differences and their origin, J. Non. Cryst. Solids. 213–214 (1997) 205–214, https://doi.org/10.1016/S0022-3093(97)00010-0.
[402] P. Ghalsasi, P.S. Ghalsasi, Single crystal X-ray structure of BeF_2: α-quartz, Inorg. Chem. 50 (2011) 86–89, https://doi.org/10.1021/ic101248g.
[403] B. Winkler, H.; Brehler, Über das alpha- und beta-Zn Cl_2, Naturwissenschaften. 46 (1959) 553–554.
[404] P.A. Kokkoros, P.J. Rentzeperis, The crystal structure of the anhydrous sulphates of copper and zinc, Acta Crystallogr. 11 (1958) 361–364, https://doi.org/10.1107/S0365110X58000955.
[405] W.H. Zachariasen, The crystal structure of potassium metaborate, $K_3(B_3O_6)$, J. Chem. Phys. 5 (1937) 919–922, https://doi.org/10.1063/1.1749962.
[406] N. Elliot, A redetermination of carbon-oxygen distance in calcite and the nitrogen-oxygen distance in sodium nitrate, J. Am. Ceram. Soc. 59 (1937) 1380–1382.
[407] H.Y. Becht, R. Struikmans, A monoclinic high-temperature modification of potassium carbonate, Acta Crystallogr. Sect. B. B32 (1976) 3344–3346, https://doi.org/10.1107/s0567740876010303.
[408] A. Le Bail, T. Hansen, W.A. Crichton, Tetrapotassium pyrophosphates γ- and δ-$K_4P_2O_7$, Powder Diffr. 28 (2013) 2–12, https://doi.org/10.1017/S0885715612000954.
[409] K. Bernet, R. Hoppe, Zur kristallstruktur von $K_4[SiO_4]$, Z. Anorg. Allg. Chem. 589 (1990) 129–138.
[410] S. Sasaki, C.T. Prewitt, Y. Sato, E. Ito, Single-crystal X-ray study of γ-Mg_2SiO_4, J. Geophys. Res. 87 (1982) 7829–7832.
[411] R.W.G. Wyckoff, Crystal Structures, 2nd ed., Interscience Publishers Inc., New York, 1951.
[412] A.G. Nord, P. Kierkegaard, The crystal structure of $Mg_3(PO_4)_2$, Acta Chem. Scand. 22 (1968) 1466–1474.
[413] S. Block, G. Burley, A. Perloff, R.D. Mason Jr., R.D. Mason, R.D. Mason Jr., Refinement of the crystal structure of triclinic magnesium pyroborate, J. Res. Natl. Bur. Stand. 62 (1959) (1934) 95, https://doi.org/10.6028/jres.062.017.
[414] H. Li, K. Lu, Z. Wu, J. Dong, EXAFS studies of molten $ZnCl_2$, RbCl and Rb_2ZnCl_4, J. Phys. Condens. Matter. 6 (1994) 3629–3640, https://doi.org/10.1088/0953-8984/6/20/003.
[415] G.W. Brady, Structure of tellurium oxide glass, J. Chem. Phys. 27 (1957) 300–303, https://doi.org/10.1063/1.1743690.
[416] A. Witkowska, B. Sikora, K. Trzebiatowski, J. Rybicki, Germanate anomaly in heavy metal oxide glasses: an EXAFS analysis, J. Non. Cryst. Solids. 352 (2006) 4356–4361, https://doi.org/10.1016/j.jnoncrysol.2006.07.033.
[417] R. Wang, M.D. Merz, Non-crystallinity and polyamorphism in elemental solids, Nature. 260 (1976) 36–38, https://doi.org/10.1038/260036a0.
[418] B. Bridge, T.E. Bavins, D. Woods, T. Woolven, On the preparation and composition of the glass formation range of the TeO_2-$ZnCl_2$, J. Non. Cryst. Solids. 88 (1986) 262–270.
[419] L. Fortes, L.F. Santos, M. Clara Gonçalves, R.M. Almeida, The effects of $ZnCl_2$ and $ErCl_3$ on the vibrational spectra and structure of tellurite glasses, J. Non. Cryst. Solids. 352 (2006) 690–694, https://doi.org/10.1016/j.jnoncrysol.2005.11.052.
[420] H.C. Hsueh, R.K. Chen, H. Vass, S.J. Clark, G.J. Ackland, W.C.K. Poon, J. Crain, Compression mechanisms in quasimolecular XI3 (X = As, Sb, Bi) solids, Phys. Rev. B. 58 (1998) 14812–14822.
[421] M. Poulain, New trends in halide glass compositions, J. Non. Cryst. Solids. 184 (1995) 103–108, https://doi.org/10.1016/0022-3093(95)80027-1.
[422] J.A. Duffy, M.D. Ingram, Use of thallium(I), lead(II), and bismuth(III) as spectroscopic probes for ionic-covalent interaction in glasses, J. Chem. Phys. 52 (1970) 3752–3754, https://doi.org/10.1063/1.1673553.

[423] J. Wong, C.A. Angell, Spectroscopic detection of the glass transition: The n→π transition in nitrate melts, J. Non. Cryst. Solids. 7 (1972) 109–112.
[424] A.S. Rao, B. Sreedhar, J.L. Rao, S.V.J. Lakshman, Electron paramagnetic resonance and optical absorption spectra of Mn^{2+} ions in alkali zinc borosulphate glasses, J. Non. Cryst. Solids. 144 (1992) 169–174.
[425] S.V. Stefanovskii, A.I. Aleksandrov, Structural features of sulfate-vanadate-phosphate glasses, J. Appl. Spectrosc. 56 (1992) 43–46.
[426] A. Mohajerani, V. Martin, D. Boyd, J.W. Zwanziger, On the mechanical properties of lead borate glass, J. Non. Cryst. Solids. 381 (2013) 29–34, https://doi.org/10.1016/j.jnoncrysol.2013.09.015.
[427] S. Striepe, N. Da, J. Deubener, L. Wondraczek, Micromechanical properties of (Na,Zn)-sulfophosphate glasses, J. Non. Cryst. Solids. 358 (2012) 1032–1037, https://doi.org/10.1016/j.jnoncrysol.2012.01.045.
[428] M. Soltwisch, J. Sukmanowski, D. Quitmann, M. Kuenzler, Brillouin scattering on noncrystalline $ZnCl_2$, Mater. Sci. Forum. 6 (1985) 571–576, https://doi.org/10.4028/www.scientific.net/MSF.5-6.571.
[429] T. Rouxel, H. Ji, T. Hammouda, A. Moréac, Poisson's ratio and the densification of glass under high pressure, Phys. Rev. Lett. 100 (2008) 1–4, https://doi.org/10.1103/PhysRevLett.100.225501.
[430] J. Das, M.B. Tang, K.B. Kim, R. Theissmann, F. Baier, W.H. Wang, J. Eckert, "Work-hardenable" ductile bulk metallic glass, Phys. Rev. Lett. 94 (2005) 1–4, https://doi.org/10.1103/PhysRevLett.94.205501.
[431] O. Benzine, S. Bruns, Z. Pan, K. Durst, L. Wondraczek, Local deformation of glasses is mediated by rigidity fluctuation on nanometer scale, Adv. Sci. 5 (2018), https://doi.org/10.1002/advs.201800916.
[432] T. Ichitsubo, H. Kato, E. Matsubara, S. Biwa, S. Hosokawa, K. Matsuda, H. Uchiyama, A.Q.R.R. Baron, Static heterogeneity in metallic glasses and its correlation to physical properties, J. Non. Cryst. Solids. 357 (2011) 494–500, https://doi.org/10.1016/j.jnoncrysol.2010.06.056.
[433] M.D. Ingram, A. Mackenzie Margaret, W. Müller, M. Torge, A. Mackenzie, W. Müller, M. Torge, Cluster and Pathways: A new approach to ion migration in glass, Solid State Ionics 28–30 (1988) 677–680.
[434] O.L. Anderson, D.A. Stuart, Calculation of Activation Energy of Ionic Conductivity in Silica Glasses by Classical Methods, J. Am. Ceram. Soc. 37 (1954) 573–580, https://doi.org/10.1111/j.1151-2916.1954.tb13991.x.
[435] O.V. Mazurin, V.O. Mazurin, Glass relaxation, J. Non. Cryst. Solids. 87 (1986) 392–407, https://doi.org/10.1016/S0022-3093(86)80013-8.
[436] M. Tatsumisago, C.A. Angell, S.W. Martin, A new problem in the correlation of nuclear-spin relaxation and ionic conductivity in superionic glasses, J. Chem. Phys. 97 (1992) 6968–6974, https://doi.org/10.1063/1.463658.
[437] K.L. Ngai, Absence of difference between ^{11}B nuclear spin relaxation and Li ion conductivity relaxation in $(LiCl)_{0.6}$-$(Li_2O)_{0.7}$-$(B_2O_3)_{1.0}$ superionic glasses, J. Non. Cryst. Solids 162 (1993) 268–274, https://doi.org/10.1016/0022-3093(93)91246-Y.
[438] C.A. Angell, Formation of glasses from liquids and biopolymers, Science (80-.) 267 (1995) 1924–1935, https://doi.org/10.1126/science.267.5206.1924.
[439] K. Matusita, M. Koide, T. Komatsu, Viscous flow mechanism of fluoride glasses over a wide temperature range, J. Non. Cryst. Solids. 140 (1992) 141–145, https://doi.org/10.1016/S0022-3093(05)80757-4.
[440] T. Yano, S. Inoue, H. Kawazoe, M. Yamane, Vicosity of the melt of ZrF4-BaF2-LaF3-AlF3-NaF (OR LiF) systems, Mater. Sci. Forum. 32–33 (1988) 583–588, https://doi.org/10.4028/www.scientific.net/MSF.32-33.583.
[441] W.C. Hasz, S.N. Crichton, C.T. Moynihan, Viscosity temperature dependence of ZrF4-based melts, Mater. Sci. Forum. 32–33 (1988) 589–594, https://doi.org/10.4028/www.scientific.net/MSF.32-33.589.
[442] W.C. Hasz, J.H. Whang, C.T. Moynihan, Comparison of physical properties of ZrF4- and HfF4-based melts and glasses, J. Non. Cryst. Solids. 161 (1993) 127–132, https://doi.org/10.1016/0022-3093(93)90683-O.
[443] C.A. Angell, C.A. Scamehorn, D.J. List, J. Kieffer, Glassforming liquid oxides at the fragile limit of the viscosity-temperature relationship, Proc. XV Int. Congr. Glas. Leningr. (1989) 204–209.
[444] M. Matecki, F. Smektala, J. Even, M. Bertault, Y. Delugeard, J. Lucas, Heat capacity measurement of a cadmium fluorochloride glass, J. Non. Cryst. Solids. 213–214 (1997) 409–412, https://doi.org/10.1016/S0022-3093(97)00019-7.
[445] A.B. Seddon, M.A. Hemingway, Thermal characterisation of infrared-transmitting Ge–S–I glasses, J. Non. Cryst. Solids. 161 (1993) 323–326, https://doi.org/10.1016/0022-3093(93)90729-H.
[446] M. Naftaly, A. Jha, E.R. Taylor, K.C. Mills, Viscosity measurement in halide melts, J. Non. Cryst. Solids. 213–214 (1997) 106–112, https://doi.org/10.1016/S0022-3093(97)00097-5.
[447] J.R. Moyer, J.C. Evans, G.Y.-S. Lo, A raman spectroscopic study of the molten salt system $ZnCl_2$-KCl, J. Electrochem. Soc. 113 (1966) 158–161.
[448] C.T. Moynihan, S. Cantor, Viscosity and its temperature dependence in molten BeF_2, J. Chem. Phys. 48 (1968) 115–119, https://doi.org/10.1063/1.1664455.
[449] R. Böhmer, K.L. Ngai, C.A. Angell, D.J. Plazek, Nonexponential relaxations in strong and fragile glass formers, J. Chem. Phys. 99 (1993) 4201–4209, https://doi.org/10.1063/1.466117.
[450] T.A. Vilgis, Strong and fragile glasses: A powerful classification and its consequence, Phys. Rev. B. 47 (1993) 2882–2885.
[451] N.A. Mauro, M. Blodgett, M.L. Johnson, A.J. Vogt, K.F. Kelton, A structural signature of liquid fragility, Nat. Commun. 5 (2014) 1–7, https://doi.org/10.1038/ncomms5616.
[452] V.N. Novikov, A.P. Sokolov, Poisson's ratio and the fragility of glass-forming liquids of glass-forming liquids, Lett. to Nat. 431 (2004) 961–963, https://doi.org/10.1038/nature02947.
[453] Y. Vaills, Y. Luspin, G. Hauret, Annealing effects in SiO_2-Na_2O glasses investigated by Brillouin scattering, J. Non. Cryst. Solids. 286 (2001) 224–234, https://doi.org/10.1016/S0022-3093(01)00523-3.
[454] J. Schröder, R. Mohr, P.B. Macedo, C.J. Montrose, Rayleigh and Brillouin Scattering in K_2O–SiO_2 Glasses, J. Am. Ceram. Soc. 56 (1973) 510–514, https://doi.org/10.1111/j.1151-2916.1973.tb12399.x.
[455] K.J. Rao, M.H. Bhat, S. Kumar, Fragility thy name is glass, J. Indian Inst. Sci. 81 (2001) 3–13.
[456] K.L. Ngai, R.W. Rendell, H. Jain, Anomalous isotope-mass effect in lithium borate glasses: Comparison with a unified relaxation model, Phys. Rev. B. 30 (1984) 2133–2139, https://doi.org/10.1103/PhysRevB.30.2133.
[457] M. Lazzari, B. Scrosati, C.A. Vincent, Solid electrolytes with a glass-like structure, J. Am. Ceram. Soc. 61 (1978) 451–455, https://doi.org/10.1111/j.1151-2916.1978.tb09358.x.
[458] C. Mai, A. Asseiro, G.P. Johari, S. Etienne, K. Abbes, Dielectric and dynamic mechanical study of ionic motions in a ZrF_4-based fluorine glass, J. Non. Cryst. Solids. 93 (1987) 35–44.
[459] C. Liu, C.A. Angell, Mechanical vs electrical relaxation in Agl-based fast ion conducting glasses, J. Non. Cryst. Solids. 83 (1986) 162–184, https://doi.org/10.1016/0022-3093(86)90066-9.
[460] O.N. Antzukin, M.J. Duer, J.W. Emsley, I. Farnan, J. Klinowski, U. Scheler, J.F. Stebbins, Solid State NMR Spectroscopy Principles and Applications, Blackwell Science Ltd., Oxford, UK, 2002.
[461] S.W. Martin, Recent advances in the study of fast ionically conducting glasses using nuclear magnetic resonance techniques, Mater. Chem. Phys. 23 (1989) 225–265.
[462] K.L. Ngai, The universal low frequency responses of ionic conductors: A review of data and a unified model interpretation, Solid State Ionics. 5 (1981) 27–33.
[463] D.R. Macfarlane, M. Matecki, M. Poulain, Crystallization in fluoride glasses I. Devitrifcation on reheating, J. Non. Cryst. Solids 64 (1984) 351–362.
[464] C. Liu, H.G.K. Sundar, C.A. Angell, All-halide superionic glasses, Solid State Ionics 18 and 19 (1986) 442–448, https://doi.org/10.1016/0167-2738(86)90157-8.
[465] S.W. Mayer, T.H. Mills, R.C. Alden, B.B. Owens, Liquidus curves for molten alkali metaphosphate-sulfate systems, J. Phys. Chem. 65 (1961) 822–825.
[466] T. Komatsu, Design and control of crystallization in oxide glasses, J. Non. Cryst. Solids. 428 (2015) 156–175, https://doi.org/10.1016/j.jnoncrysol.2015.08.017.
[467] D. Dimas, I. Giannopoulou, D. Panias, Polymerization in sodium silicate solutions: A fundamental process in geopolymerization technology, J. Mater. Sci. 44 (2009) 3719–3730, https://doi.org/10.1007/s10853-009-3497-5.
[468] K.H. Nielsen, D.K. Orzol, S. Koynov, S. Carney, E. Hultstein, L. Wondraczek, Large area, low cost anti-reflective coating for solar glasses, Sol. Energy Mater. Sol. Cells. 128 (2014) 283–288, https://doi.org/10.1016/j.solmat.2014.05.034.
[469] D.A. McKeown, I.S. Muller, H. Gan, I.L. Pegg, C.A. Kendziora, Raman studies of sulfur in borosilicate waste glasses: Sulfate environments, J. Non. Cryst. Solids. 288 (2001) 191–199, https://doi.org/10.1016/S0022-3093(01)00624-X.
[470] M. Lenoir, A. Grandjean, S. Poissonnet, D.R. Neuville, Quantitation of sulfate solubility in borosilicate glasses using Raman spectroscopy, J. Non. Cryst. Solids. 355 (2009) 1468–1473, https://doi.org/10.1016/j.jnoncrysol.2009.05.015.
[471] D. Manara, A. Grandjean, O. Pinet, J.L. Dussossoy, D.R. Neuville, Sulfur behavior in silicate glasses and melts: Implications for sulfate incorporation in nuclear waste glasses as a function of alkali cation and V_2O_5 content, J. Non. Cryst. Solids. 353 (2007) 12–23, https://doi.org/10.1016/j.jnoncrysol.2006.09.041.
[472] W.L. Konijnendijk, J.H.J.M. Buster, Raman-scattering measurements of silicate glasses containing sulphate, J. Non. Cryst. Solids. 23 (1977) 401–418, https://doi.org/10.1360/zd-2013-43-6-1064.
[473] N. Yamada, T. Matsunaga, Structure of laser-crystallized $Ge_2Sb_2+xTe_5$ sputtered thin films for use in optical memory, J. Appl. Phys. 88 (2000) 7020–7028, https://doi.org/10.1063/1.1314323.
[474] M. Wuttig, N. Yamada, Phase change materials for rewriteable data storage, Nat. Mater. 6 (2007) 824–832.
[475] A. Pirovano, A.L. Lacaita, A. Benvenuti, F. Pellizzer, R. Bez, Electronic Switching in Phase-Change Memories, IEEE Trans. Electron Devices. 51 (2004) 452–459, https://doi.org/10.1109/TED.2003.823243.
[476] S.N. Baker, G.A. Baker, M.A. Kane, V.F. Bright, F.V. Bright, The cybotactic region surrounding fluorescent probes dissolved in 1-butyl-3-methylimidazolium hexafluorophosphate: Effects of temperature and added carbon dioxide, J. Phys. Chem. B. 105 (2001) 9663–9668, https://doi.org/10.1021/jp0103528.
[477] J. Zhang, W. Wu, T. Jiang, H. Gao, Z. Liu, J. He, B. Han, Conductivities and viscosities of the ionic liquid [bmim][PF_6] + water + ethanol and [bmim][PF_6] + water + acetone ternary mixtures, J. Chem. Eng. Data. 48 (2003) 1315–1317, https://doi.org/10.1021/je034078h.
[478] R.A. Mantz, P.C. Trulove, Viscosity and Density of Ionic Liquids, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., WILEY-VCH Verlags GmbH & Co, Weinheim, 2008, pp. 73–86, , https://doi.org/10.1055/s-2003-40869.
[479] K.R. Seddon, A. Stark, M.-J.J. Torres, Viscosity and density of 1-alkyl-3-methylimidazolium ionic liquids, ACS Symp. Ser. 819 (2002) 34–49, https://doi.org/10.1021/bk-2002-0819.ch004.
[480] T.G. Coker, B. Wunderlich, G.J. Janz, Melting mechanisms of ionic salts. Tetra-n-amyl ammonium thiocyanate, Trans. Faraday Soc 65 (1969) 3361–3368, https://doi.org/10.1039/TF9696503361.
[481] C. Hardacre, Order in the Liquid State and Structure, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., WILEY-VCH Verlags GmbH & Co, Weinheim, 2008, pp. 175–203.
[482] Y.S. Badyal, R.A. Howe, D.A. Allen, R.A. Howe, M.P. Tosi, The structure of liquid $AlCl_3$ and structural modification in $AlCl_3$-MCl (M = Li, Na) molten salt mixtures, J. Phys. Condens. Matter. 6 (1994) 10193–10220.

[483] P.A. Hunt, J. Maginn, R.M. Lynden-Bell Edward, M.G. Del Pópolo, Computational Modeling in Ionic Liquids, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., WILEY-VCH Verlags GmbH & Co, Weinheim, 2008, pp. 206–244.
[484] K.R. Seddon, A. Stark, M.J. Torres, Influence of chloride, water, and organic solvents on the physical properties of ionic liquids, Pure Appl. Chem, Walter de Gruyter GmbH, 2000, pp. 2275–2287, , https://doi.org/10.1351/pac200072122275.
[485] P. Bonhôte, A.-P. Dias, N. Papageorgiou, K. Kalyanasundaram, M. Grätzel, M. Armand, N. Papageorgiou, A.-P. Dias, K. Kalyanasundaram, M. Grätzel, Hydrophobic, highly conductive ambient-temperature molten salts, Inorg. Chem. 35 (1996) 1168–1178, https://doi.org/10.1021/ic951325x.
[486] J.A. Widegren, E.M. Saurer, K.N. Marsh, J.W. Magee, Electrolytic conductivity of four imidazolium-based room-temperature ionic liquids and the effect of a water impurity, J. Chem. Thermodyn. 37 (2005) 569–575, https://doi.org/10.1016/j.jct.2005.04.009.
[487] R.A. Mantz, Electrochemical Properties of Ionic Liquids, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., WILEY-VCH Verlags GmbH & Co, Weinheim, 2008, pp. 141–165.
[488] P. Walden, Über organische Lösungs-und Ionisierung Mittel. III. Teil: Innere Reibung und deren Zusammenhang mit dem Leitvermögen, Z. Phys. Chem. 55 (1906) 207–246.
[489] J.D. Holbrey, R.D. Rogers, Physicochemical Properties of Ionic Liquids: Melting Points and Phase Diagrams, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., WILEY-VCH Verlag GmbH & Co, Weinheim, 2008, p. 57.
[490] M. Maase, Industrial Applications of Ionic Liquids, in: P. Wasserscheid, T. Welton (Eds.), Ion. Liq. Synth. 2nd ed., WILEY-VCH Verlags GmbH & Co, Weinheim, 2008, pp. 663–686.
[491] H. Tao, T.D. Bennett, Y. Yue, Melt-Quenched Hybrid Glasses from Metal–Organic Frameworks, Adv. Mater. 29 (2017) 1–6, https://doi.org/10.1002/adma.201601705.
[492] T.D. Bennett, A.L. Goodwin, M.T. Dove, D.A. Keen, M.G. Tucker, E.R. Barney, A.K. Soper, E.G. Bithell, J.C. Tan, A.K. Cheetham, Structure and properties of an amorphous metal-organic framework, Phys. Rev. Lett. 104 (2010) 2–5, https://doi.org/10.1103/PhysRevLett.104.115503.
[493] T.D. Bennett, D.A. Keen, J.C. Tan, E.R. Barney, A.L. Goodwin, A.K. Cheetham, Thermal amorphization of zeolitic imidazolate frameworks, Angew. Chemie - Int. Ed. 50 (2011) 3067–3071, https://doi.org/10.1002/anie.201007303.
[494] T.D. Bennett, A.K. Cheetham, Amorphous Metal − Organic Frameworks, Acc. Chem. Res. 47 (2014) 1555–1562, https://doi.org/10.1021/ar5000314.
[495] M. Kim, J.F. Cahill, H. Fei, K.A. Prather, S.M. Cohen, Postsynthetic Ligand and Cation Exchange in Robust Metal-Organic Frameworks, J. Am. Chem. Soc. 134 (2012) 18082–18088, https://doi.org/10.1021/ja3079219.
[496] R. Gaillac, P. Pullumbi, K.A. Beyer, K.W. Chapman, D.A. Keen, T.D. Bennett, F.-X. Coudert, Liquid Metal-Organic Frameworks, Nat. Mater. 16 (2017) 1–7, https://doi.org/10.1038/nmat4998.
[497] C. Zhou, L. Longley, A. Krajnc, G.J. Smales, A. Qiao, I. Erucar, C.M. Doherty, A.W. Thornton, A.J. Hill, C.W. Ashling, O.T. Qazvini, S.J. Lee, P.A. Chater, N.J. Terrill, A.J. Smith, Y. Yue, D.A. Keen, S.G. Telfer, T.D. Bennett, G. Mali, D.A. Keen, S.G. Telfer, T.D. Bennett, Metal-organic framework glasses with permanent accessible porosity, Nat. Commun. 9 (2018) 1–9, https://doi.org/10.1038/s41467-018-07532-z.
[498] Z. Hu, L. Zhang, J. Jiang, Development of a force field for zeolitic imidazolate framework-8 with structural flexibility, J. Chem. Phys. 136 (2012) 244703, https://doi.org/10.1063/1.4729314.
[499] E.O.R. Beake, M.T. Dove, A.E. Phillips, D.A. Keen, M.G. Tucker, A.L. Goodwin, T.D. Bennett, A.K. Cheetham, Flexibility of zeolitic imidazolate framework structures studied by neutron total scattering and the reverse Monte Carlo method, J. Phys. Condens. Matter. 25 (2013) 395403–395412, https://doi.org/10.1088/0953-8984/25/39/395403.
[500] T.D. Bennett, A.H. Fuchs, A.K. Cheetham, F.-X. Coudert, Flexibility and disorder in metal–organic frameworks, Dalt. Trans. 45 (2016) 4058–4059, https://doi.org/10.1039/C6DT90026G.
[501] T.D. Bennett, J. Sotelo, J.-C.C. Tan, S.A. Moggach, Mechanical properties of zeolitic metal-organic frameworks: Mechanically flexible topologies and stabilization against structural collapse, CrystEngComm. 17 (2015) 286–289, https://doi.org/10.1039/C4CE02145B.
[502] H. Fei, J.F. Cahill, K.A. Prather, S.M. Cohen, Tandem postsynthetic metal ion and ligand exchange in zeolitic imidazolate frameworks, Inorg. Chem. 52 (2013) 4011–4016, https://doi.org/10.1021/ic400048g.
[503] C.S. Cundy, P.A. Cox, The hydrothermal synthesis of zeolites: Precursors, intermediates and reaction mechanism, Microporous Mesoporous Mater. 82 (2005) 1–78, https://doi.org/10.1016/j.micromeso.2005.02.016.
[504] R. Morris, Parnham, Ionothermal synthesis of zeolites, metal-organic frameworks, and inorganic-organic hybrids, Acc. Chem. Res. 40 (2007) 1005–1013, https://doi.org/10.1021/ar700025k.
[505] J.B. Murdoch, J.F. Stebbins, I.S.E. Carmichael, High-resolution 29Si NMR study of silicate and aluminosilicate glasses: the effect of network-modifying cations, Am. Mineral. 70 (1985) 332–343 https://doi.org/0003-004X/85/0304-0332.
[506] J. Haines, C. Levelut, A. Isambert, P. Hebert, S. Kohara, D.A. Keen, T. Hammouda, D. Andrault, Topologically Ordered Amorphous Silica Obtained from the Collapsed Siliceous Zeolite, Silicalite-1-F : A Step toward " Perfect " Glasses, J. Am. Chem. Soc. 131 (2009) 12333–12338, https://doi.org/10.1021/ja904054v.
[507] E.F. Baxter, T.D. Bennett, C. Mellot-Draznieks, C. Gervais, F. Blanc, A.K. Cheetham, Combined experimental and computational NMR study of crystalline and amorphous zeolitic imidazolate frameworks, Phys. Chem. Chem. Phys. 17 (2015) 25191–25196, https://doi.org/10.1039/c5cp02552d.
[508] P.F. Mcmillan, B.T. Poe, P. Gillet, B. Reynard, A study of SiO_2 glass and supercooled liquid to 1950 K via high-temperature Raman spectroscopy, Geochim. Cosmochim. Acta. 58 (1994) 3653–3664, https://doi.org/10.1016/0016-7037(94)90156-2.
[509] E. Moayedi, S. Sawamura, J. Hennig, E. Gnecco, L. Wondraczek, Relaxation of scratch-induced surface deformation in silicate glasses: Role of densification and shear flow in lateral indentation experiments, J. Non. Cryst. Solids. 500 (2018) 382–387, https://doi.org/10.1016/j.jnoncrysol.2018.08.032.
[510] V. Nozari, C. Calahoo, J.M. Tuffnell, P. Adelhelm, K. Wondraczek, S.E. Dutton, T. Bennett, L. Wondraczek, Sodium Ion Conductivity in Superionic IL-Impregnated Metal-Organic Frameworks: Enhancing Stability Through Structural Disorder, Sci. Rep. accepted (2020).
[511] T.D. Bennett, J.C. Tan, Y. Yue, E. Baxter, C. Ducati, N.J. Terrill, H.H.M.M. Yeung, Z. Zhou, W. Chen, S. Henke, A.K. Cheetham, G.N. Greaves, Hybrid glasses from strong and fragile metal-organic framework liquids, Nat. Commun. 6 (2015) 1–7, https://doi.org/10.1038/ncomms9079.
[512] Q. Qin, G.B. Mckenna, Correlation between dynamic fragility and glass transition temperature for different classes of glass forming liquids, J. Non. Cryst. Solids. 352 (2006) 2977–2985, https://doi.org/10.1016/j.jnoncrysol.2006.04.014.
[513] T. Yamanaka, T. Fukuda, T. Tsuchiya, Bonding character of SiO_2 stishovite under high pressures up to 30 Gpa, Phys. Chem. Miner. 29 (2002) 633–641, https://doi.org/10.1007/s00269-002-0257-3.
[514] E.C. Spencer, R.J. Angel, N.L. Ross, B.E. Hanson, J.A.K. Howard, Pressure-Induced Cooperative Bond Rearrangement in a Zinc Imidazolate Framework : A High-Pressure Single-Crystal X-Ray Diffraction Study, J. Am. Chem. Soc. 131 (2009) 4022–4026.